

Lecture 23

Electrostatics II

Introduction to Charges:

In the last lecture, we discussed the fundamental concept of charges. Charges are entities that exert forces on each other. These forces are governed by certain principles. One key fact is that charge is conserved, meaning the total amount of charge remains constant in any closed system.

Coulomb's Law:

Coulomb's Law describes the force between two-point charges. If there are two charges, q_1 and q_2 , separated by a distance r , the force F between them is given by:

$$F = k \frac{q_1 q_2}{r^2}$$

where k is Coulomb's constant. This force is a vector quantity, meaning it has both magnitude and direction. The direction of the force is along the line connecting the two charges. Importantly, if charge q_1 exerts a force on charge q_2 , charge q_2 exerts an equal and opposite force on charge q_1 .

Electric Fields:

Previously we have discussed the concept of electric fields. When a charge is present in an electric field, it experiences a force. The strength of this force depends on the strength of the field. If a charge Q is placed in an electric field E , the force F acting on the charge is:

$$F = Q E$$

Continuous Charge Distributions:

In previous studies, the focus has been on point charges, which are idealized models where the entire charge is assumed to be concentrated at a single point. However, in many real-world scenarios, charges are distributed over a larger region of space rather than being concentrated

at a single point. Understanding how to deal with these continuous charge distributions is crucial for accurately calculating the electric fields they produce.

Summing Up Contributions from Small Elements:

When dealing with continuous charge distributions, the electric field at any point in space is the result of contributions from all infinitesimal charge elements within the distribution. This is expressed as:

$$E = \Delta E_1 + \Delta E_2 + \Delta E_3 + \cdots + \Delta E_n$$

Here, ΔE_n represents the infinitesimal electric field contribution from the n-th small charge element. Summing these contributions gives the total electric field E .

Transition to Integration:

Since the charge distribution is continuous, the sum of the contributions from all infinitesimal charge elements is represented by an integral:

$$E = \sum \Delta E_n$$

This summation can be approximated by an integral for continuous distributions:

$$E = \int dE$$

Vector Components of the Electric Field:

The electric field $\vec{E}$ is a vector quantity, meaning it has both magnitude and direction. It can be decomposed into its Cartesian components along the x, y, and z axes:

$$\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$$

Where:

- E_x , E_y and E_z are the components of the electric field in the x, y, and z directions, respectively.

- $\hat{i}$, $\hat{j}$, and $\hat{k}$ are the unit vectors in the x, y, and z directions.

Calculating Each Component:

To find the total electric field, you need to integrate the contributions to each component over the entire charge distribution:

$$E_x = \int dE_x$$

$$E_y = \int dE_y$$

$$E_z = \int dE_z$$

These integrals take into account the continuous distribution of charge and calculate the contributions to the electric field in each direction.

Linear Charge Distribution:

Consider a line of charge where charges are continuously distributed along its length. To find the electric field at a point P due to this line of charge, we break the line into small segments and sum the contributions of each segment.

Suppose we have a line of charge with charge density λ , which is charge per unit length. If we take a small segment of length ds , the charge dq on this segment is:

$$dq = \lambda ds$$

By summing the contributions of all these small segments, we can determine the total electric field at point P .

Surface and Volume Charge Distributions:

In two dimensions, we consider surface charge distributions. Here, charges are spread over a surface, such as a sheet of paper. The charge density σ is charge per unit area. For a small area element dA , the charge dq is:

$$dq = \sigma dA$$

In three dimensions, we consider volume charge distributions. Here, charges are distributed throughout a volume. The charge density ρ is charge per unit volume. For a small volume element dV , the charge dq is:

$$dq = \rho dV$$

Calculating Electric Fields for Different Distributions:

To find the electric field due to a continuous charge distribution, we integrate over the entire distribution. For a line charge, the electric field E at a point can be calculated using:

$$E = \int \frac{k dq}{r^2} \hat{r}$$

where dq is the charge element, r is the distance from the charge element to the point, and $\hat{r}$ is the unit vector pointing from the charge element to the point.

Electric Field Due to a Ring of Charge:

Now, consider a ring of charge with radius R . We want to find the electric field along the axis of the ring. By symmetry, we only need to consider the z-component of the electric field. If we break the ring into small segments, each segment contributes a small electric field dE :

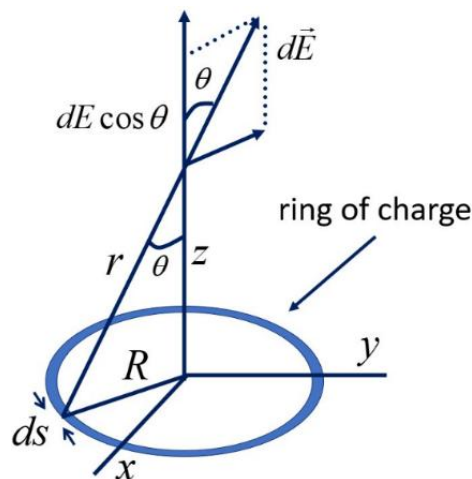


Fig 23. 1: The electric field of a charged ring points along the ring axis, since field components perpendicular to the axis cancel in pairs.

Starting with an infinitesimal electric field element dE :

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2} = \frac{\lambda ds}{4\pi\epsilon_0 (z^2 + R^2)}$$

This expression gives the infinitesimal electric field dE due to a small charge segment ds of the ring, where:

- λ is the linear charge density (charge per unit length).
- ds is the infinitesimal segment length of the ring.
- r is the distance from the charge segment to the point where the field is being calculated.
- ϵ_0 is the permittivity of free space.

The distance r is given by $r = \sqrt{z^2 + R^2}$, where z is the distance along the axis from the centre of the ring to the point of interest, and R is the radius of the ring.

Component of dE along the z-axis dE_z :

$$dE_z = dE \cos \theta$$

Here, $\cos \theta$ is the cosine of the angle between the electric field vector and the z-axis. This angle θ can be calculated using the geometry of the ring.

Expression for $\cos \theta$:

$$\cos \theta = \frac{z}{r} = \frac{z}{(z^2 + R^2)^{1/2}}$$

This step uses the definition of $\cos \theta$ in terms of the z-component and the hypotenuse r .

Substituting $\cos \theta$ into dE_z :

$$dE_z = \frac{z\lambda ds}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$

This equation represents the infinitesimal component of the electric field along the z-axis due to a small segment ds of the charged ring.

Integrating dE_z around the ring:

$$E_z = \frac{z\lambda}{4\pi\epsilon_0(z^2 + R^2)^{3/2}} \int ds$$

Since the ring is symmetrical, we integrate dE_z over the entire circumference of the ring, where $\int ds = 2\pi R$, the circumference of the ring.

Solving the integral:

$$E_z = \frac{z\lambda(2\pi R)}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}$$

Simplifying, we obtain:

$$E_z = \frac{qz}{4\pi\epsilon_0(z^2 + R^2)^{3/2}}$$

Here, $q = \lambda(2\pi R)$ is the total charge of the ring.

Electric field for $z \gg R$:

When the point of interest is far along the z-axis $z \gg R$, the expression simplifies. Under this approximation:

$$E_z \approx \frac{qz}{4\pi\epsilon_0 z^3} = \frac{q}{4\pi\epsilon_0 z^2}$$

This is the electric field along the axis of the ring when the observation point is far from the ring. It resembles the electric field of a point charge q at a distance z . The steps provide the derivation of the electric field along the axis of a charged ring, starting from the contribution of a small segment and integrating over the entire ring, leading to a simplified form for distances much larger than the ring radius.

Electric Field Due to a Disk of Charge:

Consider a uniformly charged disk with charge density σ . To find the electric field at a point along the axis of the disk, we divide the disk into concentric rings and integrate over these rings:

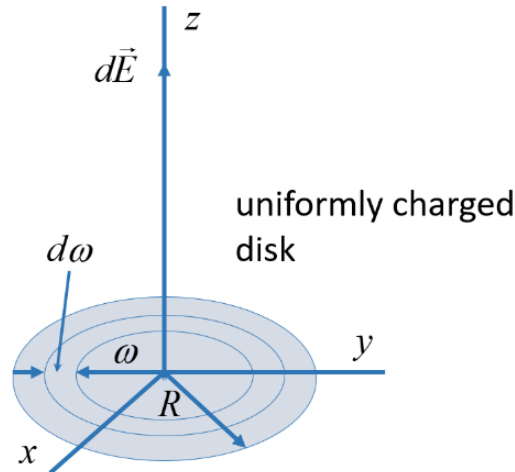


Fig 23. 2: A uniformly charged disk of radius R. The field on the z-axis due to an infinitesimally small ring element at a radius ω with thickness $d\omega$.

To find the electric field at a point along the axis of a uniformly charged disk, we divide the disk into concentric rings and integrate over these rings. Let's break down the given steps in detail:

Surface Charge Density and Elemental Charge:

- The disk has a uniform surface charge density σ , which is the charge per unit area.
- Consider an infinitesimally small ring element at a radius ω with thickness $d\omega$.
- The area of this ring element dA is given by the circumference of the ring ($2\pi\omega$) times the thickness ($d\omega$).

$$dA = 2\pi\omega d\omega$$

- The charge dq on this ring element is:

$$dq = \sigma dA = \sigma(2\pi\omega)d\omega$$

Electric Field Contribution by the Ring Element:

To find the electric field dE_z at a point along the axis (distance z from the disk centre) due to this ring element, we use Coulomb's law.

The vertical component dE_z of the electric field due to the ring element is:

$$dE_z = \frac{z\sigma 2\pi\omega d\omega}{4\pi\epsilon_0 (z^2 + \omega^2)^{3/2}}$$

Simplifying the expression:

$$dE_z = \frac{\sigma z}{4\epsilon_0} (z^2 + \omega^2)^{-3/2} (2\omega) d\omega$$

Integrating Over All Rings:

To find the total electric field E_z along the axis, integrate dE_z from $\omega=0$ to $\omega=R$ (the radius of the disk):

$$E_z = \int dE_z = \frac{\sigma z}{4\epsilon_0} \int_0^R (z^2 + \omega^2)^{-3/2} (2\omega) d\omega$$

Solving the Integral:

Evaluate the integral:

$$\int_0^R \frac{2\omega d\omega}{(z^2 + \omega^2)^{3/2}}$$

Using a suitable substitution ($\omega = z \tan \theta$), this integral evaluates to:

$$\int_0^R \frac{2\omega d\omega}{(z^2 + \omega^2)^{3/2}} = \frac{2}{z^2} \left[\sqrt{z^2 + \omega^2} \right]_0^R = \frac{2}{z^2} (\sqrt{z^2 + R^2} - z)$$

Final Electric Field Expression:

Plugging the evaluated integral back:

$$E_z = \frac{\sigma z}{4\epsilon_0} \cdot \frac{2}{z^2} \left(\sqrt{z^2 + R^2} - z \right) = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

Special Cases:**1. When $z \ll R$:**

If $z \ll R$, the distance z is much smaller than the radius R , implying the point is very close to the disk:

$$E_z = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$$

$$E_z = \frac{\sigma}{2\epsilon_0}$$

This is the same as the electric field near an infinite sheet of charge.

2. When $z \gg R$:

If $z \gg R$, the distance z is much larger than the radius R , implying the point is far from the disk:

$$E_z = \frac{\sigma}{4\epsilon_0} \frac{R^2}{z^2}$$

Since $q = \sigma\pi R^2$ (total charge on the disk):

$$E_z = \frac{q}{4\pi\epsilon_0 z^2}$$

This is the same as the electric field due to a point charge q at a distance z . This process involves breaking the disk into infinitesimal rings, calculating the electric field contribution from each ring, and integrating these contributions to find the total electric field along the axis of the disk. The results match known cases for very close and very far distances, providing confidence in the calculations.

Electric Field Due to infinite Line of Charge:

To calculate the electric field due to an infinite line of charge, consider a line of charge distributed along the z -axis. We want to find the electric field at a point P on the y -axis. Here's a step-by-step explanation of the process:

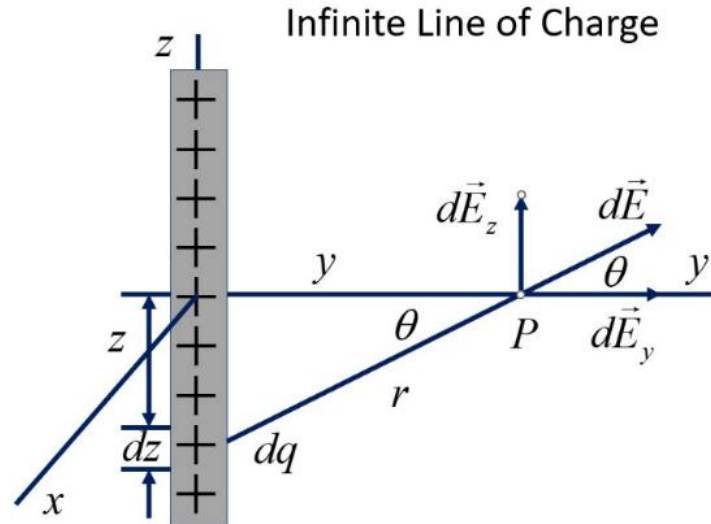


Fig 23. 3: The field of a charged line is the vector sum of the fields from all the individual charge elements dq along the line.

Linear Charge Density:

Let the linear charge density be λ , which is the charge per unit length.

Electric Field Contribution by an Elemental Segment:

Consider a small segment of the line of charge dz at a distance z from the origin. The charge dq on this segment is:

$$dq = \lambda dz$$

The distance r from this segment to the point on the y -axis at distance y is given by:

$$r = \sqrt{y^2 + z^2}$$

The electric field dE due to this charge segment at the point is:

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dz}{y^2 + z^2}$$

Components of the Electric Field:

The electric field dE can be decomposed into two components: dE_y along the y-axis and dE_z along the z-axis. Due to symmetry, the z-components from opposite segments cancel out, leaving only the y-components. The y-component dE_y is:

$$dE_y = dE \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{\lambda dz}{y^2 + z^2} \frac{y}{\sqrt{y^2 + z^2}} = \frac{1}{4\pi\epsilon_0} \frac{\lambda y dz}{(y^2 + z^2)^{3/2}}$$

Integration Over the Entire Line:

To find the total electric field E_y , integrate dE_y from $z = -\infty$ to $z = \infty$:

$$E_y = \int_{z=-\infty}^{z=\infty} dE_y = \frac{\lambda y}{4\pi\epsilon_0} \int_{z=-\infty}^{z=\infty} \frac{dz}{(y^2 + z^2)^{3/2}}$$

Simplifying the Integral:

Use the substitution $z = y \tan \theta$, hence $dz = y \sec^2 \theta d\theta$:

$$E_y = \frac{\lambda y}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \frac{y \sec^2 \theta d\theta}{y^3 \sec^3 \theta} = \frac{\lambda y}{4\pi\epsilon_0 y^2} \int_{-\pi/2}^{\pi/2} \cos \theta d\theta$$

Simplifying further:

$$E_y = \frac{\lambda}{4\pi\epsilon_0 y} [\sin \theta]_{-\pi/2}^{\pi/2} = \frac{\lambda}{4\pi\epsilon_0 y} (1 - (-1)) = \frac{\lambda}{2\pi\epsilon_0 y}$$

Final Result and Symmetry:

The electric field due to an infinite line of charge at a point on the y-axis is given by:

$$E = \frac{\lambda}{2\pi\epsilon_0 y}$$

Given the cylindrical symmetry of the problem, this result can be generalized to any point at a perpendicular distance r from the line of charge:

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

By breaking the infinite line of charge into infinitesimal segments, calculating their contributions to the electric field, and integrating over the entire length of the line, we find that the electric field at a perpendicular distance r from an infinite line of charge is inversely proportional to the distance r and depends on the linear charge density λ .

Corona Discharge:

Corona discharge is an electrical discharge brought about by the ionization of a fluid such as air surrounding a conductor that is electrically charged. It occurs when the electric field around the conductor becomes sufficiently strong to ionize the surrounding air or other gaseous medium, but not strong enough to cause a complete electrical breakdown or spark.

Conditions for Corona Discharge:

- **High Voltage:** The conductor must be at a high voltage relative to its surroundings.
- **Sharp Points or Edges:** Corona discharge is more likely to occur at sharp points or edges on the conductor because the electric field is concentrated in these areas.
- **Gas Type and Pressure:** The surrounding gas and its pressure influence the onset of corona discharge. Air is the most common medium, but other gases like SF_6 (sulphur hexafluoride) are used in high-voltage equipment to suppress corona.

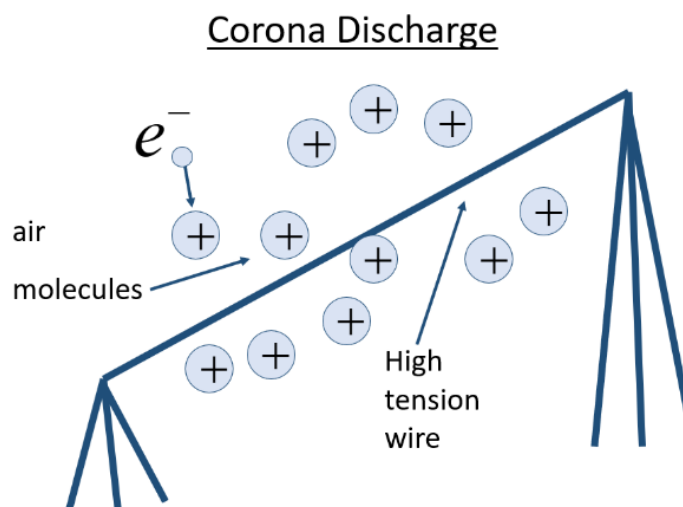


Fig 23. 4: The hissing noise at high voltage transmission lines is due to corona discharge. Corona discharge occurs due to high voltage which causes surrounding medium (air) to ionized.

Electric Flux:

Electric flux is a measure of the number of electric field lines passing through a surface. If we have an electric field E passing through a surface area A at an angle θ , the flux Φ_E is given by:

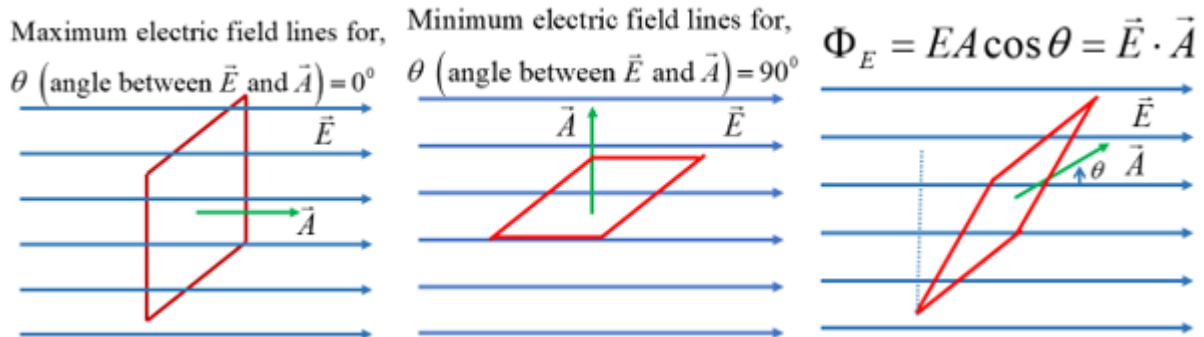


Fig 23. 5: Field lines for a uniform electric field through an area A for different angles to the field. Because the number of lines that go through the shaded area A is the same as the number that go through A , we conclude that the flux through A is equal to the flux through A and is given by $\Phi_E = EA \cos \theta$.

$$\Phi_E = EA \cos \theta$$

For a curved surface, we sum the flux through small area elements dA :

$$\Phi_E = \int E \cdot dA$$

$$\Phi_E = \sum \vec{E}_i \cdot \Delta \vec{A}_i$$

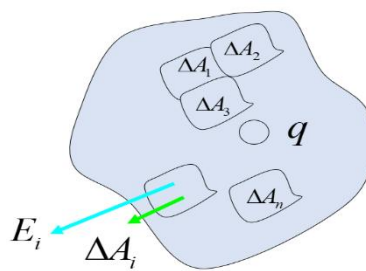


Fig 23. 6: Flux for a curved surface will be the sum of the flux through small area elements.

Gauss's Law:

Gauss's Law is a fundamental principle in electromagnetism that relates the electric flux through a closed surface to the charge enclosed by that surface. It is one of Maxwell's four equations, which form the foundation of classical electromagnetism.

Gauss's Law states that the net electric flux through any closed surface is equal to the charge enclosed divided by the permittivity of the medium. Mathematically, it is expressed as:

$$\Phi_E = \int E \cdot dA = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

where:

- $\int E \cdot dA$ is the surface integral of the electric field E over a closed surface S .
- dA is a vector representing an infinitesimal area on the surface S , pointing outward.
- Q_{enclosed} is the total electric charge enclosed within the surface S .
- ϵ_0 is the permittivity of free space (vacuum permittivity).

Explanation:

Gauss's Law essentially states that the total electric flux through a closed surface is proportional to the total charge enclosed within that surface. The electric flux is a measure of the number of electric field lines passing through a given area. If more charge is enclosed, more field lines emanate from or terminate at the charges, resulting in greater flux.

1. Electric Field of a Point Charge:

Consider a point charge q located at the origin. The electric field $\vec{E}$ at a distance r from the charge is given by Coulomb's Law:

$$E = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

where $\hat{r}$ is the unit vector pointing radially outward from the charge.

2. Flux Through a Spherical Surface:

Imagine a spherical surface of radius r centered around the charge. The surface area of the sphere is $4\pi r^2$. The electric field is radially symmetric and has the same magnitude at all points on the surface.

The flux Φ_E through the spherical surface is:

$$\Phi_E = \int E \cdot dA$$

Since E and dA are parallel and $|E|$ is constant over the sphere, this simplifies to:

$$\Phi_E = E \int d\vec{A} = E \times 4\pi r^2$$

Substituting $E = \frac{q}{4\pi\epsilon_0 r^2}$:

$$\Phi_E = \left(\frac{q}{4\pi\epsilon_0 r^2} \right) \times 4\pi r^2 = \frac{q}{\epsilon_0}$$

3. Generalization:

For a general distribution of charges, consider a closed surface S enclosing a total charge Q_{enclosed} . By the superposition principle, the total flux through S is the sum of the fluxes due to each individual charge. Hence:

$$\Phi_E = \int E \cdot dA = \frac{Q_{\text{enclosed}}}{\epsilon_0}$$

This concludes the derivation, showing that the electric flux through any closed surface is proportional to the enclosed charge, as stated by Gauss's Law.

Applications:

1. **Symmetry:** Gauss's Law is particularly powerful for calculating electric fields of symmetric charge distributions (e.g., spherical, cylindrical, or planar symmetry).
2. **Conductors:** Helps determine the distribution of charge on conductors and the resulting electric field.
3. **Capacitors:** Used to analyse the electric field and potential in capacitors.
4. **Electrostatics:** Fundamental in solving electrostatic problems and understanding the behaviour of electric fields in various configurations.

Electric Field Due to infinite Parallel Plates:

application of Gauss's Law in finding the electric field between two infinite parallel plates with uniform surface charge densities:

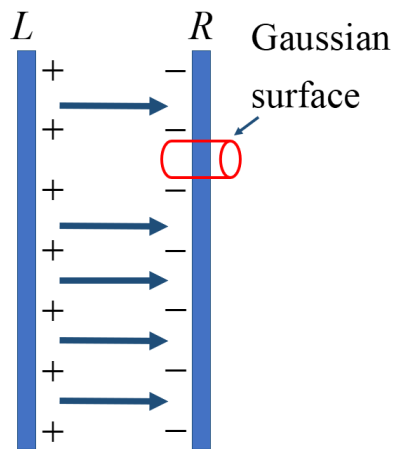


Fig 23. 7: Electric field $\vec{E}$ between two infinite parallel plates with uniform surface charge densities. A cylindrical Gaussian surface surrounds a parallel plate.

1. Electric Flux Definition:

$$\Phi = \int \vec{E} \cdot d\vec{A}$$

This integral defines the electric flux Φ through a closed surface, where $\vec{E}$ is the electric field and $d\vec{A}$ is a differential area vector on the surface.

2. Decomposing the Flux:

$$\Phi = \int_{\text{outer cap}} \vec{E} \cdot d\vec{A} + \int_{\text{inner cap}} \vec{E} \cdot d\vec{A} + \int_{\text{side walls}} \vec{E} \cdot d\vec{A}$$

This equation breaks down the total electric flux through the Gaussian surface into three parts:

- The flux through the outer cap.
- The flux through the inner cap.
- The flux through the side walls.

3. Evaluating Each Component:

$$\Phi = 0 + EA + 0$$

For the chosen Gaussian surface (a cylindrical "surface" that straddles one of the plates):

- The flux through the outer cap is zero because the electric field outside the plates cancels out.
- The flux through the inner cap is EA because the electric field E is perpendicular to the surface and uniform over the area A of the cap.
- The flux through the side walls is zero because the electric field is parallel to these surfaces and doesn't contribute to the flux.

4. Relating Flux to Charge:

$$\epsilon_0 \Phi = q$$

According to Gauss's Law, the electric flux Φ through a closed surface is equal to the enclosed charge q divided by the permittivity of free space ϵ_0 .

5. Enclosed Charge in Terms of Surface Charge Density:

$$\epsilon_0 EA = \sigma A$$

The charge q enclosed by the Gaussian surface is the surface charge density σ times the area A of the inner cap.

6. Solving for the Electric Field:

$$E = \frac{\sigma}{\epsilon_0}$$

Simplifying this equation gives the electric field E between the plates.

Electric Field Due to a long, straight wire:

Consider a straight wire with a constant charge distributed along its length with charge density λ , we use a cylindrical Gaussian surface with radius r and length L .

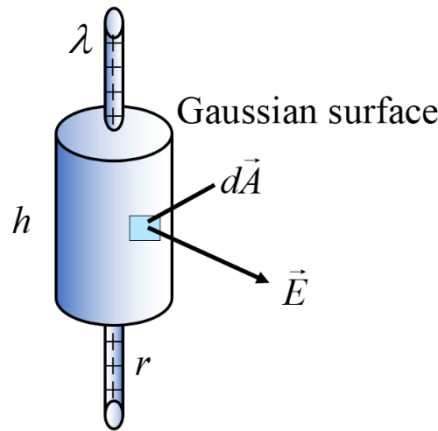


Fig 23. 8: A cylindrical Gaussian surface surrounds a line charge.

1. **Gaussian Surface:** We create an imaginary cylindrical Gaussian surface around the wire to simplify our calculations. The surface is closed and we can choose its shape to our advantage.
2. **Calculating the Electric Flux:**
 - The flux through the cylindrical surface is the product of the electric field and the surface area of the cylinder.

1. **Gauss's Law:**

$$\epsilon_0 \int \vec{E} \cdot d\vec{A} = q$$

This is the integral form of Gauss's Law, where ϵ_0 is the permittivity of free space, $\vec{E}$ is the electric field, $d\vec{A}$ is a differential area element, and q is the charge enclosed by the Gaussian surface.

2. **Choosing the Gaussian Surface:** For a long, straight wire with a uniform linear charge density λ (charge per unit length), we choose a cylindrical Gaussian surface with:
 - Radius r
 - Height h This surface is coaxial with the wire.
3. **Calculating the Electric Flux:** The flux through the Gaussian surface is:

$$\epsilon_0 \int \vec{E} \cdot d\vec{A}$$

Since the electric field $\vec{E}$ is radial and perpendicular to the surface of the cylinder, and is uniform at a distance r from the wire, this simplifies to:

$$\epsilon_0 E (2\pi r h)$$

Here, $2\pi r$ is the circumference of the cylinder's base, and h is the height of the cylinder.

4. **Enclosed Charge:** The charge enclosed by the Gaussian surface is:

$$q = \lambda h$$

Here, λ is the linear charge density, and h is the height of the cylinder.

5. **Applying Gauss's Law:** According to Gauss's Law:

$$\epsilon_0 E (2\pi r h) = \lambda h$$

Simplifying this equation to solve for $\vec{E}$:

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

For a point charge q at the centre of a sphere of radius r , the electric field is radial and has the same magnitude at every point on the surface. The flux through the sphere is:

$$\oint \vec{E} \cdot d\vec{A} = E \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$$

Solving for E , we get:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

This result is consistent with Coulomb's Law.

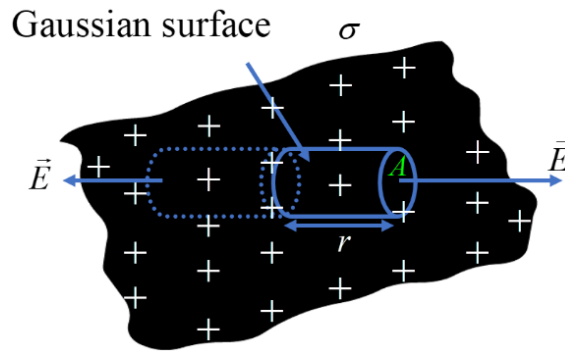
Example.3: Electric Field Due to Infinite Sheet of Charge:

Fig 23. 9: A cylindrical Gaussian surface penetrating an infinite sheet of charge. (b) A cross section of the same Gaussian cylinder. The flux through each end of the Gaussian surface is EA . There is no flux through the cylindrical surface.

We use Gauss's Law to determine the electric field near an infinite plane of charge.

- Gauss's Law:**

$$\epsilon_0 \int \vec{E} \cdot d\vec{A} = q$$

This is the integral form of Gauss's Law, where ϵ_0 is the permittivity of free space, $\vec{E}$ is the electric field, $d\vec{A}$ is a differential area element on the Gaussian surface, and q is the charge enclosed by the surface.

- Choosing the Gaussian Surface:** For an infinite plane of charge with a surface charge σ (charge per unit area), we choose a Gaussian surface in the shape of a rectangular box (a "surface") that extends symmetrically on both sides of the plane.
- Calculating the Electric Flux:** The total electric flux through the Gaussian surface is the sum of the flux through the two caps (top and bottom) of the surface.

$$\epsilon_0 (EA + EA)$$

Since the electric field $\vec{E}$ is perpendicular to the plane and uniform over each cap with area $\vec{A}$, the flux through each cap is EA . There are two such caps, so the total flux is $EA + EA$.

- Enclosed Charge:** The charge enclosed by the Gaussian surface is:

$$q = \sigma A$$

Here, σ is the surface charge density, and $\vec{A}$ is the area of the plane enclosed by the Gaussian surface.

5. **Applying Gauss's Law:** According to Gauss's Law:

$$\epsilon_0 (EA + EA) = \sigma A$$

Simplifying this equation to solve for $\vec{E}$:

$$2\epsilon_0 EA = \sigma A$$

$$E = \frac{\sigma}{2\epsilon_0}$$

This derivation shows how Gauss's Law can be used to find the electric field near an infinite sheet of charge. The result is that the electric field is uniform and directed perpendicular to the

plane, with a magnitude given by $\frac{\sigma}{2\epsilon_0}$.

Electric Field Due to a Spherical Shell:

We consider a spherical shell with charge distributed only on its surface and calculate the electric field both inside and outside the shell using Gauss's Law.

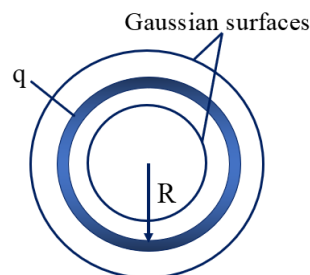


Fig 23. 10: A spherical Gaussian surface surrounds a spherical shell of radius R and charge q .

1. Inside the Shell:

- **Gauss's Law:** Gauss's Law states that the total electric flux through a closed surface is proportional to the charge enclosed by that surface.

$$\int \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

- **Applying Gauss's Law Inside the Shell:**

- For a point inside the spherical shell (at radius r where $r < R$, with R being the radius of the shell), consider a Gaussian surface that is a sphere of radius r .
- Since the charge is distributed only on the surface, the enclosed charge q_{enc} within this Gaussian surface is zero.

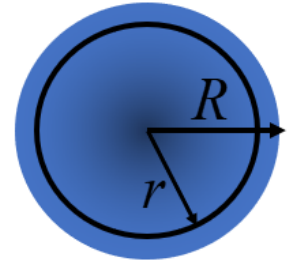


Fig 23. 11: A spherical Gaussian surface that is a radius r inside a spherical shell of radius R and charge q .

$$q_{\text{enc}} = 0$$

- Therefore, the electric flux is zero:

$$\int \vec{E} \cdot d\vec{A} = E \cdot 4\pi r^2 = 0$$

- This implies that the electric field inside the shell is zero:

$$E = 0 \quad (\text{spherical shell, } r < R)$$

2. Outside the Shell

- **Applying Gauss's Law Outside the Shell:**

- For a point outside the spherical shell (at radius r where $r > R$), consider a Gaussian surface that is a sphere of radius r .

- The enclosed charge q_{enc} is the total charge q on the spherical shell.

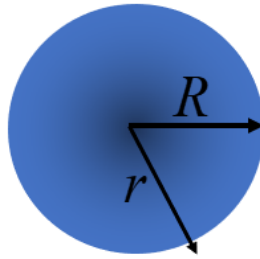


Fig 23. 12: A spherical Gaussian surface that is a radius r surrounds a spherical shell of radius R and charge q .

$$q_{\text{enc}} = q$$

- According to Gauss's Law, the electric flux through this surface is:

$$\int \vec{E} \cdot d\vec{A} = 4\pi r^2 E = \frac{q}{\epsilon_0}$$

- Solving for the electric field E :

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

- Thus, the electric field outside the shell behaves as if all the charge were concentrated at the centre of the shell:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad (\text{spherical shell, } r > R)$$

This result shows the electric field due to a spherical shell of charge follows the same pattern as the electric field due to a point charge for points outside the shell, and is zero inside the shell.

Conductors and Electric Fields:

In conductors, the electric field inside is zero because the charges redistribute to cancel any internal fields. When a charge is placed inside a conducting cavity, the charges on the conductor rearrange to neutralize the effect, resulting in no electric field within the cavity.

1. Conducting Sphere:

- If a charge is placed inside a hollow conducting sphere, the charges on the inner surface of the conductor will rearrange to cancel the field within the cavity.

2. Conductor Surface:

- If a charge is placed on a conductor, it will move to the surface to minimize the repulsive forces between like charges. The electric field just outside the surface of a conductor is perpendicular to the surface and proportional to the surface charge density.

Discontinuity of the Electric Field at a Charged Surface:

When there is a surface charge density σ on a boundary, the electric field experiences a discontinuity as you move across the surface. Gauss's Law can be used to quantify this discontinuity.

Applying Gauss's Law to a Surface Charge:

Consider a small Gaussian surface that straddles the surface with area A . The surface has one face just above the surface and one face just below the surface. The electric field on each side of the surface is represented as E_{n1} (above the surface) and E_{n2} (below the surface).

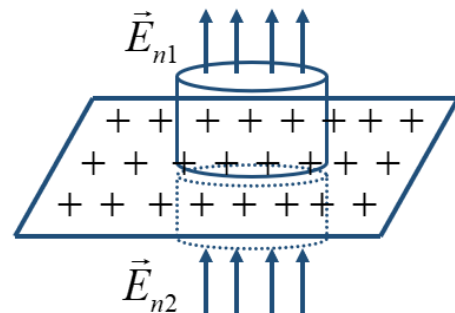


Fig 23. 13: An infinite sheet of charge, with a Gaussian surface straddling the sheet.

Gauss's Law states:

$$\int \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

For the surface, the only contribution to the flux is through the flat faces of the surface (since the sides are perpendicular to the electric field and contribute nothing to the flux). The charge enclosed q_{enc} is the surface charge density σ times the area A .

$$q_{\text{enc}} = \sigma A$$

The electric flux through the surface is the sum of the flux through the top and bottom faces:

$$\int \vec{E} \cdot d\vec{A} = E_{n2}A - E_{n1}A$$

By Gauss's Law:

$$E_{n2}A - E_{n1}A = \frac{\sigma A}{\epsilon_0}$$

Dividing by the area A :

$$E_{n2} - E_{n1} = \frac{\sigma}{\epsilon_0}$$

This equation tells us that the difference between the electric field just outside (E_{n1}) and just inside (E_{n2}) a charged surface is directly proportional to the surface charge density σ , with the proportionality constant being $\frac{1}{\epsilon_0}$.

Potential Energy and Electric Fields:

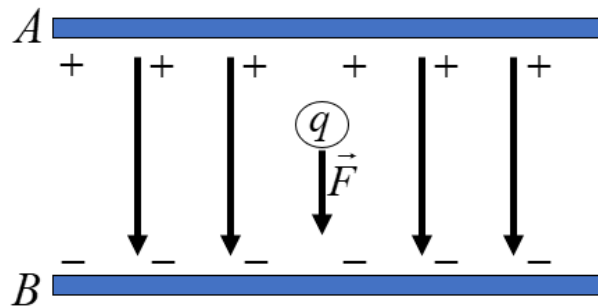


Fig 23. 14: Electric field (E) between two infinite parallel plates with uniform surface charge densities ($+\sigma$ and $-\sigma$). A charge q moves from the positively charged plate to the negatively charged plate, experiencing a force F due to the electric field.

Electric potential energy is the energy stored in an electric field due to the positions of charges. Moving a charge against an electric field requires work, which increases the potential energy of the system. This concept is analogous to gravitational potential energy.

1. Relationship Between Potential Energy and Work:

The change in electric potential energy (ΔU) between two points A and B is related to the work done (W_{ab}) to move a charge from point A to point B:

$$U_B - U_A = W_{ab}$$

$$\Delta U = -W_{ab}$$

2. Work Done by Electrostatic Force:

The work done by an electrostatic force $\vec{F}$ when moving a charge from point A to point B is given by:

$$U_b - U_a = -\int_a^b \vec{F} \cdot d\vec{s}$$

3. Electrostatic Force and Potential Energy:

The electrostatic force is conservative and can be represented by a potential energy function. Comparing with gravitational force:

$$F = G \frac{m_1 m_2}{r^2} \quad \text{and} \quad F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

Where G is the gravitational constant and ϵ_0 is the permittivity of free space.

4. Electric Field and Force:

The relationship between the electric field $\vec{E}$ and the force $\vec{F}$ on a charge q is:

$$\vec{E} = \frac{\vec{F}}{q}$$

For a gravitational field $\vec{g}$ and a mass m :

$$\vec{g} = \frac{\vec{F}}{m}$$

5. Calculation of Potential Energy:

Consider a charge q_2 in the electric field created by another charge q_1 . The work done to move q_2 from distance r_a to r_b is:

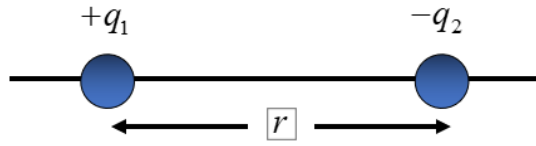


Fig 23. 15: Electric field E experiencing by charge q_2 due to charge q_1 at a distance r .

$$U_b - U_a = -q_2 \int_{r_a}^{r_b} E_x dr$$

For a point charge, the electric field E due to q_1 at a distance r is:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r^2}$$

Thus,

$$U_b - U_a = -\frac{1}{4\pi\epsilon_0} q_1 q_2 \int_{r_a}^{r_b} \frac{dr}{r^2}$$

$$U_b - U_a = \frac{1}{4\pi\epsilon_0} q_1 q_2 \left(\frac{1}{r_b} - \frac{1}{r_a} \right)$$

6. Reference Point and Potential Energy:

If we take the reference point $r_a \rightarrow \infty$ where $U_a \rightarrow 0$, then at a distance $r_b = r$, the potential energy $U_b = U(r)$ is:

$$U(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

7. Comparison with Gravitational Potential Energy:

The electric potential energy $U(r)$ is analogous to the gravitational potential energy:

$$U_{\text{electric}}(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$U_{\text{gravitational}}(r) = -G \frac{m_1 m_2}{r}$$

Electric potential energy is the energy stored due to the relative positions of charges. Moving a charge in an electric field requires work, changing the potential energy. This work is path-independent because the electrostatic force is conservative. The potential energy between two-point charges is inversely proportional to the distance between them, analogous to gravitational potential energy.

Summary

In this lecture, we have explored the foundational concepts of electrostatics, including charges, electric fields, continuous charge distributions, and Gauss's Law. We have seen how these principles are applied in real-world problems and practical applications. Mastery of these topics is essential for a deeper understanding of electromagnetism and its applications in technology and nature.

Lecture_24

Electric Potential

Let's start understanding the forces between charges and how they relate to electric potential energy.

The Basics of Electric Charges and Forces

In electrostatics, we deal with charges that can be either positive or negative. When there are two positive or two negative charges, a force arises between them. This force is repulsive, meaning the charges push away from each other. Conversely, a positive and a negative charge will attract each other. The force between these charges is crucial in understanding electric potential energy.

Work and Potential Energy

When we bring one charge close to another, we must exert a force to overcome the repulsive, as shown in fig. 24.1, or attractive force between them. This exertion of force is called work. As we do work to move a charge, potential energy is added to the system. This potential energy is what gives rise to the concept of electric potential.

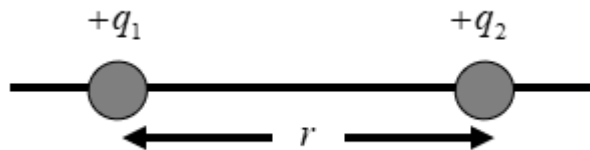


Figure 24.1: Two opposite charges repel each other.

To illustrate this, imagine pressing a spring. Pressing the spring requires work against the force it exerts, storing potential energy in the compressed spring. Similarly, when we bring two charges close together, we do work against the electric force, thereby increasing the electrical potential energy.

Visualizing Electric Potential Energy

Consider two charges being brought close to each other. A simple animation, given in ppt file, can help visualize this process. Imagine a monkey holding one charge and bringing it close to another fixed charge. The monkey must do work to bring the charges closer, but if the monkey releases the charge, all the work done is recovered as kinetic energy. This demonstrates that no energy is lost, and all the energy stored in the system can be recovered.

Calculating Electric Potential Energy

Let's imagine one charge is fixed in place, and we want to bring another charge, a unit charge ($q = 1$), from a far distance to a specific distance R . We need to calculate the total work required for this process. This calculation is divided into two parts.

1. **Starting from Infinity:** We begin by considering the unit charge at infinity, where the electric potential energy is considered zero.
2. **Moving to Distance R:** We move the charge to a distance R, calculating the work done step by step.

The work done in moving a unit charge from infinity to a distance R is given by the integral of the force over the distance:

$$W = \int_{\infty}^R E dr$$

Here, E represents the electric field, and dr is the infinitesimal distance. The electric field E is derived from Coulomb's law:

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Substituting this into our integral:

$$W = \frac{1}{4\pi\epsilon_0} q \int_{\infty}^R \frac{dr}{r^2}$$

This integral simplifies as:

$$\begin{aligned} W &= \frac{1}{4\pi\epsilon_0} q \int_{\infty}^R r^{-2} dr \\ W &= \frac{1}{4\pi\epsilon_0} q \left[\frac{r^{-2+1}}{-2+1} \right]_{\infty}^R \\ W &= \frac{1}{4\pi\epsilon_0} q \left[\frac{r^{-1}}{-1} \right]_{\infty}^R \\ W &= \frac{1}{4\pi\epsilon_0} q \left[-\frac{1}{r} \right]_{\infty}^R \end{aligned}$$

Evaluating this from infinity to R:

$$\begin{aligned} W &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R} - \frac{1}{\infty} \right) \\ W &= \frac{q}{4\pi\epsilon_0} \frac{1}{R} \quad \because \frac{1}{\infty} = 0 \end{aligned}$$

Potential Energy Between Two Charges

If we have two charges, q_1 and q_2 , the potential energy U between them when separated by a distance r is:

$$U(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

We assume the potential energy is zero when the charges are infinitely far apart. This simplifies our calculations and provides a reference point.

Units of Electric Potential

The unit of electric potential is the volt (V), which is equivalent to one joule per coulomb (J/C). When you multiply the electric potential (in volts) by the charge (in coulombs), you get the energy (in joules).

Electron Volt

A useful unit in atomic and nuclear physics is the electron volt (eV). One electron volt is the energy gained or lost by an electron when it moves through an electric potential difference of one volt. This is equal to 1.6×10^{-19} joules.

Applications of Electric Potential

In atomic physics, the electron volt is particularly significant. For instance, to ionize a hydrogen atom (i.e., to remove an electron from a proton), you need to provide 13.6 electron volts of energy. In nuclear physics, the binding energy of protons and neutrons in a nucleus is often measured in million electron volts (MeV).

Movement of Charges in Electric Potential

A positive charge naturally moves from a region of higher potential to a region of lower potential, whereas a negative charge moves in the opposite direction. This behaviour can be visualized with a simple experiment. If you place a charge in an electric field, the charge will move according to the potential difference between two points.

Work Done by Electric Field

Consider a charge moving from point A to point B in an electric field. The work done by the electric field on the charge can be calculated. For instance, if the electric field does 8 joules of work on a charge of 2 coulombs moving from point A to point B, as shown in fig. 24.2, the potential difference is:

$$\Delta V = -\frac{W}{q}$$
$$\Delta V = -\frac{8}{2} = -4V$$

The negative sign indicates a decrease in potential.

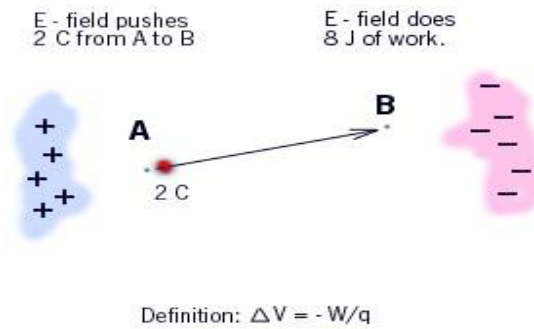


Figure 24.2: Change in potential as charge moves from A to B.

Conservative and Non-Conservative Forces

The force between two charges is conservative, meaning the work done in moving a charge between two points is path independent. This property allows us to define potential energy and potential differences accurately. In contrast, non-conservative forces, such as friction, depend on the path taken.

Electrostatic Potential Energy in Multi-Charge Systems

Understanding the electrostatic potential energy in systems of multiple point charges is fundamental in electrostatics. Consider a system of three-point charges q_1 , q_2 , and q_3 , as shown in fig. 24.3. The total electrostatic potential energy $U(r)$ of this system is given by

$$U(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}}$$

$$U(r) = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

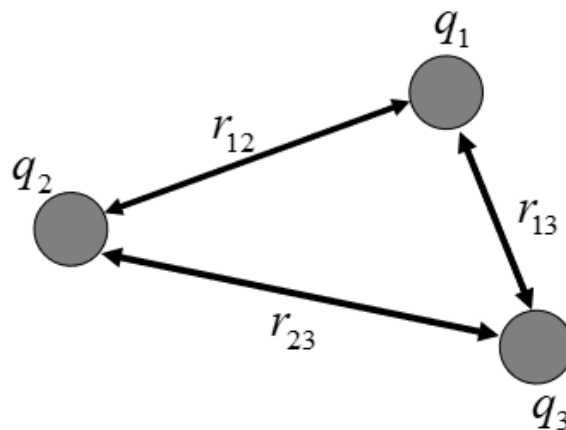


Figure 24.3: Electrostatic potential energy for three charges.

Dipole

A dipole consists of two equal but opposite charges separated by a distance d , as shown in fig. 24.4. The dipole moment p is defined as:

$$P = q \cdot d$$

Dipoles are significant in many molecules where there is a natural separation of charge, such as in water (H_2O).

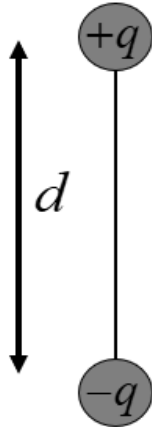


Figure 24.4: A dipole separated by distance d .

To calculate the potential V at a point P due to a dipole, consider the distances from the charges to the point P , as shown in fig. 24.5:

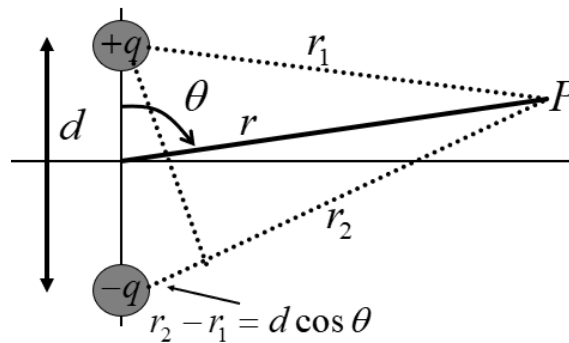


Figure 24.5: Potential at point P due to a dipole.

$$\begin{aligned}
 V_P &= V_1 + V_2 \\
 &= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_1} + \frac{-q}{r_2} \right) \\
 &= \frac{q}{4\pi\epsilon_0} \frac{r_2 - r_1}{r_1 r_2}
 \end{aligned}$$

For points far from the dipole, where r is the distance from the centre of the dipole to the point P and θ is the angle between the dipole axis and the line joining the centre of the dipole to the point P:

For $r \gg d$

$$r_2 - r_1 \approx d \cos \theta$$

$$r_1 r_2 \approx r^2$$

$$V \approx \frac{q}{4\pi\epsilon_0} \frac{d \cos \theta}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \quad \because p = q \cdot d$$

Note that $V = 0$ at $\theta = \frac{\pi}{2}$

Ring of Charges

Consider a uniformly charged ring lies in xy-plan with its center at origin, as shown in fig. 24.6, have a total charge q and radius R . To find the potential at a point P along the axis of the ring, use the symmetry of the problem.

The potential due to a point charge dq at a distance r is given by:

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} \quad \text{eq(1)}$$

For a small element of charge dq on the ring, at coordinates $(R \cos \theta, R \sin \theta, 0)$, the distance to point P is:

$$r = \sqrt{R^2 + z^2} \quad \text{eq(2)}$$

Since the ring is uniformly charged, the charge density λ (charge per unit length) is:

$$\lambda = \frac{q}{2\pi R}$$

An infinitesimal charge element is:

$$dq = \lambda R d\theta = \frac{q}{2\pi R} R d\theta = \frac{q}{2\pi} d\theta \quad \text{eq(3)}$$

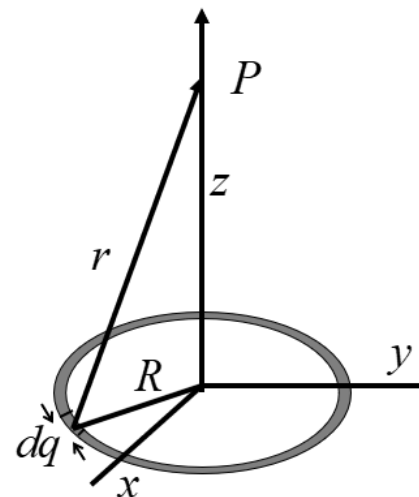


Figure 24.6: Potential due to a charged ring at perpendicular axis.

Integrate dV over the entire ring:

$$V = \int dV$$

using eq(1) and eq(2), we get

$$V = \int_0^{2\pi} \frac{1}{4\pi\epsilon_0} \frac{dq}{\sqrt{R^2 + z^2}}$$

using eq(3), we get

$$V = \int_0^{2\pi} \frac{1}{4\pi\epsilon_0} \frac{\frac{q}{2\pi} d\theta}{\sqrt{R^2 + z^2}}$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{2\pi\sqrt{R^2 + z^2}} \int_0^{2\pi} d\theta$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{2\pi\sqrt{R^2 + z^2}} 2\pi \quad \because \int_0^{2\pi} d\theta = 2\pi$$

$$V = \frac{q}{4\pi\epsilon_0\sqrt{R^2 + z^2}}$$

So, the potential at a point P on the axis of the ring is:

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{R^2 + z^2}}$$

Disk of Charges

For a uniformly charged disk lies in the xy -plane with its centre at the origin have surface charge density σ and radius R , as shown in fig. 24.7, we want to find the electric potential V at a point P on the axis perpendicular to the disk at a distance z from the centre of the disk.

Consider an infinitesimal ring of radius ω and thickness $d\omega$ within the disk. The charge dq on this ring is:

$$dq = \sigma \cdot 2\pi\omega \cdot d\omega$$

The distance from any point on this ring to point P is:

$$r = \sqrt{\omega^2 + z^2}$$

The potential dV due to this ring at point P is:

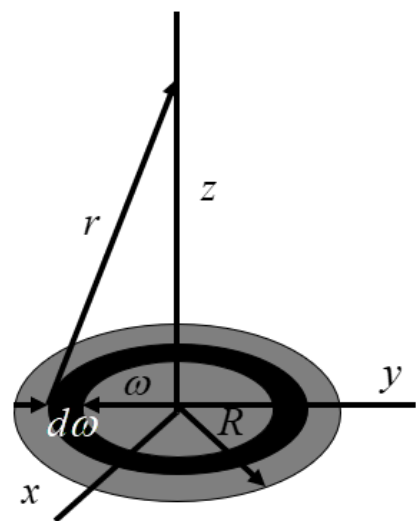


Figure 24.7: Potential due to a disk of uniform surface charge density at perpendicular axis.

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{\sqrt{\omega^2 + z^2}}$$

$$dV = \frac{1}{4\pi\epsilon_0} \frac{\sigma \cdot 2\pi\omega \cdot d\omega}{\sqrt{\omega^2 + z^2}}$$

$$dV = \frac{\sigma}{2\epsilon_0} \frac{\omega d\omega}{\sqrt{\omega^2 + z^2}}$$

$$V = \int_0^R \frac{\sigma}{2\epsilon_0} \frac{\omega d\omega}{\sqrt{\omega^2 + z^2}}$$

$$\text{let } u = \omega^2 + z^2 \text{ then } (du = 2\omega d\omega) \text{ and } \omega d\omega = \frac{du}{2}$$

$$V = \frac{\sigma}{2\epsilon_0} \int_{z^2}^{R^2+z^2} \frac{du}{2\sqrt{u}}$$

$$V = \frac{\sigma}{4\epsilon_0} \left[2\sqrt{u} \right]_{z^2}^{R^2+z^2} = \frac{\sigma}{2\epsilon_0} (\sqrt{R^2 + z^2} - z)$$

So, the potential at point P on the axis perpendicular to the disk is:

$$V = \frac{\sigma}{2\epsilon_0} (\sqrt{R^2 + z^2} - z)$$

Understanding Electric Potential:

Imagine a uniform electric field. This means the strength of the electric field doesn't change as you move through it. Now, consider a scenario where we have two points, a and b, in this field. If we move a charge (let's call it q) from point b to point a, as shown in fig. 24.8, we have to do work against the electric field. This work done by the electric field is the force pushing the charge multiplied by the distance it moves.

The electric potential difference ΔV between points a and b is given by: $V_b - V_a = EL$ and work done $W_{ab} = q_0 EL$ where:

- E is the strength of the electric field,
- L is the distance between points a and b,
- W_{ab} is the work done.

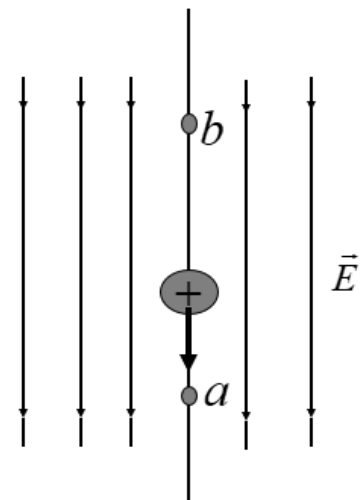


Figure 24.8: Motion of a charge in an equipotential surface from point a to b.

Equipotential Surfaces:

As we move around in space, the potential (electric potential energy per unit charge) changes. However, there are surfaces where the electric potential is the same everywhere. These are called equipotential surfaces, as shown in fig. 24.9. For instance, around a single point charge or between two oppositely charged plates, every point on an equipotential surface has the same electric potential.

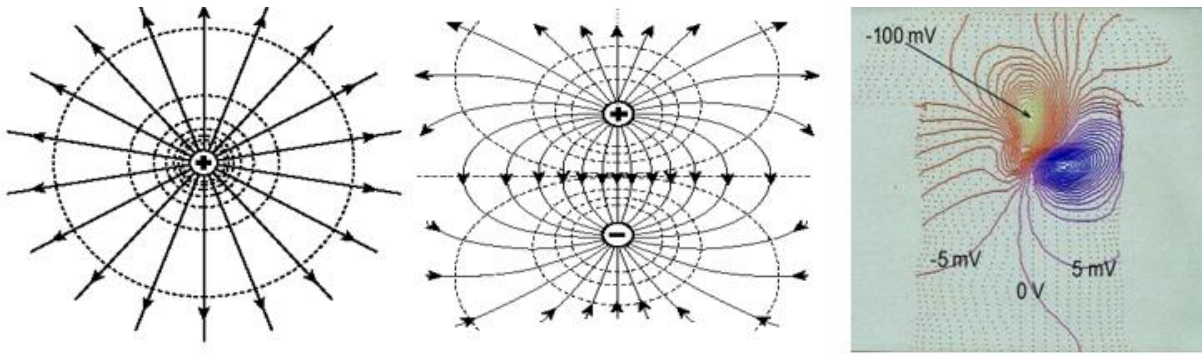


Figure 24.9: Equipotential surfaces for three different situations.

Creating Potential Differences:

To create a potential difference between two points, we need a device like a battery. Batteries use chemical reactions (like those between carbon and zinc electrodes in sulfuric acid, as shown in fig. 24.10) to generate a potential difference. This difference in potential allows us to move charges and generate electric current.

The potential difference ΔV produced by a battery can be calculated based on the chemistry involved and the physical setup of the battery.

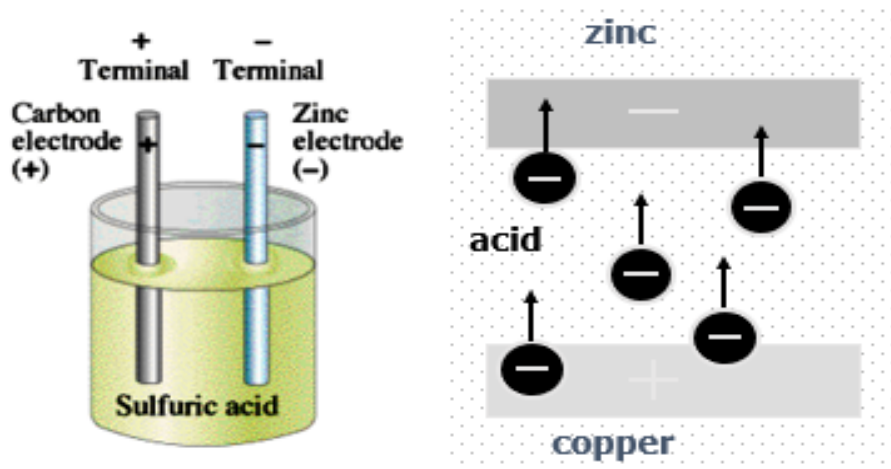


Figure 24.10: Carbon and zinc electrodes in sulfuric acid generates potential differences.

Various Methods to Create Potential Difference:

Potential differences can also be created by mechanical means, such as in the Van de Graaff generator, where charges are moved using belts and rollers against electric forces. Similarly, natural processes like ionization due to rising air currents can create potential differences, leading to phenomena like lightning.

Problem:

Consider two conducting balls with different charges. When connected by a wire, as shown in fig. 24.11, charge redistributes between them until they reach the same potential. The charge Q on each ball is proportional to its radius r and the potential V it reaches.

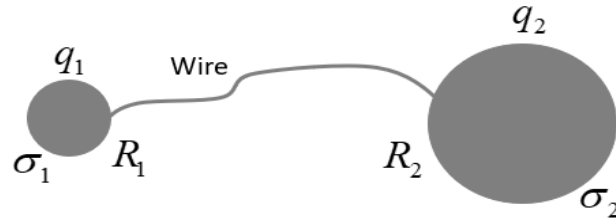


Figure 24.11: Sphere of different radii having different surface charge densities are connected with conducting wire.

Solution:

as we know, surface charge density for a sphere is

$$\sigma = \frac{q}{4\pi R^2}$$

for each sphere

$$\sigma_1 = \frac{q_1}{4\pi R_1^2} \text{ and } \sigma_2 = \frac{q_2}{4\pi R_2^2}$$

$$\Rightarrow q_1 = \sigma_1 4\pi R_1^2 \text{ and } q_2 = \sigma_2 4\pi R_2^2$$

using it in eq(4)

$$\frac{\sigma_1 4\pi R_1^2}{\sigma_2 4\pi R_2^2} = \frac{R_1}{R_2}$$

$$\frac{\sigma_1}{\sigma_2} = \frac{R_1 R_2^2}{R_2 R_1^2}$$

$$\Rightarrow \frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$$

as given $R_2 > R_1$ so we can say, $\sigma_1 > \sigma_2$

Thus, the electric field at the surface of the smaller sphere is greater than that at the surface of the larger sphere:

$$E_{\text{small sphere}} > E_{\text{large sphere}}$$

PHY101-Lecture# 25

Capacitors and Current

Capacitors and Energy Storage

In this lecture, we discuss **capacitors**, which store electrical energy. Capacitors are used in various devices like camera flashes, where they store energy and release it quickly. They consist of two conducting plates, and when a potential difference is created between them, charge gets stored.

Basics of Capacitors

1. Structure and Working:

- Capacitors consist of two plates with a potential difference between them. When charge is transferred from one plate to the other, an **electric field** is created between the plates.
- This results in energy being stored in the capacitor as charges are moved against the electric field.

2. Charge and Potential Relationship:

- The amount of **charge** q stored in a capacitor is proportional to the **potential difference** V between the plates:

$$Q \propto V$$

- Introducing a constant of proportionality C , we get the formula for **capacitance**:

$$Q = CV$$

- Capacitance C is defined as “the amount of charge stored per unit potential difference”.

$$C = \frac{Q}{V}$$

Parallel Plate Capacitor

1. Capacitance Calculation:

- Consider a parallel plate capacitor with plates of area A and separated by a distance d .
- Using **Gauss's law**, the **electric flux** Φ through the surface is:

$$\Phi = E \cdot A$$

- According to Gauss's law:

$$\begin{aligned}\Phi &= \int \vec{E} \cdot d\vec{A} \\ \Phi &= \frac{Q_{\text{enclosed}}}{\epsilon_0} \quad \therefore \Phi = E \cdot A \\ E &= \frac{Q}{\epsilon_0 A}\end{aligned}$$

where ϵ_0 is the permittivity of free space.

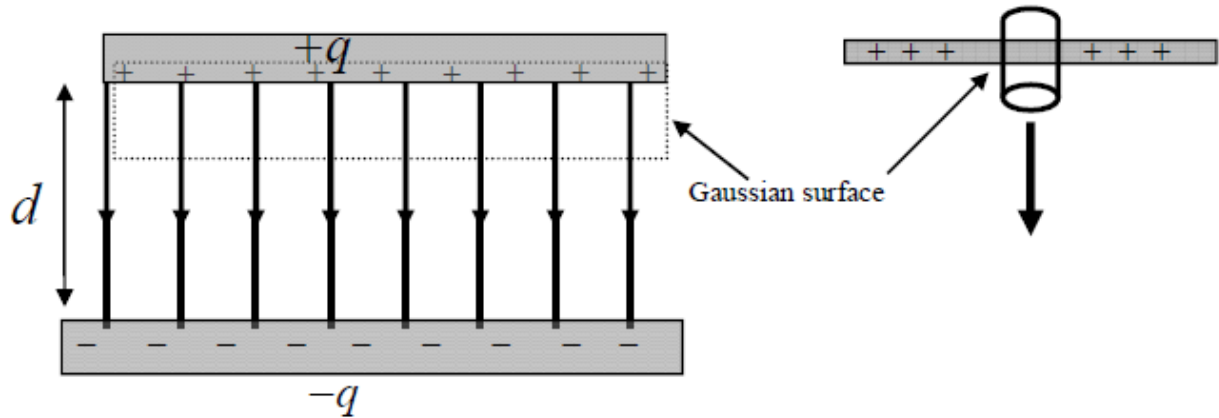
- The **electric field** E is related to the potential difference V given by:

$$V = Ed$$

- Substituting the value of E into this equation, the capacitance C of a parallel plate capacitor is:

$$\begin{aligned}V &= \frac{Q}{\epsilon_0 A} d \\ C &= \frac{Q}{V} = \epsilon_0 \frac{A}{d}\end{aligned}$$

- This shows that capacitance increases with the plate area A and decreases with the distance d between the plates.



Units and Practical Examples

- **Units of Capacitance:**

- The unit of capacitance is the **farad** (F), defined as:

$$1 \text{ farad} = 1 \frac{\text{coulomb}}{\text{volt}}$$

- The farad is a large unit, so capacitors are commonly measured in microfarads (μF), nano-farads (nF), or picofarads (pF).

$$1 \text{ microfarad} = 10^{-6} \text{ farad}$$

$$1 \text{ nanofarad} = 10^{-9} \text{ farad}$$

$$1 \text{ picofarad} = 10^{-12} \text{ farad}$$

- **Example Calculation:**

How big is 1 Farad?

- For a **1 farad** capacitor with plates 1 mm apart, the required area A is approximately:

Choose $d = 1\text{mm}$

$$C = \epsilon_0 \frac{A}{d}$$

Calculate area,

$$A = \frac{Cd}{\epsilon_0}$$

$$A = \frac{1\text{F} \times 0.001\text{m}}{8.85 \times 10^{-12}} = 1.1 \times 10^8 \text{m}^2$$

- This large value is impractical for everyday capacitors.

Types of Capacitors

Capacitors are in various shapes and sizes, such as **spherical** or **cylindrical** capacitors, but the basic principle of charge storage and potential difference remains the same.

- **Air Capacitors:** These consist of two plates separated by air, where changing the area of the plates (by moving them closer or farther apart) changes the capacitance. This type was traditionally used for tuning radios.
- **Electrolytic Capacitors:** These are much larger in size, measured in microfarads (μF), and are used for storing larger amounts of charge.

Charging and Discharging Capacitors:

- A battery is used to charge capacitors by connecting the positive terminal to one plate and the negative to the other. Once charged, a capacitor can store energy that can be discharged later.
- **Discharging:** The stored energy can be discharged by connecting the plates to a circuit. A spark occurs as the charge flows back through the wires.

Analogy of Storing Charge:

- The process of charging and discharging a capacitor is compared to filling and emptying a water tank. A pump (battery) fills the tank (capacitor), and water (charge) can be released (discharged) when needed.

Capacitor Charging Behavior:

- During charging, the current decreases as the capacitor becomes fully charged. The current also depends on the resistance in the circuit, and if a large resistance is present, the capacitor will discharge slowly.

Parallel and Series Connections:

- Parallel Connection:**

If capacitors are connected in parallel, the total capacitance increases. The formula for total capacitance is the sum of individual capacitances as given:

For each capacitor,

$$q_1 = C_1 V, \quad q_2 = C_2 V$$

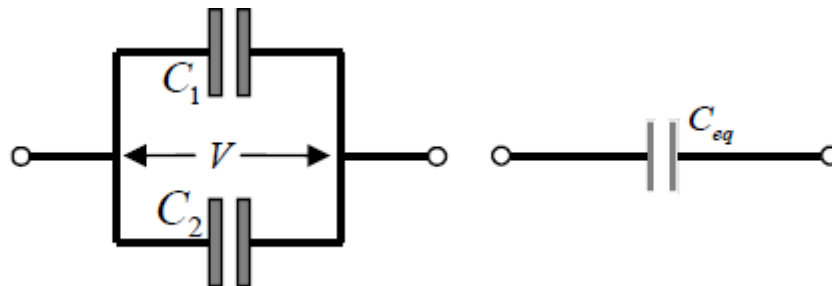
Where V is the potential between terminals a and b

$$q = q_1 + q_2$$

$$C_{eq} V = C_1 V + C_2 V$$

$$\Rightarrow C_{eq} = C_1 + C_2$$

$$C_{eq} = \sum C_n \quad (\text{For } n \text{ capacitors})$$



- Series Connection:** In a series connection, the total capacitance is calculated differently, as the voltage is divided among the capacitors.

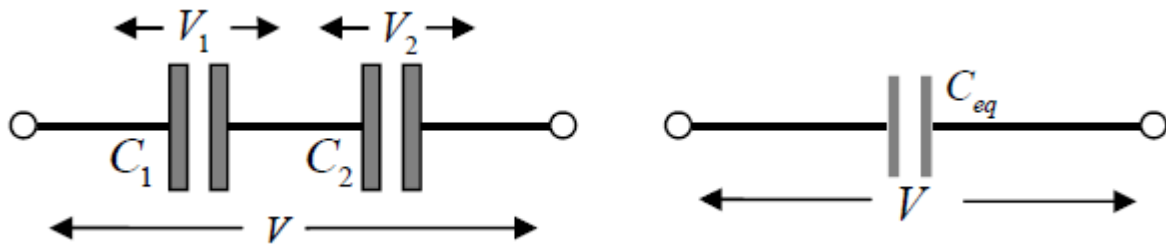
$$V_1 = \frac{q}{C_1}, \quad V_2 = \frac{q}{C_2}$$

$$V = V_1 + V_2$$

$$\frac{q}{C_{eq}} = \frac{q}{C_1} + \frac{q}{C_2}$$

$$\Rightarrow \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C_{eq}} = \sum \frac{1}{C_n} \quad (\text{For } n \text{ capacitors})$$



Energy Stored in Capacitors:

When a battery is connected to a capacitor, positive and negative charges appear on the opposite plates. Some energy has been transferred from the battery to the capacitor and now been stored in it. When the capacitor is discharged, the energy is recovered. Now let us calculate the energy required to charge a capacitor from zero to volts.

The amount of energy required to transfer a small charge dq to the plates is

$$dU = Vdq$$

where V is the voltage at a time when the charge is $q = CV$. As time goes on, the total charge increases until it reaches the final charge Q (at which point the voltage becomes V). So,

$$\begin{aligned} dU &= Vdq = \frac{q}{C} dq \\ U &= \int dU = \int_0^Q \frac{q}{C} dq \\ U &= \frac{Q^2}{2C} \end{aligned}$$

where C is the capacitance and V is the voltage across the capacitor. The energy increases quadratically with voltage. This energy is analogous to work done in transferring charge between the plates of a capacitor.

Energy Density:

Where in the capacitor is the energy stored?

Answer:

It is present in the electric field in the volume between the two plates.

The energy stored per unit volume (energy density) in a capacitor is given by:

$$U = \frac{1}{2} CV^2$$

$$u = \frac{U}{Ad} = \frac{\frac{1}{2} CV^2}{Ad}$$

$$C = \frac{\epsilon_0 A}{d}$$

$$u = \frac{\epsilon_0}{2} \left(\frac{V}{d} \right)^2 = \frac{1}{2} \epsilon_0 E^2$$

where E is the electric field between the plates, and ϵ_0 is the permittivity of free space. The energy density is proportional to the square of the electric field.

Electric Field and Energy:

- The energy stored in a capacitor is proportional to the square of the electric field.
- When a dielectric is added, it reduces the electric field, decreasing the potential difference and increasing capacitance.

Electric Current:

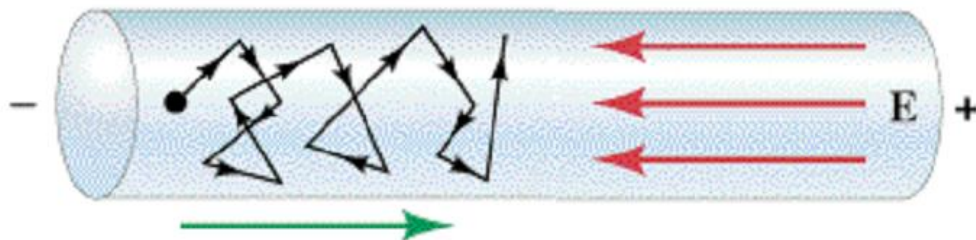
- Current is the flow of charge, typically electrons in a wire.

$$i = \frac{dq}{dt}$$

constant current: $q = i \times t$

else: $q = \int i dt$

- **Drift velocity** of electrons is a key factor in current, and the flow of electrons is affected by collisions within the conductor.



- The **current** is measured in amperes (A).
- **Current: units and sizes:**

$$1 \text{ ampere} = \frac{1 \text{ coulomb}}{\text{second}}$$

$$1 \text{ milliampere} = 1 \text{ ma} = 10^{-3} \text{ A}$$

$$1 \text{ microampere} = 1 \mu\text{A} = 10^{-6} \text{ A}$$

$$1 \text{ nanoampere} = 1 \text{ nA} = 10^{-9} \text{ A}$$

$$1 \text{ picoampere} = 1 \text{ pA} = 10^{-12} \text{ A}$$

- Currents can vary widely, from large currents in car batteries to tiny currents (nano or pico-amperes) in integrated circuits.

Conservation of Current:

- Current is conserved in a circuit: if it splits, the total current before and after the split remains the same.
- The current flowing through a resistor depends on the applied voltage and the resistance (Ohm's Law): $I = V/R$.
- This law holds true within certain limits, and excessive current can cause a resistor to overheat or break.

Capacitor's Function:

- The result concludes with an analogy to a spring, where energy is stored in the capacitor due to the electric field, just like energy is stored in a spring due to its extension.

Coulomb's Law with Dielectrics:

Coulomb's Law with dielectrics states that the electrostatic force between two charges is reduced by a factor of the dielectric constant (k) of the medium. The force is given by,

$$F = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q_1 q_2}{r^2}$$

$$\epsilon_r = \text{relative permittivity}$$

$$E_{\text{medium}} = \frac{1}{4\pi\epsilon_0\epsilon_r} \frac{q}{r^2}$$

$$E_{\text{medium}} = \frac{E_{\text{vacuum}}}{\epsilon_r}$$

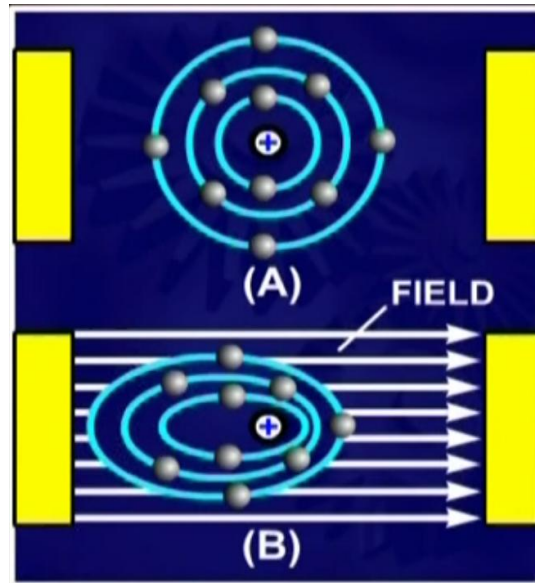


Fig. Distortion of electron orbital paths due to electrostatic force.

Effect of Dielectric Materials on Capacitance:

- The capacitance of a capacitor increases with the insertion of dielectric materials, which reduce the electric field and potential difference between plates.
- Dielectric materials, like water or plastic, polarize when an electric field is applied, affecting the electric field inside the material and influencing the capacitance.
- The **relative permittivity** of materials (also called the dielectric constant) determines how much the dielectric material can reduce the electric field, increasing capacitance.

Dielectric Constant and Polarization:

- The dielectric constant ϵ_r is greater than 1 and depends on the material. It determines how much the material can reduce the electric field.
- When a dielectric material (like water or plastic) is placed between plates, the molecules within it align with the electric field, reducing the overall electric field between the plates.

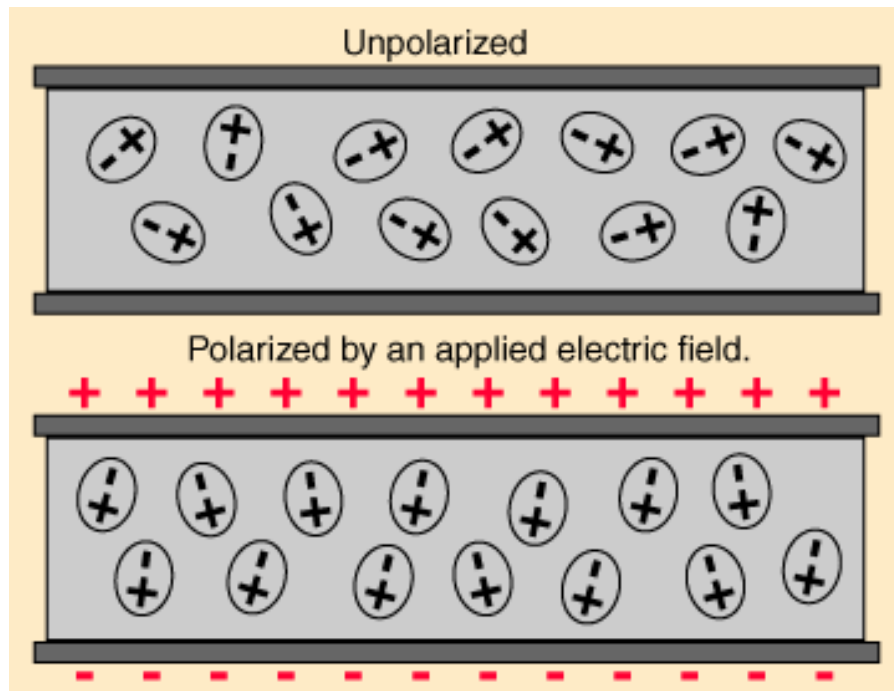


Fig. Electric field decreases in dielectrics due to polarization.

Practical Use of Dielectrics:

- Capacitors with high dielectric constants (such as electrolytic capacitors) can store much more charge in a smaller volume compared to air-based capacitors, but these capacitors may not always be perfect due to the small conductivity of the dielectric, which can lead to leakage of charge.

Lecture#26

ELECTRIC POTENTIAL ENERGY - II

Electric Current and Electric Circuits

Electric current and electric circuits are fundamental concepts in the study of electricity. Electric current refers to the flow of electric charge, and the magnitude of this current depends on the amount of charge passing through a given point in a conductor over time. The greater the movement of charge, the higher the current in the system. An electric circuit is a complete and closed path through which electric charge flows. In a broader sense, the term "circuit" can refer to any closed pathway through which objects, such as vehicles, can move. However, in the context of electricity, an electric circuit specifically refers to a pathway that allows charged particles to move and eventually return to their starting point. The significance of electric circuits and current in modern life is profound, as most technological advancements rely on a thorough understanding and application of these principles.

To better understand electric current, it can be compared to the flow of water in a closed-loop system. Consider a pump connected to a network of pipes. The pump exerts a force that drives the water through the pipes, eventually returning it to the starting point. If the pipes are wide, water can flow with ease, but if they are narrow, greater force is required to push the same amount of water through in a given time. A similar concept applies to electricity. In an electrical circuit, if the conducting wire has a smaller cross-sectional area, the flow of charge encounters greater resistance, making it more difficult for current to pass through. Thus, narrowing a wire increases its electrical resistance. Electric current I is formally defined as the amount of electric charge dq that passes through a conductor in a given period of time dt .

$$I = \frac{dq}{dt} = \frac{q}{t}$$

Thus in terms of units

$$1 \text{ Ampere} = \frac{1 \text{ Coulomb}}{1 \text{ second}}$$

Resistance, on the other hand, is defined as the ratio of the potential difference V across a resistor to the current I flowing through it as shown in figure 26.1. In an electrical circuit, if

one end of a resistor is at a higher electric potential compared to the other end, a potential difference is established, driving the flow of current. For current to flow from one end of the resistor to the other, the potential at the starting point 'a' must be greater than the potential at the endpoint 'b', similar to how water naturally flows from a higher elevation to a lower elevation due to gravitational potential.

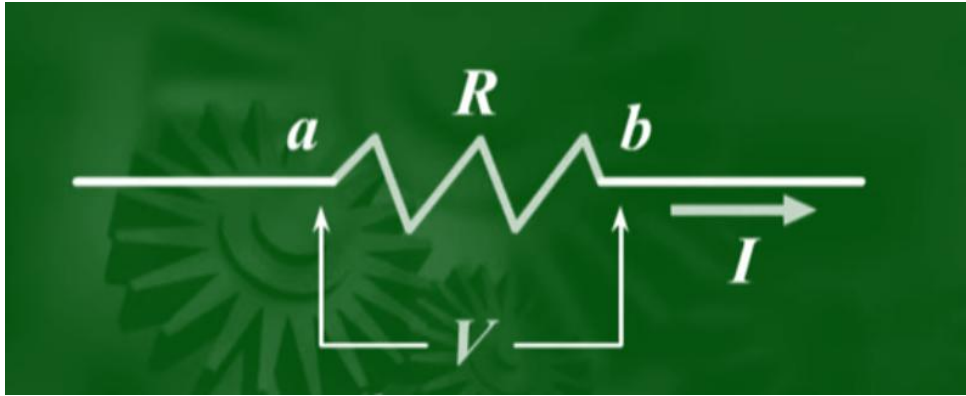


Figure 26.1: Current I flowing through Resistor R because of potential difference V across resistor. a is at higher potential than b so current I is flowing from a toward b .

The direction of electric current is conventionally considered to be from a region of higher potential to a region of lower potential. It is important to note that this conventional current flow assumes the movement of positive charges, even though in many practical scenarios, such as in metallic conductors, the actual charge carriers are negatively charged electrons moving in the opposite direction.

Resistance is commonly expressed as the ratio of the potential difference across a resistor to the current passing through it.

$$R = \frac{V}{I}$$

However, this relationship alone does not define Ohm's law. Ohm's law specifically applies only when the resistance of the material remains constant under varying potential differences and resulting currents. In other words, if the current through a material increases linearly with an applied potential difference, the material is said to obey Ohm's law.

In practical applications, the resistance of a material may change with factors such as temperature. For example, when a current is passed through a wire, it generates heat due to resistance. As the potential difference across the wire increases, more current flows, and the wire's temperature rises. If excessive potential difference is applied, the wire may overheat and

eventually burn out, preventing any further current flow. In such cases, the current is no longer proportional to the applied potential difference, meaning that Ohm's law no longer holds under these extreme conditions.

Equivalent Resistance in Series and Parallel Circuits

The concept of equivalent resistance is crucial in analyzing electrical circuits containing multiple resistors. When resistors are connected in different configurations, such as series or parallel, their combined effect can be represented by a single equivalent resistor that simplifies circuit analysis.

Resistors in Series

When two resistors are connected in series as shown in figure 26.2, the total or equivalent resistance is the sum of the individual resistances. In a series configuration, the same current I flow through all resistors since there is only a single path for the charge to travel. The total voltage V across the series combination is equal to the sum of the voltage drops across each individual resistor.

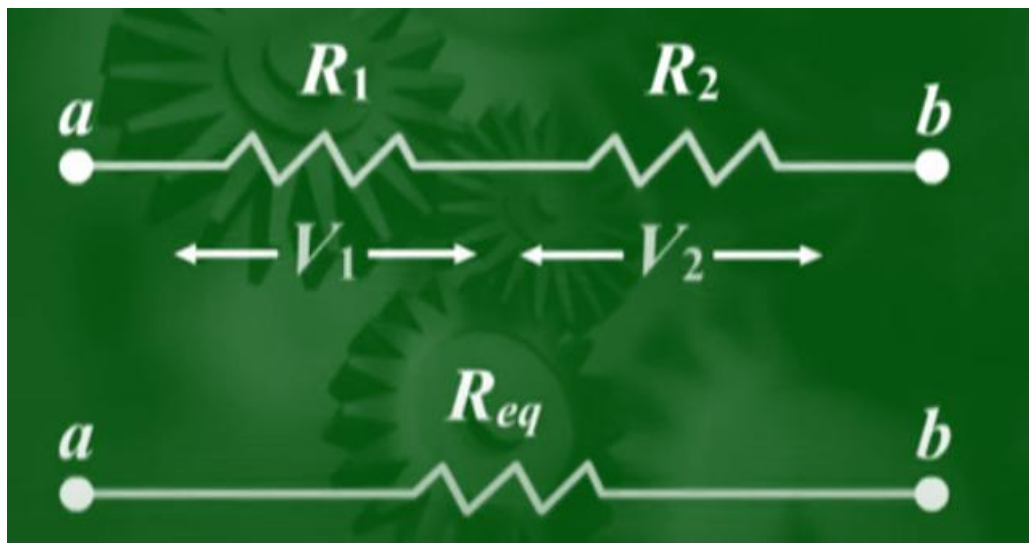


Figure 26.2: Resistor in Series

$$V_1 = IR_1, \quad V_2 = IR_2$$

$$V = V_1 + V_2$$

$$IR_{eq} = IR_1 + IR_2$$

I can be cancelled from both sides of equation so,

$$R_{eq} = R_1 + R_2$$

The equivalent resistance in a series circuit can be determined by adding the resistance values of the individual resistors. This means that if two resistors with resistances R_1 and R_2 are connected in series, their equivalent resistance will be the sum of the two, resulting in a total resistance that is greater than either of the individual resistances.

Resistors in Parallel

When resistors are connected in parallel, the situation differs significantly from the series configuration. In a parallel circuit, the resistors provide multiple paths for current to flow, meaning that the total current entering the combination is divided among the parallel branches. However, the voltage across each resistor remains the same. The conservation of charge principle applies here, ensuring that the total current flowing into the parallel network is equal to the sum of the currents flowing through each individual resistor. The equivalent resistance in a parallel circuit is determined using the reciprocal relationship of the individual resistances.

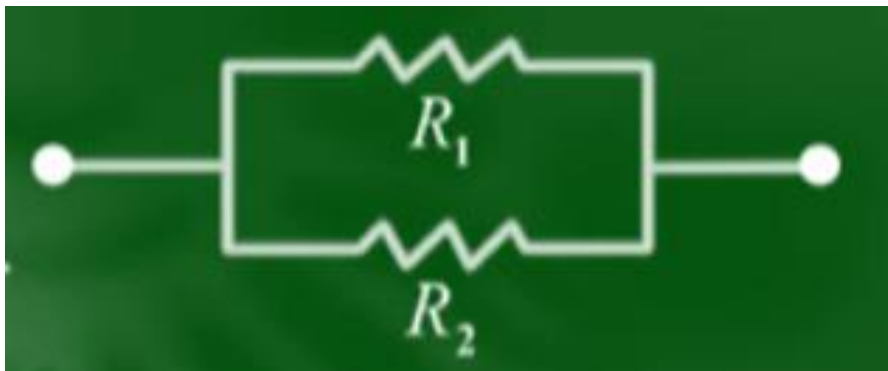


Figure 26.3: Resistors in Parallel

$$I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}$$

$$I = I_1 + I_2$$

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

For two resistors R_1 and R_2 connected in parallel, the reciprocal of the equivalent resistance is equal to the sum of the reciprocals of the individual resistances. This results in an equivalent resistance that is always less than the smallest resistor in the parallel network.

In practical terms, when resistors are connected in parallel, the total resistance decreases because the current has multiple paths to flow through. This reduced resistance allows more current to pass through the circuit compared to a single resistor of the same value.

Comparison with Capacitors

The behavior of resistors in series and parallel can be contrasted with that of capacitors. When capacitors are connected in parallel, their capacitance increases, as the total charge-storing capability of the system is enhanced. In contrast, when capacitors are connected in series, the total capacitance decreases. This behavior is opposite to that of resistors, where resistance increases in series and decreases in parallel. The reduction in resistance in parallel circuits occurs because the presence of multiple paths allows charge to flow more easily, thereby reducing the overall opposition to current flow.

Electromotive Force

The movement of electric charge within a circuit is a fundamental concept in physics, leading to the question of what drives this movement. Understanding the driving force behind the motion of electrons helps in comprehending the working principles of electrical circuits. Historically, certain terms used in physics can be misleading or inaccurately named, and one such example is the term Electromotive Force (EMF). Despite its name, EMF is not actually a force in the conventional sense but rather a measure of energy supplied to charge carriers in a circuit.

To illustrate this concept, consider an analogy with water flowing through a pipe. Water moves through the pipe because of an applied force, typically provided by a pump. Similarly, in an electrical circuit, electrons move through a conductor due to the presence of electromotive force. However, the term force in EMF can be misleading, as it does not represent a physical force but rather the energy required to move charge through the circuit.

A useful analogy to understand EMF is to imagine a person lifting balls from the ground and placing them onto a table. The balls roll off the table, experience friction, lose energy, and eventually reach the ground again. The person then lifts them back onto the table, repeating the process. In this analogy, the person providing energy to lift the balls acts as the source of EMF,

supplying the necessary energy to keep the process going. Similarly, in an electrical circuit, the EMF provides the energy required to move charge through various components, compensating for energy losses due to resistance and other factors.

Formally, electromotive force is defined as the energy provided per unit charge to move it around a closed circuit. This means that if twice the amount of charge is moved, twice the amount of energy is required. The source of EMF in an electrical circuit can vary, with different physical processes generating the required potential difference.

The sources of EMF can vary, depending on different physical processes that generate the required potential difference. One of the most common sources is chemical reactions, as seen in batteries where interactions between electrodes and electrolytes create a potential difference that drives electrons from one electrode to another. Another source of EMF is magnetism, where a changing magnetic field induces an electromotive force in a conductor, forming the basis of electrical generators and transformers. Light can also generate EMF through the photovoltaic effect, where certain materials, when exposed to light, produce electrical energy. This principle is used in solar panels, which convert sunlight into electricity to power satellites and other remote applications. Furthermore, heat energy can generate EMF through the thermoelectric effect, where differences in temperature across certain materials induce a potential difference. This phenomenon is widely applied in thermocouples for temperature measurement and power generation in space applications. Regardless of the source, the conversion of one form of energy into electrical energy results in the generation of electromotive force, which then drives current through the circuit.

Kirchhoff's Voltage Law (KVL)

Once charge carriers begin to move through the circuit under the influence of electromotive force, they encounter various components such as resistors and capacitors. As the charge moves through these elements, energy is dissipated or stored. A fundamental principle governing this behavior is Kirchhoff's Voltage Law (KVL), which states that the sum of all potential differences around a closed loop in a circuit is equal to zero. This law can be understood by considering a circuit where charge starts its journey at a certain potential and moves through different circuit elements such as resistors and capacitors. As it progresses, it experiences drops in potential across each component. However, by the time it completes the loop and returns to the starting point, the total potential must sum to zero. This means that the energy supplied by

the EMF source is entirely accounted for by the energy dissipated or stored in the circuit elements. Mathematically, Kirchhoff's Voltage Law is expressed as the sum of all voltage drops and EMF sources in a closed loop being equal to zero. In practical terms, this law ensures that no energy is lost or created within an electrical circuit, aligning with the principle of conservation of energy.

In the simplest case of an electrical circuit as shown by figure 26.4, consider a battery that provides an electromotive force (EMF) and a single resistor connected in series with it. The circuit consists of four points: a, b, c and d, forming a closed loop.

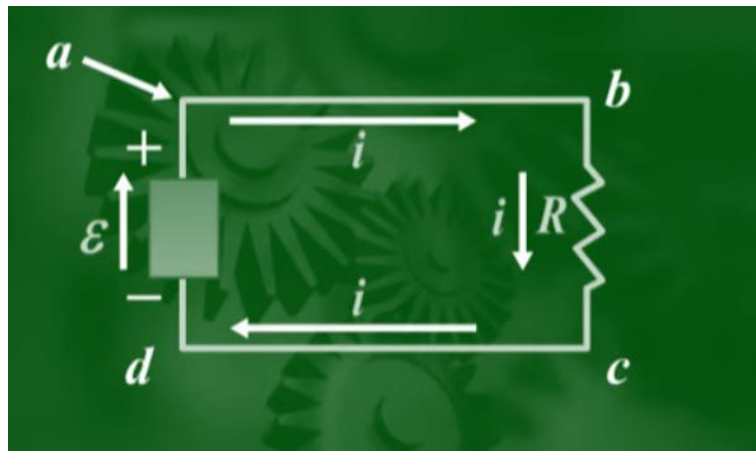


Figure 26.4: A simple circuit as an example for Kirchhoff's law

Starting from point a and moving towards point b, the wire is considered to have negligible resistance, resulting in no potential difference between these points. As we move from point b to point c, the resistor comes into play, and a voltage drop occurs due to the resistance. Since the resistor opposes the flow of current, the potential at point b is higher than at point c, and the voltage drop across the resistor is given by the product of the current and resistance. Continuing from point c to point d, since there is no additional resistance in the connecting wire, there is no change in potential. Finally, moving from d back to point a, the battery provides the electromotive force, which increases the potential by an amount equal to the EMF. Applying Kirchhoff's Voltage Law (KVL), which states that the sum of potential differences in a closed loop must equal zero, we find that the sum of the voltage drop across the resistor and the EMF of the battery results in a balanced equation.

$$V_a - iR + \varepsilon = V_a$$

$$iR + \varepsilon = 0$$

$$\varepsilon = iR$$

Simplifying this equation leads to the well-known relation that the EMF of the battery is equal to the voltage drop across the resistor, confirming Ohm's Law. This simple example demonstrates the fundamental principle of Kirchhoff's Voltage Law, which is applicable to more complex electrical circuits. The law asserts that if we start from any point in the circuit and traverse through all the components to return to the same point, the algebraic sum of all potential differences encountered must equal zero. This principle is rooted in the conservative nature of the electric field, meaning that the work done in moving a charge around a closed loop is zero. In physics, forces can be classified as conservative or non-conservative. Conservative forces, such as gravitational force, allow the definition of potential energy as a function of position. For a conservative force, if an object is moved around a closed path and returns to its starting position, no net work is done. In electrical circuits, the electric force behaves conservatively, which is why Kirchhoff's Voltage Law holds true.

The figure 26.5 illustrates another application of Kirchhoff's Voltage Law (KVL) in a simple electrical circuit. In this setup, there is a source of electromotive force (EMF) connected to two resistances.

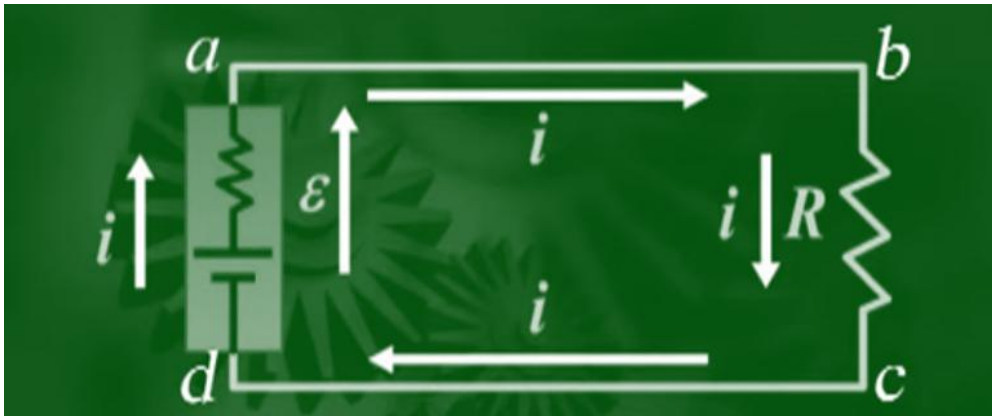


Figure 26.5: A circuit where EMF source has internal resistance r .

The first resistance is intrinsic to the EMF source, often referred to as the internal resistance, while the second is an external resistance connected in the circuit. A current flows through the entire loop, and the objective is to determine its magnitude. The internal resistance of the EMF source arises from the physical properties of its components. For example, in a typical battery (such as a 1.5V or 9V battery), the internal resistance is due to the materials and electrolytes present between its terminals. Even though there is no visible resistor inside the battery, this inherent resistance prevents an infinite current from flowing when the terminals are directly

connected without an external load. Therefore, in any circuit containing a real-world EMF source, the internal resistance must be accounted for when analyzing the current flow. To solve for the current, Kirchhoff's Voltage Law is applied to the circuit. According to KVL, the sum of potential differences across all elements in a closed loop is equal to zero. Here, the total EMF of the source is opposed by the voltage drops across the internal resistance and the external resistance. This relationship can be expressed mathematically to calculate the current as follows.

$$V_a + \varepsilon - ir - iR = V_a$$

$$\varepsilon - ir - iR = 0$$

$$\varepsilon = i(r + R)$$

$$i = \frac{\varepsilon}{r + R}$$

While this example deals with a simple, single-loop circuit, Kirchhoff's laws can be extended to more complex configurations. In circuits with multiple loops, KVL can be applied individually to each loop to analyze current flow and potential differences. Similarly, resistors, capacitors, or other circuit elements can be incorporated into the loop, and Kirchhoff's laws will still hold true for the system.

RC Circuit as an example of KVL

The figure 26.6 illustrate the behavior of a capacitor in a circuit and its relation to current flow, potential drops, and time dependence. The circuit components include a battery (as a source of EMF), a capacitor, a resistor, and, in some cases, a light-emitting diode (LED).

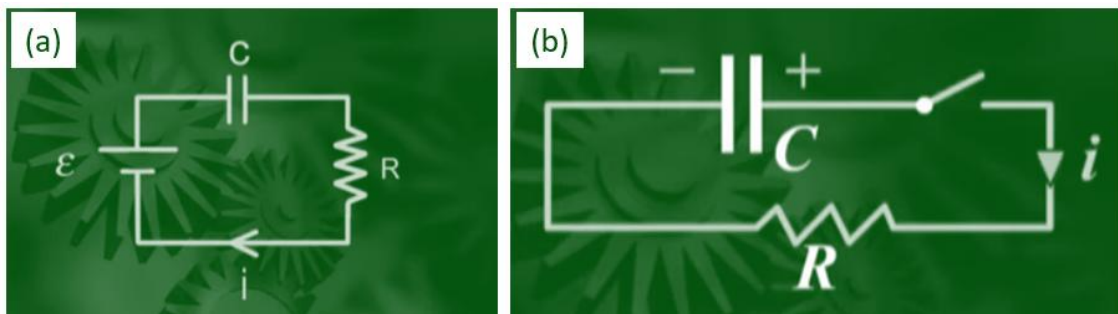


Figure 26.6: (a) charging and (b) discharging of capacitor

When the battery is connected to the circuit containing the capacitor, current initially flows, allowing the capacitor to charge. This charging process is analogous to filling a water tank: as

water is pumped into the tank, the flow continues until the tank is full. Similarly, once the capacitor is fully charged, the current stops flowing. This is because the potential difference across the capacitor becomes equal to the EMF of the battery, leaving no net driving force for further current. Kirchhoff's Voltage Law (KVL) governs this process. According to KVL, the sum of potential drops in a closed circuit loop is zero. In this circuit, the potential drops occur across the capacitor and the resistor. The voltage across the capacitor is proportional to the charge stored on it, while the voltage drop across the resistor is proportional to the current. These relationships lead to a mathematical equation describing the circuit, which can be solved to determine how the charge and current vary over time. For circuit depicted by figure 26.6(a)

$$\varepsilon - \frac{q}{C} - iR = 0$$

If the charged capacitor is disconnected from the battery and connected to an LED, the stored charge creates a current that briefly powers the LED. The LED lights up momentarily, as the current diminishes with time until the capacitor is fully discharged. This process illustrates that the charge on the capacitor decreases exponentially, governed by the time constant of the circuit. For circuit depicted by figure 26.6(b)

$$\frac{q}{C} + iR = 0$$

Differentiating whole equation with respect to t i.e. d/dt

$$\frac{1}{C} \frac{dq}{dt} + \frac{di}{dt} R = 0$$

As

$$\frac{dq}{dt} = i$$

Equation becomes

$$\frac{i}{C} + \frac{di}{dt} R = 0$$

$$\frac{di}{dt} R = -\frac{i}{C}$$

$$\frac{di}{dt} = -i \frac{1}{RC}$$

$$i = i_0 e^{-\frac{t}{RC}}$$

The time constant (denoted by RC) is a critical parameter that defines how quickly the capacitor charges or discharges. Physically, it depends on the product of the resistance and capacitance. A larger capacitance (analogous to a larger water tank) or higher resistance (analogous to a narrower pipe or a partially opened tap) results in a longer time constant, meaning the capacitor will take more time to charge or discharge. This behavior of the capacitor can be described using differential equations. The solution to these equations shows that the current decreases exponentially over time, starting at an initial value determined by the circuit configuration. This mathematical model aligns with the physical intuition of how charge and current evolve in RC circuits, offering insights into both transient and steady-state behavior.

Kirchhoff's Current Law (KCL)

In electrical circuit analysis, there are two fundamental laws that govern the behavior of currents and voltages. The first law, commonly referred to as Kirchhoff's Voltage Law (KVL), states that the sum of all potential drops around a closed loop in a circuit is equal to zero. This principle is based on the conservation of energy, ensuring that the total energy gained and lost by charges in a complete loop balances out.

The second law, known as Kirchhoff's Current Law (KCL), states that the total current entering a junction in a circuit is equal to the total current leaving the junction. This is a direct consequence of the conservation of electric charge, which implies that charge cannot accumulate or disappear at any point in the circuit. Together, these laws provide a comprehensive framework for analyzing and understanding electrical circuits.

KVL and KCL example

This circuit analysis shown below (for circuit in figure 26.7) demonstrates the use of Kirchhoff's laws to calculate current distribution in a circuit containing resistors and voltage sources. The circuit includes three resistors and two batteries providing electromotive force (emf). The current in the circuit splits into three paths: one through the first resistor, another through the second resistor, and a third through the resistor between the branches.

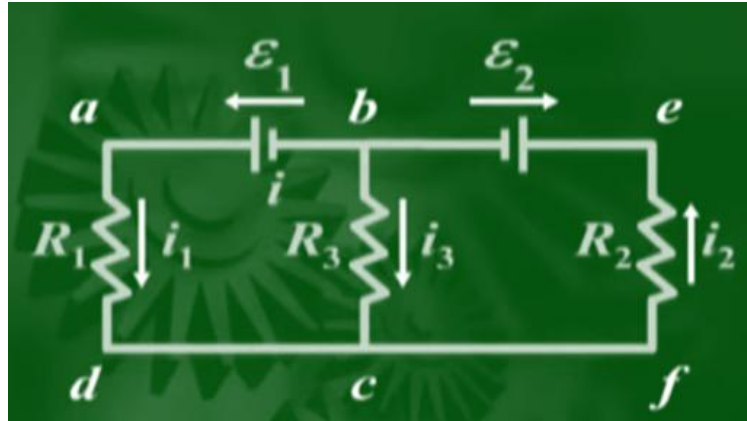


Figure 26.5: A circuit as an example of KVL and KCL

To analyze the circuit, Kirchhoff's Current Law (KCL) is first applied. This law states that the total current entering a junction equals the total current leaving it. At the key junction in this circuit, the current entering divides into two parts, ensuring that the sum of these two currents is equal to the current exiting the junction.

When analyzing electrical circuits using Kirchhoff's laws, it is crucial to carefully apply the rules for assigning signs to voltage drops and rises. While traversing a circuit loop in the direction of the current, the voltage drop across a resistor should be considered negative. This is because the current flows from a higher potential to a lower potential across the resistor. On the other hand, when you encounter an electromotive force (EMF) source, the sign depends on the direction of traversal. If you move from the negative terminal to the positive terminal of the EMF source, it represents a gain in potential, and the EMF should be assigned a positive value.

Applying Kirchhoff's laws is fundamentally straightforward, relying more on systematic practice than on memorization of formulas. These laws, based on the principles of energy conservation and charge conservation, provide an intuitive method for analyzing even complex circuits. With consistent practice, one can confidently solve various types of intricate circuit problems, ranging from basic setups to advanced configurations. The key to mastering these laws is a clear understanding of the physical principles they represent and a structured approach to solving circuit loops.

Next, Kirchhoff's Voltage Law (KVL) is used. This law states that the sum of all voltages around a closed loop in a circuit must be zero. By applying KVL to different loops in the circuit, equations are written to relate the emf of the batteries to the voltage drops across the resistors.

The voltage drops are proportional to the currents flowing through the resistors, according to Ohm's law.

$$i_1 + i_3 = i_2$$

$$\varepsilon_1 - i_1 R_1 + i_3 R_3 = 0$$

$$-i_3 R_3 - i_2 R_2 + \varepsilon_2 = 0$$

Value of three variables (i_1 , i_2 and i_3) can be determined as we have three equations. By combining the principles of KCL and KVL, the relationships between the currents and resistances in the circuit can be determined, enabling the calculation of the individual current values through each resistor. This approach is fundamental in circuit analysis and illustrates the power of Kirchhoff's laws in solving complex electrical networks.

Power in Circuits

When applying Kirchhoff's laws, which are closely related to the concept of potential, one can explore how much work is done by a source of electromotive force (EMF) when charges move through a circuit. To illustrate this, consider a scenario where a person lifts a heavy object repeatedly, using energy to move it against gravity. Ultimately, the work done in this process transforms into another form of energy, such as heat. In electrical circuits, a similar phenomenon occurs, but it involves charges. Charges do work as they move through the circuit, and the amount of work done depends on two factors: the magnitude of the EMF provided by the source and the quantity of charge involved.

When a charge moves through a potential difference created by an EMF source, work is performed. The work done on the charge is directly proportional to both the value of the EMF and the amount of charge. For instance, when a charge is raised through a potential difference, the work done can be calculated by multiplying the EMF by the charge. To understand this concept, consider a resistor within the circuit. Inside the resistor, charges experience a potential difference, requiring work to move from one end to the other. This work is done by the EMF source, which provides the energy required to push the charges through the resistor.

The relationship between the work done, EMF, and charge can be further refined. Work is defined in terms of the small charge transferred across the potential difference. The potential difference, expressed in volts, measures the energy transferred per unit charge. For instance, if one coulomb of charge passes through a potential difference of one volt, the work done is equal

to one joule. This relationship defines a volt as one joule per coulomb. When current flows through a resistor, the potential difference across the resistor causes energy dissipation in the form of heat. The energy dissipation depends on both the current flowing through the resistor and its resistance.

$$\varepsilon = \frac{dW}{dq} \quad \left(1 \text{ volt} = \frac{1 \text{ joule}}{1 \text{ coulomb}} \right)$$

$$dW = \varepsilon \cdot dq = \varepsilon \cdot i \cdot dt \quad \left(\text{as } i = \frac{dq}{dt} \right)$$

$$dW = \varepsilon \cdot i \cdot dt = (i \cdot R) \cdot i \cdot dt = i^2 \cdot R \cdot dt \quad \left(\text{as } i = \frac{\varepsilon}{R} \text{ or } \varepsilon = iR \right)$$

As

$$P = \frac{dW}{dt} = \frac{i^2 \cdot R \cdot dt}{dt} = i^2 R$$

$$P = \left(\frac{V}{R} \right)^2 \cdot R = \frac{V^2}{R}$$

$$P = i^2 R = i(iR) = iV$$

The power dissipated in a resistor, which represents the rate of energy conversion, can be calculated using multiple relationships. Power is the work done per unit time. It is proportional to the square of the current multiplied by the resistance or, alternatively, to the square of the potential difference divided by the resistance. Another expression for power involves the product of current and potential difference. These relationships highlight that the energy consumed in a resistor per unit time is directly connected to the electrical properties of the circuit. In practical applications, the units used to measure this energy dissipation are joules for work and watts for power. One watt is defined as one joule of work done per second. Power dissipation in devices such as home appliances is often measured in kilowatts (thousand watts). Understanding these fundamental principles is essential for analyzing electrical circuits and quantifying energy usage effectively.

How Electric Current Flows

In understanding the flow of current through a material, several fascinating phenomena are revealed. If it were possible to observe the inner workings of a conductor, one would see that the charge carriers responsible for current are primarily negatively charged electrons. While

there are no visible moving positive charges in typical conductors, the electrons themselves do not move in a smooth, straight path. Instead, they constantly experience collisions and scatter in various directions as they make their way through the material.

A closer examination, perhaps using animation shown in video lecture, illustrates how charge carriers move within a conductor, such as a copper wire. When a current flows through the wire, it causes devices like bulbs to light up as the moving electrons transfer energy. If the wire is interrupted, the current flow stops, and the bulb goes out. Observing at a microscopic level, one can see electrons colliding with atoms and other electrons as they move through the conductor. This erratic movement leads to what is known as the drift velocity, which is the average speed at which electrons move in response to an electric field.

To understand how much current flows through a material when an electric field is applied, consider a small piece of wire with length L , cross-sectional area A , and a specific number density of electrons n . The total charge q in the wire can be calculated by considering the charge of individual electrons and the number of electrons in a given volume $V=AL$. The current, defined as the flow of charge q per unit time t , depends on the number of charge carriers, their density, and their drift velocity V_d . The current density j , a measure of the current per unit area, is directly proportional to the product of the charge carrier density and the drift velocity. This means that materials with a higher density of carriers n or faster drift speeds V_d will exhibit higher current densities.

$$q = (nV)e = (nAL)e$$

$$V_d = \frac{L}{t} \quad \text{or} \quad t = \frac{L}{V_d}$$

$$i = \frac{q}{t} = \frac{nALe}{L/V_d} = nAeV_d$$

$$V_d = \frac{i}{nAe} = \frac{j}{ne} \quad \left(\text{as } j = \frac{i}{A} \right)$$

$$j = neV_d$$

Drift velocity V_d , though central to understanding current flow, is surprisingly small. Electrons in a conductor move very slowly on average due to constant collisions. For instance, an electron in a wire may take hours or even days to travel a relatively short distance. Despite this slow individual movement, electricity appears to travel almost instantaneously. This phenomenon is

explained by the collective motion of electrons. When a switch is closed, electrons near the switch push those ahead of them, setting off a chain reaction that transmits energy across the entire circuit at nearly the speed of light.

In subsequent lectures, we will explore topics such as resistivity and conductivity, which depend on factors like the material's structure and the type of charge carriers present. While most discussions of conductivity focus on electrons, there are cases, such as in semiconductors, where positive charge carriers (holes) also contribute to current flow. This interplay of positive and negative carriers can be observed more closely when studying phenomena like the Hall effect, which provides insight into the behavior of holes as current carriers. These topics will be discussed in greater detail in future lectures.

LECTURE # 27_ THE MAGNETIC FIELD

Discovery of Magnets

The phenomenon of magnetism was first observed in ancient times. It is said that a shepherd named Magnus discovered stones that attracted pieces of iron. The term "magnet" is derived from his name. Initially, magnets were used for navigation in shipping, but today, they are integral to various technologies.

Applications of Magnetism

Magnets are present in various aspects of daily life and advanced technology. For instance:

- **Magnetic Levitation (Maglev) Trains:** These trains float above tracks, lifted by powerful magnetic forces, reducing friction.
- **Magnetic Resonance Imaging (MRI):** This technology allows detailed imaging of internal organs without physical contact, leveraging magnetic fields to interact with cellular structures.

Properties of Magnets

Magnetic Poles

Magnets have two poles: North and South. **Opposite poles attract, while like poles repel.** This interaction can be demonstrated using bar magnets or other magnetic materials.

Magnetic Fields

When iron filings are scattered around a magnet, they form patterns indicating the magnetic field's strength and direction. The concentration of filings is highest near the poles, where the field is strongest.

Differences Between Magnetism and Electrostatics

Magnetism differs fundamentally from electrostatics:

- Unlike electrostatics, where positive and negative charges exist, magnetostatics (study of stationary magnets) does not have isolated magnetic poles. Instead, every magnet always has a North and South Pole.
- In electrostatics, a stationary charge experiences a force in an electric field, whereas in magnetostatics or magnetism, a force acts on a moving charge in a magnetic field.

Force on a Charge in a Magnetic Field (Magnetic Force)

A stationary charge in a magnetic field experiences no force. When the charge moves, the force F is given by:

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Where q is the charge, $\vec{v}$ is the velocity of the charge, $\vec{B}$ is the magnetic field. $\vec{v} \times \vec{B}$ denotes the cross product, which ensures the force is perpendicular to both $\vec{v}$ and $\vec{B}$, as determined by the **right-hand rule** shown in figure 27.1.

The Right-Hand Rule

To determine the direction of the magnetic force, use the right-hand rule:

1. Forefinger (Pointer Finger): Point your forefinger in the direction of the velocity $\vec{v}$ of the positive charge.
2. Middle Finger: Point your middle finger in the direction of the magnetic field $\vec{B}$.
3. Thumb: Your thumb will point in the direction of the magnetic force $\vec{F}$ acting on the charge.

For a negative charge, like an electron, the force direction will be opposite to that indicated by your thumb.

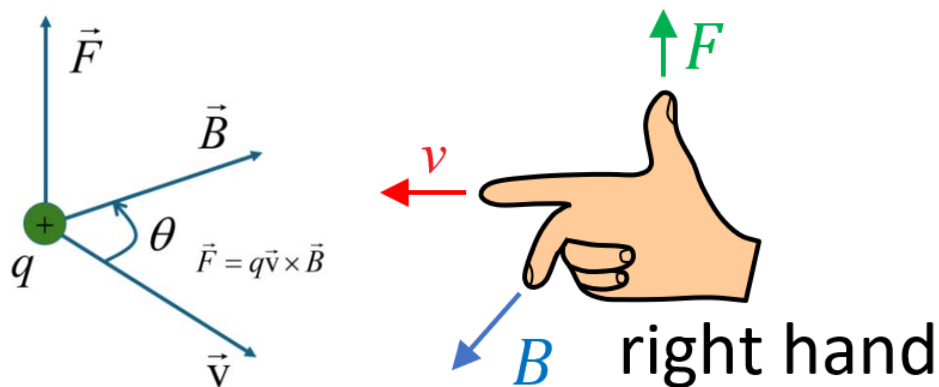


Figure 27. 1. Right hand rule to find the direction of force on a charged particle in a magnetic field.

Magnetic Units and Field Strength

Tesla and Gauss

The unit of magnetic field strength is the Tesla (T). One Tesla is defined as:

$$\vec{F} = q\vec{v} \times \vec{B}$$

$$F = qvB \sin \phi$$

$$B = \frac{F}{qv} \quad (\text{if } \phi = \frac{\pi}{2})$$

$$1 \text{ tesla} = 1 \frac{\text{newton}}{\text{coulomb} \cdot \text{meter/second}}$$

$$1 \text{ tesla} = 1 \frac{\text{newton}}{\text{ampere} \cdot \text{meter}} = 10^4 \text{ gauss}$$

Examples of Magnetic Field Strengths

- | | |
|---------------------------|---------------------|
| 1. Earth's surface | 10^{-4} T |
| 2. Bar magnet | 10^{-2} T |
| 3. Powerful electromagnet | 1 T |
| 4. Superconducting magnet | 5 T |

The Lorentz Force

When both electric and magnetic fields are present, the total force on a charge is the Lorentz Force:

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B}) = q\vec{E} + q\vec{v} \times \vec{B}$$

Here E is the electric field and $\vec{v} \times \vec{B}$ is the magnetic component.

This relationship is pivotal in understanding the interplay of electric and magnetic forces, which can act in different directions.

Magnetic and Electric Forces in Opposing Directions

Consider a scenario where magnetic and electric forces are aligned in opposite directions such that they cancel each other. This setup can be visualized with two parallel plates: one negatively charged and the other positively charged. A positively charged particle placed between the plates experiences a downward electric force due to the electric field.

If a positively charged particle enters the region between these plates from one side, it is pushed downward by the electric field. Now, suppose a magnetic field is introduced perpendicular to the plane of motion. The magnetic field can exert an upward force on the particle, opposing the electric force. By adjusting the potential difference between the plates, the electric field can be fine-tuned so that the electric and magnetic forces exactly cancel each other.

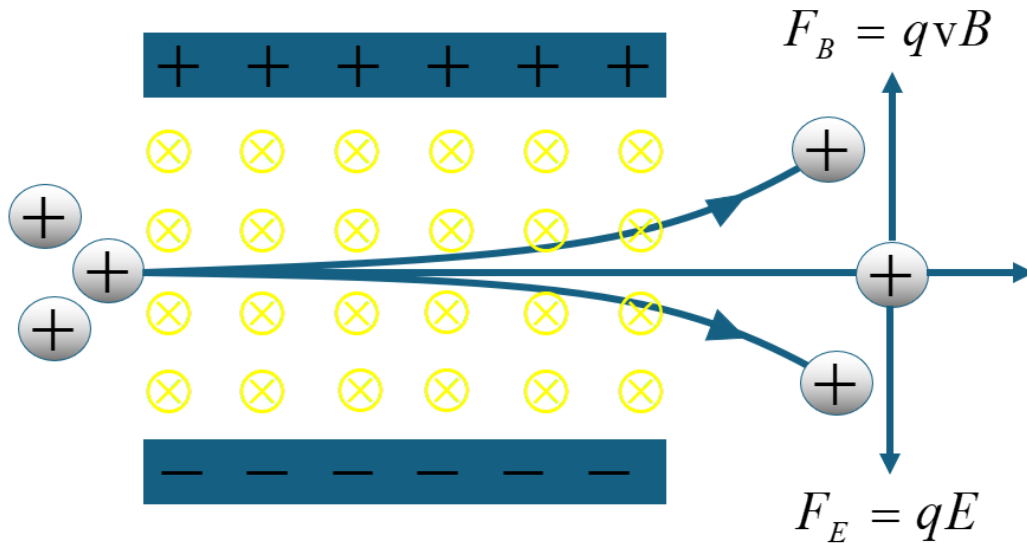


Figure 27. 2. Working mechanism of velocity selector.

The condition for this balance can be mathematically expressed as:

$$qE = qvB \Rightarrow v = \frac{E}{B} \quad \therefore \vec{v} \times \vec{B} = vB \sin \theta = vB \sin 90 = vB$$

This arrangement acts as a **velocity selector**, allowing particles with a specific velocity to pass through undeflected (see figure 27.2).

Work Done by Electric and Magnetic Fields

- The electric field does work on a charged particle by exerting a force that moves the particle along the direction of the field.
- In contrast, the magnetic field does not perform work on a charged particle. The force exerted by a magnetic field is always perpendicular to the velocity of the particle, resulting in no displacement in the direction of the force. Hence, there is no work done by the magnetic force.

Mathematically:

$$\begin{aligned}
 dW &= \vec{F} \cdot d\vec{r} = q(\vec{v} \times \vec{B}) \cdot d\vec{r} & \therefore \text{putting the value of } \vec{F} \\
 &= q(\vec{v} \times \vec{B}) \cdot \frac{d\vec{r}}{dt} dt \\
 &= q(\vec{v} \times \vec{B}) \cdot \vec{v} dt & \therefore \frac{d\vec{r}}{dt} = \vec{v} \\
 &= 0
 \end{aligned}$$

This is because $\vec{v} \times \vec{B}$ is perpendicular to $\vec{v}$, and the dot product of perpendicular vectors is zero.

Motion of Charged Particles in Magnetic Fields

When a charged particle enters a magnetic field, it experiences a force that changes its direction but not its speed. This results in circular motion, with the **magnetic force acting as the centripetal force**. The radius r of this circular path is determined by:

$$qvB = m \frac{v^2}{r}$$

$$r = \frac{mv}{qB}$$

$$\omega = \frac{v}{r} = \frac{qB}{m}$$

where m is the mass of the particle, v is its velocity, q is its charge, and B is the magnetic field strength. *Larger values of m or v result in a larger radius, while stronger magnetic fields (higher B) lead to a smaller radius.*

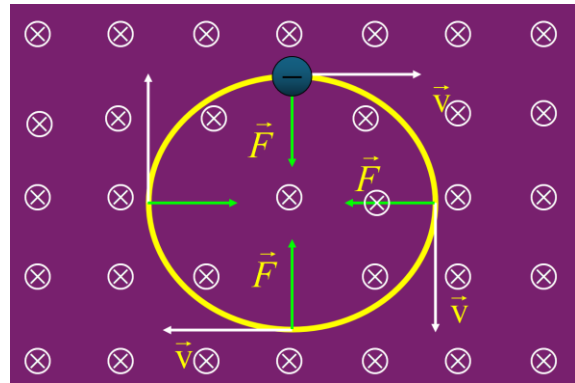


Figure 27. 3. Circulating charges in a magnetic field.

Applications of Magnetic Fields

Mass Spectrometer

A mass spectrometer is a device that uses the principles of electric and magnetic fields to measure the mass of atoms. In a mass spectrometer:

1. Charged particles are first passed through a velocity selector, ensuring they have a specific velocity.
2. These particles enter a region with a magnetic field, where they follow circular paths as shown in figure 27.4.
3. The radius of the path depends on the particle's mass, velocity, charge, and the magnetic field strength. As the particles enter the magnetic field, those with a higher mass-to-charge ratio will have a larger radius of curvature and will be deflected less. Conversely, lighter particles will be deflected more and follow a smaller radius.

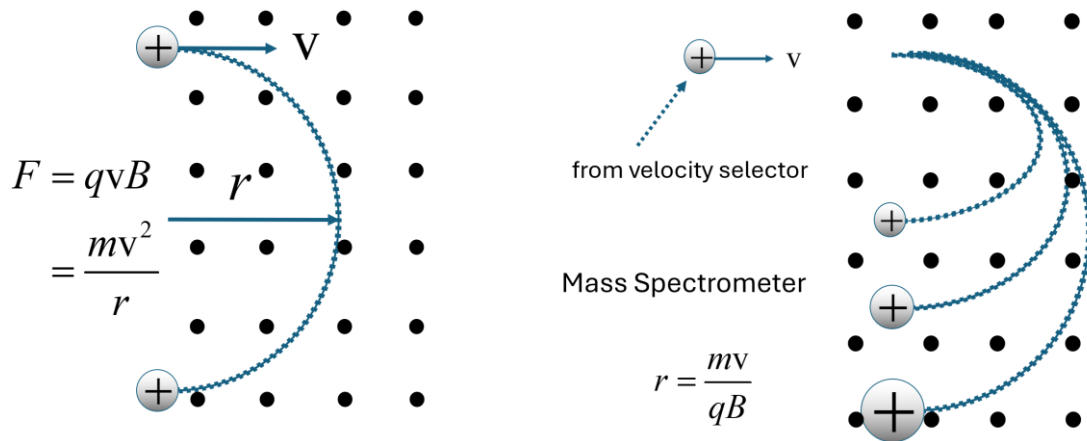


Figure 27. 4. Working mechanism of mass spectrometer.

By measuring the radius of the path, the mass of the particle can be calculated. This technique separates particles like lithium, sodium, oxygen, and lead, enabling precise mass measurements.

The Protective Role of Earth's Magnetic Field

The Earth's magnetic field plays a crucial role in protecting life. Charged particles from the Sun, such as protons and alpha particles, are deflected by the Earth's magnetic field. Without this deflection, these high-energy particles would collide with the Earth, causing significant damage.

Particle Accelerators

Magnetic fields are also used in particle accelerators to study the fundamental properties of matter. In these accelerators, charged particles are made to travel in circular paths within magnetic fields. With each cycle, their energy is increased until they reach speeds close to the speed of light. By colliding these high-energy particles, scientists can investigate the basic constituents of nature. Facilities like CERN in Geneva utilize this principle to conduct cutting-edge research in physics.

Magnetic Force on Current-Carrying Conductors

When many charges move together, as in an electric current, the collective force can be calculated. The motion of electrons within the conductor is responsible for the current, and their interactions with the magnetic field result in the observed force.

When electrons move through a wire, they do not move at a uniform velocity but rather exhibit an average velocity known as **drift velocity** $\vec{v}_d$. This drift velocity is used to calculate the force on a current-carrying wire in a magnetic field.

$$\vec{F} = Ne\vec{v}_d \times \vec{B}$$

The **number density of electrons** in the wire is crucial for this calculation, which is the number of electrons per unit volume ($n = N / V$). The total number of electrons in the wire can be determined by multiplying this number density by the volume of the wire, which is the cross-sectional area multiplied by the length ($N = nV = nAL$).

$$\begin{aligned}\vec{F} &= nALe\vec{v}_d \times \vec{B} \\ nALe\vec{v}_d &= \vec{I} \\ \vec{F} &= \vec{I} \times \vec{B}\end{aligned}$$

This shows that the force increases as the current and magnetic field strength increase.

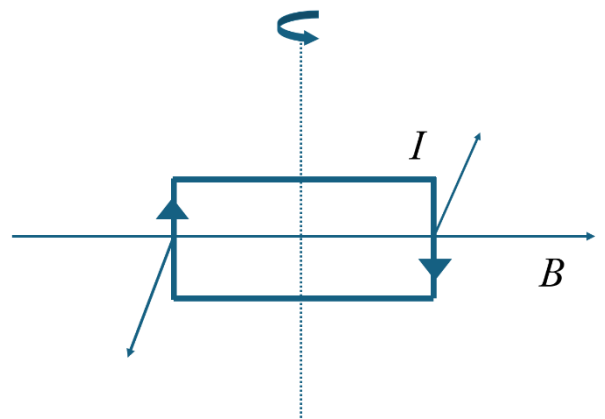
When a straight wire carrying current is placed inside a magnetic field, a force is exerted on it. The direction of the force is given by the **cross product** of the current and magnetic field vectors. If the wire is in the shape of a rectangular loop, with current flowing through it, the force acts differently on various parts of the loop:

- The part of the wire that is parallel to the magnetic field experiences no force, as the cross product of the current and magnetic field vectors results in zero force.
- The parts of the wire that are perpendicular to the magnetic field experience forces that act inward or outward, creating a **torque** that causes the loop to rotate.

$$\begin{aligned}\tau &= F \frac{a}{2} + F \frac{a}{2} \\ &= IBb \frac{a}{2} + IBb \frac{a}{2} \\ &= I \times (\text{area of loop}) \times B \\ &= \mu B\end{aligned}$$

$\mu = IA$ is called the MAGNETIC MOMENT

Magnetic Moment



The **magnetic moment** of a current loop is a vector quantity, and it represents the strength and orientation of the magnetic effect generated by the current. The direction of the magnetic moment is perpendicular to the plane of the loop, and its magnitude is given by the product of the current and the area of the loop.

When a magnetic moment is placed in an external magnetic field, it experiences a torque that tries to align it with the magnetic field. The energy of the system is minimized when the magnetic moment is aligned with the magnetic field. If the magnetic moment is reversed, its energy is maximized.

The behaviour of magnetic moments is critical in various fields such as atomic physics, nuclear physics, and material science, where currents circulate at the atomic or molecular level. The smallest magnetic moment is associated with an electron, which can be considered as the smallest magnet. The electron's magnetic moment arises due to its charge and its rotational motion.

Ampere's Law

Ampere's Law describes the relationship between a current and the magnetic field it produces. Ampere's Law states that the integral of the magnetic field around a closed loop is proportional to the total current enclosed by the loop. The mathematical expression for Ampere's Law is:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{\text{enclosed}}$$

Where B is the magnetic field, ds is a differential element of the area around the loop, I_{enc} is the enclosed current, μ_0 is the permeability of free space.

Ampere's Law is often applied to a hypothetical loop, called an **Amperian loop**, that encloses a current. By integrating the magnetic field along the loop and calculating the enclosed current, we can determine the magnetic field in the region surrounding the current. In the loop shown in figure 27.5, total current $I = i_1 + i_2$. Here i_3 is excluded as it lies outside.

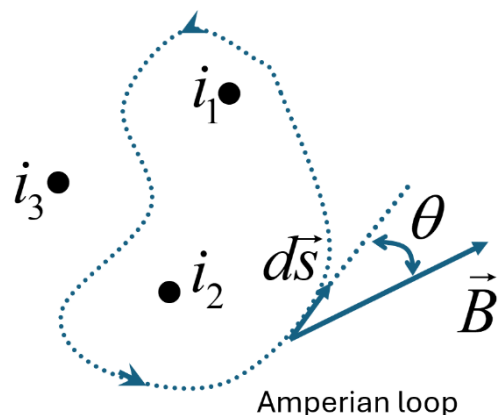


Figure 27. 5. Amperian loop

Application of Ampere's Law

The law is particularly useful for calculating the magnetic field in cases of symmetry, such as around straight wires or circular current loops.

Magnetic Field Around an Infinite Straight Wire

Consider an infinitely long straight wire carrying current I as shown in figure 27.6. To calculate the magnetic field at a point around the wire, we can apply Ampere's Law. The magnetic field produced by the current flows in circular loops around the wire, and at every point on the loop, the magnetic field is tangent to the circle. To calculate the magnetic field at a distance r from the wire, we apply **Ampere's Law**.

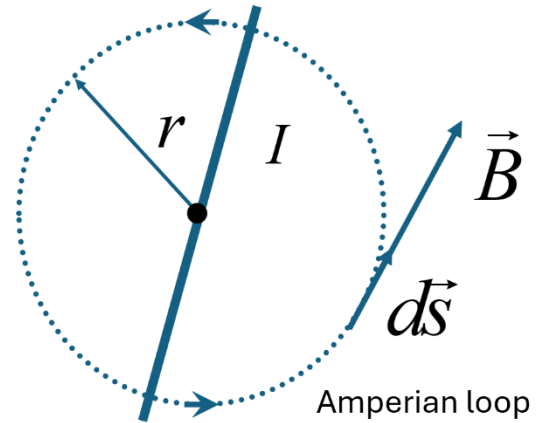


Figure 27. 6. Amperian loop around an infinite straight wire.

Ampere's Law states that the integral of the magnetic field along the loop is equal to the current enclosed by the loop. Mathematically,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{\text{enclosed}} \quad \text{.....(1)}$$

Since the magnetic field and the differential area element ds are parallel, we can write:

$$\begin{aligned} \oint \vec{B} \cdot d\vec{s} &= \int B ds \cos \theta = B \int ds & \therefore \cos \theta &= 1 \\ \oint \vec{B} \cdot d\vec{s} &= B(2\pi r) & \text{.....(2)} & \therefore \int ds = \text{total length of loop} = \text{circumference of circle} \end{aligned}$$

where r is the distance from the wire. Substituting eq (2) in (1), and solving for B , we get:

$$B = \frac{\mu_0 I}{2\pi r}$$

This result shows that the magnetic field strength is directly proportional to the current and inversely proportional to the distance from the wire. As the distance increases, the magnetic field strength decreases.

Magnetic Field Inside a Solenoid

Consider a solenoid, which is a coil of wire with many turns. If current flows through this coil, a magnetic field is generated inside it. The magnetic field inside a solenoid is approximately uniform (see figure 27.7), and it is directed along the axis of the solenoid. To calculate the magnetic field, we use Ampere's Law again. We draw an **Amperian loop** abcd inside the solenoid.

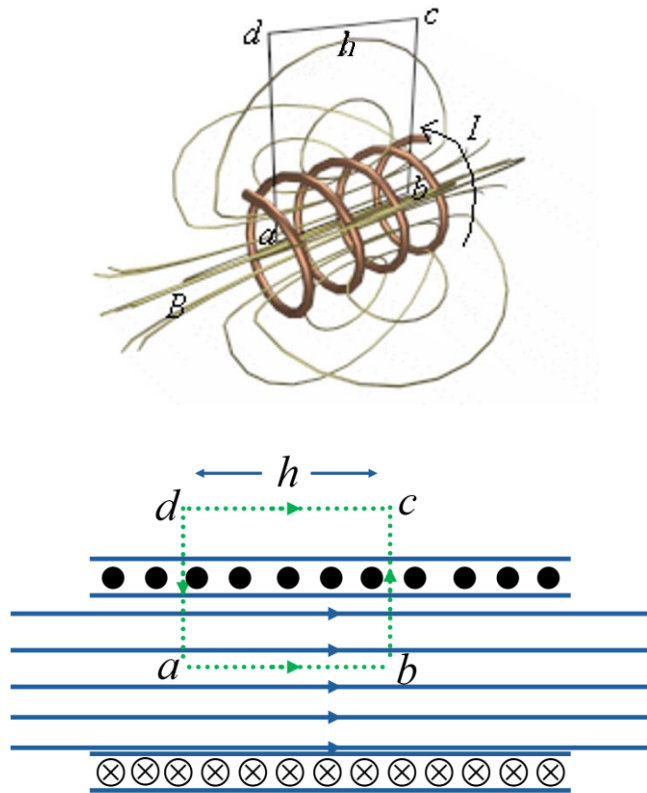


Figure 27. 7. Uniform magnetic field inside a solenoid.

Inside the solenoid, along path ab, the magnetic field is uniform and parallel to the path of integration. The other three sides are outside or perpendicular to the solenoid, where the magnetic field is negligible or perpendicular to the path ($\cos 90 = 0$). Thus, for a solenoid with n turns per unit length and a current I , the magnetic field inside the solenoid is given by:

$$\int \vec{B} \cdot d\vec{s} = \int_a^b \vec{B} \cdot d\vec{s} + \int_b^c \vec{B} \cdot d\vec{s} + \int_c^d \vec{B} \cdot d\vec{s} + \int_d^a \vec{B} \cdot d\vec{s}$$

$$\int \vec{B} \cdot d\vec{s} = \int_a^b \vec{B} \cdot d\vec{s} + 0 + 0 + 0$$

$$I = I_0 n h$$

Applying Ampere's law

$$B \cancel{h} = \mu_0 I_0 n \cancel{h}$$

$$B = \mu_0 I_0 n$$

This equation shows that the magnetic field inside the solenoid is directly proportional to the number of turns per unit length and the current.

Magnetic Field Inside an Infinite Wire with Finite Thickness

Consider a thick, infinite wire and calculate the magnetic field inside it. Assume the current density is uniform across the cross-section of the wire, meaning it is the same everywhere within the wire's radius R . The total current I is related to the current density J by:

$$I = J.A_{total}$$

$$J = \frac{I}{A_{total}} = \frac{I}{\pi R^2}$$

Draw an **Amperian loop** inside the wire as shown in figure 27.8. The magnetic field at a distance r inside the wire depends on the amount of current enclosed within that radius. For a radius r (where $r < R$), the area of the cross-section enclosed by the Amperian loop is:

$$A_{enclosed} = \pi r^2$$

The current enclosed within this smaller cross-section is proportional to the enclosed area, so:

$$I_{enc} = J.A_{enclosed}$$

Substitute $J = \frac{I}{\pi R^2}$ and $A_{enclosed} = \pi r^2$

$$I_{enc} = \frac{I}{\pi R^2} \cdot \pi r^2 = I \left(\frac{\pi r^2}{\pi R^2} \right)$$

Applying Ampere's law to find the magnetic field at a distance r from the centre of the wire and substituting the value of I_{enc} from the above equation we get:

$$B(2\pi r) = \mu_0 I \left(\frac{\pi r^2}{\pi R^2} \right)$$

$$B = \frac{\mu_0 I r}{2\pi R^2}$$

This result shows that the magnetic field inside the wire increases linearly with the distance r from the centre. At the centre of the wire, the magnetic field is zero, and it increases as we move outward, reaching a maximum when $r = R$ (see graph in fig 27.9).

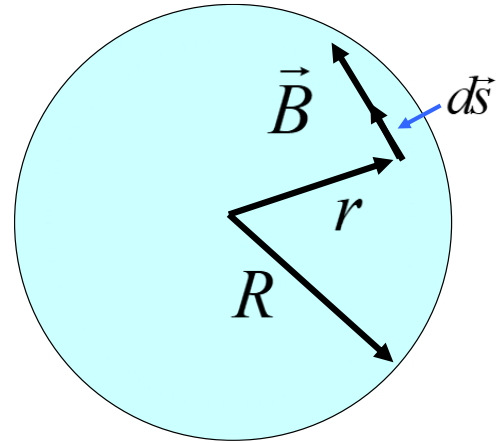


Figure 27. 8. Amperian loop inside an infinite wire with finite thickness

Magnetic Field Outside the Wire

For a point outside the wire, the magnetic field decreases with the distance from the wire (figure 27.9).

At a distance r from the centre of the wire (where $r > R$), the magnetic field is calculated using the total current I flowing through the wire, as no further current is enclosed in the loop.

Applying Ampere's Law for $r > R$:

$$B \cdot 2\pi r = \mu_0 I$$

Solving for B , we get:

$$B = \frac{\mu_0 I}{2\pi r}$$

This is the same formula as for the infinite straight wire, which shows that the magnetic field outside the wire decreases inversely with the distance from the wire.

Magnitude of Magnetic Fields

The magnetic field produced by a current-carrying wire depends on the geometry of the wire:

- For an **infinite straight wire**, the magnetic field decreases inversely with the distance from the wire.
- For a **solenoid**, the magnetic field is uniform inside the solenoid and proportional to the number of turns per unit length and the current.
- For a **thick wire**, the magnetic field inside increases linearly with the distance from the centre, while outside the wire, the field decreases inversely with the distance.

Electron as a Magnet

Electrons themselves behave as tiny magnets due to their rotational motion (spin), and this is the basis for the smallest magnetic moment in nature. On the other hand, objects like **pulsars**, which rotate rapidly, generate extremely powerful magnetic fields.

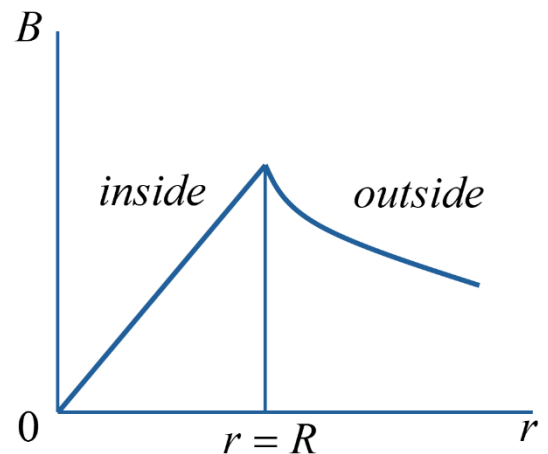


Figure 27. 9. Graph showing the variation of magnetic field B with radius r inside and outside the thick wire.

Lecture 28

Electromagnetic Induction

Introduction to Electromagnetism

When an electric current flows through a conductor, a magnetic field is created around it. Similarly, when a charge moves, it generates electricity and a magnetic field simultaneously. This interconnection between electricity and magnetism forms the basis of electromagnetism. Now, an important question arises: Can electricity be generated from a magnetic field?

The answer is yes, and this principle forms the foundation for devices like generators. Without this understanding, it would have been impossible to create generators or electrical appliances. Hence, this relationship is not just a scientific curiosity but a cornerstone of modern life.

Basic Properties of Magnets

In our previous lectures, we discussed magnets and their poles. Magnets have two poles: north (Shumaal) and south (Junoob). Opposite poles attract each other, while like poles repel. If you examine a magnet closely, you'll observe that its internal structure is arranged in a specific order: north-south, north-south, north-south. This regular alignment generates the magnet's overall magnetic field.

In non-magnetic materials, however, the tiny magnetic domains are disordered, cancelling each other out and resulting in no net magnetism. For instance, in a piece of iron that isn't magnetized, the magnetic domains are scattered in various directions.

Magnetic Lines of Force

The magnetic field around a magnet is represented by lines of force. These lines originate from the north pole, curve around the magnet, and re-enter at the south pole, forming closed loops. Unlike the electric field lines in electrostatics, magnetic field lines have no starting or ending points. This is because there is no equivalent of a "magnetic charge" (like electric charges). Magnetic field lines circulate continuously, without a source or sink. This fundamental difference between electricity and magnetism is a key concept in electromagnetism.

Properties of Magnetic Lines of Force

1. **Closed Loops:** Magnetic lines of force form continuous loops without a source or endpoint.
2. **Non-Intersecting Lines:** Magnetic lines of force never cross each other. To understand why, imagine holding a compass near a magnet. As you move the compass along the magnetic field, its needle always points in one direction, indicating the local direction of the magnetic field. If two lines of force were to intersect, the compass needle would point in two different directions at the same point, which is impossible.

Definition of Magnetic Flux

Magnetic flux is defined as the product of the perpendicular magnetic field and the area through which it passes. This definition holds when the magnetic field is constant and the area is fixed, with the magnetic field being perpendicular to the surface.

$$\phi_B = B.A = BA \cos\theta$$

Where,

- Φ_B is the magnetic flux.
- B is the magnetic field
- A is the area
- θ the angle at which the field lines pass through the given surface area

Magnetic Flux Unit

Magnetic flux is usually measured with a flux meter. The SI and CGS unit of magnetic flux is given below:

- The SI unit of magnetic flux is **Weber (Wb)**.
- The fundamental unit is **Volt-seconds**.
- The CGS unit is **Maxwell**.

Concept of Flux through a Small Area

Consider a small area, denoted as dA , through which a magnetic field passes. If the magnetic field is perpendicular (normal) to this area, the flux through this small piece is the magnetic field B multiplied by the area dA .

Summing Contributions for Total Flux

To calculate the total flux through a surface, the area can be divided into smaller pieces. The flux through each surface element is the magnetic field perpendicular to the surface element multiplied by its area. The total flux is the sum of these contributions, represented by the formula:

$$\phi_B = B_1.dA_1 + B_2.dA_2 + B_3.dA_3 = \sum B_i.dA_i$$

where θ is the angle between the magnetic field and the normal to the surface. When the field is perpendicular, $\theta = 0$, so the flux is simply $B.dA$. If the field is slanted, θ increases, and the flux reduces. When the angle reaches 90° , the flux is zero. The net flux on a closed surface is always zero.

Differences Between Electrostatics and Magnetism

In electrostatics, a closed surface can have a net flux due to the charge enclosed within it, according to Gauss's Law. However, in magnetism, no net flux can emerge from a closed surface, as there are no magnetic charges. This is why a sphere around a current-carrying wire does not have any magnetic flux passing through it. The net flux on a closed surface is always zero.

$$\phi_B = \int \vec{B}.d\vec{A} = 0$$

A simple experiment, conducted by Michael Faraday around 300 years ago, illustrates the relationship between electricity and magnetism. The experiment involves a coil with wire wrapped around it, connected to a meter to measure electromotive force (voltage). When a magnet is moved through the coil, the meter registers a voltage. This experiment demonstrates the fundamental connection between electricity and magnetism.

Demonstrating Electromagnetic Induction

In this experiment, when a magnet is moved through the coil, an EMF is generated, causing the meter's needle to move. As the magnet moves, the voltage is generated, appearing and disappearing in cycles. This effect occurs because the changing magnetic field induces a current in the coil.

The same effect can be achieved by keeping the magnet stationary and moving the coil instead. This experiment, shown in animation, demonstrates that the movement of a magnet through the coil induces electricity, generating a current. This process is known as **induced current**, and it occurs due to the change in the magnetic field.

Electromotive Force (EMF)

Induced current arises from a change in magnetic flux. Although the term "electromotive force" suggests a force, it is not a force in the traditional sense. Rather, it refers to the energy required to move a unit charge through a circuit. The units of EMF are energy, not force.

Faraday's Law of Induction

Faraday's Law explains that the induced electromotive force is proportional to the rate of change of the magnetic flux:

$$\mathcal{E} = -\frac{d\phi_{\mathbf{B}}}{dt}$$

This means that when the magnetic flux changes, EMF is generated. The faster the change in flux, the greater the induced EMF. The negative sign in the equation indicates the direction of the induced current, which is explained by Lenz's Law.

Methods of Changing Magnetic Flux

Three ways to change the magnetic flux and induce EMF are:

1. **Movement of the Magnet:** When the magnet moves through the coil, the flux changes due to the movement of the magnetic field lines.

2. **Changing the Area of the Coil:** If the area of the coil changes (e.g., it becomes larger or smaller), the flux changes, even if the magnet remains fixed.
3. **Rotation of the Magnet:** If the magnet rotates, the flux through the coil changes, even if the magnet does not move closer or farther from the coil.

These three scenarios all lead to a change in flux, inducing EMF in the coil.

Practical Applications

The principle of electromagnetic induction is fundamental to modern technology. Electric generators operate by rotating coils within magnetic fields to produce electricity. Transformers, electric motors, and wireless charging systems are all based on similar principles.

Problem 1: Calculating Induced EMF in a Collapsing Loop

Scenario:

A functional loop of radius **12 cm** is in a uniform magnetic field of strength **0.15 T**. The loop is initially open but is closed at points A and B, collapsing the area to zero within **0.20 seconds**. We need to calculate the induced EMF.

Solution:

Initial Magnetic Flux ($\phi_{initial}$):

The magnetic flux is the product of the magnetic field (B) and the area (A):

$$\phi = B \times A = B \times (\pi r^2)$$

$$\text{Substituting } B = 0.15 \text{ T and } r = 0.12 \text{ m:}$$

$$A = \pi (0.12)^2 \approx 0.0452 \text{ m}^2$$

$$\phi_{initial} = 0.15 \times 0.0452 \approx 0.00678 \text{ Wb (Weber)}$$

Final Magnetic Flux (ϕ_{final}):

When the loop is closed, the area becomes zero. Hence:

$$\phi_{final} = 0$$

Rate of Change of Flux:

The change in flux is given by:

$$\Delta\phi = \phi_{\text{final}} - \phi_{\text{initial}}$$

Substituting the values :

$$\Delta\phi = 0 - 0.00678 = -0.00678 \text{ Wb}$$

Over a time interval $\Delta t = 0.20 \text{ s}$, the rate of change of flux is :

$$\Delta\phi / \Delta t = -0.00678 / 0.20 \approx -0.034 \text{ Wb/s}$$

Induced EMF:

Using Faraday's law:

$$EMF = -\Delta\phi / \Delta t = 0.0339 \text{ V}$$

Thus, the induced EMF is approximately 0.034 V .

Problem 2: EMF with Changing Magnetic Field Strength**Scenario:**

A circular wire loop with a radius of **0.3 m** is placed in a magnetic field. Initially, the field strength is **0.3 T**, perpendicular to the loop. The field changes to **-0.2 T** in **1.5 seconds**. Find the induced EMF.

Solution:**Initial and Final Flux:**

$$\phi = B \cdot A$$

$$A = \pi r^2 = \pi (0.3)^2 \approx 0.2827 \text{ m}^2$$

$$\phi_{\text{initial}} = 0.3 \cdot 0.2827 \approx 0.0848 \text{ Wb}$$

$$\phi_{\text{final}} = -0.2 \cdot 0.2827 \approx -0.0565 \text{ Wb}$$

Change in Flux:

$$\Delta\phi = \phi_{\text{final}} - \phi_{\text{initial}} = -0.0565 - 0.0848 = -0.1413 \text{ Wb}$$

Induced EMF:

$$EMF = -\Delta\phi / \Delta t = -(-0.1413) / 1.5 \approx 0.0942 \text{ V}$$

The induced EMF is approximately 0.095 V.

Flux as a Function of Time

In the graph below, the flux is at its maximum at time $t = 0$, then decreases to zero after about one second, reverses to become negative, and continues to oscillate between positive and negative values. This periodic behavior repeats over time.

Identifying Positive Induced Electromotive Force (EMF)

In the graph, there are three intervals: A, B, and C. The question is: in which interval is the induced electromotive force (EMF) positive?

- In **interval B**, the flux does not change, so the rate of change of flux is zero, meaning the induced EMF is also zero.
- In **interval C**, the flux is increasing, which means the induced EMF is at its maximum and negative.
- In **interval A**, the flux is decreasing. Since the rate of change of flux is negative, the induced EMF is positive.

Thus, the induced EMF is positive in **interval A**.

Induced Electric Field

Electromotive force (EMF) is related to the electric field by the equation:

$$\mathcal{E} = \int \vec{E} \cdot d\vec{s} = -\frac{d\phi_B}{dt}$$

Faraday's Law states that the induced EMF is the negative rate of change of magnetic flux. The question arises: what is the induced electric field? The electric field is responsible for driving the current, and the induced EMF is the result of the changing magnetic flux.

Example: Moving Rod in a Magnetic Field

Consider a scenario where a metal rod is placed between two parallel conducting rails, forming a complete circuit. As the rod moves, the magnetic flux changes. This causes an induced EMF, which drives current through the circuit. When the current flows, it lights up a bulb.

In this case, the change in flux occurs due to the rod's motion. The induced EMF is proportional to the rate at which the flux changes, which depends on the speed of the rod and the area of the circuit.

Electromagnetic Induction and Current Flow

When a rod moves through a magnetic field, the area of the rod relative to the magnetic field changes. According to Faraday's Law, this change in area results in a change in magnetic flux, which induces an electromotive force (EMF). This EMF pushes the electrons inside the wire, causing current to flow and light up a bulb. The question now arises: How much current is induced, and how can it be calculated?

Using Faraday's Law

To calculate the induced current, we first calculate the flux and then determine how fast it changes. In a typical setup, a loop is being pulled through a magnetic field with speed v . The magnetic field is confined to a region, and as the loop moves, it cuts the magnetic flux lines, resulting in a change in flux.

The induced EMF and the resulting current can be calculated using the formula derived from Faraday's Law:

$$\mathcal{E} = -\frac{d\phi_B}{dt} = -\frac{d(BDx)}{dt} = BDv$$

where:

- ϕ_B is the magnetic flux
- B is the magnetic field strength
- D is the width of the loop
- v is the velocity at which the loop moves

The current I induced in the loop is then given by:

$$I = \frac{\varepsilon}{R} = \frac{BDv}{R}$$

Force and Power Dissipation

The force acting on the loop due to the induced current is calculated by:

$$\vec{F} = I\vec{L} \times \vec{B}$$

which, for a rectangular loop, results in:

$$F_1 = IBD = \frac{B^2 D^2 v}{R^2}$$

The power dissipated due to this force is:

$$P = F_1 v = \frac{B^2 D^2 v^2}{R^2}$$

Alternatively, the power dissipated can also be expressed as:

$$P = I^2 R = \left(\frac{BDv}{R} \right)^2 R$$

where both expressions yield the same result.

Energy Conversion

The mechanical energy used to pull the loop is converted into electrical energy due to the induced EMF in the magnetic field. This process demonstrates the relationship between magnetic fields, induced currents, and energy dissipation, as governed by Faraday's Law of Induction.

Applications of Magnetic Induction

Magnetic induction has numerous applications in the world around us. Let's explore a few examples through an experiment and practical devices.

Experiment with Two Coils

In the experiment, two coils are involved: a primary coil and a secondary coil. When current flows through the primary coil, a magnetic field is generated, which causes a change in the magnetic flux. This change induces an electromotive force (EMF) in the secondary coil. The EMF is measured across the coil using a meter, demonstrating how a changing flux induces an EMF, as stated in Faraday's Law.

Telephone Cards with Magnetic Stripes

A practical example of magnetic induction is the magnetic stripe on telephone cards. The magnetization of the stripe varies across different areas. As the card moves through a reader, the changing magnetic flux induces an EMF in the reader, which reads the information encoded on the card, such as call duration or remaining balance.

VHS Cassettes and Magnetic Tapes

Another example is the VHS cassette, which contains information recorded on a magnetic tape. The playback head of a VCR detects changes in the magnetic flux as it moves over the tape, inducing an EMF. This allows the VCR to retrieve the recorded information, such as video and audio signals.

Magnetic Induction in Rotating Magnets

In a rotating device, such as a motor or generator, a rotating magnet causes changes in the magnetic flux. This induces an EMF, which powers the device. A battery inside the device supplies current to a coil, and the rotation of the magnet continuously changes the flux, inducing a current. This process does not violate conservation of energy, as energy is supplied by the battery.

These examples illustrate the widespread use of magnetic induction in daily life, from telephone cards to VCRs and motors. The key principle in all of these applications is Faraday's

Law, which states that a change in magnetic flux induces an electromotive force, powering various devices.

Lenz's Law and the Direction of Induced Current

After understanding the applications of Faraday's Law, one crucial question remains: in which direction will the induced current flow? This was first addressed by Lenz's Law, which states that:

"Whenever a current is induced, its magnetic field opposes the agent that caused the induction."

In other words, the direction of the induced magnetic field always works against the cause of its induction.

Example of Lenz's Law

In the following animation, you can observe the behaviour of a magnet as it moves towards a coil. When the magnet approaches the coil, a current is induced, and the magnetic field generated by the current opposes the approach of the magnet. When the magnet moves in the opposite direction, the current reverses, and the induced magnetic field opposes this motion as well.

Ring Example with a Magnet

Consider a ring made of wire. When a magnet is brought closer to the ring, an induced current flows, creating a magnetic field. As the magnet gets closer, the magnetic field generated by the induced current increases, making it more difficult to bring the magnet near the ring. This opposing force arises due to the induced current's magnetic field.

Superconducting Wire and High Induced Currents

If the ring were made of a superconducting material, there would be no resistance to the current. Even a small induced electromotive force would produce a very high current, creating a powerful magnetic field that strongly resists the magnet's approach. This shows that Lenz's Law not only governs the direction of induced currents but also explains energy conservation—the induced field always works to prevent further changes.

Energy Conservation and Lenz's Law

The direction of the induced current is governed by Lenz's Law, which ensures energy conservation. The magnetic field generated by the induced current always opposes the change in the magnetic flux that caused it. This principle can be observed in various practical situations, highlighting the fundamental nature of Lenz's Law.

Eddy Currents and Energy Loss

In some cases, induced currents result in energy loss rather than benefits. For instance, in a motor, when the armature rotates, induced currents generate heat in the resistance, leading to energy dissipation. This is referred to as **eddy currents**. Eddy currents are also generated when magnets induce currents in conductors, resulting in energy loss.

Example of Eddy Currents in Motion

Consider a system with two magnets. If one magnet rotates within a conductor, an electromotive force (EMF) is induced. This induced current causes energy loss in the form of heat, as the conductor resists the flow of current. This energy loss due to the induced currents is an example of eddy currents.

Practical Uses of Eddy Currents

While eddy currents often result in unwanted energy loss, they can also be harnessed for useful applications. For example, when detecting metals underground, eddy currents can help identify the presence of metal objects. A magnetic field is applied, and the resulting current is measured to determine if the material is metal. Eddy currents help distinguish between metal and non-metallic substances like wood or stone.

Eddy Currents in Metal Detection

In a mine detector, a pulse of current generates a magnetic field. As this magnetic field changes, an induced current is created in nearby conductive materials, such as metal. According to Faraday's Law, the rate of change in the magnetic flux induces a current. The induced current generates a magnetic field that opposes the applied magnetic field, causing a decrease in the total magnetic field. This difference allows for the detection of metal objects underground.

The Relationship Between Electricity and Magnetism

This lecture highlights the deep and fundamental relationship between magnetic fields and electricity. A changing magnetic field induces a current, while a steady current generates a magnetic field. Understanding these interactions is crucial for various technological applications, from metal detection to electrical engineering.

The Role of Steady Currents

A steady current in a wire generates a magnetic field, but this magnetic field does not affect a stationary charge. This is due to the fact that the magnetic force on a stationary charge is zero, as the velocity vector of the charge is zero. Therefore, a magnetic field alone does not exert force on a charge at rest.

A Thought-Provoking Question

At this point, a thought-provoking question arises: Imagine a wire with positive and negative charges moving through it. If you are standing next to the wire, when the current starts to flow, you would not feel the magnetic field since the charge in your hand is at rest. However, if you were moving with the charge, you would observe the magnetic force acting on you, leading to a contradiction.

The Relativity of Electric and Magnetic Fields

This paradox is a serious question that was first addressed by Einstein. The magnetic field appears differently depending on the frame of reference. Similarly, the electric field also transforms between frames of reference. This transformation of electric and magnetic fields is a key topic in the study of Relativity. While the question will not be answered here, it is worth exploring when studying the theory of relativity in detail.

Lecture 29

Alternating Current

Introduction:

You must have opened or closed a switch many times and whenever you open or close a switch, the circuit is complete. Current flows in this circuit and generally in our homes and in factories, this current is alternating current or AC. The theme of today's lecture is Alternating Current and Alternating Current Circuits.

Alternating current is very useful for us because we can produce it very easily but what I mean by AC current is that it is the current that first goes into one direction then into the other direction and it keeps oscillating or alternating, so if you Look at the electron inside of a wire, sometimes it goes in one direction and sometimes it goes in another direction but on the average, it stays at one place.

First of all, it comes to the question of how AC can be produced.

In the last lecture, we discussed about the Faraday's Law of Induction, based on which “if a magnetic field or say a magnetic flux gets changed, then as a result, an electromotive force is generated”. A force is generated which is called EMF and EMF is equal to minus the rate of change of flux ($EMF = -d\phi/dt$). Using this principle, all the generators in the world produce AC or any current, though it is a big generator of tarbela dam or a small electric motor or a motor or a generator which is attached to your bicycle, which is called a dynamo. The basic principle of all this is that electromotive force is induced by changing magnetic fields or a magnetic flux.

Let us calculate if a coil is inside a magnetic field. If it rotates, how much electromotive force is produced by it?

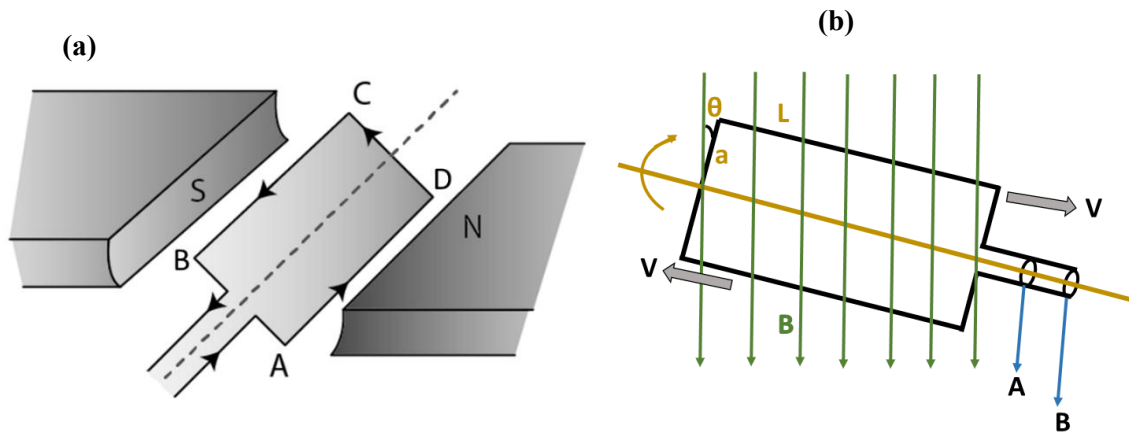


Fig. 29.1: A simple AC generator (a) It represents the rotation of a rectangular coil ABCD between two magnets (b) Representation of the magnetic flux through the rectangular coil having length L.

Here you see that a coil is rotating between two magnets. The length of this coil is L and width is A. Here the EMF is produced only in the parts BC and DA. However, sides AB and CD, being parallel to the magnetic field lines, do not cut across the magnetic field as the coil rotates. Therefore, there is no change in magnetic flux through these sides, resulting in no induced EMF along AB and CD.

Both the parts, BC and DA are equal in length and an equal electromotive force is also produced in them because the electromotive force is the rate of change of magnetic flux, hence the electromotive force in BC and DA is,

$$\varepsilon_{BC} = \varepsilon_{DA} = Blv_{\perp}$$

$$\varepsilon_{total} = 2Blv_{\perp} = 2Blv \sin \theta$$

If the loop is rotating with angular velocity ω , then

$$\varepsilon = 2Blv \sin \omega t$$

$$\because v = r\omega = a/2 \cdot \omega$$

$$\varepsilon = 2Bl\left(\frac{a}{2}\omega\right) \sin \omega t$$

When the loop is rotating with an angular velocity ω , the linear velocity v of any point on the loop can be expressed in terms of the distance from the axis of rotation ($a/2$) and the angular velocity (ω).

The linear velocity v of a point rotating at a distance r from the axis of rotation with angular velocity ω is given by $v = r\omega$.

In the case of a rectangular loop, if we consider a point at a distance $a/2$ from the center of rotation, its linear velocity v can be expressed as:

$$V = a/2 \cdot \omega$$

Here, $a/2$ represents the distance from the center of rotation to the midpoint of one side of the rectangular loop. Multiply this by ω , the angular velocity, to get the linear velocity of a point on the loop.

Here you see the $\sin\omega t$ appears as if a sine wave is produced as a result of this rotation and this is the reason why it is so easy to produce alternating current. Only if there is a coil in between two magnets, it will be easy to generate AC.

Now let us want to understand AC better. The first thing is that if the current flows in one direction for a period of time, then it flows in the opposite direction for the same period of time as shown in fig 29.2. Therefore, the average value of the current or voltage is zero.

$$\begin{aligned}\mathcal{E} &= \mathcal{E}_m \sin\omega t \\ \mathcal{E}_{average} &= \langle \mathcal{E} \rangle = 0 \quad \dots\dots(1)\end{aligned}$$

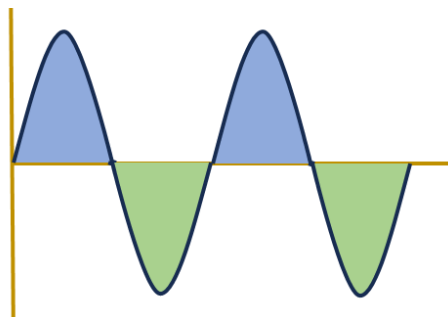


Fig. 29.2: Representation of a sin wave.

According to eq. (1) as much time as the sin wave remains positive, the same time it remains negative; hence the average value of this EMF is zero. But if the average is zero then this does not mean that there is no electricity or that current does not flow.

Remember that power is proportional to v squared divided by r ($P = v^2/r$). It means that you square the electromotive force and then divide it by the resistance r . The basic thing is that a resistor does not care in which direction the current is going; the power dissipation will be in the same direction.

Now the question is that, how can we measure the AC?

To measure a DC, one can use a common meter because its needle stops at one place but if you want to see AC then we have to measure, how many times the needle will have to move back and forth. To measure the AC, we use Oscilloscope and AC signal is identified from two aspects, one is “what is its amplitude”, or “what is its maximum voltage or maximum current”. And second is how fast it goes up and down or in other words “what is its frequency”, or “how many cycles does it complete within a second”. A signal generator is attached with an oscilloscope and the signal generator produces AC voltage and that voltage is going to oscilloscope which then shows the AC signal. We can also decrease or increase its amplitude and frequency from the signal generator to adjust AC signal.

We can measure AC current or voltages either by their maximum current or maximum voltage or there is another quantity called root mean square (RMS) value as shown in fig. 29.3. In RMS, we first square the current or voltage and then take its average and its average will not be zero, because whenever you square a quantity, it becomes positive again.

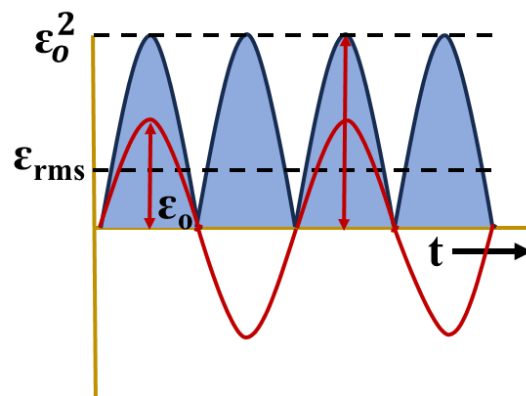


Fig. 29.3: Representation of a root mean square and average of the sin wave.

See here we want to find the average of sine square ωt which I have defined as,

$$\langle \sin^2 \omega t \rangle = \frac{1}{T} \int_0^T dt \sin^2 \omega t$$

$$\therefore \omega = \frac{2\pi}{T}$$

$\therefore$ Using the identity, $\sin^2 x = \frac{1 - \cos 2x}{2}$, if $x = \omega t$ then,

$$\langle \sin^2 \omega t \rangle = \frac{1}{T} \int_0^T dt \left(\frac{1 - \cos 2\omega t}{2} \right)$$

$$\langle \sin^2 \omega t \rangle = \frac{1}{2T} \left(\int_0^T dt - \int_0^T \cos(2\omega t) dt \right)$$

Integrating "dt" over the interval $[0, T]$ gives T:

$$\langle \sin^2 \omega t \rangle = \frac{1}{2T} \left(T - \int_0^T \cos(2\omega t) dt \right)$$

The integral of $\cos(2\omega t)$ over one period is 0, since it completes one full cycle:

$$\langle \sin^2 \omega t \rangle = \frac{1}{2T} (T - 0) = \frac{1}{2}$$

Here, there is only one term left and this tells us that the average value of $\sin^2 \omega t$ is half. So, whenever you see a quantity which is proportional to $\sin \omega t$ or $\cos \omega t$, its average will definitely be zero and the average of its square will be half.

If it is for voltage (ε) and current (I) then the root mean square of them will be,

$$\varepsilon_{rms} = \sqrt{\frac{\varepsilon_m^2}{2}} = \frac{\varepsilon_m}{\sqrt{2}} = 0.707 \varepsilon_m$$

$$I_{rms} = \sqrt{\frac{I_m^2}{2}} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

ε_m , I_m are the peak value of the voltage and current.

Here, one has talked about sin and cosine but alternating current wave can be triangular or square wave. However here also the average will be zero but not the root mean square value. And the root mean square value will not be $1/\sqrt{2}$. We will have to calculate this by squaring it, then integrate it from zero to t, and divide by $1/t$. However, in this lecture we will deal only with sine and cosine functions because they can be very easily produced. A very important thing about AC is that we can increase or decrease its voltage as per our wish and for this we only need a transformer.

Transformer:

A transformer has two coils which are called primary and secondary, then the voltage that is put inside the primary coil gets transformed into AC voltage and then reaches inside the secondary coil. There is a core in the transformer which is usually made of iron or some magnetic material as shown in fig. 29.4. The primary coil that is wrapped around the core is responsible to generate a magnetic field inside this core. This field reverses its direction after some time. If it completes 50 cycles within a second, then the magnetic field first occurs in one direction and then after one second it will change its direction. In this way, a changing magnetic flux is generated inside it.

Also, in secondary coil, a changing magnetic field is passing through the secondary coil and due to this, a current and emf is induced inside it according to the faraday's law.

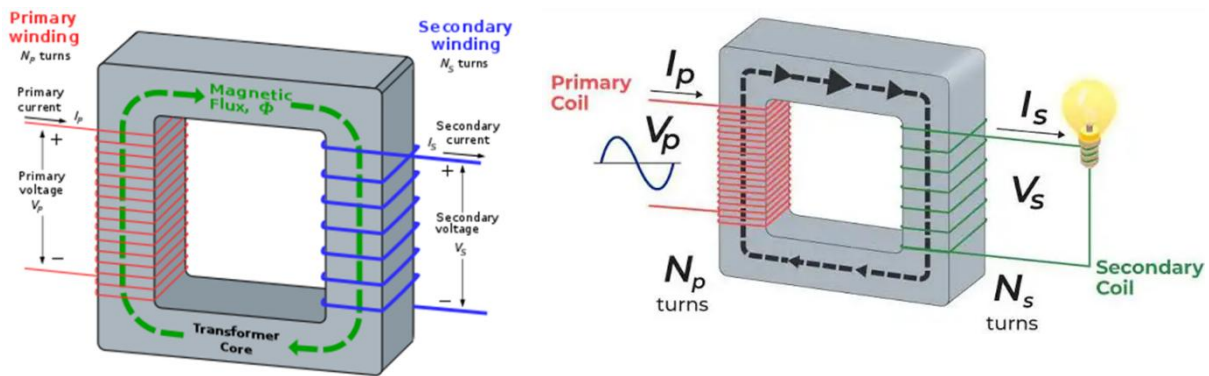


Fig. 29.4: Schematic representation of a transformer.

Flux in transformer:

To calculate the flux in the core, we have:

ϕ = Flux in core

$$\text{EMF (Primary coil)} = \varepsilon_p = -N_p \frac{\Delta\phi}{\Delta t}$$

$$\text{EMF (Secondary coil)} = \varepsilon_s = -N_s \frac{\Delta\phi}{\Delta t}$$

$$\frac{\varepsilon_p}{\varepsilon_s} = \frac{-N_p \frac{\Delta\phi}{\Delta t}}{-N_s \frac{\Delta\phi}{\Delta t}} = \frac{N_p}{N_s}$$

According to the energy conservation,

Primary power = Secondary power

$$\varepsilon_p I_p = \varepsilon_s I_s$$

$$\frac{I_p}{I_s} = \frac{\varepsilon_s}{\varepsilon_p} = \frac{N_s}{N_p}$$

Well, now you must have seen that wapda has very big transformers and the power transmission lines as shown in fig. 29.5 which can be up to 512 kilovolts where electricity is generated.

Electricity generation, transmission, and distribution

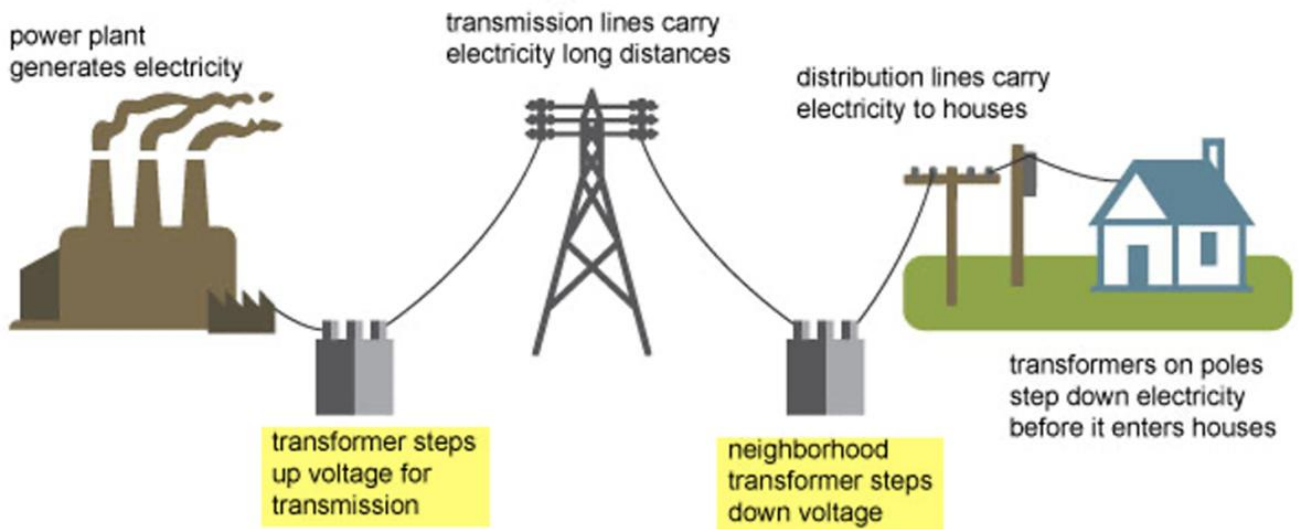


Fig. 29.5: Schematic representation of an electricity transfer from power plant to cities.

In tarbela/mangla dam or in any power station, the voltage is first increased through transformers, then when this electricity reaches the cities, it is decreased through step down transformers. Why does this happen? Why we are not kept this to the same level? And another question is that why we don't use DC?

The reason for this is that power loss in electrical transmission lines is directly proportional to the square of the current (I^2) and the resistance (R) of the transmission lines. By increasing the

voltage and thereby decreasing the current, the power loss in the transmission lines can be minimized. This is described by the formula,

$$P_{\text{loss}} = I^2 R.$$

By reducing the current, the amount of power lost as heat due to resistance in the wires decreases significantly but the product of both the voltage and current remains constant ($VI = \text{constant}$).

Lower power loss means more of the generated electricity reaches its destination. This is particularly important for large-scale power stations like Tarbela and Mangla dams, where huge amounts of electricity are generated and distributed over extensive networks.

Higher voltages used in long-distance transmission lines reduce the need for thicker, more expensive cables, making the system more economically viable. Additionally, lower currents associated with higher voltages are safer for workers and the public, reducing the risk of electrical hazards. In this way the electricity generated at the source in a generator or in a power station reaches the city. And perhaps there would have been a loss of 2, 4 or maximum 5%.

If we had used DC, then about 90% of the electricity would have been wasted. Transformer work on the principle of electromagnetic induction, which relies on the changing magnetic field produced by alternating current (AC). DC (direct current) does not produce a changing magnetic field, so traditional transformers do not work with DC.

Self-inductance:

Self-inductance is a property of every coil and every wire. We define it in such a way that when any current passes through that coil or that wire, as a result a magnetic field (B) is generated which is proportional to the current (I). So, B is proportional to I . This proportional constant will be more or less depending on the number of turns and on the thickness of this coil etc. Phenomenon of self-inductance is shown in fig 29.6.

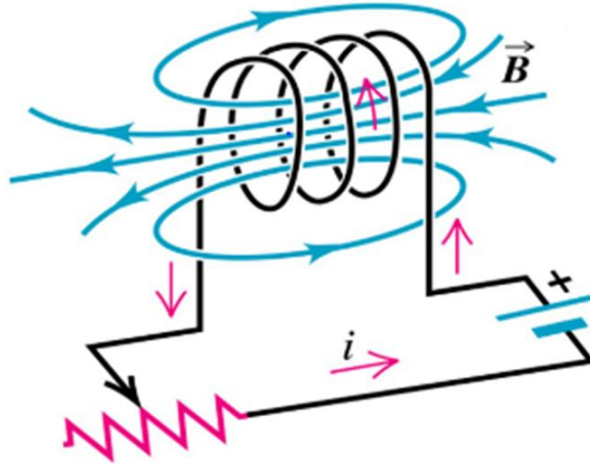


Fig. 29.6: Constructional representation of a self-inductance of a solenoid.

The definition of inductance is,

For any geometry of loop, the magnetic flux through the loop produced by the current in the loop is proportional to the current. The inductance L is the constant of proportionality. So,

$$\Phi_B = LI$$

The unit of inductance is called henry:

$$1\text{H} = 1\text{T}\cdot\text{m}^2/\text{A}$$

$$1\text{H} = 1(\text{T}\cdot\text{m}^2/\text{s}) (\text{s}/\text{A})$$

$$1\text{H} = 1\text{V}\cdot\text{s}/\text{A}$$

Here note that the inductance, like capacitance is a geometric quantity, which means that its size depends on, how many turns a coil have, its thickness, and the area of cross section.

Calculation of inductance for a solenoid:

Consider a solenoid having N number of turns of wire in the solenoid, A is the cross-sectional area of the solenoid (in square meters, m^2), and l is the length of the solenoid (in meters, m).

Magnetic field inside a solenoid is given as,

$$B = \mu_o n I$$

$$\therefore n = N / l$$

$$B = \mu_o \frac{N}{l} I$$

Magnetic flux for a solenoid having N number of turns is:

$$N\phi_B = NBA = (nl)BA$$

$$N\phi_B = (nl)\mu_o n I A = \mu_o n^2 l I A$$

Self-inductance for N number of turns is given as,

$$\phi = N I$$

$$L = \frac{N\phi_B}{I} = \frac{\mu_o n^2 l I A}{I} = \mu_o n^2 l A$$

L is the self – inductance of the solenoid (in Henries, H)

μ_o is the permeability of free space, approximately $4\pi \times 10^{-7} \text{ H / m}$

Let's solve a problem here.

Problem:

A coil whose radius is 5 cm and the length is 10 cm and there are 3500 turns on it. Find the self-inductance of the coil.

Solution:

Here,

$$\text{number of turns} = N = 3500$$

$$\text{radius} = r = 5 \text{ cm} = 0.05 \text{ m}$$

$$\text{length} = l = 10 \text{ cm} = 0.10 \text{ m}$$

$$\text{As, } A = \pi r^2$$

$$L = \frac{\mu_o N^2 A}{l} = \frac{\mu_o N^2 \pi r^2}{l}$$

$$L = 4\pi \times 10^{-7} \frac{\text{T.m}^2}{\text{A}} \cdot 3500^2 \cdot \frac{\pi(0.05 \text{ m})^2}{0.10 \text{ m}}$$

$$L = 1.21 \frac{\text{T.m}^2}{\text{A}} = 1.21 \text{ H}$$

Faraday's Law and Inductance:

Self-inductance is that when current passes through this coil, a magnetic flux is generated and from Faraday's law when this flux changes, an EMF is induced as a result.

$$\begin{aligned}\mathcal{E} &= -\frac{d\phi_B}{dt} \\ \therefore \phi_B &= LI \\ \mathcal{E} &= -\frac{d(LI)}{dt} \\ \mathcal{E} &= -L\frac{dI}{dt}\end{aligned}$$

This equation shows the induced electromotive force is equal to the negative of the inductance to the rate of change of current. Now we want to calculate the energy required to reach to that factor of current.

Energy Stored in a Magnetic Field:

Suppose we have an inductor, so when we connected it to the battery, a current flowed in it. When the current flowed, it generates a magnetic field. The magnetic field prevented this current from exceeding, so we had to work out how much energy is there to fill the magnetic energy inside it. Now let us calculate it.

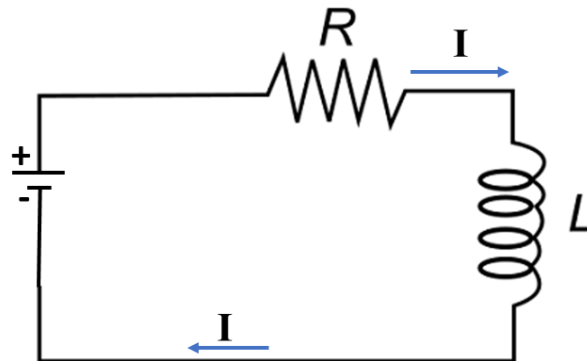


Fig. 29.7: Constructional representation of a resistor with inductor to a battery.

Now imagine that we connect a battery with a resistor and an inductor and close the switch, then current start flowing in it as shown in fig 29.7. Now as soon as the current passes through it, the factor ‘ LdI/dt ’ become non-zero, which means that there will be an emf in the opposite direction, a voltage will be generated which will try to ensure that current does not flow in it. Mathematically, it is represented as:

$$\varepsilon = IR + L \frac{dI}{dt}$$

$$\varepsilon I = I^2 R + LI \frac{dI}{dt}$$

Where,

εI = Power expended by battery

$I^2 R$ = Power dissipated in resistor

$LI \frac{dI}{dt}$ = Work done on inductor

The factor $LI \frac{dI}{dt}$ shows the battery did some work for the movement of current and the amount of that work is εI , and $I^2 R$ power is lost in the resistor so whatever is left remains in the form of magnetic energy in this inductor. Just as we had charged the capacitor by connecting it with the battery, here we have charged the inductor to achieve the energy balance. Mathematically, it is given as:

$$\frac{dU_B}{dt} = LI \frac{dI}{dt}$$

$$\int_0^{U_B} dU_B = L \int_0^I IdI$$

$$U_B = \frac{1}{2} LI^2$$

As energy stored in a capacitor is $U_E = \frac{1}{2} CV^2$

Now this is exactly the same formula as we drew once before for the capacitor. For capacitor, all energy is stored in the electric field and for the inductor, all the energy is stored in magnetic field.

Now what will happen when the magnetic field becomes zero?

It will be happened only in such a situation, if you connect this inductor with a resistor then a current will flow and all the energy that was present inside the magnetic field will be lost in a resistor.

Energy Density of Magnetic Field:

$$U_B = 1/2 LI^2 \dots\dots\dots(1)$$

As,

$$L = \mu_0 n^2 l A \text{ and } B = \mu_0 n I \Rightarrow I = B / \mu_0 n$$

Use solenoid results in equation (1):

$$U_B = \frac{1}{2} (\mu_0 n^2 l A) (B / \mu_0 n)^2$$

$$\therefore l A = V(\text{volume})$$

$$U_B = \frac{1}{2} (\mu_0 n^2 V) (B^2 / \mu_0^2 n^2)$$

$$\frac{U_B}{V} = \frac{B^2}{2\mu_0}, \quad \therefore u_B = \frac{U_B}{V}$$

$$u_B = \frac{B^2}{2\mu_0}$$

Units of energy density:

Energy density = Magnetic energy per unit volume (J/m^3)

$$\text{Energy density} = B^2 / 2\mu_0$$

Unit conversion for 1 tesla field:

$$\text{Energy density} = 1\text{T}^2 / 8\pi * 10^{-7} \text{ T.m/A}$$

$$\text{Energy density} = 3.98 * 10^5 \text{ T.A/m}$$

$$\text{Energy density} = 3.98 * 10^5 \text{ T.Am/m}^2$$

$$\text{Energy density} = 3.98 * 10^5 \text{ N/m}^2 = 3.98 * 10^5 \text{ J/m}^3$$

Problem:

Let's solve a problem in which both a resistor and an inductor are used. And it has a battery which is connected with an inductor and a resistor. As you know that whenever we connect it, a back emf will be generated which will prevent the current from increasing but eventually the current will reach its maximum value. And this maximum value is the voltage of the battery or its EMF divided by r. We would like to solve its equation now,

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$$\text{Voltage across resistor} = V_R = IR$$

$$\text{Voltage across inductor} = V_L = L \frac{dI}{dt}$$

$$\text{Electromotive force} = V_L + V_R$$

$$\mathcal{E} = L \frac{dI}{dt} + IR$$

The solution of the equation is:

$$I(t) = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau})$$

$$\frac{dI}{dt} = \frac{\mathcal{E}}{R} \frac{1}{\tau} e^{-t/\tau}$$

$$\tau = \frac{L}{R} = \text{Time constant}$$

Time constant tells the time after which the current has reached or is close to its maximum value. Let's prove how time constant is equal to second by following:

$$[\tau] = \frac{[L]}{[R]} = \frac{\text{henry}}{\text{ohm}} = \frac{\text{volt.second/ampere}}{\text{ohm}}$$

$$[\tau] = \left(\frac{\text{volt}}{\text{ampere.ohm}} \right) \text{second}$$

$$\therefore \frac{\text{volt}}{\text{ampere.ohm}} = \frac{IR}{IR} = 1$$

$$[\tau] = \text{second} = t$$

$$I = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau}) = \frac{\mathcal{E}}{R} (1 - e^{-t/t})$$

$$I = \frac{\mathcal{E}}{R} (1 - e^{-1}) = \frac{\mathcal{E}}{R} \left(1 - \frac{1}{2.718} \right)$$

The natural exponential function $e = 2.718$

$$I = \frac{\mathcal{E}}{R} (1 - 0.37)$$

$$I = \frac{\mathcal{E}}{R} 0.63$$

Conclusion:

In an RL circuit, at $t = \tau$, the current through the inductor has reached 63% of its final steady-state value $\mathcal{E}/R$. This means that after one-time constant τ , the inductor current has risen to 63% of the value.

Similarly, we have solved the case of a resistor and a capacitor. For a capacitor,

$$C = \frac{Q}{V}$$

And energy stored in a capacitor is given as:

$$U = \frac{Q^2}{2C} = \frac{(CV)^2}{2C} = \frac{C^2V^2}{2C}$$

$$U = \frac{1}{2}CV^2$$

When we connect the capacitor with the resistor and the battery, the current flows for some time and then it ends. After how much time does it end, this is based on the time constant which is there in the resistor and the capacitor and this time constant is equal to RC .

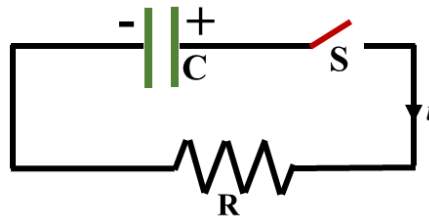


Fig. 29.8: Representation of a circuit diagram of a resistor and a capacitor to a battery.

Let us solve the equation for capacitor here:

$$\text{Voltage across resistor} = V_R = iR$$

$$\text{Voltage across capacitor} = V_C = \frac{q}{C}$$

$$\text{Electromotive force} = V_C + V_R$$

$$\mathcal{E} = \frac{1}{C} \frac{dq}{dt} + \frac{di}{dt} R$$

$$\frac{1}{C} \frac{dq}{dt} + \frac{di}{dt} R = 0$$

As there is no battery attached to it hence, $\mathcal{E} = 0$

$$\text{and } \frac{dq}{dt} = i$$

$$\frac{di}{dt} = -\frac{1}{RC} i$$

The solution of the equation is:

$$i = i_0(e^{-t/\tau}) = i_0(e^{-t/RC})$$

$$\therefore \tau = RC = \text{Time constant}$$

Now let us understand that if only capacitors and inductors are present and if there is no resistor, then what will happen? This is a very interesting thing to know.

As, power is spent inside the resistor, power is never wasted in capacitor, it is exactly like a tank, you can fill water inside it but it itself does not absorb water from it and the same thing applies to an inductor. You fill magnetic energy inside the inductor as when current passes through it, a magnetic field is produced. As magnetic field is energy.

When an inductor is connected to a capacitor, they form an LC circuit which exhibits oscillatory behavior. Here's a summary of the process:

Initial State: The capacitor is fully charged, storing energy in its electric field, while the inductor has no current flowing through it, storing zero energy in its magnetic field. The energy is entirely within the electric field of the capacitor.

Initiation of Current Flow: When the circuit is closed, current begins to flow. As current flows, the magnetic field inside the inductor starts to increase, while the electric field across the capacitor starts to decrease.

Equalization of Energies: As current continues to flow, the magnetic energy stored in the inductor increases, while the electric energy stored in the capacitor decreases. Eventually, the magnetic energy becomes equal to the initial electric energy.

Reversal of Energies: Further flow of current causes the magnetic energy to exceed the initial electric energy. At this point, the capacitor is discharged, and all the energy is stored in the magnetic field of the inductor.

Oscillatory Behavior: The cycle repeats as the stored energy oscillates between the electric field of the capacitor and the magnetic field of the inductor. The total energy, however, remains constant, as it is conserved in the ideal case with no resistive losses.

Electromagnetic oscillations:

$$U = U_B + U_E$$

$$U = \frac{1}{2}LI^2 + \frac{1}{2}\frac{q^2}{C}$$

$$U = \text{constant so } dU/dt = 0$$

$$\frac{dU}{dt} = \frac{d}{dt}\left(\frac{1}{2}LI^2 + \frac{1}{2}\frac{q^2}{C}\right)$$

$$0 = \frac{d}{dt}\left(\frac{1}{2}LI^2 + \frac{1}{2}\frac{q^2}{C}\right)$$

$$I = \frac{dq}{dt}, \quad \frac{dI}{dt} = \frac{d^2q}{dt^2}$$

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$

$$\therefore \text{Oscillation frequency} = \omega^2 = \frac{1}{LC}$$

$$\frac{d^2q}{dt^2} + \omega^2q = 0$$

The solution of the equation is:

$$q = q_m \cos \omega t$$

The frequency of oscillation determines how quickly the energy exchanges between the electric and magnetic fields in the LC circuit. Higher frequencies correspond to faster oscillations, while lower frequencies correspond to slower oscillations.

RLC driven by an alternating voltage:

When analysing circuits with alternating current (AC) sources, such as sinusoidal voltage $V = V_0 \sin(\omega t)$, the behaviour of resistors, capacitors, and inductors differs due to their inherent electrical properties. Let's delve into these differences:

Behaviour of Resistors (R):

For resistors, the voltage V and the current I are in phase. This means that both the voltage across and the current through a resistor reach their maximum and minimum values at the same time during each cycle of the AC waveform. The relationship between voltage, current, and resistance for a resistor is described by Ohm's Law:

$$V = I \cdot R$$

Behaviour of Capacitors (C):

For capacitors, the relationship between voltage and current is characterized by a phase difference of 90° . When the voltage across a capacitor reaches its maximum (or minimum), the current through the capacitor is zero, and vice versa. This phase difference leads the current and voltage lag the current by 90° . The relationship between voltage and current for a capacitor in an AC circuit is given by:

$$I = C dV/dt$$

Behaviour of Inductors (L):

For inductors, the relationship between voltage and current also exhibits a 90° phase difference, but in this case, the current lags the voltage. When the voltage across an inductor reaches its maximum (or minimum), the current through the inductor is zero, and vice versa. This phase difference causes the current to lag the voltage by 90° as shown in fig 29.9. The relationship between voltage and current for an inductor in an AC circuit is given by:

$$V = L \, dI/dt$$

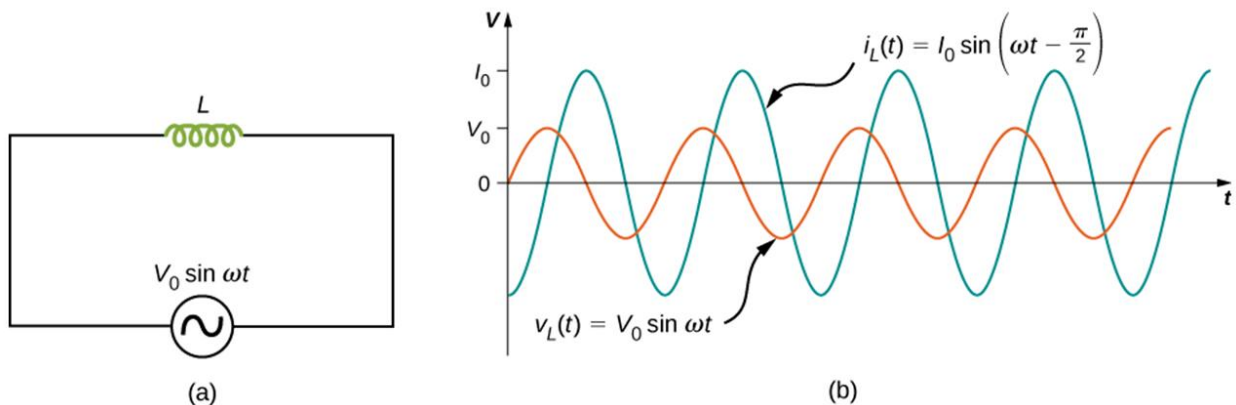
An inductive element:

Fig. 29.9: (a) It represents an AC inductive circuit (b) Representation of the waveform where voltage leads the current by 90 degrees.

$$L \frac{dI}{dt} = V_0 \sin \omega t$$

$$\frac{dI}{dt} = \frac{V_0}{L} \sin \omega t$$

On integrating both sides:

$$I = -\frac{V_0}{\omega L} \cos \omega t$$

$$I = \frac{V_0}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right)$$

This equation describes the current (I) flowing through an inductor in an AC (alternating current) circuit. Where:

- I represent the instantaneous current flowing through the inductor.
- V_0 represent the peak voltage of the AC source.
- ω is the angular frequency of the AC source, measured in radians per second ($\omega = 2\pi f$, where f is the frequency in Hertz).
- L is the inductance of the inductor, measured in Henry.
- $\cos(\omega t)$ represent the time-varying behavior of the current, where t is time.

Now, let's discuss the physical significance of this equation:

Current Variation with Time:

The term $\cos(\omega t)$ represents a sinusoidal variation in current with time. As time progresses, the current oscillates sinusoidally between positive and negative values.

The cosine function describes the oscillation pattern, indicating that the current reaches its peak at different times depending on the phase angle ωt .

Peak Current:

The peak current is given by $V_o/\omega L$. This term signifies that the peak current is directly proportional to the peak voltage (V_o) of the AC source and inversely proportional to both the angular frequency (ω) and the inductance (L) of the inductor.

A higher peak voltage or lower inductance would result in a higher peak current, assuming the angular frequency remains constant.

Similarly, a higher angular frequency or higher inductance would result in a lower peak current, assuming the peak voltage remains constant.

Phase Relationship:

The phase angle ωt determines the phase relationship between the current and the voltage in the circuit.

For a purely inductive circuit, the current lags the voltage by a phase angle of 90° (or $\pi/2$ radians). This lag is due to the behavior of inductors, which oppose changes in current.

The cosine function captures this phase relationship, indicating the time-varying behavior of the current with respect to the voltage.

In conclusion, this equation describes how the current flowing through an inductor in an AC circuit varies sinusoidally with time, depending on the peak voltage of the source, the angular frequency, and the inductance of the inductor. It also indicates the phase relationship between the current and voltage in the circuit.

A capacitive element:

If a capacitor is connected to an alternating source, then current will also flow in it. Here we get current from the charge. The behavior of a capacitor with AC is shown in fig 29.10.

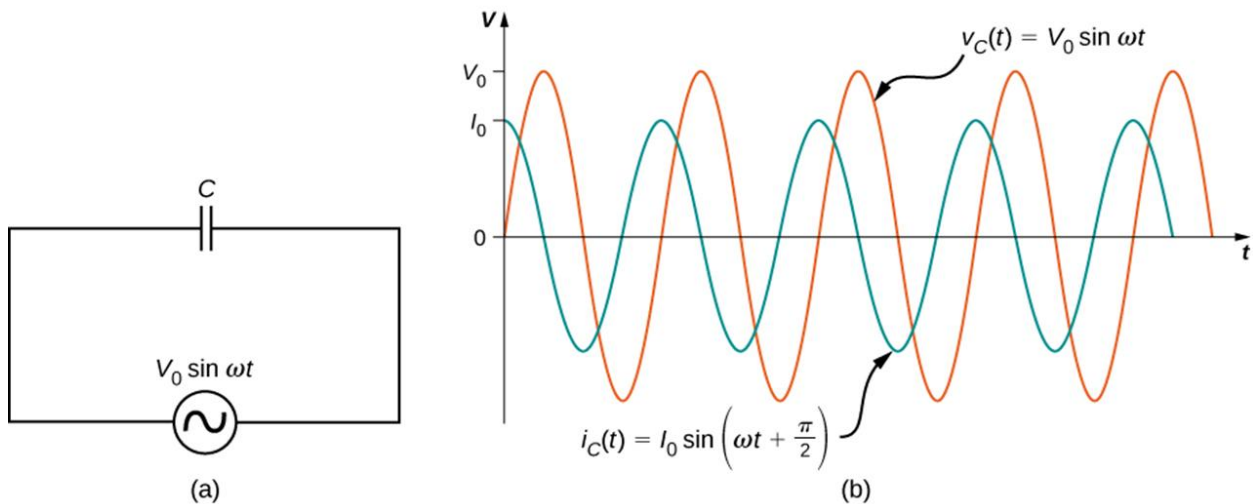


Fig. 29.10: (a) It represents an AC capacitive circuit (b) Representation of the waveform where voltage lags the current by 90 degrees.

What is the charge for a capacitor? Current is the rate of change of charge.

$$q = CV_0 \sin \omega t$$

$$I = \frac{dq}{dt} = \frac{d}{dt}(CV_0 \sin \omega t)$$

On differentiating:

$$I = \omega CV_0 \cos \omega t$$

$$I = \omega CV_0 \sin\left(\frac{\pi}{2} - \omega t\right)$$

The equations are related to the behavior of a capacitor in an AC (alternating current) circuit.

Where:

- q represents the charge stored in the capacitor at any given time t .
- ω_0 is the angular frequency associated with the capacitor's behavior. It depends on the capacitance (C) and inductance (L) of the circuit ($\omega_0 = 1/LC$).
- $\sin(\omega t)$ represents a sinusoidal variation of charge with time. As time progresses, the charge oscillates sinusoidally between positive and negative values, following a sine wave pattern.
- The maximum charge ($q_{\max}$) is reached when $\sin(\omega t) = 1$ or -1 , and it is proportional to ω_0 . The larger the capacitance (C), the larger the maximum charge.
- I represent the current flowing through the circuit at any given time t .
- dq/dt represent the rate of change of charge with respect to time, which is the definition of current.
- ω is the angular frequency of the AC source.
- C is the capacitance of the capacitor.
- V_0 is the peak voltage of the AC source.

- $\cos(\omega t)$ represents the variation of current with time. It's a cosine function because the current is leading the voltage by 90° due to the behavior of capacitors.
- $\sin(90 - \omega t)$ also represents the variation of current with time, but it's expressed in terms of a sine function. This equation emphasizes the phase relationship between current and voltage: the current leads the voltage by 90° .

In conclusion, these equations describe the behavior of a capacitor in an AC circuit. The charge stored in the capacitor oscillates sinusoidally with time, and the current flowing through the circuit follows a sinusoidal pattern as well. The phase relationship between current and voltage is such that the current leads the voltage by 90° .

Comparison between RLC results:

$$I = \frac{V_o}{R} \sin \omega t \quad \text{For resistor}$$

$$I = \frac{V_o}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right) \quad \text{For inductor}$$

$$I = \frac{V_o}{1/\omega C} \sin\left(\frac{\pi}{2} - \omega t\right) \quad \text{For capacitor}$$

That's why:

$$Z_L = \omega L \quad \text{is called inductive reactance}$$

$$Z_C = 1/\omega C \quad \text{is called capacitive reactance}$$

Here, we came to know that how R, L and C respond when an alternating voltage applied to them. One special thing about resistor, capacitor and inductor is that the energy inside the inductor and capacitor remains filled in them and does not go away. They are storages and storehouses for energy. If you keep energy in them, the energy does not lose. But resistor is not like this. When current passes through a resistor, power is then lost to heat.

Lecture#30

Electromagnetic waves

The purpose of this lecture is to present you the great achievement of James Clark Maxwell's which he had accomplished in the year 1865. The reason why this achievement is so famous is because Maxwell unifies the two different forces i.e., electric and magnetic force.

From the previous studies, one can define the **electric field** as, “the force F experienced by the charge q ”. Since the force is a vector quantity hence electric field is also a vector quantity. The electric force is given as:

$$\vec{F} = q\vec{E}$$

The more powerful the electric field, the greater its strength, the force will also be greater accordingly, so this is the identity of the electric force. Here, an important thing is that, it does not matter whether the charge is standing or at rest or whether it is moving, the force will always be in the direction of the electric field.

The magnetic force is completely different from this.

The **magnetic field** is the region surrounding the particle in which the magnetic force is felt. **Magnetic force** refers to the force exerted between magnets or between a magnet and a moving charged particle. The force on a moving charged particle in a magnetic field is given by,

$$\vec{F} = q(\vec{v} \times \vec{B})$$

In this formula, B is the magnetic field and see that $(\vec{v} \times \vec{B})$ means that the force that acts on it is perpendicular to both $\vec{v}$ and $\vec{B}$.

Direction of the magnetic force:

The direction of the magnetic force can be determined using the **right-hand rule**: if you point the fingers of your right hand in the direction of $\vec{v}$ and curl them towards $\vec{B}$, your thumb points in the direction of the force on a positive charge.

Lorentz force:

The Lorentz force is the force exerted on a charged particle moving through electric and magnetic fields. It is a fundamental concept in electromagnetism and is described by the Lorentz force law. The total Lorentz force $\vec{F}$ experienced by a particle with charge q in the presence of an electric field $\vec{E}$ and a magnetic field $\vec{B}$ is given by:

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

Now the question arises here is that, where does the electric field come from? What produces the electric field? And the answer is that “the charge” and from the charge the electric field lines are produced. If you place a positive charge, then that charge is the source of the electric field and the direction of the field lines will be outward. And if the charge is negative then the electric field lines will be inward.

Basic Equations of Electromagnetism:

The fundamental equations of electromagnetism are following:

1. Gauss' law of electricity

$$\int \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

Where E is the electric field, ϵ_0 is the permittivity of free space (8.854×10^{-12} F/m (farads per meter)). Illustration of an electric flux through a closed surface is shown in fig.1.

2. Gauss' law of magnetism

$$\int \vec{B} \cdot d\vec{A} = 0$$

Gauss's law for magnetism states that the net magnetic flux through any closed surface is zero as shown in fig. 2. And hence gauss's law for electric fields and gauss's law for magnetic fields are completely different from each other.

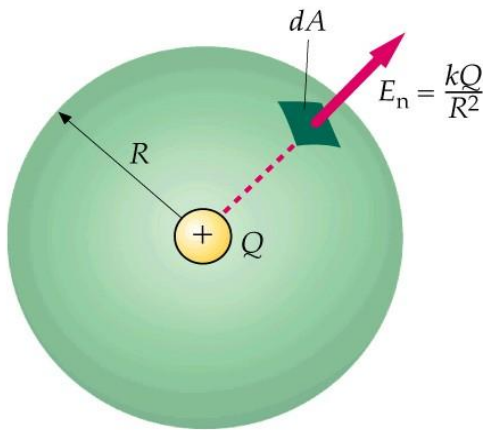


Fig. 1: Illustration of an electric flux through a closed surface (Gaussian surface), with field lines emanating from a positive charge inside the surface.

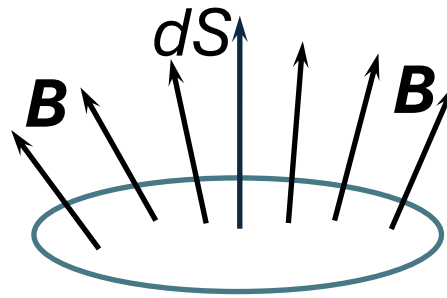


Fig. 2: An illustration of magnetic field lines passing through a closed surface (ds). Unlike electric field lines, magnetic field lines form closed loops, indicating that there are no magnetic monopoles.

3. Faraday's law of induction

$$\int \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

$$\Phi_B = \oint \vec{B} \cdot d\vec{S}$$

This is Faraday's Law which tells us that when a magnetic flux changes with time, as a result, an electromotive force comes into existence and that motive force then produce the current in a wire or in a circuit.

4. Ampere's law

$$\int \vec{B} \cdot d\vec{s} = \mu_0 I$$

Ampère's Law relates the circulating magnetic field in a closed loop to the electric current passing through the loop. Its states that the integral of the magnetic field B around a closed loop is proportional to the total current I passing through the loop as shown in fig. 3.

From here it is clear that the source of the magnetic field is electric current, just as a charge produces an electric field, similarly current produces a magnetic field.

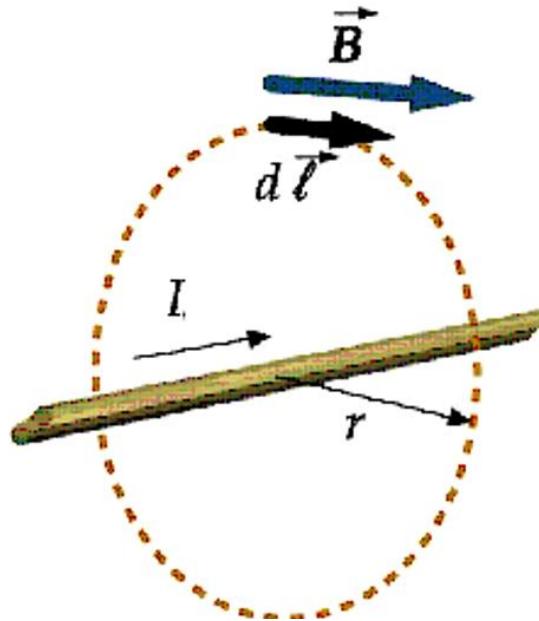


Fig. 3: Illustration of a circulating magnetic field around a boundary path produced by an electric current or changing electric flux through a closed surface.

Maxwell's Displacement Current:

250 years ago, James Clark Maxwell Started paying attention to these four laws. Maxwell was a resident of Scotland and he has interest in all fields of physics but was very interested in electricity and magnetism. He saw that there was a problem in Ampere's law and then he created an imaginary example to point out the mistake. He used two plates of capacitors and connected these plates to an alternate current source. Now if it is an AC source then when the electric field changes between these plates, a current will definitely flow as a result. Then he thought to make different surfaces to see how Ampere's law applies for capacitors as shown in fig. 4.

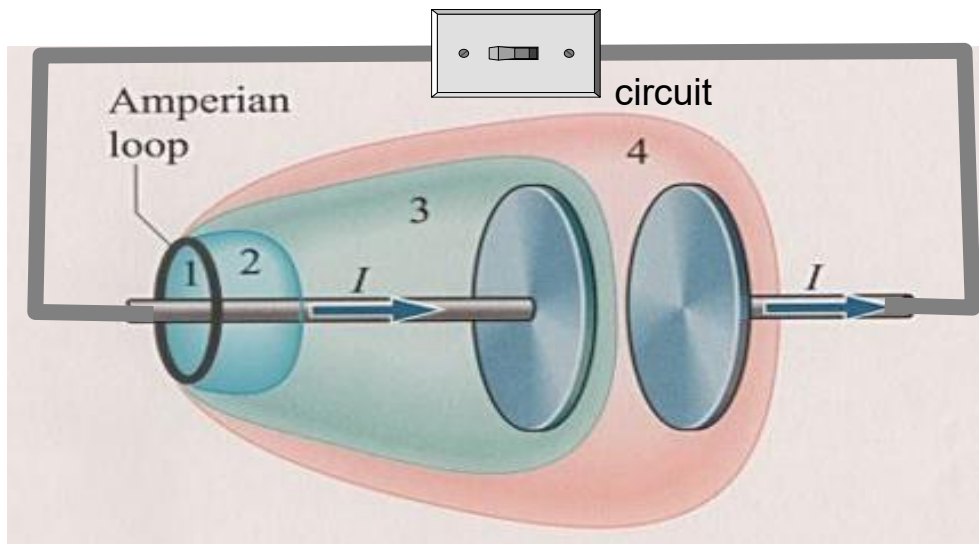


Fig. 4: Illustration of Ampère's Law with Maxwell's addition. This figure shows a closed loop encircling a current-carrying wire and a capacitor connected to an alternating current (AC) source. As the electric field between the capacitor plates changes, Maxwell realized that a displacement current $\epsilon_0 d\Phi_E/dt$ must be added to account for the changing electric field.

From the fig. 4 we can see that there are four different surfaces in it, which is named as one, two, three and four. There is only one emperian loop in it. On this loop, we will calculate the integral according to Ampere's law as:

$$\int \vec{B} \cdot d\vec{s} = \mu_0 I$$

So now if you choose surface one, two and four then since only one current can pass through this wire, so here the enclosed current will be I. Now suppose if we will choose the number three surface instead of one, two and four then the enclosed current is zero and for this the integral is equal to zero.

$$\int \vec{B} \cdot d\vec{s} = 0$$

Hence, James Clerk Maxwell realized around 1865 that the equations of electricity & magnetism were inconsistent.

$$\left[\int \vec{B} \cdot d\vec{s} \right]_{(1,2,4)} \neq \left[\int \vec{B} \cdot d\vec{s} \right]_3$$

Then he thought, there must be another effective current which can produce magnetic field and he called it a **displacement current**. Maxwell generalized Ampere's law like this and he said that there is such a thing as a displacement current which is added on to the other current and its generalization is:

$$\int \vec{B} \cdot d\vec{s} = \mu_0 (I + I_D)$$

$$I_D = \epsilon_0 \frac{d\Phi_E}{dt}$$

The current (rate of charges) that flows in the circuit is:

$$I = \frac{dQ}{dt}$$

But the charge on the capacitor plate is: $Q = \epsilon_0 EA$

Hence,

$$I = \frac{d}{dt}(\epsilon_0 EA)$$

$$I = \epsilon_0 \frac{d(EA)}{dt}$$

$$I = \epsilon_0 \frac{d\phi_E}{dt} \equiv I_D$$

In other words, we can also say that if Maxwell didn't add the displacement current I_D , then the current would not have been conserved which means that the charges would have been lost while passing through a wire and they would have disappeared somewhere and conservation of charge will become violate. Thus, in order to maintain the conservation of charge, Maxwell was forced to invent this additional term and this is a great miracle of mathematical physics. The Maxwell equations of

electromagnetism are now completed, which will not change anymore as these have been tested many times and now there is not even a single flaw in it. So, now the four Maxwell equations are:

1. $\oint \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon_0}$
2. $\oint \vec{B} \cdot d\vec{S} = 0$
3. $\oint \vec{E} \cdot d\vec{\ell} = -\frac{d\phi_B}{dt}$
4. $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \left(I + \epsilon_0 \frac{d\phi_E}{dt} \right)$

So, Maxwell actually discovered that when an electric field changes, its effect is exactly the same as that of a current. The flow of current produces a magnetic field and similarly, due to the changing electric field, a magnetic field is generated. Hence, whether there is a current of an electron or a displacement current (remember that it is proportional to the rate of change of electric flux), the effect of both is the same. So, whenever the field changes its value, a magnetic field is produced as a result.

This brings to light a very important new fact that electromagnetic waves that can pass through space, can exist as a result of these equations. And the basic reason for the existence of electromagnetic waves is that the changing electric flux produces a magnetic field.

Up till now, according to Ampere's law, we saw that the magnetic field was produced by the current. But now, the current may not be there, we may be in free space where there are no electrons or nothing, but if there is a changing electric field in the free space then it results a magnetic field.

Introduction to electromagnetic waves:

The basic idea behind electromagnetic waves is similar to other types of waves. Understanding these waves requires familiarity with concepts like wavelength, frequency, and amplitude also shown in fig. 5.

Wavelength (λ):

Wavelength is the distance between two successive points that are in phase on a wave. This means it is the distance over which the wave's shape repeats.

Frequency (f):

The number of cycles of displacement that pass a point in one second. It is measured in hertz (Hz). Frequency represents how many times the wave oscillates per second.

Relationship Between Wavelength, Frequency, and Speed:

Wave Speed (v):

The product of wavelength (λ) and frequency (f) is equal to the speed of wave propagation.

$$V = \lambda \cdot f$$

Amplitude (A):

The maximum magnitude of displacement in a wave. Higher amplitude means greater energy in the wave. Amplitude indicates how much the medium is displaced from its rest position.

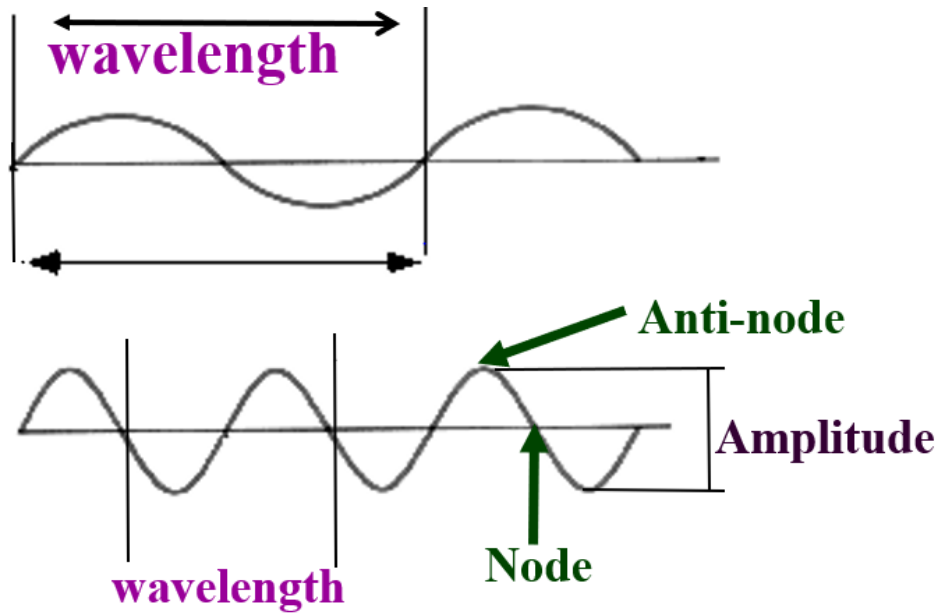


Fig. 5: Illustration of a standing wave with labeled features: amplitude, nodes, anti-nodes, and wavelength. The **amplitude** is the maximum displacement from the rest position. **Nodes** are points where there is no displacement, while **anti-nodes** are points of maximum displacement. The **wavelength λ** is the distance between two consecutive nodes or anti-nodes.

Electromagnetic Waves:

These waves have no restrictions on frequency or amplitude. The electromagnetic spectrum ranges from zero frequency to extremely high frequencies, although generating or detecting zero or infinite frequencies is not feasible.

Example of Electromagnetic Waves:

Red Light: It has a wavelength of approximately 700 nanometers (nm) as shown in fig. 6.

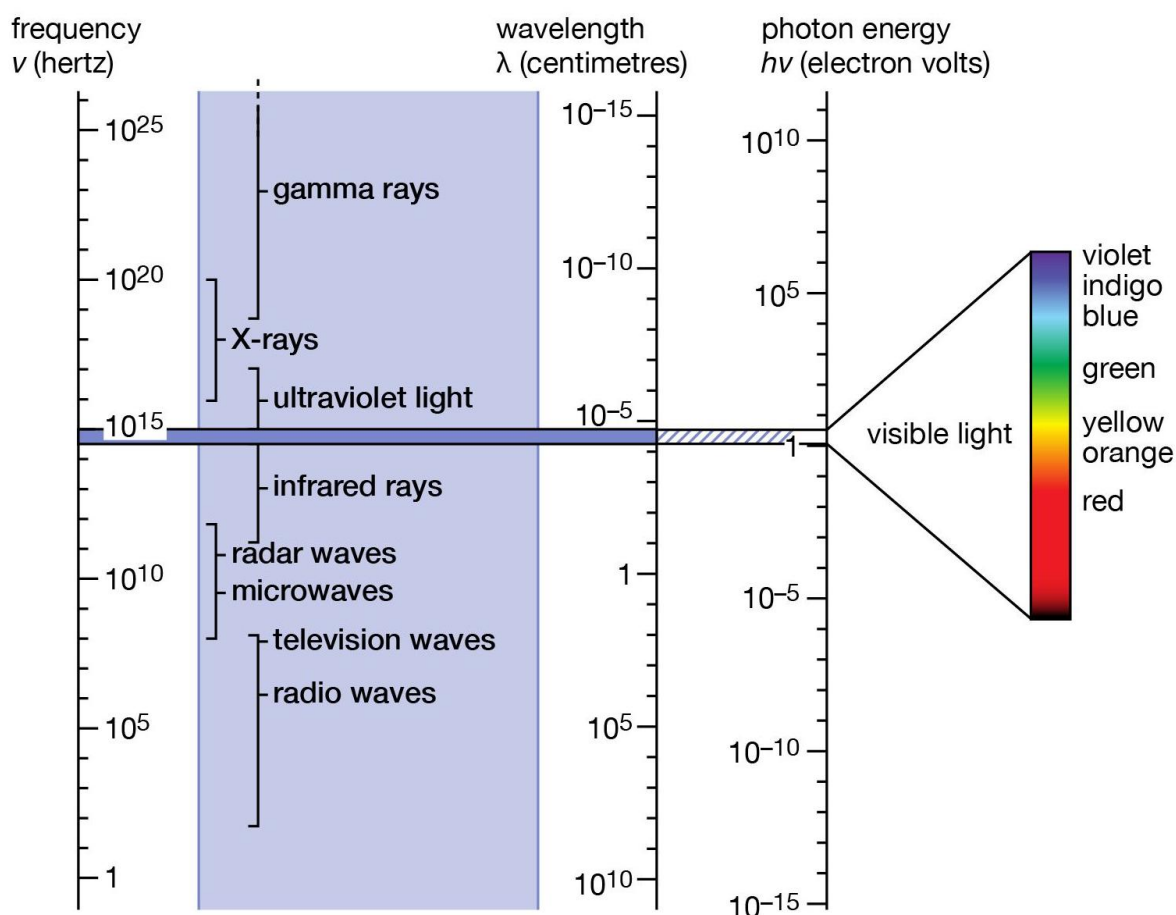


Fig. 6: A representation of the electromagnetic spectrum, showing the range of all possible electromagnetic radiations. It spans from low-frequency radio waves to high-frequency gamma rays, including microwaves, infrared, visible light, ultraviolet, and X-rays.

Speed of Light

Light travels at a speed of 3×10^8 meters per second (or 3×10^{10} centimeters per second).

Example Calculation:

Suppose a red-light having wavelength 7×10^{-7} meters (700 nm). Calculate its frequency.

Calculation:

Speed of light (c) = 3×10^8 meters per second

Wavelength (λ) = 7×10^{-7} meters

$$V = \lambda \cdot f$$

$$f = V / \lambda$$

$$f = 3 \times 10^8 / 7 \times 10^{-7} \text{ m} = 4.29 \times 10^{14} \text{ Hz}$$

Result: The frequency of red light is approximately 4.29×10^{14} hertz (Hz). 1 Hertz (Hz) equals one cycle per second. Thus, 10^{14} Hz means 10^{14} cycles per second.

Electromagnetic Spectrum:

Range: The electromagnetic spectrum is extremely wide, with no upper limit on frequency. The lower limit approaches zero but can never actually reach zero.

Examples:

1. **Radio Waves:** Frequencies range from 10^4 to 10^{10} cycles per second.
2. **Amplitude Modulation (AM) Radio waves:** Frequencies range from 10^6 to 10^7 Hz.
3. Radio Pakistan operates around 800 kilocycles (800 kHz).

Frequency Modulation (FM) Radio:

FM radio is another type of radio transmission, differing from AM radio in terms of frequency modulation.

Frequency: FM radio operates at around 100 megahertz (MHz), which equals 100 million cycles per second.

Television Frequencies Range: Television frequencies range from 10^8 to 10^9 hertz (Hz).

Microwave Ovens: Common household microwave ovens heat food using microwaves without the need for fire.

Frequency: Microwaves have a frequency of approximately 10^{12} Hz.

Wavelength: The wavelength of microwaves is about 5 centimeters; hence they are also called centimeter waves.

Cosmology: Microwaves are significant in cosmology and astrophysics, particularly in the study of black body radiation, such as the 3-degree black body radiation from the Big Bang.

Light Spectrum:

Visible Light: Light follows microwaves in the electromagnetic spectrum. For example, red light has a frequency of about 10^{14} Hz.

Ultraviolet Light: Beyond visible light is ultraviolet (UV) light.

X-Rays: X-rays have frequencies higher than UV light.

Speed of Light in Different Media:

Vacuum: The speed of light in a vacuum is constant for all frequencies.

Materials: When light passes through a medium, different frequencies travel at different speeds. This means some frequencies pass through the medium faster than others.

Traveling Electromagnetic Fields:

Traveling electromagnetic fields refer to the propagation of electromagnetic waves through space. These fields consist of oscillating electric and magnetic fields that move together at the speed of light.

Electromagnetic Wave Generation:

Electromagnetic waves are generated because a changing electric field creates a magnetic field and vice versa. To understand this in more detail, consider an example of a long wire with a current passing through it.

Current and Magnetic Field:

When a current passes through the wire, it is due to a potential difference, causing charges to move, which generates the current. As a result of the current, a magnetic field is produced around the wire as shown in fig. 7. According to Ampere's Law, the strength of this magnetic field is determined by the integral of the magnetic field over a closed loop around the wire.

Ampere's Law is given by the integral form $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$, where $\vec{B}$ is the magnetic field, $d\vec{l}$ is the differential length element of the loop, and I is the current through the loop.

Application: For a long straight wire, the magnetic field $\vec{B}$ at a distance r from the wire is:

$$B = \mu_0 I / 2\pi r.$$

An electric field is also present due to the potential difference along the wire. Current cannot flow unless there is an electric field driving the charges from a higher potential to a lower potential. When the current in the wire changes direction, both the electric and magnetic fields change direction. The electric field and magnetic field are perpendicular to each other and change rapidly over time as the current alternates.

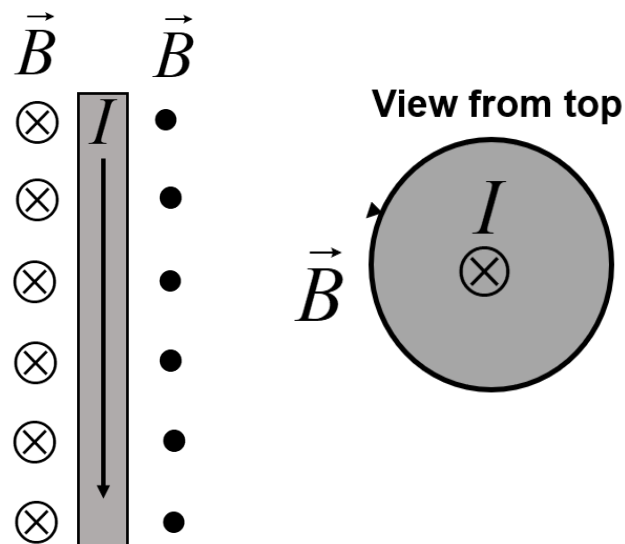


Fig. 7: Illustration of the magnetic field $\vec{B}$ produced around a long straight wire due to an electric current I . The dot convention indicates the current is considered to flow into the wire (entering the wire) while the cross convention indicates that the current is flowing out of the wire (leaving the wire).

Faraday's Law and Maxwell's Displacement Current:

Faraday's Law: A time-varying magnetic field induces an electromotive force (EMF) and, consequently, an electric field.

Maxwell's Addition: A changing electric field induces a magnetic field, a concept known as the Maxwell Displacement Current.

Electromagnetic Waves Propagation: The changing electric field creates a changing magnetic field, and the changing magnetic field creates a changing electric field. This continuous process allows an electromagnetic wave to propagate through space.

Field directions in an electromagnetic wave:

An electromagnetic wave consists of an electric field component and a magnetic field component. Both fields are perpendicular to each other and to the direction of wave propagation.

Visualization

Imagine a positively charged particle placed in the path of the electromagnetic wave:

Electric Field Direction: Along the z-axis.

Magnetic Field Direction: Along the y-axis.

Wave Propagation: Along the x-axis, all three vectors (electric field, magnetic field, and direction of propagation) are perpendicular to each other as shown in fig. 8. If the direction of propagation changes, the electric and magnetic fields adjust accordingly to remain perpendicular.

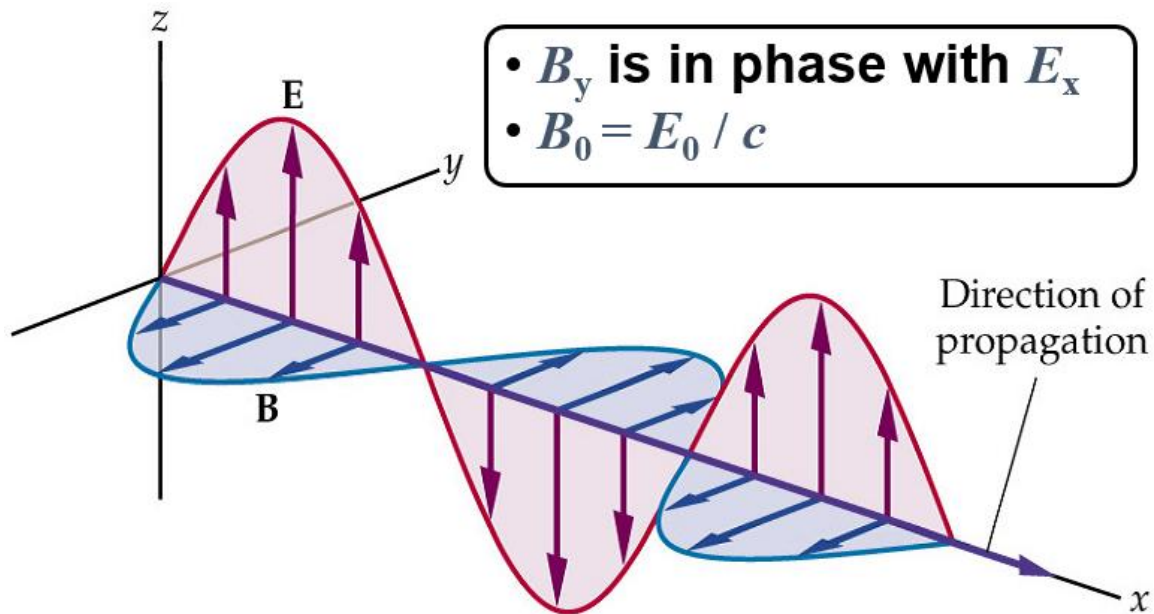


Fig. 8: Illustration of the orientation of an electromagnetic wave with a positively charged particle in its path. The electric field (z-axis), magnetic field (Y-axis), and wave propagation direction (x-axis) are shown as mutually perpendicular vectors.

Plane Harmonic Wave:

The electric field can be represented sinusoidally, varying with time and distance as shown in fig. 9. The wave moves in the z-direction at the speed of light. The wave equation is typically expressed as,

$$E_x = E_0 \sin(kz - \omega t)$$

where, $\omega = kc$, $E_0 = cB_0$

$$B_y = B_0 \sin(kz - \omega t)$$

Note: the direction of propagation is given by,

$$\hat{s} = \hat{e} \times \hat{b}$$

where $(\hat{e}, \hat{b})$ are the unit vectors in the (E, B) directions.

Where k is the wave number and ω is the angular frequency.

To observe the wave's movement, one must travel at the speed of light. Gauss's Law indicates that both electric and magnetic flux through a closed surface is zero because there are no net charges inside the wave. Historically, the concept of 'ether' was proposed as a medium for wave propagation, but it has been disproven.

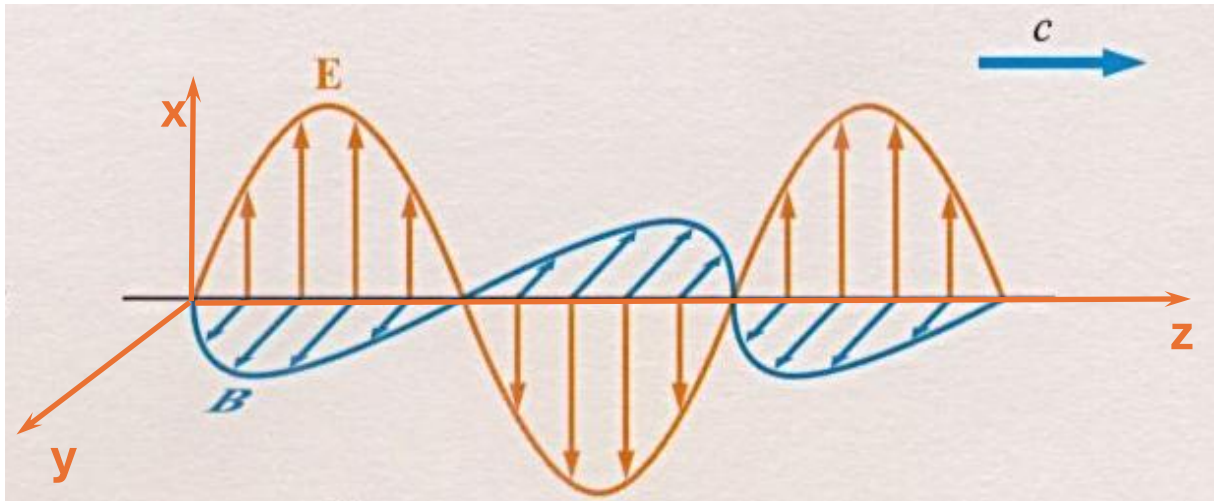


Fig. 9: Representation of the sinusoidal variation of the electric field over time and distance for an electromagnetic wave propagating in the z -direction at the speed of light. The electric field (x -axis), magnetic field (Y -axis), and wave propagation direction (z -axis) are shown as mutually perpendicular vectors.

Production of Electromagnetic waves:

Wave Generation Mechanism:

- Electromagnetic waves are generated where charges accelerate and currents change.
- A typical setup involves a transmitter that sends a current through a transformer to an antenna.
- The antenna contains electrons that move back and forth, creating an oscillating current.
- This oscillating current generates electromagnetic waves that radiate outward as shown in fig. 10.

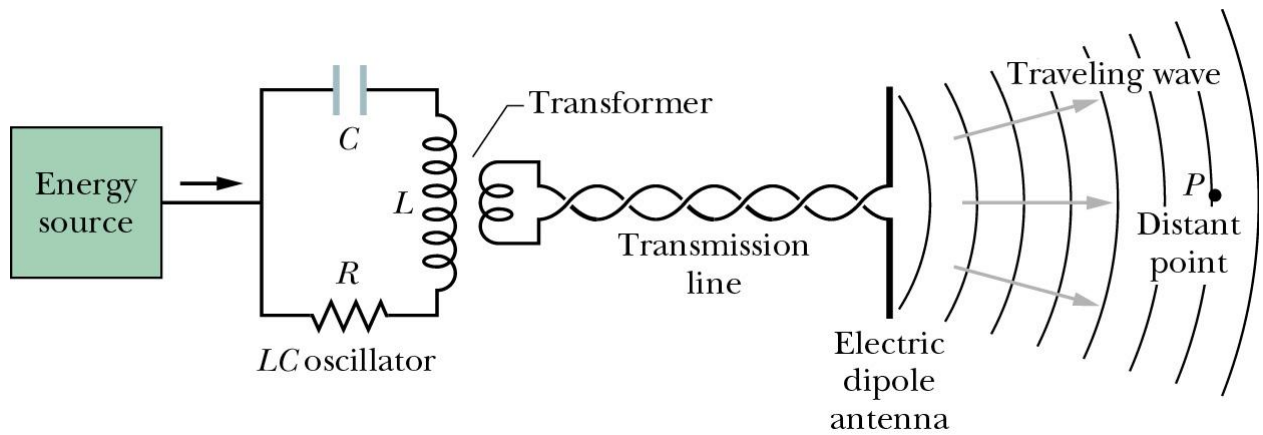


Fig. 10: Illustration of the production mechanism of electromagnetic waves: electromagnetic waves are generated when charges accelerate or currents change. Typically, a transmitter sends a current through a transformer to an antenna. Electrons in the antenna move back and forth due to the oscillating current. This oscillation of electrons creates electromagnetic waves that propagate outward from the antenna.

Types of Antennas and Their Function

Loop Antenna:

- It consists of a ring where current circulates rapidly as shown in fig. 11.
- Changing current produces a changing magnetic field, inducing an electromotive force (EMF).
- This process creates an oscillating electric field, generating electromagnetic waves.

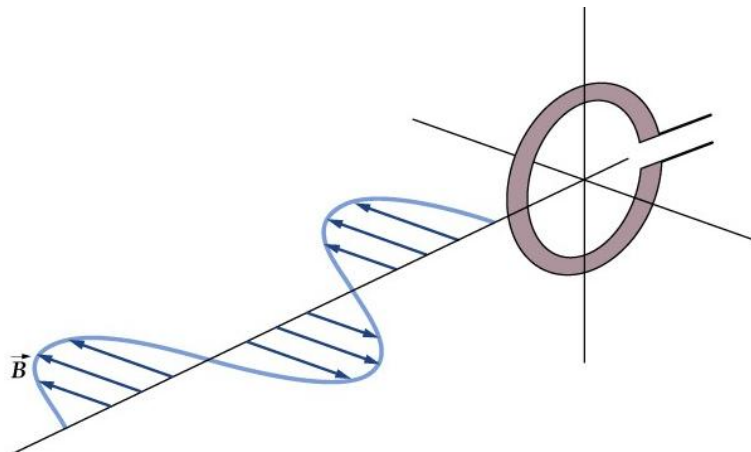


Fig. 11: Illustration of a circular loop with circulating current. Changing current induces EMF via magnetic field change, generating oscillating electric field and electromagnetic waves.

Dipole Antenna:

- Composed of two wires with an alternating current source as shown in fig. 12.
- Creates an electric field that changes direction with the current cycle.
- Changing electric fields induce magnetic fields, producing electromagnetic waves.

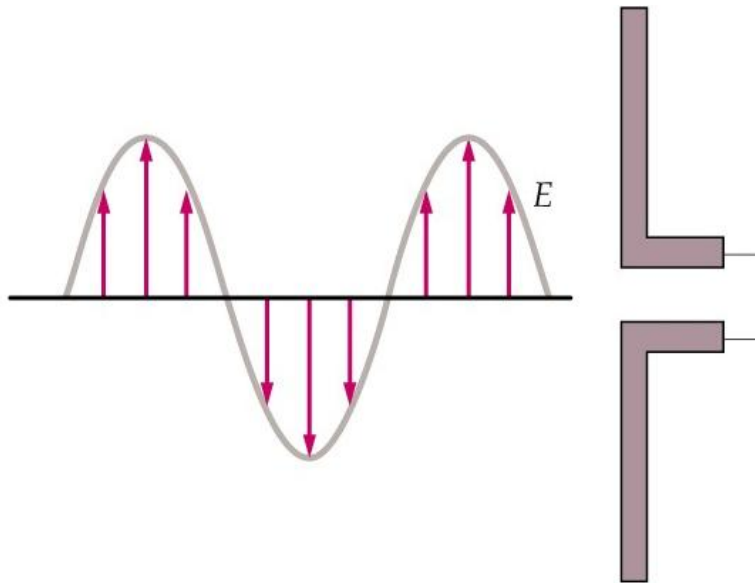


Fig.12. Representation of a dipole antenna, generating changing electric fields inducing magnetic fields, and emitting electromagnetic waves.

Electromagnetic Wave Radiation:

- Waves radiate outward, forming plane waves at a distance.
- The dipole antenna's radiation pattern is strongest perpendicular to the wires and zero along the axis.
- Power radiated is proportional to $\sin^2(\theta)$, where θ is the angle from the axis.
- Intensity decreases with distance, following an inverse square law ($1/r^2$) as shown in fig. 13.
- $$I(\theta) \propto \frac{\sin^2 \theta}{r^2}$$
- Electromagnetic waves from the dipole antenna are polarized along the direction of the electric field.

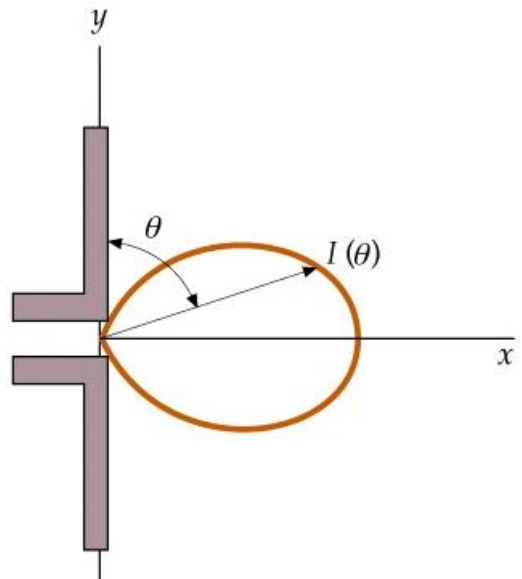


Fig.13: Representation of the electromagnetic waves radiated from a dipole antenna propagate outward, forming plane waves at a distance. The antenna exhibits a radiation pattern where the strongest emission occurs perpendicular to its wires, with zero radiation along the axis.

Practical Implementation of Antennas

Transmitter and Receiver:

- A transmitter generates electromagnetic waves using an antenna.
- A receiver antenna, often on a vehicle or another device, captures these waves, completing the communication process.

Function of Transmitter:

- Inside the transmitter, electronic components like transistors, capacitors, inductors, and resistors, powered by a battery.
- The battery supplies energy, converted into electromagnetic radiation when signals are sent out.
- Moving the lever changes the current, creating a changing magnetic field, which generates a changing electric field, resulting in electromagnetic radiation.

Example of radio frequency (RF) Generator:

- An RF generator can produce different frequencies, controlled by many dials.
- It can generate frequencies ranging from kilocycles to megacycles.
- A simple antenna can be used to emit and receive these frequencies.

Types of Modulation

Amplitude Modulation (AM):

- Amplitude Modulation (AM) uses changes in the signal *strength* to convey information.
- The amplitude of the electromagnetic waves varies according to the sound signal's amplitude.
- The electromagnetic wave (carrier) is modulated by the sound (signal), changing its amplitude to match the sound's intensity as shown in fig. 14.

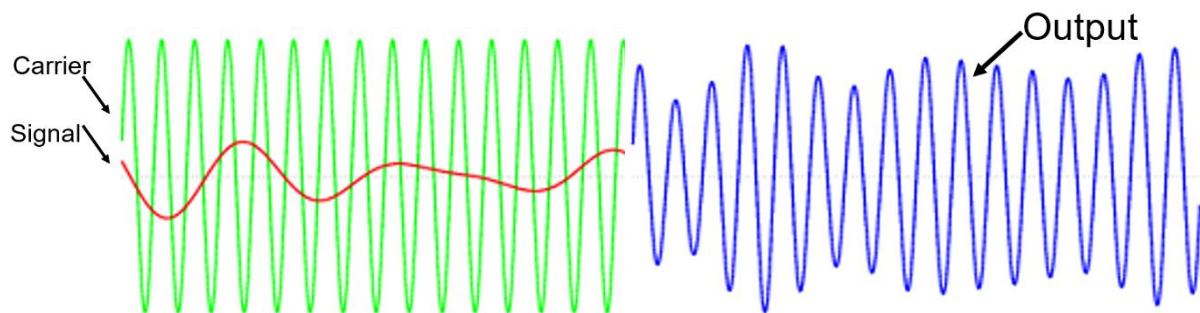


Fig. 14: Illustration of an Amplitude Modulation (AM) technique in signal processing and transmission. The waveform on the left depicting the unmodulated carrier wave and the waveform on the right representing the AM signal, where the amplitude of the carrier wave varies in accordance with the amplitude of the modulating signal, while the frequency remains constant.

Frequency Modulation (FM):

- Frequency Modulation (FM) uses changes in the wave's *frequency* to convey information.
- The frequency of the electromagnetic wave varies while the amplitude remains constant as shown in fig. 15.
- FM is more complex to produce and detect compared to AM, but it provides better sound quality.

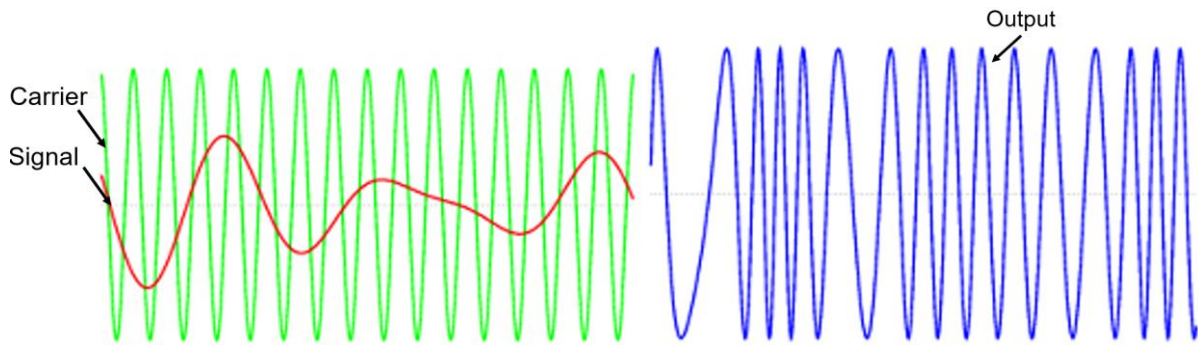


Fig. 15: Figure illustrates Frequency Modulation. The waveform on the left-hand side represents the unmodulated carrier wave with a constant frequency. The waveform on the right-hand side depicts output of the FM signal, where the carrier frequency varies according to the amplitude of the modulating signal, while its amplitude remains constant.

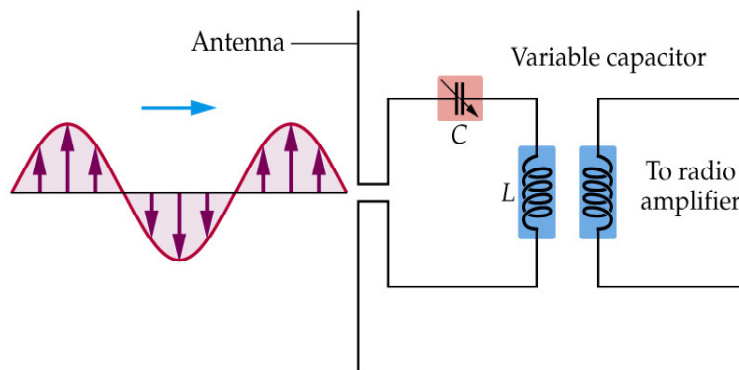
Detection of Modulated Signals

AM Detection:

- AM signals are easier to detect using simple circuits.
- The received AC current is passed through a diode or diode bridge and smoothed with a capacitor to extract the original sound signal as shown in fig. 16.

FM Detection:

- FM signals require more complex circuitry for frequency demodulation, which adjusts the frequency according to the sound signal's intensity.



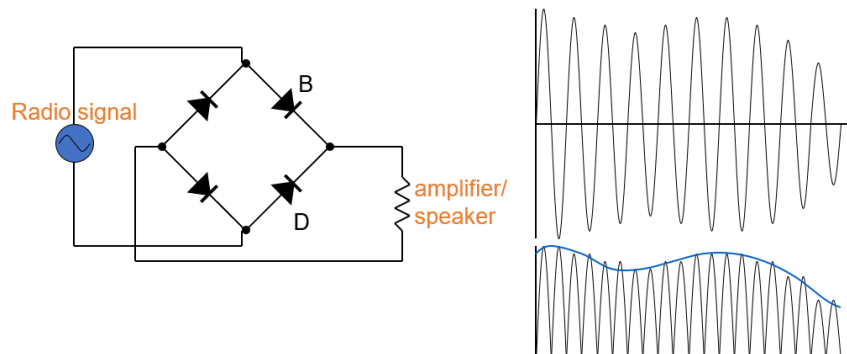


Fig. 16: Illustration of AM detection with a circuit involving a diode or diode bridge and capacitor to recover the original signal.

Polarization:

Electromagnetic Wave Polarization:

Electromagnetic Waves from an antenna are polarized along the antenna's direction. And other electromagnetic waves, such as sunlight, can have different polarizations.

Unpolarized Radiation:

When electromagnetic waves reach the ground, their polarization can be in any direction, resulting in unpolarized radiation. For a wave traveling in the z direction, if it is polarized along one direction, then the electric field is defined, and the magnetic field is perpendicular to it.

Polarized Waves:

The direction of the electric field determines the wave's polarization. Polarized radiation can be achieved from specific sources like antennas, while natural sources like the sun or light bulbs emit unpolarized radiation. Polarizers can select a particular polarization from unpolarized light by allowing only the light with an electric field aligned in a certain direction to pass through.

A typical plane electromagnetic wave equation is given by,

$$\begin{aligned} E_x &= E_0 \sin(kz - \omega t), & E_y &= 0, & E_z &= 0 \\ B_x &= 0, & B_y &= B_0 \sin(kz - \omega t), & B_z &= 0 \\ \therefore B_0 &= \frac{E_0}{c} \end{aligned}$$

This wave is an example of a *linearly* polarized wave. The direction of polarization is the direction of the oscillation of the electric field vector.

If a wave traveling in the z -direction and polarized at an arbitrary angle θ in the x - y plane, then the electric field vector be written as;

$$E_x = E_0 \cos \theta \cdot \sin(kz - \omega t)$$

$$E_y = E_0 \sin \theta \cdot \sin(kz - \omega t)$$

$$E_z = 0$$

Linear Polarization:

Absorption/reflection of vector component of wave $\perp$ to polarizer can produce linear polarization as shown in fig. 17.

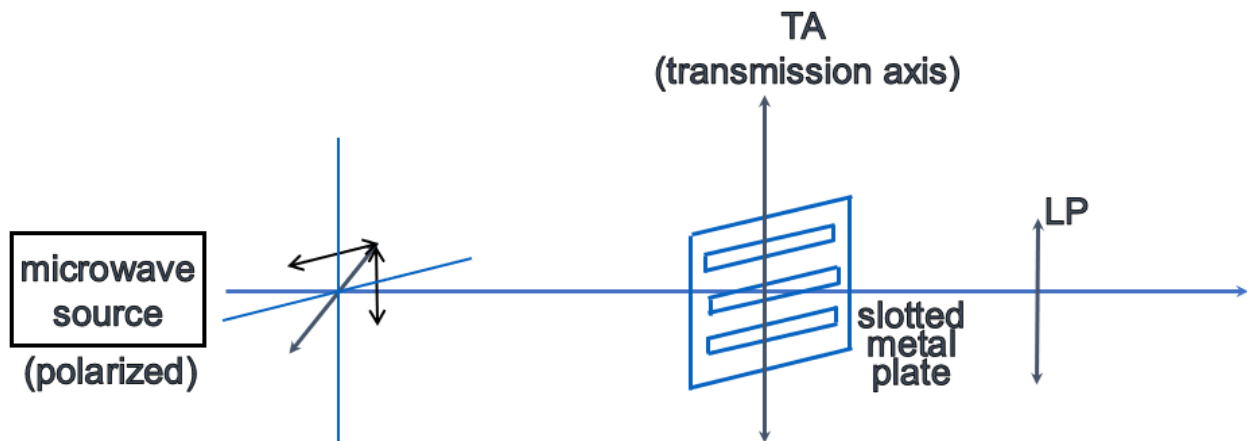


Fig.17: This figure depicts the concept of light polarization. On the left, unpolarized light is represented, showing light waves vibrating in multiple planes perpendicular to the direction of propagation. On the right, polarized light is shown, where light waves are restricted to vibrate in a single plane.

- The E-field component parallel to the slots is absorbed and/or reflected.
- The E-field component perpendicular to the slots is transmitted.
- Absorption produces linearly polarized waves but also reduces intensity.

Mechanism of Polarizers:

- Simple polarizers can be made from finely scratched iron sheets or long molecules that absorb electric fields aligned with them, blocking perpendicular fields.
- Polarized sunglasses (ID glasses) reduce glare by selecting a specific polarization of light.

Historical Aspect:

- James Clark Maxwell's mathematical work over 250 years ago, led to the understanding of electromagnetic waves.
- Experimental confirmation of these waves came 20-22 years later through Hertz's experiments.
- Later, the identification of the three basic forces of nature included electromagnetism, tying in with the weak force.

Recognition:

- Professor Abdus Salam, a Pakistani physicist, contributed significantly to this field and was honoured for his work on the death anniversary of James Clark Maxwell in Scotland.

Lecture#31

The Reality of Light

Understanding the nature of light has been a long journey, requiring the collective efforts of thousands of people over centuries. Only about 250 years ago, we began to understand its true nature. Today, we know that light is actually electromagnetic waves, produced when a charge moves. This movement generates a current, which in turn produces these waves. The importance of light in our lives drives our desire to know everything about it.

The Speed of Light

One of the first questions people asked about light was about its speed. Early studies attempted to measure this, with Galileo being one of the pioneers in this field.

Galileo's Experiment

Galileo conducted an experiment in 1667 where he asked one of his assistants to stand on a mountain with a lantern. He stood on another mountain with his lantern. The idea was that when Galileo uncovered his lantern, the assistant would uncover his as soon as he saw the light, and they would measure the time difference. However, this method proved to be impractical due to the incredibly high speed of light.

Roemer's Discovery

- In 1675, the Danish astronomer Ole Roemer made a significant contribution to understanding the speed of light.
- He observed the moons of Jupiter and noticed that the eclipses of these moons occurred at different times depending on the Earth's position relative to Jupiter as shown in fig.1.
- When the Earth was closer to Jupiter, the eclipses happened sooner, and when it was further, the eclipses were delayed by about 16 minutes and 6 seconds (16.6 min. late).
- Roemer used this time difference to estimate the speed of light (estimated $c = 2.3 \times 10^8$ m/s), concluding it takes this amount of time for light to travel the additional distance across the Earth's orbit.

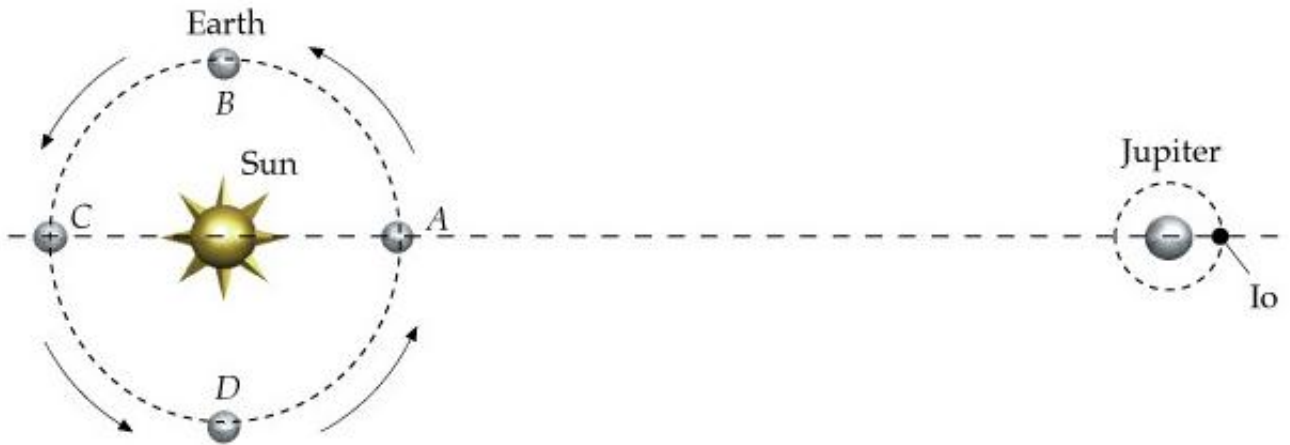


Fig. 1: Ole Roemer's observations of Jupiter's moons' eclipses in 1675 showed time variations based on Earth's position, leading to an estimate of the speed of light at 2.3×10^8 m/s.

Fizeau's Experiment

Overview

Hippolyte Fizeau, a French scientist, conducted a pioneering experiment in 1849 to measure the speed of light. He used a rotating cogwheel apparatus as shown in the fig. 2.

1. A beam of light was directed through the gaps between the teeth of a rotating wheel.
2. The light travelled a known distance L to a distant mirror.
3. The mirror reflected the light back through the rotating wheel.
4. By adjusting the rotational speed of the wheel, Fizeau determined the speed of light with remarkable accuracy.

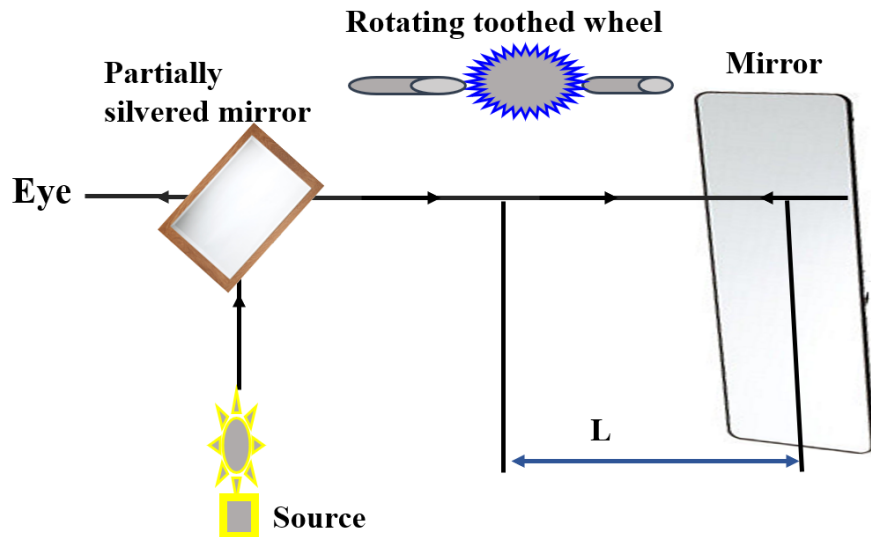


Fig. 2: Fizeau's 1849 experiment used a rotating cogwheel and a distant mirror to measure the speed of light by timing the light's round trip and adjusting the wheel's rotation speed.

Derivation of the Formula to calculate the speed of light:

The light beam travels a total distance of $2L$ (to the mirror and back).

The time taken for the light to travel this distance is given by:

$$t = \frac{2L}{C}$$

The cogwheel has N teeth and an equal number of gaps.

The angular distance between consecutive teeth (or gaps) is:

$$\theta = \frac{2\pi}{2N} = \frac{\pi}{N}$$

The wheel rotates with an angular velocity ω (radians per second).

In time t , the wheel rotates through an angle ωt .

For the returning light beam to be observed, it must pass through the next gap.

Therefore, the angle rotated by the wheel in time t should equal to:

$$\theta = \omega t$$

$$\text{Using, } t = \frac{2L}{C}$$

$$\theta = \omega \left(\frac{2L}{C} \right)$$

$$\frac{2L}{C} = \frac{\theta}{\omega}$$

$$C = \frac{2L\omega}{\theta} \approx 3 \times 10^8 \text{ m/s}$$

Where,

L : Distance to the mirror.

c : Speed of light.

θ : Angular distance between the teeth of the cogwheel (in radians).

ω : Angular velocity of the rotating wheel.

Foucault's Experiment

In 1860, Léon Foucault improved upon Fizeau's work using a rotating mirror system. This method involved:

1. A light source directed towards an eight-sided rotating mirror as shown in the fig.3.
2. The light reflected from the mirror to a fixed mirror and back.
3. The rotation speed was adjusted so the light returned to the observer through the same path.
4. He found no signal unless mirror rotates $1/8$ turn in ΔT
5. This experiment further confirmed the high but finite speed of light, $c = 2.98 \cdot 10^8$ m/s

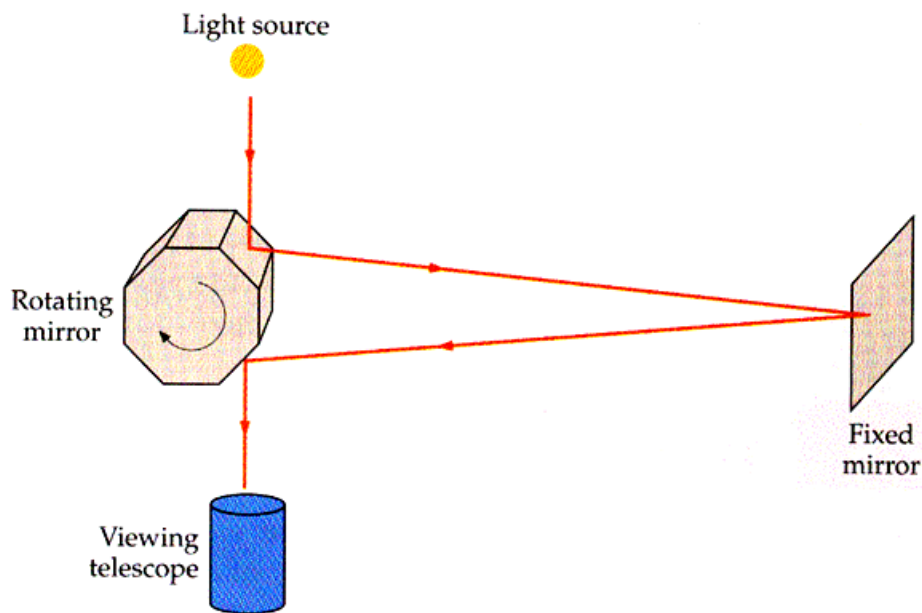


Fig. 3: Foucault's 1860 experiment used an eight-sided rotating mirror and a fixed mirror to measure the speed of light, refining the estimate to 2.98×10^8 m/s by adjusting the rotation speed for precise light reflection.

The Nature of Light and Electromagnetic Waves

All these experiments show that light travels incredibly fast around 300,000 kilometres per second, completing multiple orbits (7-8) around the Earth in one second. This explains why we

see lightning before hearing thunder since sound waves travel much slower, at 1100 meters per second.

Light travels much faster than sound. For example:

1. Thunder and lightning start at the same time, but we will see the lightning first.
2. When a starting pistol is fired, we see the smoke first and then hear the bang.

The Visible Spectrum

Light and sound are waves, but light is part of a vast electromagnetic spectrum. The visible range for humans is from about 400 nanometers (deep blue) to 700 nanometers (deep red) as shown in fig.4. Different light sources emit varying wavelengths within this range.

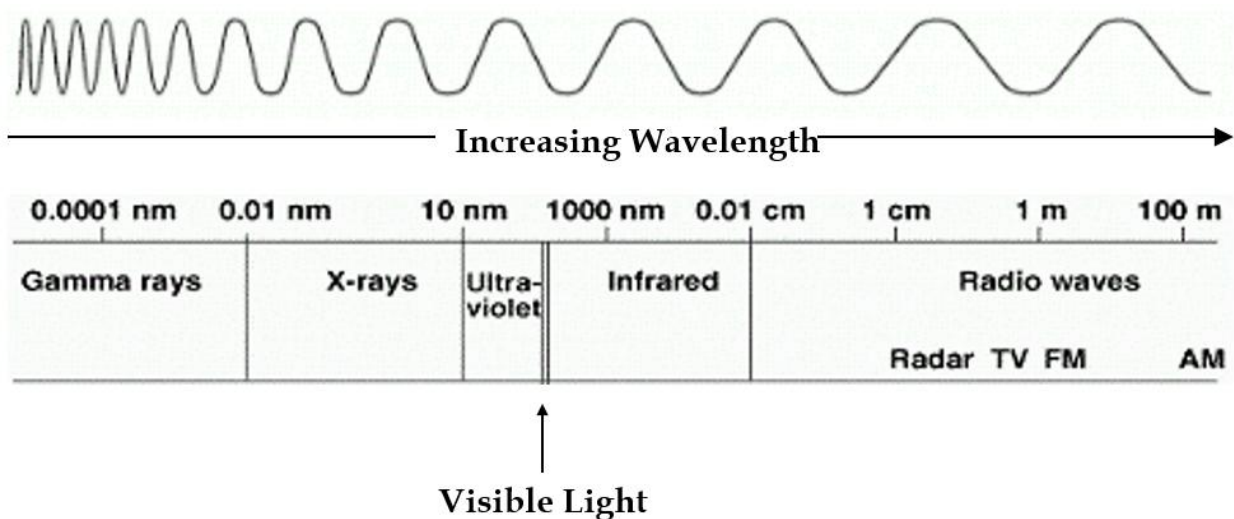


Fig. 4: The visible spectrum, ranging from 400 nm (deep blue) to 700 nm (deep red), represents the portion of the electromagnetic spectrum perceivable by the human eye.

White Light and Color Perception

White light contains all colours in the visible spectrum. It is perceived as white when all the frequencies are approximately equal. Combining different colors can create other colors:

- Blue and red make turquoise.
- Blue and green make light blue.
- Red and green make yellow.

- Combining red, blue, and green in equal amounts produces white as shown in fig. 5.

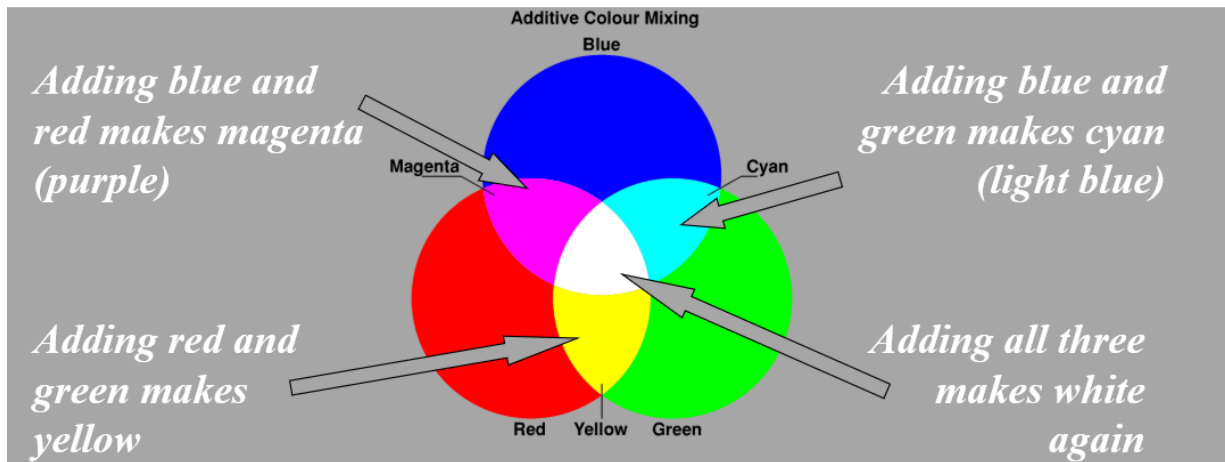
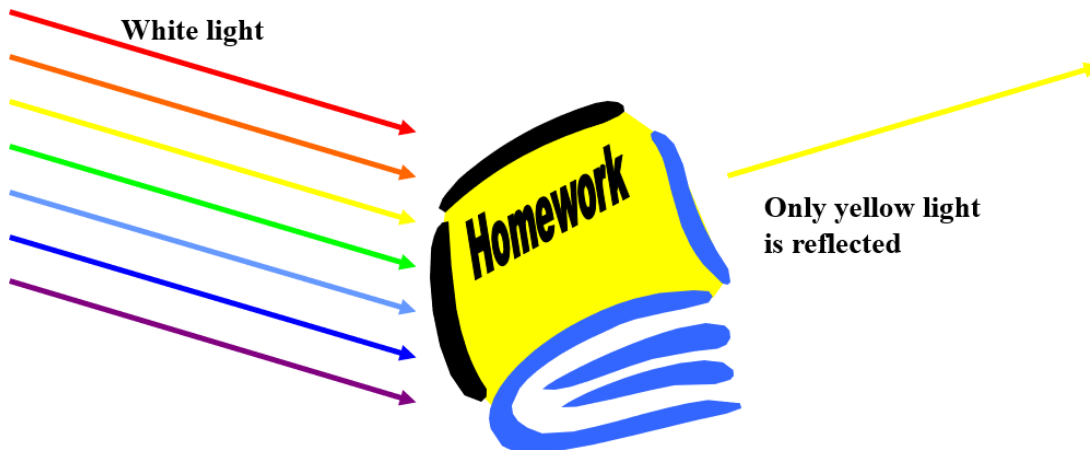


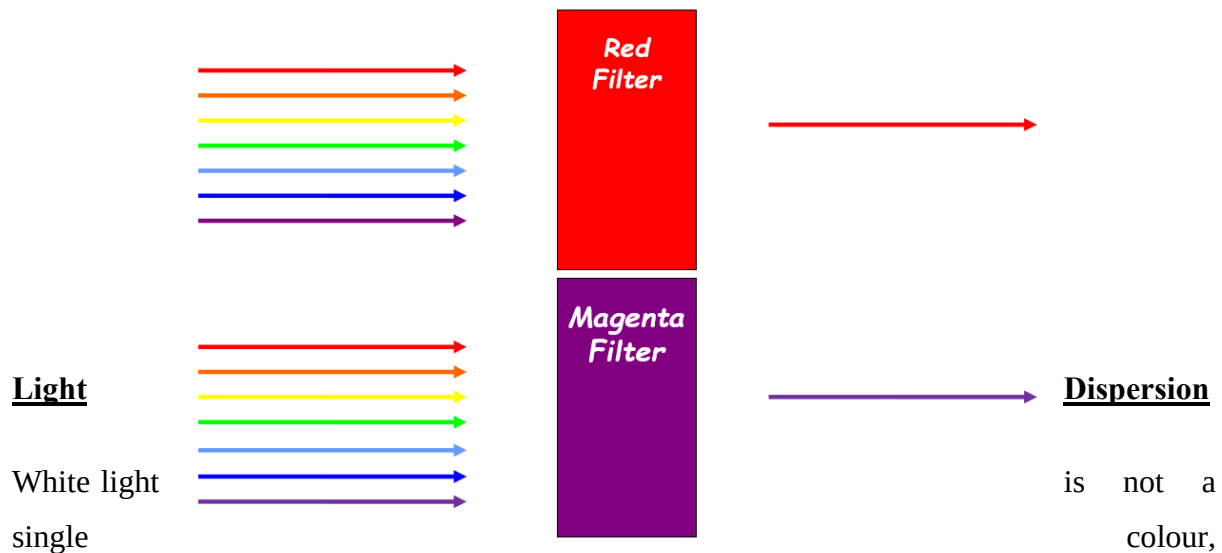
Fig. 5: White light, containing all colors of the visible spectrum, can be split into individual colors. Combining different colors like blue and red, blue and green, and red and green can create turquoise, light blue, and yellow, respectively, while equal amounts of red, blue, and green produce white.

Objects appear coloured based on the light they reflect. For example, a yellow object reflects yellow light, while a white object reflects all colours.

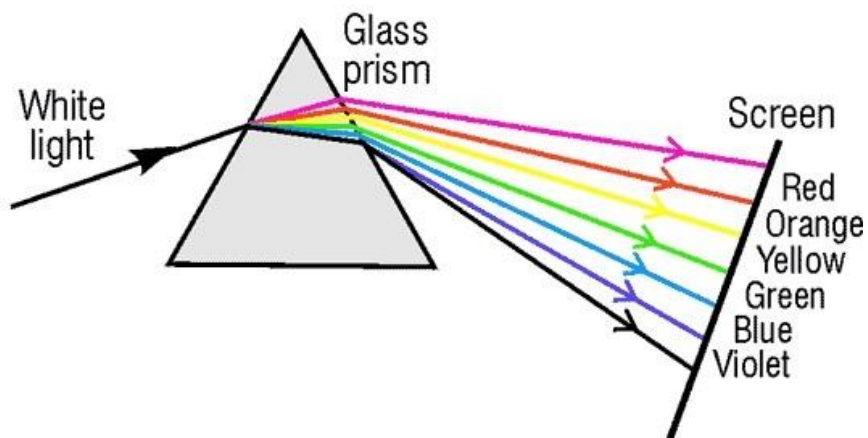


Filters and Color Selection

Colour filters can select specific colours from white light by absorbing other colours. For example, a red filter allows only red light to pass through.



when white light passes through a prism, it disperses into its component colours due to different wavelengths traveling at different speeds through the medium. This results in a spectrum of colours from blue to red.



Exploring Light: Reflection and Refraction

Understanding the behaviour of light through simple experiments can illuminate the principles of reflection and refraction.

Reflection of Light

Experiment with a Laser and Mirror

Using a laser, we can conduct experiments to explore how light reflects off surfaces. When a laser beam hits a mirror:

1. Part of the light passes through the mirror.
2. Another part of the light is reflected.

Key Concepts of Reflection

- **Incident Ray:** The incoming light ray that strikes the mirror.
- **Reflected Ray:** The light ray that bounces off the mirror.
- **Angle of Incidence:** The angle between the incident ray and the normal (a line perpendicular to the surface).
- **Angle of Reflection:** The angle between the reflected ray and the normal.

In a smooth mirror, the angle of incidence is equal to the angle of reflection as shown in fig. 6.

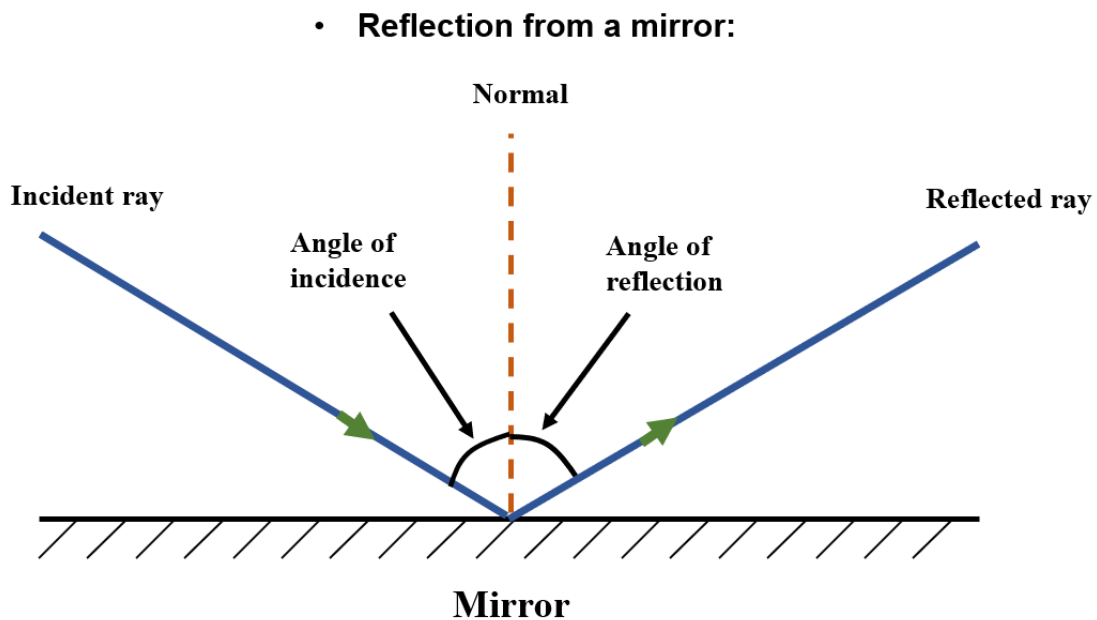


Fig. 6: demonstrating the law of reflection in a smooth mirror, where the angle of incidence is equal to the angle of reflection.

Diffuse Reflection

When light hits a rough surface, such as a coat, it scatters in many directions. Each small element of the surface reflects light according to the law of reflection, but the overall effect is diffuse reflection, where light is reflected in multiple directions as shown in fig. 7.

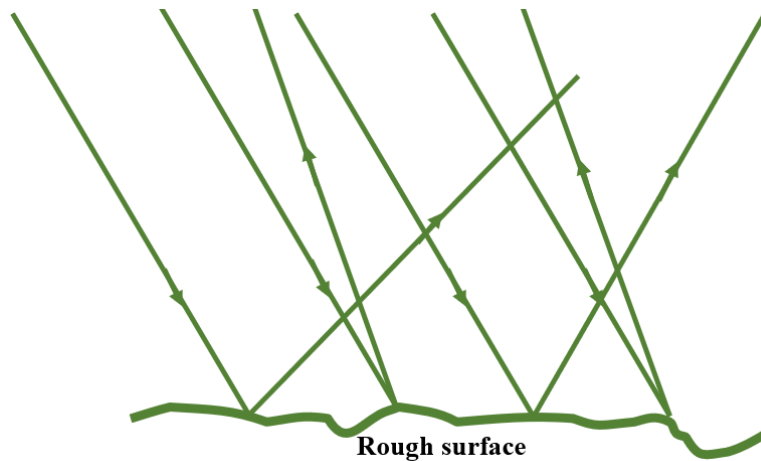


Fig. 7: Diffuse reflection occurs when light strikes a rough surface, causing it to scatter in various directions due to the irregularities on the surface.

Refraction of Light

Experiment with a Laser and Glass

When a laser beam passes through a glass surface:

1. The light enters the glass and bends towards the normal.
2. As it exits the glass and re-enters the air, it bends away from the normal.

Key Concepts of Refraction

- **Refraction:** The bending of light as it passes from one medium to another.
- **Normal:** A line perpendicular to the surface at the point of incidence.
- **Angle of Incidence:** The angle between the incident ray and the normal.
- **Angle of Refraction:** The angle between the refracted ray and the normal.

When light travels from air to glass, it bends towards the normal as shown in fig. 8. Conversely, when it exits the glass and enters air, it bends away from the normal. The incident light and transmitted light are parallel but displaced.

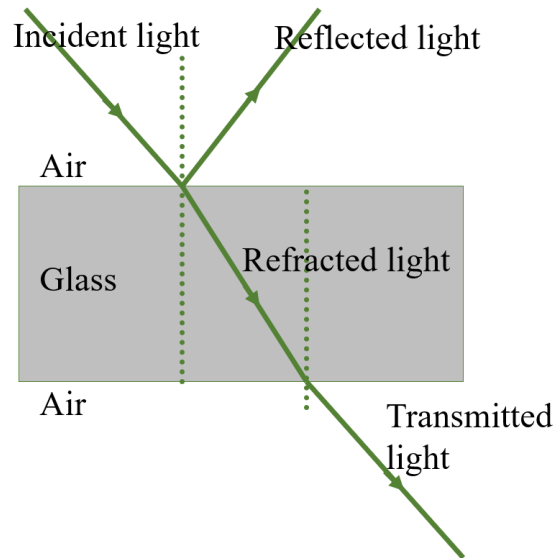


Fig. 8: Refraction of light at the interface between air and glass, demonstrating bending towards the normal upon entering glass and away from the normal upon exiting, with incident and transmitted light paths shown as parallel but displaced.

Understanding Refractive Index and Light Behavior

Definition of Refractive Index

The refractive index (n) is defined as the speed of light in a vacuum (c) divided by the speed of light in a medium (v):

$$\text{Refractive index} = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in material}}$$

$$n = \frac{c}{v}$$

When light enters a medium, it usually slows down due to interactions with the medium's particles, causing a change in speed.

Reasons for Light Slowing Down

When light enters a medium, the electrons within the medium interact with the electromagnetic field of the light, causing them to vibrate and emit electromagnetic radiation. This process results in the formation of a new wave that moves forward but at a slower speed than the original wave.

Refractive Index of Common Materials:

- **Air:** 1.0003
- **Water:** 1.33
- **Alcohol:** 1.36
- **Oil:** 1.5
- **Diamond:** 2.42

Light Refraction Experiment with Water

1. Shine a laser through an empty glass.
2. Gradually fill the glass with water.
3. Observe the laser beam bending towards the normal as water is added to the glass due to the higher refractive index of water (1.33).

Effects of Refraction

Objects submerged in water appear bent or displaced due to the change in light direction as it passes from water (higher refractive index) to air (lower refractive index).

Fermat's Principle

Historical Context

Fermat's Principle, which is fundamental in the study of optics, states that light takes the path that requires the least time when traveling from one point to another. This principle was also mentioned by Ibn Al-Haytham (Ibne Haim) in his significant work "Kitab al-Manazir" (The Book of Optics), though the mathematics behind it was not fully developed during his time. Fermat's Principle provides an easier and more straight forward understanding of the behavior of light compared to the methods available in earlier periods.

Fermat's Principle Explained Light in Free Space:

In free space or a vacuum, light travels in straight lines. According to Fermat's Principle, this is because the straight-line path is the shortest and therefore the quickest route for light to travel between two points.

To mathematically represent this, consider two points A and B in free space as shown in fig. 9. The time t taken for light to travel this distance L is given by:

$$t = L / c$$

where c is the speed of light in a vacuum. Since the distance L in a straight line is the shortest possible distance, the time t is minimized.

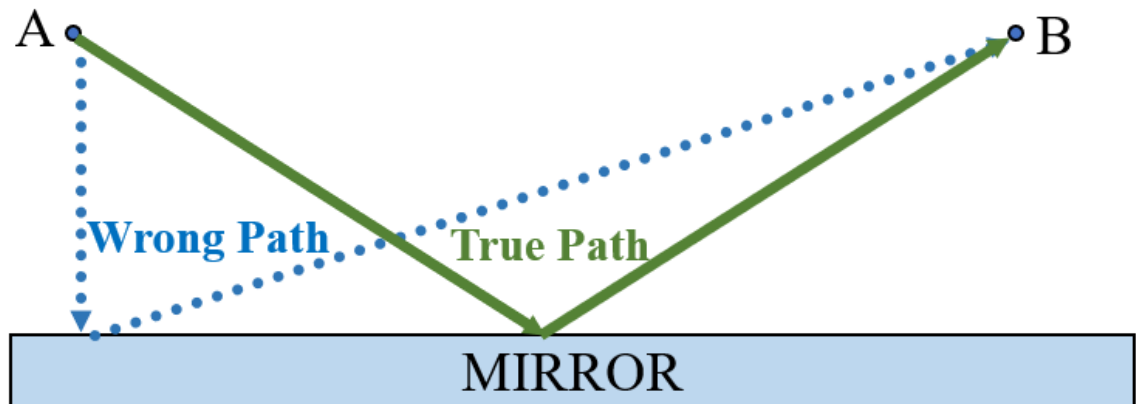
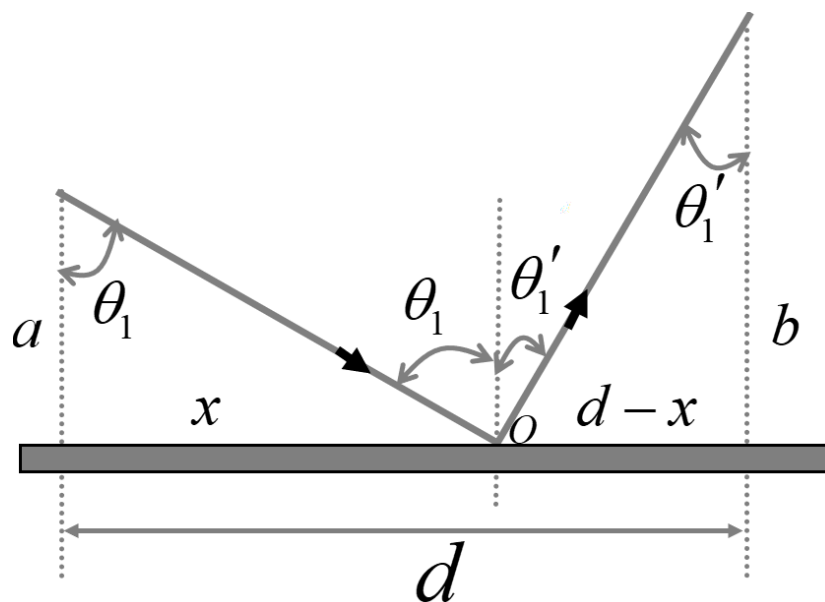


Fig. 9: Fermat's Principle dictates that light travels in straight lines in free space or a vacuum, following the path that minimizes the time between two points A and B.

A light ray traveling from one fixed point to another fixed point follows a path such that, compared with nearby paths, the time required is minimum.

For a case of Reflection

Consider a light wave traveling from point A to point B, reflecting off a mirror at point O. Fermat's Principle can be used to determine the path of the light.



The total distance traveled by the light is,

$$L = \sqrt{a^2 + x^2} + \sqrt{b^2 + (d - x)^2}$$

$$t = L / c$$

Taking derivative w.r.t 'x',

$$\frac{dt}{dx} = \frac{1}{c} \frac{dL}{dx}$$

where:

- a is the vertical distance from A to the mirror.
- b is the vertical distance from B to the mirror.
- x is the horizontal distance from A to O.
- d is the total horizontal distance between A and B.

To minimize the travel time, take the derivative of t with respect to x and set it to zero.

$$\frac{dt}{dx} = 0$$

$$\begin{aligned} 0 &= \frac{1}{c} \frac{dL}{dx} = \frac{1}{c} \frac{d}{dx} (\sqrt{a^2 + x^2} + \sqrt{b^2 + (d - x)^2}) \\ &= \frac{1}{2c} (a^2 + x^2)^{-1/2} (2x) + \frac{1}{2c} [b^2 + (d - x)^2]^{1/2} (2)(d - x)(-1) \end{aligned}$$

On simplifying,

$$\frac{x}{\sqrt{a^2 + x^2}} = \frac{d - x}{\sqrt{b^2 + (d - x)^2}}$$

$$\sin \theta_1 = \sin \theta'_1$$

$$\theta_1 = \theta'_1$$

This results in the equality of angles of incidence and reflection, confirming that the angle of incidence θ_1 is equal to the angle of reflection θ_2 .

Refraction and Snell's Law

When light travels from one medium to another, its speed changes due to different refractive indices. This change causes the light to bend, a phenomenon known as refraction.

Derivation Using Fermat's Principle

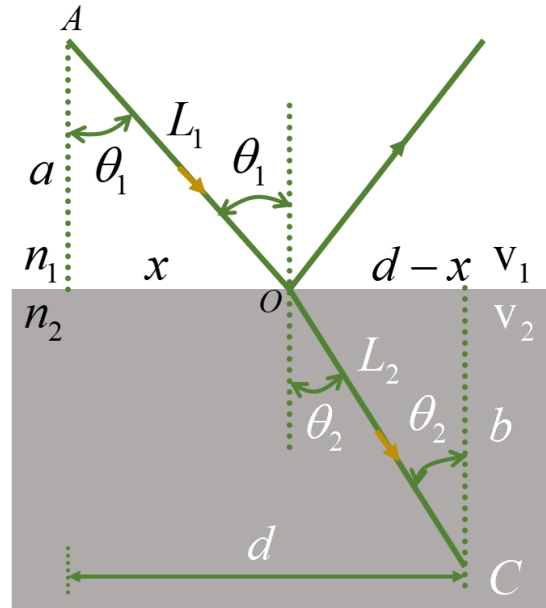
Consider a light ray traveling from point A in medium 1 (with refractive index n_1) to point C in medium 2 (with refractive index n_2). The light refracts at the boundary between the two mediums at point O.

Let:

- θ_1 be the angle of incidence.
- θ_2 be the angle of refraction.
- x be the horizontal distance from A to O.
- L_1 be the distance from A to O.
- L_2 be the distance from O to C.

Using Fermat's Principle, the total time t taken by light to travel from A to C is the sum of the time taken in each medium:

$$t = \frac{L_1}{v_1} + \frac{L_2}{v_2}$$



where v_1 and v_2 are the speeds of light in mediums 1 and 2, respectively. The speeds of light in the mediums are related to the refractive indices by:

$$n = \frac{c}{v}, \Rightarrow v_1 = \frac{c}{n_1}, \Rightarrow v_2 = \frac{c}{n_2}$$

$$t = \frac{n_1 L_1 + n_2 L_2}{c} = \frac{L}{c}$$

$$L = n_1 L_1 + n_2 L_2 \text{ (Optical path length)}$$

$$L = n_1 \sqrt{a^2 + x^2} + n_2 \sqrt{b^2 + (d-x)^2}$$

$$0 = \frac{dt}{dx} = \frac{1}{c} \frac{dL}{dx} = \frac{n_1}{2c} (a^2 + x^2)^{-1/2} (2x) + \frac{n_2}{2c} [b^2 + (d-x)^2]^{1/2} (2)(d-x)(-1)$$

$$n_1 \frac{x}{\sqrt{a^2 + x^2}} = n_2 \frac{d-x}{\sqrt{b^2 + (d-x)^2}}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \text{ (called snell's law of refraction)}$$

Total Internal Reflection

- When light travels from a medium with a higher refractive index to one with a lower refractive index, it can be totally internally reflected if the angle of incidence exceeds a certain critical angle.

- If the angle of incidence is greater than θ_c , total internal reflection occurs, and the light does not pass into the second medium.

Critical Angle and Total Internal Reflection

- Consider a laser emitting light at different angles. When the angle of incidence reaches the critical angle (θ_c), the light travels along the surface of the medium. If the angle of incidence exceeds θ_c , total internal reflection occurs, and the light remains within the medium as shown in fig. 10.
- One can calculate the critical angle by following:

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

$$\theta_c = \sin^{-1} \frac{n_2}{n_1}$$

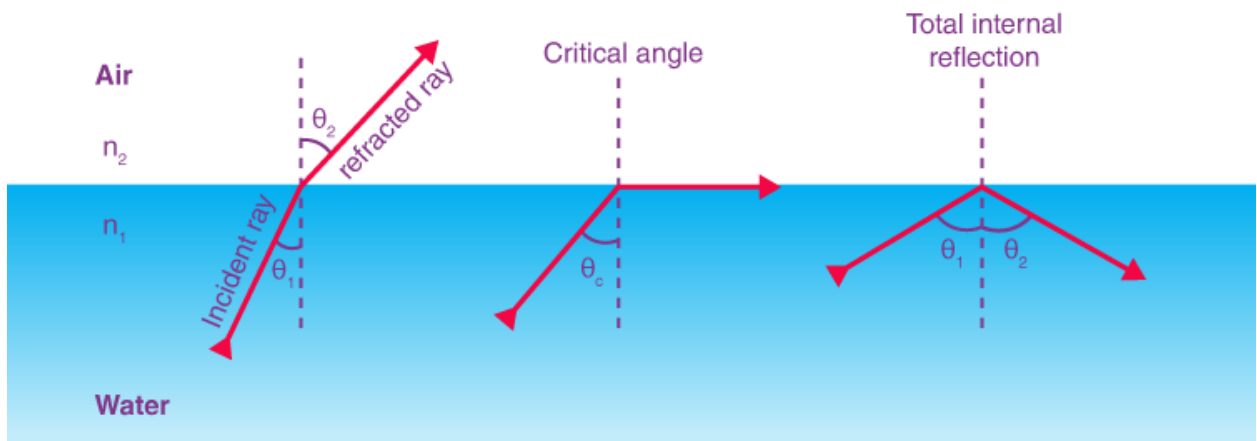


Fig. 10: Total internal reflection occurs when light within a medium strikes the boundary at an angle exceeding the critical angle θ_c , causing it to reflect back into the medium rather than refracting out.

Importance of Total Internal Reflection

- **Critical Angle:** The angle beyond which light is totally internally reflected within the medium.
- **Applications:**
 - Optical fibers for communication.
 - Diamond brilliance due to a low critical angle and high refractive index.

Optical Fibers:

Optical fibers use total internal reflection to transmit light signals over long distances with minimal loss. An optical fiber consists of a core with a high refractive index surrounded by cladding with a lower refractive index. Light entering the fiber is reflected internally, ensuring it travels through the core without escaping.

For effective transmission, the optical fiber must be very thin, and the refractive index must be controlled to ensure total internal reflection.

Optical Fiber Communication and the Nature of Light

Structure of Optical Fiber:

- **Core:** Thin glass thread, thinner than a human hair, with a high refractive index as shown in fig. 11.
- **Cladding:** Surrounds the core and has a lower refractive index to ensure total internal reflection.
- **Protective Covering:** Encases the cladding for protection.

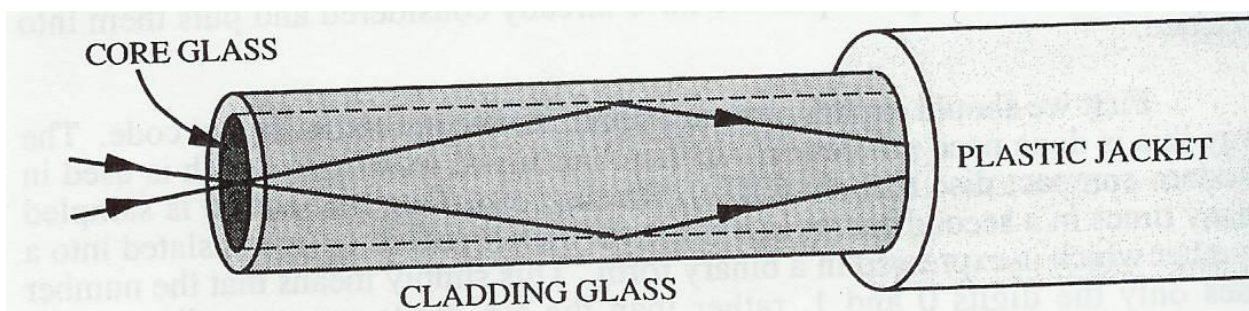


Fig. 11: Representation of an optical fibre, comprising a core with a high refractive index, surrounded by cladding with a lower refractive index, and a protective covering, essential for efficient light transmission through total internal reflection.

Optical Fiber in Communication

1. Historical Context:

- 100 years ago, cables could only support 50-60 telephone calls simultaneously.
- Modern optical fibers can handle millions of calls due to total internal reflection.

2. Mechanism:

- Light travels through the core, reflecting internally due to the difference in refractive indices.
- This prevents signal loss, even if the fiber is bent or under slight pressure.

3. Advantages:

- High capacity for data transmission.
- Minimal signal loss.

The Nature of Light: Particle or Wave?

Historical Debate

- **Newton's Particle Theory:** Light is made of particles.
- **Wave Theory:** Light behaves as waves.

Modern Understanding

- Light exhibits both particle and wave properties, known as wave-particle duality.
- A *wave* is a moving pattern that is revealed by its interaction with particles

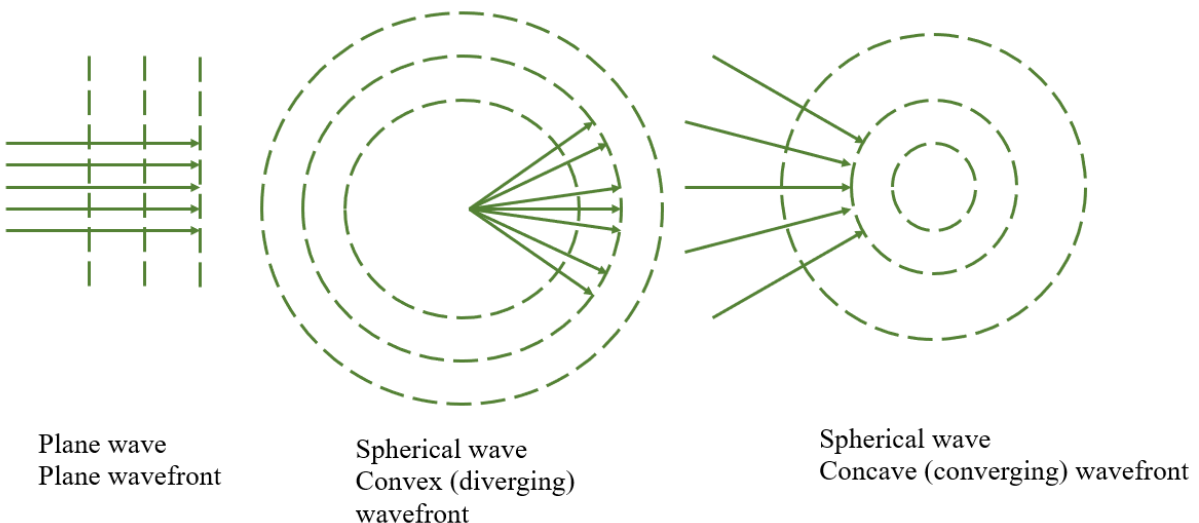
Concept of Wavefronts

1. Wavefronts:

- **Circular Wavefront:** Originates from a point source, spreads out in circles.
- **Plane Wavefront:** Forms when circular wavefronts travel far and flatten.

2. Interaction with Obstacles:

- When waves hit an obstacle, they can change direction and create new wavefronts.
- This results in regions of light and dark, demonstrating wave behavior.



Huygens' Principle

- **Basic Concept:**

- When light encounters an obstacle, each point on the wavefront acts as a source of new wavelets.
- These wavelets spread out and form new wavefronts as shown in fig. 12.
- **Implications:**
 - Light can bend around corners.
 - This bending and spreading of light waves are due to their wave nature.

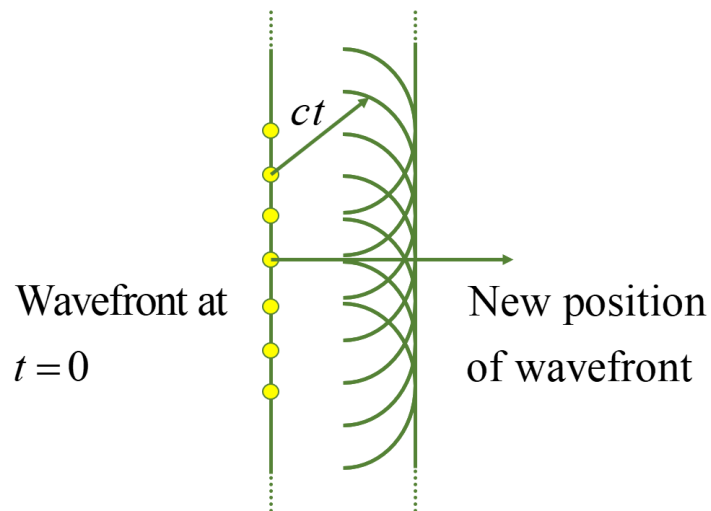


Fig. 12: Huygens' Principle illustrates how each point on a wavefront acts as a source of secondary wavelets, which propagate and form new wavefronts, allowing light to bend around corners and demonstrating its wave nature.

Light as Electromagnetic Radiation

- **Generation of Light:**
 - Light is generated by accelerating charges.
 - Examples: Heated filament in a light bulb, or electrons moving in atoms.
 - Power radiated by a single charge is:

$$P = \frac{2}{3} \frac{e^2 a^2}{c^3}$$

- Electrons in atoms emit radiation when they accelerate.
 - Classical physics could not explain why electrons don't lose energy and spiral into the nucleus.
- **Quantum Mechanics:**
 - Introduced to explain atomic behavior and energy quantization.

- **Electromagnetic Waves:**

- Composed of electric and magnetic fields oscillating perpendicular to each other.
- Energy carried by electromagnetic waves depends on the amplitude of these fields.
- Electromagnetic waves transport linear momentum as well as energy
- For complete absorption of energy U ,

$$p = \frac{U}{c}$$

- For complete reflection of energy U ,

$$p = \frac{2U}{c}$$

- Radiation pressures can be determined experimentally

$$\frac{E_{\max} B_{\max}}{2\mu_0 c} = \frac{E_{\max}^2}{2\mu_0 c^2} = \frac{B_{\max}^2}{2\mu_0}$$

Colors and Temperature

- **Color of Light:**

- The color of light from an object indicates its temperature.
- Hotter objects emit light of shorter wavelengths (e.g., blue stars).
- Cooler objects emit light of longer wavelengths (e.g., our sun appears yellow).

Radiation Pressure

- **Concept:**

- The electric field of an electromagnetic wave that strikes a surface act on an electron, giving it a velocity v as shown in fig. 13. The magnetic field then exerts a force on the moving charge in the direction of propagation of the incident light. That is the origin of the “light pressure”.
- Light exerts pressure on objects when it is absorbed or reflected.
- This radiation pressure is typically very small and not noticeable in everyday life.

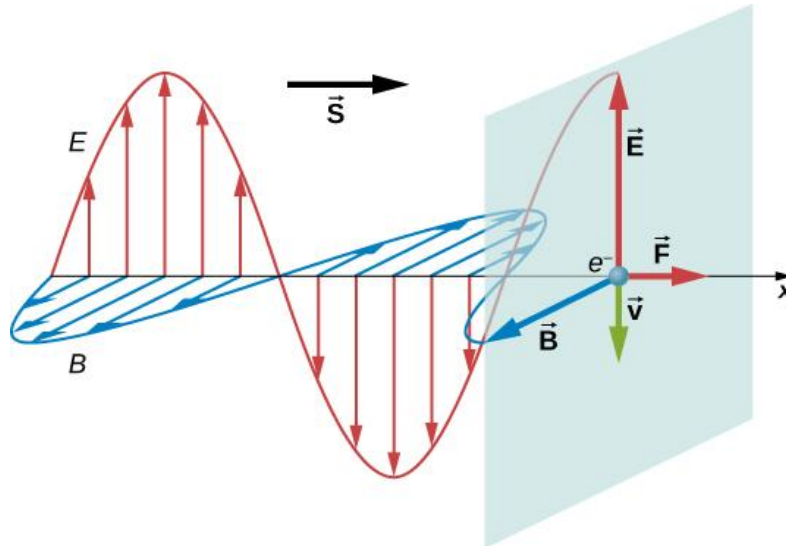


Fig. 13: Radiation pressure arises from the momentum carried by electromagnetic waves. When light is absorbed or reflected, it exerts a small but measurable pressure on surfaces, stemming from the interaction of the electric and magnetic fields with the surface of electrons.

- **Solar Radiation Pressure:**

- Within the sun, radiation pressure counteracts gravitational pull, maintaining equilibrium.

Radiant Flux:

- Total amount of energy passing through a surface or region of space *per unit time*
 - Typically denoted by Φ (Phi)
 - Also called *power*
 - Measured in watts (W) or joules/second (J/s)
 - These are metric quantities (rather than e.g., calories/second)
- Typically used for a light's total output.

Lecture#32

Introduction:

In this lecture, we will focus on the interaction between matter and light. Understanding this interaction helps us answer questions such as:

- Why do objects have different colors?
- Why is the sky blue?
- How does light pass through, refract, or reflect off objects?

Spectroscopy:

To study light properly, we use a spectrograph. The process involves:

- Light passing through a slit.
- Entering a tube where the waves become parallel.
- Entering a dispersive element (prism/mirror).
- Different colors separating and reaching an imaging lens.
- Recording the light on a photographic film or another device as shown in fig. 1.

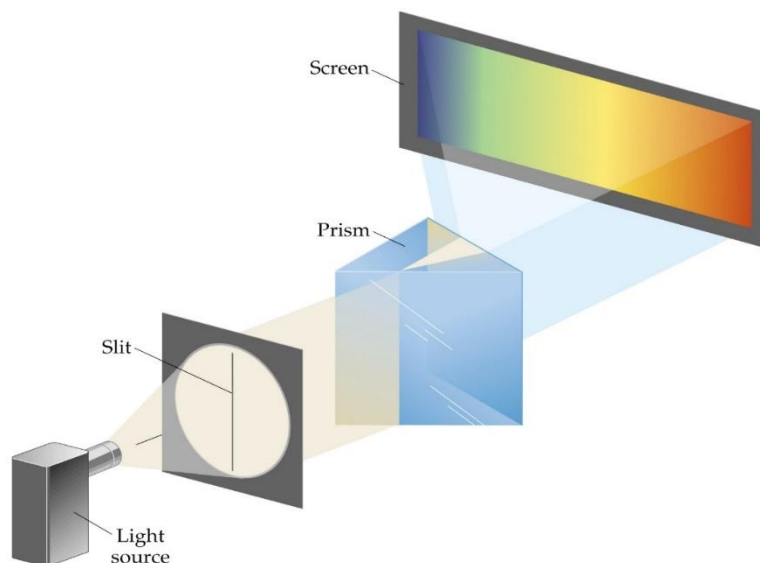


Fig. 1: Representation of a spectrograph, where light passes through a slit, is collimated, dispersed into colors by a prism, and appeared on a screen.

Light as an Electromagnetic Wave:

Light is an electromagnetic wave that does not need a medium to travel but can also pass through various mediums. Each source of light emits different frequencies and wavelengths. For example:

Red light: Frequency of 4.29×10^{14} Hz, wavelength of approximately 700 nm.

Blue light: Frequency of 7.50×10^{14} Hz, wavelength of approximately 400 nm.

Basic Spectrograph:

A basic spectrograph is an essential tool used to study the interaction between matter and light, particularly how light is emitted, absorbed, transmitted, and reflected by different substances.

Basic Spectrograph Components and Function

A spectrograph is an instrument designed to disperse light into its constituent wavelengths, creating a spectrum that can be analyzed as shown in fig. 2. Here's how it works:

- **Light Source:** The journey begins with a light source, such as a torch. The light emitted by the torch contains a wide range of frequencies and wavelengths.
- **Slit:** The light enters the spectrograph through a narrow slit. This helps to produce a precise, collimated beam of light.
- **Collimating Tube:** The light beam passes through a collimating tube, which makes the light rays parallel.
- **Dispersive Element:** The parallel light rays then hit a dispersive element, such as a prism or a diffraction grating. This element separates the light into its component colors (wavelengths).
- **Imaging Lens:** After dispersion, the separated light passes through an imaging lens, which focuses the different colors.
- **Recording Device:** Finally, the focused light reaches a recording device, such as a photographic film or a digital sensor. This device captures the spectrum, allowing for detailed analysis.

Function of the Spectrograph

- **Separation of Light:** The spectrograph separates light into its component wavelengths. This separation is crucial for studying the properties of the light.
- **Recording Spectra:** The recording device captures the spectrum, which can then be analyzed to determine the light's characteristics and the properties of the source emitting it.

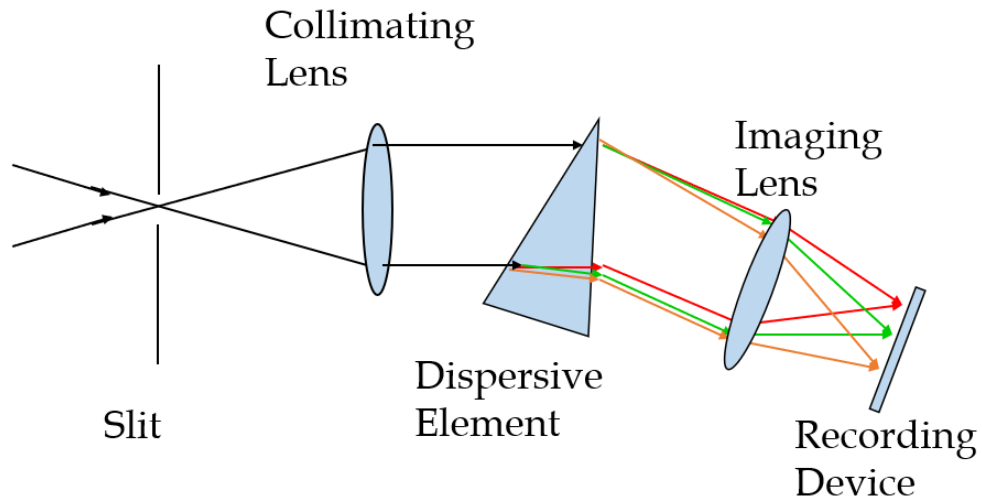


Fig. 2: Basic components of a spectrograph, illustrating the process from light entry through a slit to spectrum and recorded by a detector.

Eye Sensitivity to Color:

Our eyes are most sensitive to green light but can see from deep blue to deep red. Sensitivity varies with wavelength, peaking in the red range as shown in fig. 3.

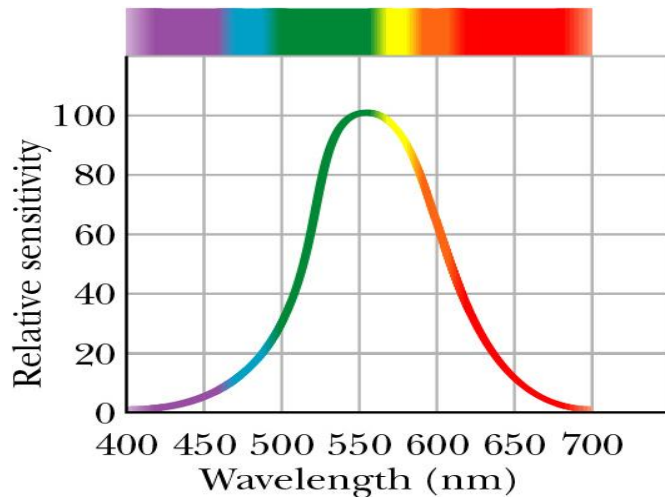


Fig. 3: A plot between wavelength and relative eye sensitivity to color, illustrates the peak sensitivity in the red range and visible spectrum extending from deep blue to deep red.

Light and Matter Interaction:

The interaction between light and matter can be divided into four parts:

- 1) **Emission:** How light is emitted from matter.
- 2) **Absorption:** How light is absorbed by matter.
- 3) **Transmission:** How light passes through matter.
- 4) **Reflection:** How light is reflected by matter.

Emission:

When a filament in a bulb is heated by an electric current, it emits light. The process involves:

- Atoms in the filament moving and accelerating electrons, causing electromagnetic radiation.
- The temperature of a filament in a common light bulb is around 100 degrees centigrade, emitting white light. As it cools, it emits red light, and if cooled further, only heat without visible light.

Black Body Radiation:

Blackbody radiation is a term used to describe the relationship between an object's temperature, and the wavelength of electromagnetic radiation it emits. A black body is an idealized object that absorbs all electromagnetic radiation it comes in contact with.

In a welding shop, as metal is heated, its color changes due to black body radiation. The sequence of colors is:

- Dark red
- Red
- Yellow
- Pure white

When metal cools, it reverses this color sequence. Black body radiation occurs because the atoms and molecules inside the heated object move, causing the emission of light.

Box Example

Imagine a closed box with a small hole. When the box is heated, light waves inside become excited and exit through the hole, demonstrating black body radiation. This process occurs due to the movement of atoms, molecules, and electrons within the heated substance.

Black body radiation is produced by the movement and acceleration of electrons within atoms and molecules, emitting electromagnetic radiation. The color of this radiation varies with temperature.

Temperature and Wavelength

The relative brightness of radiation changes with temperature as shown in fig. 4. For example:

- At 8000 K, radiation peaks around 300 nm, appearing blue.
- At 6000 K, radiation peaks around 500 nm.
- At 4000 K, radiation peaks around 700 nm, appearing reddish.

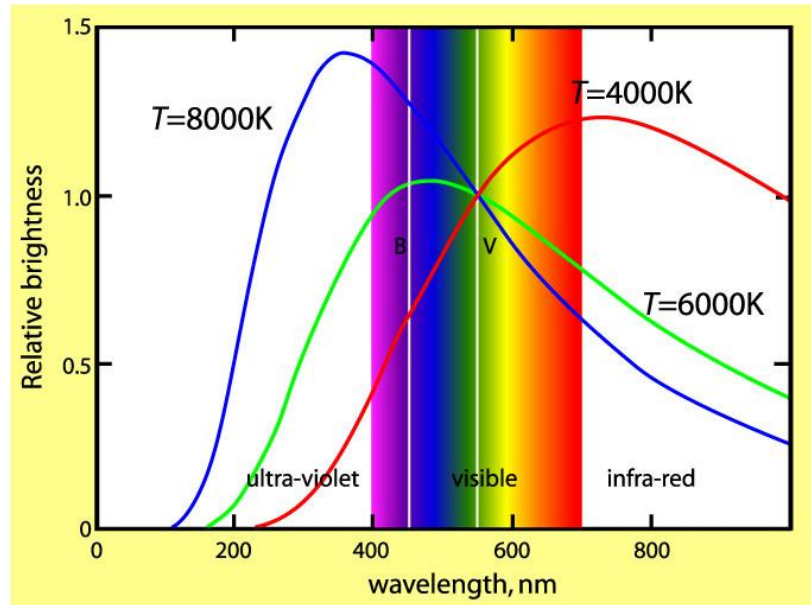


Fig. 4: A plot between wavelength and relative brightness illustrates the relative brightness of radiation at different temperatures, with peaks at 300 nm (blue) for 8000 K, 500 nm for 6000 K, and 700 nm (reddish) for 4000 K.

Wien's Law:

Wien's Law relates the temperature of an object to the wavelength at which it emits the maximum radiation: $\lambda_{\text{max}} \times T = 2.90 \times 10^{-3} \text{ m K}$ as shown in fig. 5. This law applies when a substance is in thermal equilibrium.

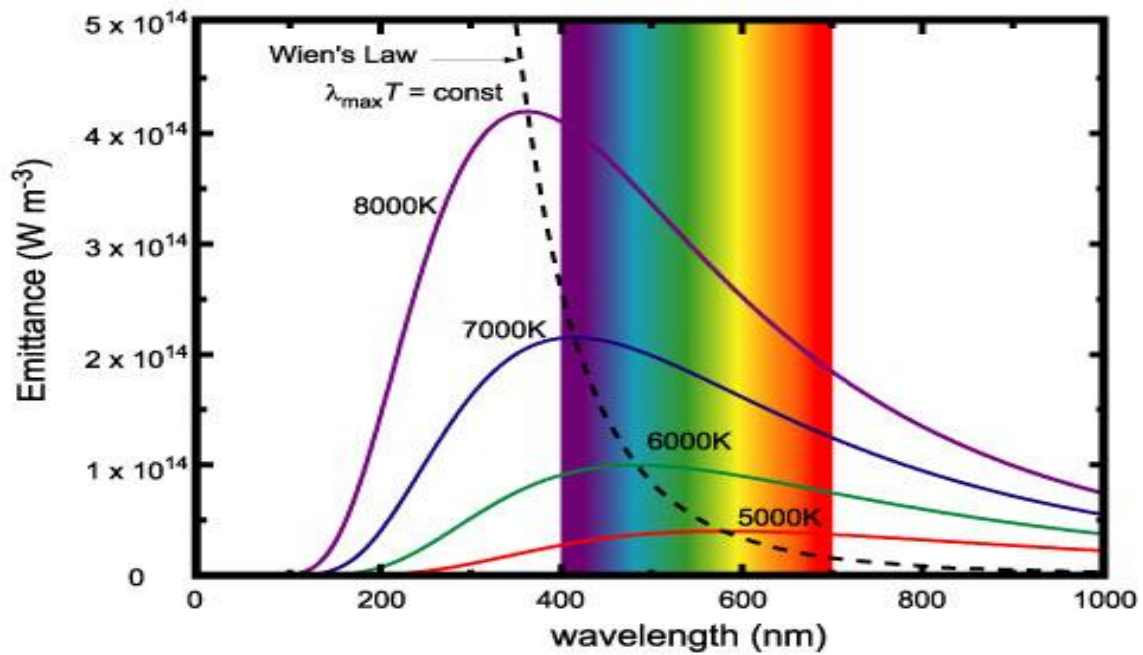


Fig. 5: Wien's Law, illustrating the relationship between an object's temperature and the wavelength of maximum radiation emission, given by $\lambda_{\text{max}} \times T = 2.90 \times 10^{-3} \text{ m K}$, applicable to substances in thermal equilibrium.

When can you use Wien's Law?

Application of Wien's Law:

- Only for objects that emit light not for those that reflect light.
- Light emitted by hot, solid objects obey Wien's Law.
- The Sun and other stars obey Wien's Law since the gases they are composed of remain at a high density (at least up to the outermost layers of the star).

Wien's Law can be used to determine the temperature of the Sun and stars, which are in thermal equilibrium. For example:

- Stars that appear blue have temperatures around 15000 K.
- The Sun's surface temperature is around 5800 K.
- Cooler stars appear red as shown in fig. 6.

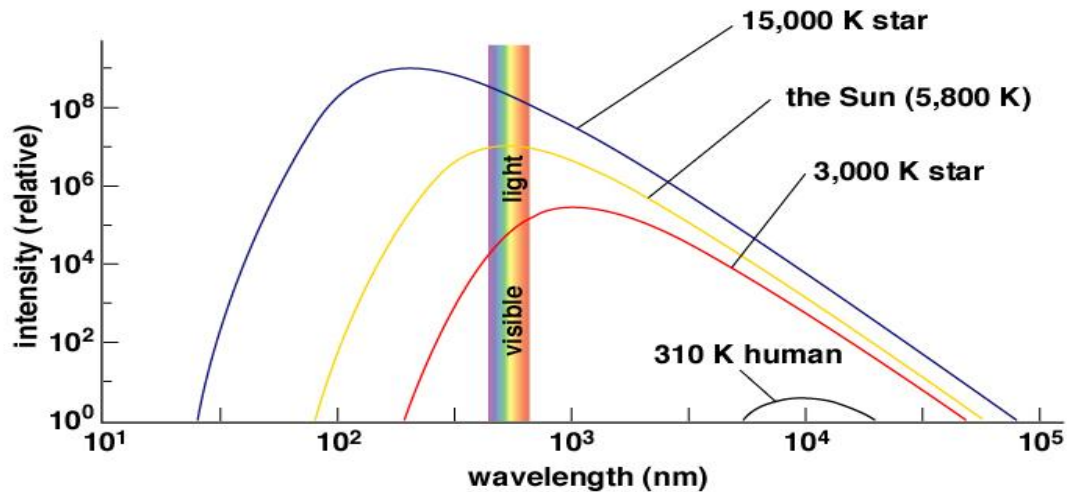


Fig. 6: A plot between the wavelength and intensity to illustrate the temperature of stars, with blue stars at approximately 15000 K, the Sun at 5800 K, and cooler stars appearing red.

Stefan-Boltzmann Law:

The Stefan-Boltzmann Law calculates the power radiated per unit area of a black body given as: $P = \sigma T^4$

where σ is the Stefan-Boltzmann constant ($5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$). All heated matter radiates energy and hotter objects radiate more energy.

Problem Example:

A planet is at distance R from the sun. The sun has temperature T_0 and radius R_{sun} . Find the temperature T of the planet.

Solution:

Calculating the Temperature of a Planet:

Consider a planet orbiting a star (like the Sun). The planet's temperature can be determined by equating the energy it receives from the Sun to the energy it radiates.

According to Stefan-Boltzmann Law: Energy radiated $P = \sigma T^4$, where T is the planet's temperature.

Energy received from the sun:

The sun's energy distributed over a sphere hence,

$$\text{Energy received} = \frac{\sigma T_{\text{sun}}^4 \times 4\pi R_{\text{sun}}^2}{4\pi R^2}$$

R is the distance from the sun, R_{sun} is the radius of the sun, and T_{sun} is the sun's temperature.

In thermal equilibrium, heat received per unit area is equal to power radiated per unit area.

$$\sigma T^4 = \frac{\sigma T_{\text{sun}}^4 \times 4\pi R_{\text{sun}}^2}{4\pi R^2}$$

Solving for T:

$$\Rightarrow T = T_{\text{sun}} \sqrt{\frac{R_{\text{sun}}}{R}}$$

Conclusion:

- Planets closer to the Sun will be hotter.
- The temperature decreases with the square root of the distance from the Sun.

Emission Spectrum:

When atoms are heated, they emit light at specific wavelengths, creating an emission spectrum unique to each element. This spectrum serves as the element's identity.

Example of Emission Spectrum

- **Copper Sulphate:** When heated, it emits a green color.
- **Sodium Chloride (Common Salt):** Emits a red color when heated.
- **Hydrogen:** Emits three distinct lines in its spectrum, known as H-alpha, H-beta, and H-gamma, corresponding to wavelengths of 6565, 4863, and 4342 Å respectively.
- **Helium:** Emits multiple lines, differing from hydrogen due to having two protons as shown in fig. 7.

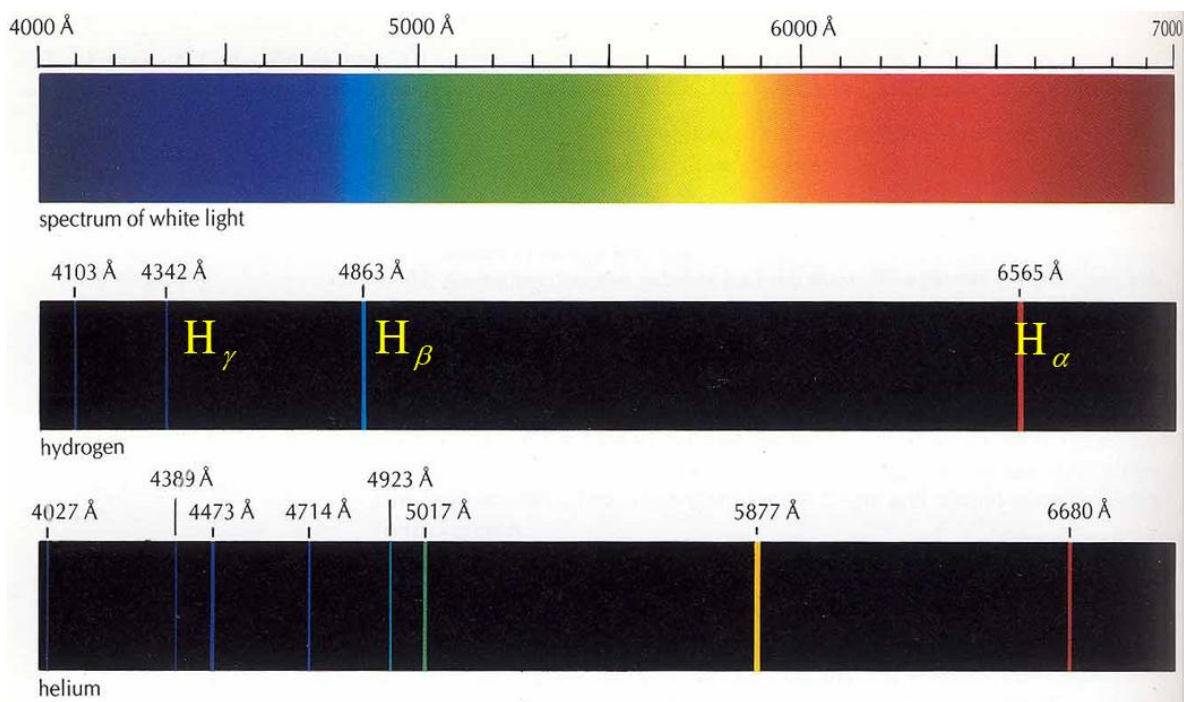


Fig. 7: Representation of emission spectra for various elements: Copper Sulphate emits green light, Sodium Chloride emits red, Hydrogen shows H-alpha, H-beta, and H-gamma lines at 6565 Å, 4863 Å, and 4342 Å respectively, and Helium emits multiple distinct lines.

Practical Application

Using a spectrograph, scientists can identify elements by their emission spectra, similar to identifying a person by their fingerprint.

Spectra:

- Each chemical element produces a very distinct pattern of colors called a spectrum
- Spectra that have all colors = Continuous spectra
- Spectra that have only certain frequencies = Line Spectra of which there are 2 types:
 - Absorption spectrum: Some material in front of a continuous source absorbs some energy.
 - Emission (line spectrum): Some material giving off energy.

Absorption Spectrum:

When light from a hot object, such as a star, passes through a cooler gas, specific wavelengths are absorbed, creating dark lines in the spectrum. This absorption spectrum can indicate the presence of certain elements between the observer and the light source.

Example

- **Hydrogen Cloud:** Light passing through a hydrogen cloud will show dark lines at H-alpha, H-beta, and H-gamma wavelengths, indicating the presence of hydrogen.

Emission spectrum:

The line spectrum, also known as the emission spectrum, occurs when atoms or molecules emit light at specific, discrete wavelengths. This happens when atoms are heated or otherwise excited, and they release energy in the form of light as they return to their ground state. Each element has a unique emission spectrum, which acts as a fingerprint for identifying the element.

Explanation of Line Spectrum

- **Excitation of Atoms:** When an atom absorbs energy, its electrons move to higher energy levels. This energy can come from various sources such as heat or electrical energy.
- **Emission of Light:** As the electrons return to their lower energy levels, they emit light. This light consists of specific wavelengths unique to the element, creating the line spectrum.
- **Observation of Emission Spectrum:** For example, when different salts are exposed to a flame, the resulting colors are characteristic of the elements within those salts. Copper sulfate produces a green color, while sodium chloride (common salt) produces a red color.

Spectral Lines:

- **Hydrogen Spectrum:** The emission spectrum of hydrogen is particularly well-known. When hydrogen gas is excited, it emits light at specific wavelengths, creating distinct lines in the spectrum. The three primary lines in the visible spectrum are:
 - **H-alpha:** 656.5 nm (red)
 - **H-beta:** 486.3 nm (blue-green)
 - **H-gamma:** 434.2 nm (violet)

These lines correspond to specific transitions of electrons between energy levels in hydrogen atoms.

- **Helium Spectrum:** Helium emits light at different wavelengths compared to hydrogen. The spectrum of helium shows multiple lines, reflecting the various possible electron transitions in helium atoms. This is due to helium having more electrons and more complex electronic structure as shown in fig. 8.

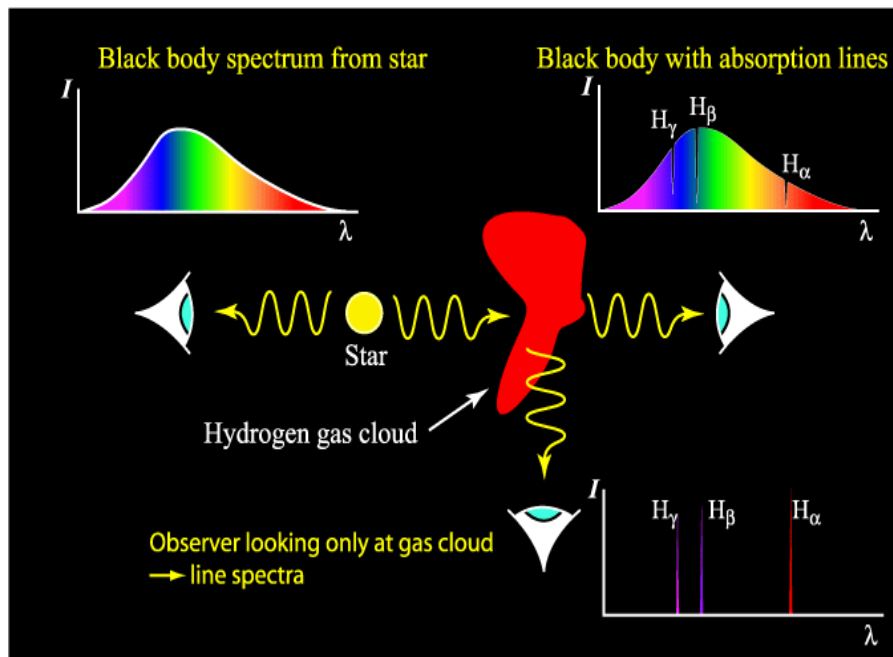


Fig. 8: Representation of spectral lines of Hydrogen and Helium. The Hydrogen spectrum shows distinct lines at 656.5 nm (H-alpha, red), 486.3 nm (H-beta, blue-green), and 434.2 nm (H-gamma, violet), corresponding to electron transitions. Helium's spectrum displays multiple lines due to its more complex electronic structure.

Why is the Sky Blue?

The blue color of the sky is due to the scattering of sunlight by small particles in the atmosphere. This scattering, known as **Rayleigh scattering**, is more effective at shorter wavelengths (blue light) than at longer wavelengths (red light).

Explanation

Air contains small particles that scatter blue light more effectively. The scattering of sunlight by these particles causes the sky to appear blue.

Conclusion

The emission and absorption spectra of elements, governed by their unique frequencies and wavelengths, help identify substances and understand celestial phenomena. Additionally, the scattering of light in the atmosphere explains the blue color of the sky.

Scattering:

Scattering is defined as:

“A change in the direction of motion of a particle because of a collision with another particle” as shown in fig. 9.

- **Atmospheric Molecules and Infrared Absorption:** Carbon dioxide (CO_2) and other atmospheric molecules like water vapor and nitrogen can absorb specific frequencies of light. When sunlight reaches Earth, it includes infrared radiation (heat). CO_2 absorbs this infrared radiation, preventing it from escaping back into space, thereby trapping heat and contributing to global warming.
- **Greenhouse Effect:** The absorption of infrared radiation by CO_2 traps heat near the Earth's surface making it harder for the heat to escape and contributing to the greenhouse effect, which leads to global warming.
- **Global Warming:** Increasing CO_2 levels from industrial emissions lead to higher global temperatures, causing polar ice to melt and other environmental changes.
- This effect is similar to **why the sky appears blue**. The sky is blue because there are fine particles in the air, such as dust. When sunlight hits these particles, the light scatters. The smaller particles scatter shorter wavelengths of light, like blue, more effectively than longer wavelengths like red. This scattering causes the sky to appear blue to us. If these small particles were removed, the sky would appear black, as there would be no scattering of light to produce the blue colour.

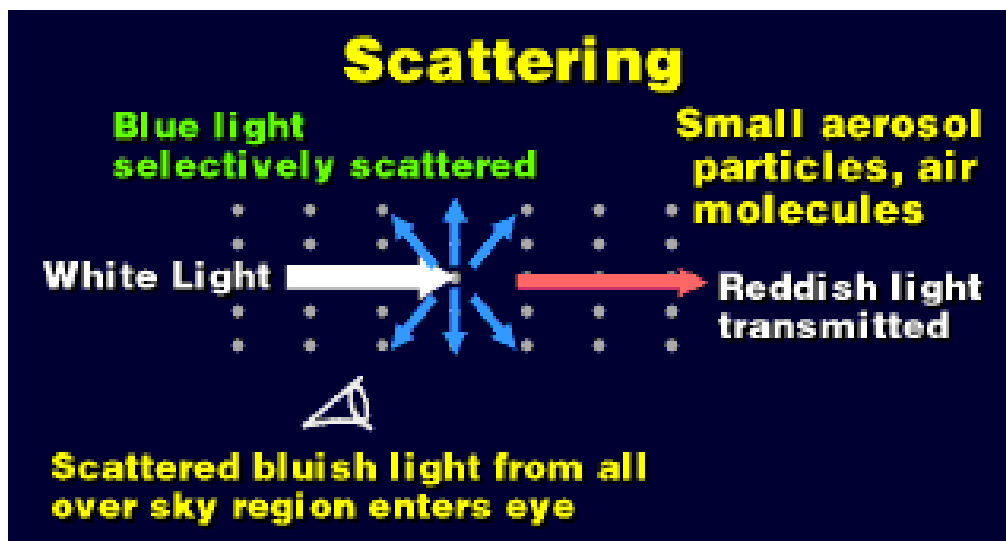


Fig. 9: When white light enters the atmosphere, it scatters upon colliding with air particles, with shorter blue wavelengths scattering more than longer red wavelengths, making the sky appear blue.

Light and Colour:

Objects appear colored due to their interaction with light at specific frequencies.

- **Blue Sky:** Due to Rayleigh scattering, small particles in the atmosphere scatter blue light more than other colors.
- **Colorless Sky:** Without particles, the sky would appear black.
- **Red Rose:** Contains carotene, which absorbs all other frequencies except red.
- **Green Grass:** Contains chlorophyll, which absorbs blue and red light, reflecting green.

Resonance in Molecules:

Molecules resonate at specific frequencies, absorbing and emitting particular wavelengths of light.

- **Carbon Dioxide (CO₂):** Resonates in the infrared region, absorbing heat.
- **Oxygen Molecule (O₂):** Acts as a simple harmonic oscillator with a specific resonance frequency.

Transparency and Opacity:

The transparency or opacity of an object depends on how its molecules interact with light.

- **Transparent Materials:** Molecules are not significantly excited by light, allowing it to pass through.
- **Opaque Materials:** Molecules are excited and absorb light, preventing it from passing through.
- Metals reflect light due to free electrons that can move easily and re-emit the light.
- **Reflection:** Electrons in metals move in response to light, causing it to reflect.

Polarization:

- Polarization occurs when the electric field vector of a light wave oscillates in a particular direction. Reflected light is polarized, with applications in reducing glare (e.g., polarized sunglasses).
- **Unpolarized Light:** Contains electric field vectors in all directions.
- **Polarized Light:** Contains electric field vectors in a single direction as shown in fig. 10.

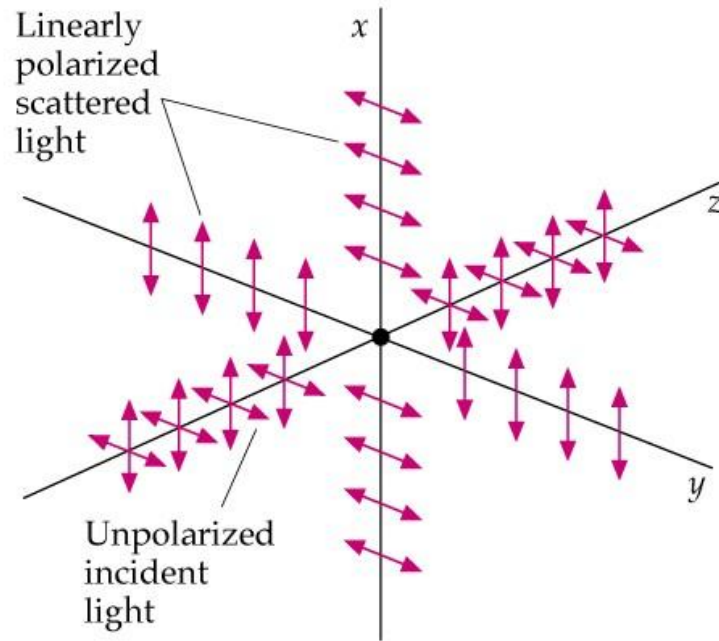


Fig. 10: Representation of Polarized and Unpolarized light. Polarized light has electric field vectors aligned in a single direction, while unpolarized light has electric field vectors in all directions.

- **Aligned Polarizers:** When two polarizers are aligned, they allow maximum light intensity to pass through as shown in fig. 11.
- **Perpendicular Polarizers:** When two polarizers are at a 90-degree angle to each other, they block most of the light, resulting in minimum intensity.

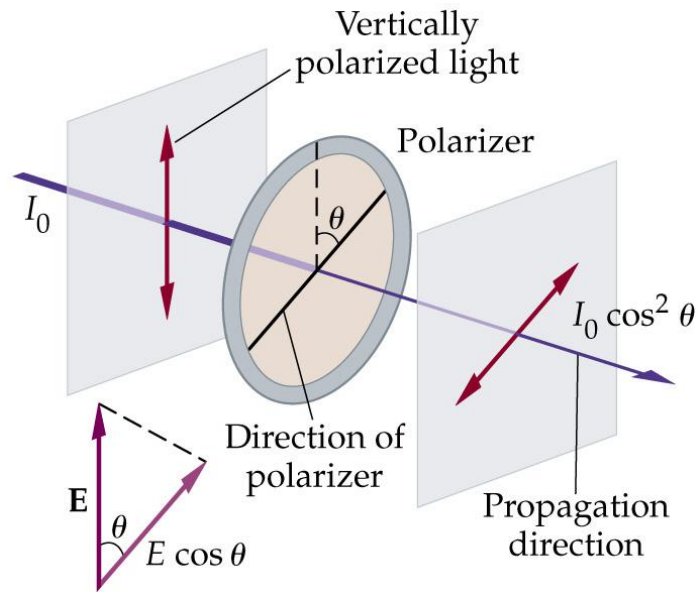


Fig. 11: Effect of polarizer alignment on light intensity. Aligned polarizers allow maximum light to pass, while perpendicular polarizers block most light, demonstrating the principle of attenuation for unpolarized light, with an average attenuation factor of $\langle \cos^2 \theta \rangle = 1/2$.

- Unpolarized light is the superposition of many waves, each with random polarization direction q , relative to fixed axis of polarizer.
- Each wave is attenuated (loss of energy) by factor $\cos^2 q$.
- Average attenuation is $\langle \cos^2 q \rangle = 1/2$
- **Applications:** Polarizing filters and sunglasses reduce glare by allowing only light polarized in a certain direction to pass through.
- **Example of Polarization:** A polarizer can be made using a metal plate with a slit. Light passing through the slit is polarized along the slit. If a second polarizer is placed in front of the first, rotating it will decrease the light intensity passing through.

Conclusion

Understanding the emission, absorption, and interaction of light with various substances provides insight into phenomena like the greenhouse effect, the colors of objects, and the polarization of light. These principles have practical applications in environmental science, optics, and everyday technologies like sunglasses.

Birefringence:

- Birefringent materials (crystals or stressed plastics) have the property that the speed of light is different in two transverse dimensions.
- Since the frequency of the wave is constant as the wave passes through the material, the wavelength is different in the two dimensions, producing a phase change.

- Birefringence can occur in any material that possesses some asymmetry in its structure, so that the material is springier in one direction than another.

Liquid Crystal Displays (LCDs):

Liquid crystal displays utilize the polarization of light and the properties of liquid crystals to create visual displays.

- **Liquid Crystals:** Contain long molecules whose alignment can be controlled by an electric field.
- **Polarization in LCDs:** The light passing through the liquid crystal is polarized along the direction of the molecules. By applying an electric field, we can change the orientation of the molecules, thus controlling the light passing through and creating images or numbers.
- **Refractive Index:** Materials can have different refractive indices based on the polarization of light.
- **Different Refractive Indices:** If light is polarized in one direction, it might have a different refractive index compared to light polarized perpendicularly. This property can be used in various optical devices to manipulate light.

Lecture#33

Interference and Diffraction of Light

Introduction to Light Interference:

It is often said that "iron cuts iron," but you have likely never heard that "light cuts light" or that light reduces the intensity of light. When you enter a room and switch on a bulb, the light from the first bulb combines with the second, resulting in a brighter room. You have never seen darkness created by adding more light sources. It seems contrary to our everyday experiences that light can cancel out light. However, this phenomenon does occur, and it is a fundamental aspect of the behaviour of waves, including light waves. In this lecture, we will explore the fascinating concept of light interference.

Interference of Light:

Interference occurs when two light waves overlap, resulting in an increase (constructive interference) or decrease (destructive interference) in intensity.

The natural consequence of light's wave nature reveals that, just like sound waves, light waves can interact in ways that either amplify or diminish their overall intensity. As we have learned in the study of sound, two sound waves can either increase each other's intensity or reduce it, depending on their phase relationship. Similarly, light waves can interfere constructively or destructively.

Constructive interference occurs when two light waves start simultaneously and their amplitudes combine, effectively doubling the wave's intensity. In contrast, **destructive interference** happens when two light waves are 180 degrees out of phase, meaning one wave is at its peak while the other is at its trough. When these waves intersect, they can cancel each other out, resulting in reduced or nullified intensity as shown in fig.1.

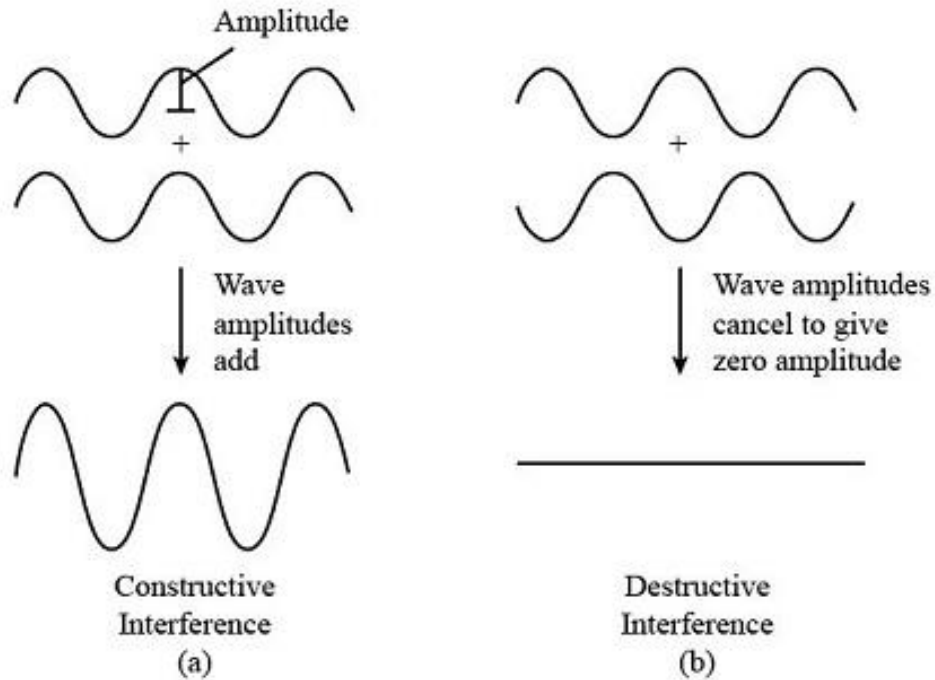


Fig. 1: Representation of the constructive and destructive interference of light waves. (a) Constructive interference results in increased intensity, while (b) destructive interference leads to reduced or nullified intensity.

Conditions for Interference:

- The two light waves must have the same frequency, If the frequencies differ, the waves will sometimes amplify and sometimes diminish each other, but not in a consistent manner.
- The phase difference between the waves must be constant, a condition known as coherence.
- It can't produce coherent light from separate sources ($f \approx 10^{14}$ Hz).
- Need two waves from single source taking two different paths having;
 - ❖ Two slits
 - ❖ Thin films for reflection as shown in fig.2.

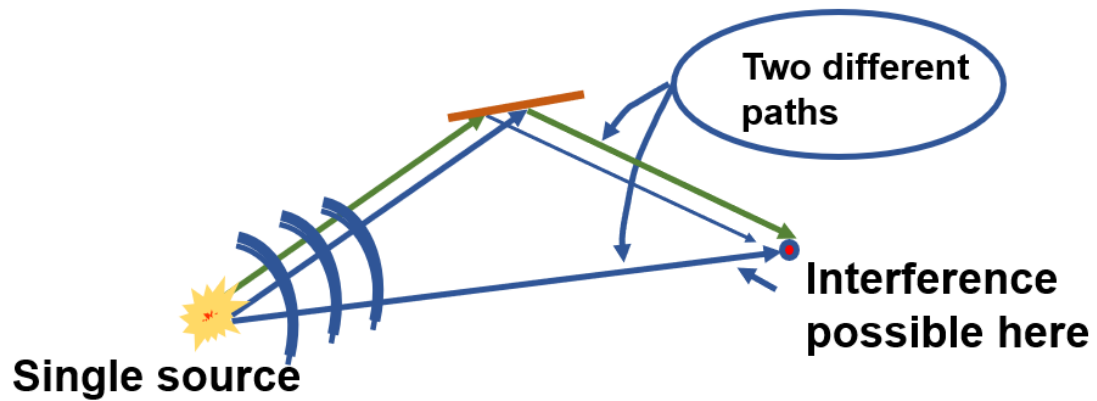


Fig. 2: Representation of the different paths to produce interference from a single source, including the use of two slits, and a thin film.

Coherent and Incoherent Light:

- Natural light sources like bulbs and the sun emit incoherent light, meaning the frequency and phase are not consistent.
- To observe interference, a single light source can be split into two coherent parts using slits or mirrors.

Double-Slit Experiment:

- Light passing through two slits produces an interference pattern of alternating bright and dark bands on a screen.
- The path difference between light from the two slits determines the interference pattern.
- An incoherent light source illuminates the first slit as shown in fig.3.
- This creates a uniform but coherent illumination of the second screen
- (from side-to-side on the screen, the light wave has the same oscillating phase).
- The two waves from the two slits S_1 and S_2 create a pattern of alternating light and dark fringes on the third screen.

Path Difference and Phase Difference:

- The path difference ($d\sin\theta$) between the two slits affects whether constructive or destructive interference occurs.
- For constructive interference, the path difference must be an integer multiple of the wavelength.

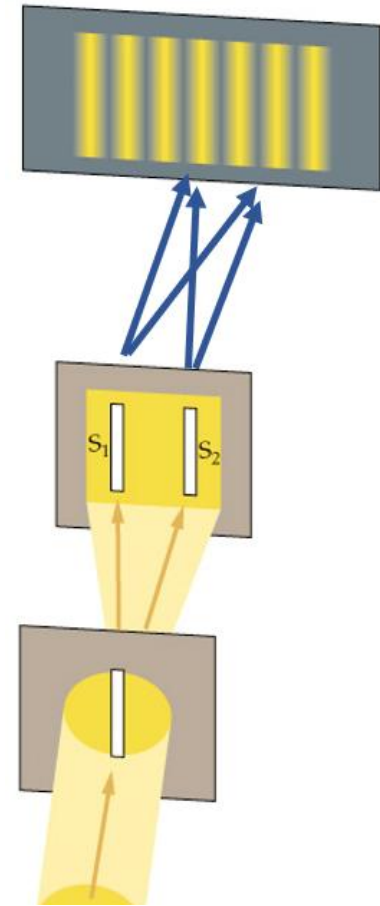


Fig. 3: Representation of the Interference pattern created by light passing through two slits. An incoherent light source illuminates the first slit, creating a uniform, coherent illumination on the second screen. The waves from slits S_1 and S_2 form alternating light and dark fringes on the third screen.

- For destructive interference, the path difference must be an odd multiple of half the wavelength.

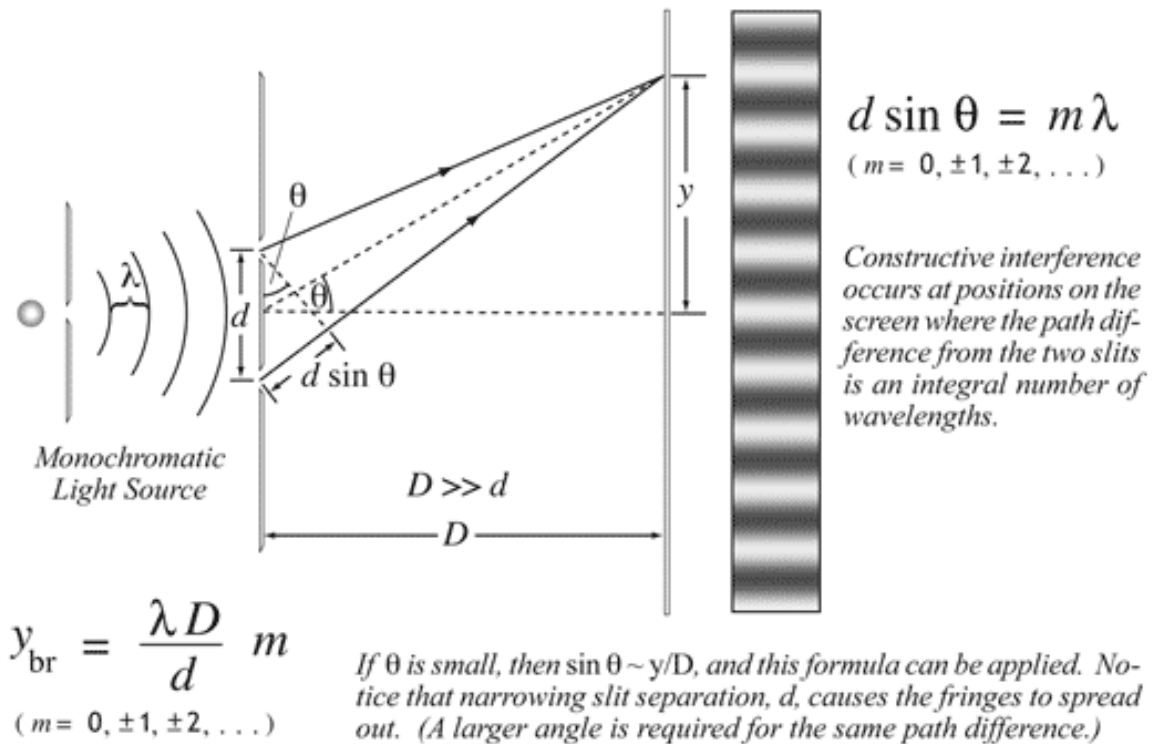


Fig. 4: Representation of the Path difference ($d \sin \theta$) determines interference: constructive interference occurs with integer multiples of the wavelength, and destructive interference occurs with odd multiples of half the wavelength.

Conditions for bright fringes (constructive interference):

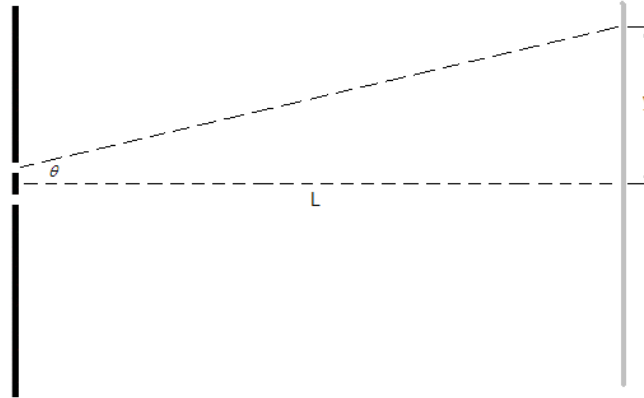
$$d \sin \theta = m \lambda \quad (m = 0, \pm 1, \pm 2, \dots)$$

Conditions for dark fringes (destructive interference):

$$d \sin \theta = (m - \frac{1}{2}) \lambda \quad (m = 0, \pm 1, \pm 2, \dots), m \text{ is called the order number.}$$

Problem:

Two slits with a separation of $8.5 \times 10^{-5} \text{ m}$ create an interference pattern on a screen 2.3 m away. If the tenth bright fringe above the central is a linear distance of 12 cm from it, what is the wavelength of light used in the experiment?



Solution:

1. Calculate the angle to the tenth bright fringe:

$$y = L \tan \theta$$

$$\theta = \tan^{-1} \left(\frac{y}{L} \right) = \tan^{-1} \left(\frac{0.12 \text{ m}}{2.3 \text{ m}} \right) = 3.0^\circ$$

2. Use $\sin \theta = m\lambda/d$ to find the wavelength:

$$\lambda = \frac{d}{m} \sin \theta = \left(\frac{8.5 \times 10^{-5} \text{ m}}{10} \right) \sin(3.0^\circ)$$

$$\lambda = 4.4 \times 10^{-7} \text{ m} = 440 \text{ nm}$$

Phase of wave reflected by interface between two media:

- When light encounters a boundary between two media, it can be reflected or refracted.
- The phase change upon reflection depends on whether the light moves from a medium with a higher to a lower refractive index or vice versa.

Conditions for Phase Change

- **Reflection from a Denser Medium:**
 - When light reflects off a boundary with a medium that has a higher refractive index (denser medium), the reflected wave undergoes a phase change of π radians (180 degrees) as shown in fig.5.
 - Example: Light reflecting off a glass surface from air.

- This is due to the wave experiencing a "hard" boundary, causing an inversion of the wavefront.
- **Reflection from a Less Dense Medium:**
 - When light reflects off a boundary with a medium that has a lower refractive index (less dense medium), there is no phase change as shown in fig.5.
 - Example: Light reflecting off an air surface from water or glass.
 - This is because the wave encounters a "soft" boundary, and the wavefront is not inverted.

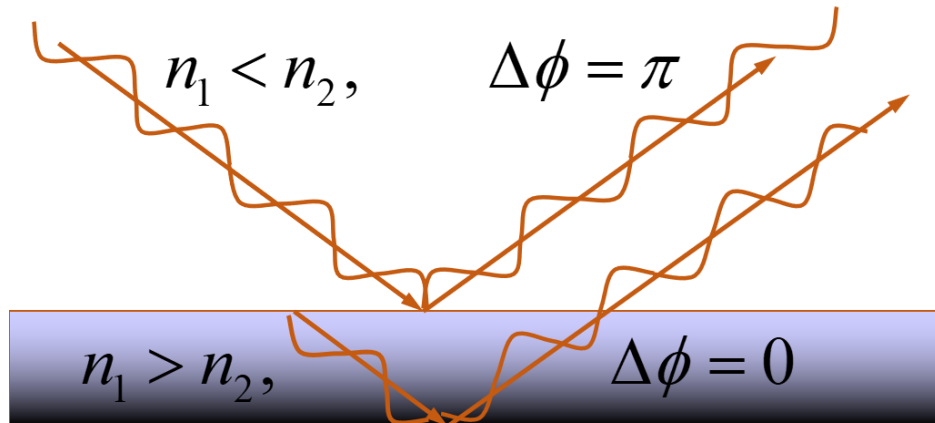
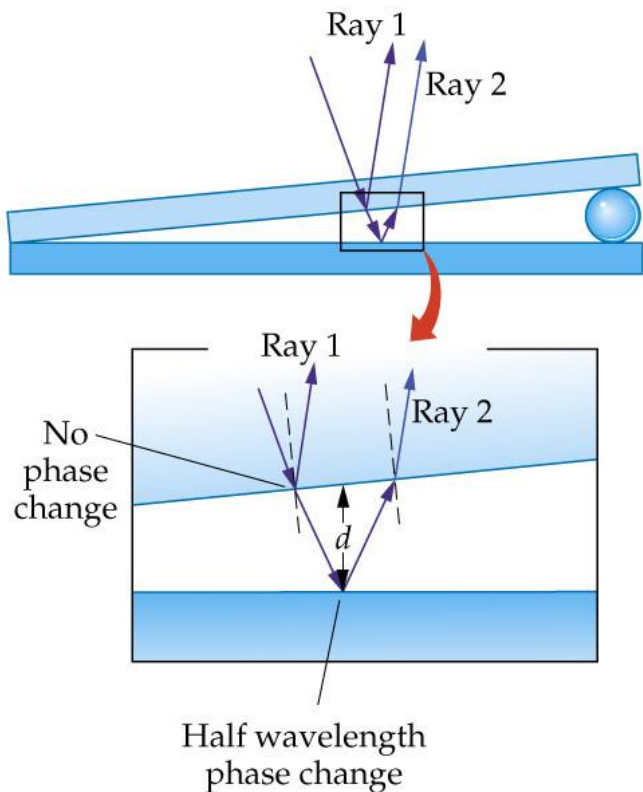


Fig. 5: Representation of the Phase change conditions upon reflection: a π radians (180 degrees) phase change occurs when light reflects from a denser medium, whereas no phase change occurs when reflecting from a less dense medium.

Interference in an Air Wedge:

Interference in an air wedge is a phenomenon observed when light waves reflect off the two surfaces of a thin film of air, often formed between two glass plates that are slightly angled to each other as shown in fig.6. This creates a wedge-shaped air film, leading to an interference pattern of bright and dark fringes.

Constructive interference:



$$\frac{1}{2} + \frac{2d}{\lambda} = m \quad (m=1,2,3...)$$

Destructive interference:

$$\frac{1}{2} + \frac{2d}{\lambda} = m + \frac{1}{2} \quad (m=0,1,2...)$$

Fig. 6: Representation of the interference in an air wedge formed between two angled glass plates, producing interference fringes due to reflection from the upper and lower surfaces of the air film.

Thin Film Interference:

- Thin films (like oil on water) create colourful interference patterns due to varying thicknesses.
- Light reflecting from different layers of the film interferes, resulting in constructive or destructive interference.

Phase Change upon Reflection:

- When light reflects off a surface, it can undergo a phase change depending on the refractive indices of the media involved.
- If light reflects from a medium with a higher refractive index to a medium with a lower refractive index, there is no phase change.
- If light reflects from a medium with a lower refractive index to a medium with a higher refractive index, there is a phase change of π (180 degrees) as shown in fig.7.

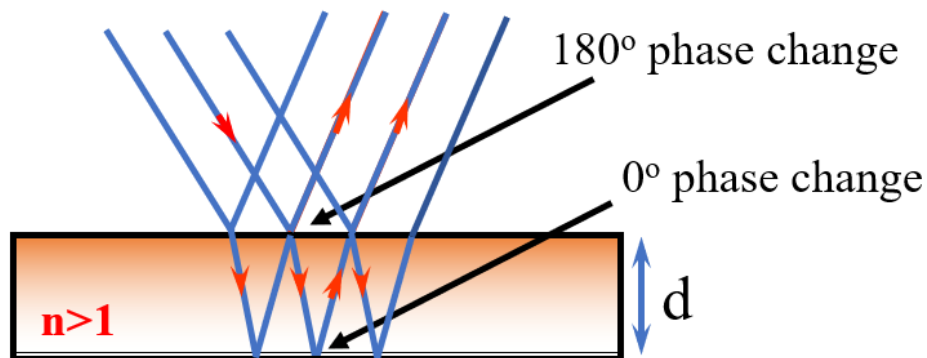


Fig. 7: Representation of the Reflection phase change: Light undergoes a π radians (180 degrees) phase change when reflecting from lower to higher refractive index media, and no change when reflecting from higher to lower index media.

Constructive interference:

$$2nd = m\lambda \quad (m=0,1,2,...)$$

Destructive interference:

$$2nd = (m + \frac{1}{2})\lambda \quad (m = 0, 1, 2, \dots)$$

Where, n is the index of refraction for the medium.

Thickness of film selects wavelength:

In the study of light interference, one important aspect is how the thickness of a film can select specific wavelengths of light. When light waves encounter a thin film, such as a soap bubble or an oil slick on water, the interaction between the light waves reflected from the top and bottom surfaces of the film can lead to interference patterns. The thickness of the film plays a crucial role in determining which wavelengths of light will constructively or destructively interfere, thereby affecting the colors we observe.

When a light wave hits a thin film, part of the wave is reflected off the top surface, while the rest enters the film and is reflected off the bottom surface. The two reflected waves can then interfere with each other. For constructive interference (which results in a bright fringe), the path difference between the two waves must be an integer multiple of the wavelength. For destructive interference (which results in a dark fringe), the path difference must be a half-integer multiple of the wavelength.

If the film thickness changes, the specific wavelengths also change, leading to different colors being observed. This is why thin films (soap bubble) often display a spectrum of colors, as different thicknesses within the film selectively amplify or diminish different wavelengths of light.

For Example:

- Thick portions of thin film appear blue, since the long-wavelength red light experiences destructive interference as shown in fig.8.

- Thinner portions appear red because the short-wavelength blue light interferes destructively.

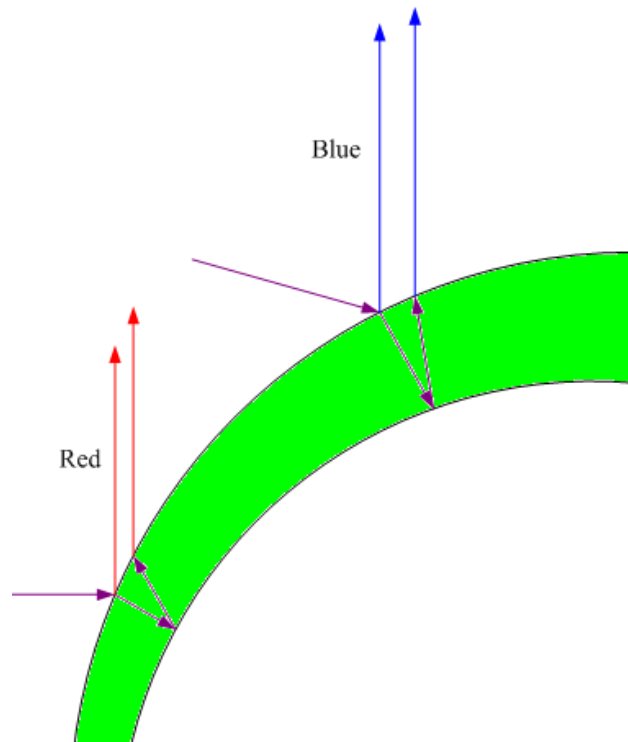


Fig. 8: Representation of the thin film interference causes thick portions to appear blue due to destructive interference of long-wavelength red light, while thinner portions appear red due to destructive interference of short-wavelength blue light.

Newton's Rings:

Newton's rings are a classic demonstration of the wave nature of light, particularly the phenomenon of interference. They are named after Sir Isaac Newton, who first studied and explained this interference pattern. The rings are formed by the interference of light waves reflected from two surfaces—a spherical surface and an adjacent flat surface.

Formation of Newton's Rings

Newton's rings are typically observed when a plano-convex lens is placed on top of a flat glass surface as shown in fig.9. The air film between the curved surface of the lens and the flat glass plate varies in thickness from zero at the point of contact to some maximum value at the edge of the lens. When monochromatic light (light of a single wavelength) is shone onto this arrangement, the light waves reflected from the top and bottom surfaces of the air film interfere with each other.

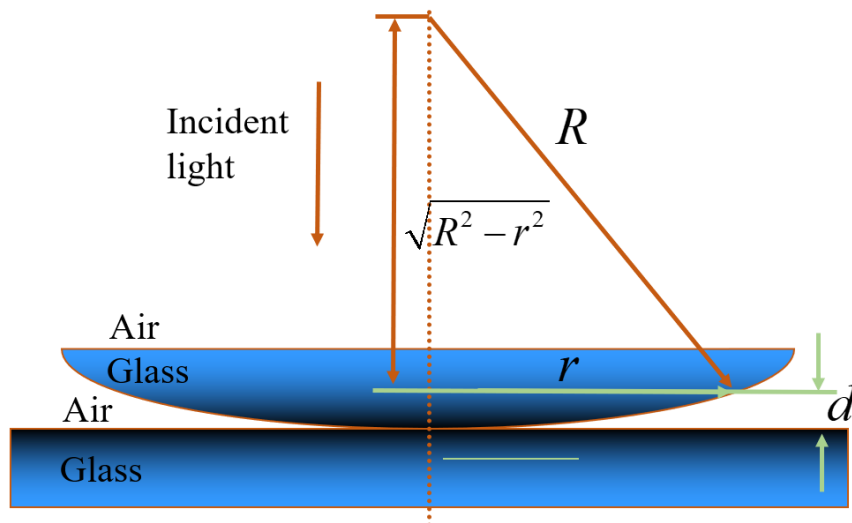


Fig. 9: Representation of the Newton's rings phenomenon observed with a plano-convex lens placed on a flat glass surface. Interference occurs between light waves reflected from the air film's top and bottom surfaces, producing concentric bright and dark rings due to varying air film thickness.

Constructive and Destructive Interference

The interference of the reflected light waves produces a pattern of concentric bright and dark rings as shown in fig.10. These rings are formed due to the constructive and destructive interference of the light waves.

- An example of interference patterns formed by placing a convex lens on a flat glass surface and observing concentric rings.
- The radius of the rings depends on the wavelength of light and the curvature of the lens.

Condition for constructive interference,

$$2d = \left(m + \frac{1}{2}\right)\lambda, \quad m = 0, 1, 2, \dots \text{(maxima)}$$

$$d = R - \sqrt{R^2 - r^2}$$

$$d = R - R \left[1 - \left(\frac{r}{R}\right)^2 \right]^{1/2}$$

Using Binomial theorem,

$$\therefore (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$$

$$d = R - R \left[1 - \frac{1}{2} \left(\frac{r}{R}\right)^2 + \dots \right]$$

$\therefore \frac{r}{R} \approx$ small quantity, hence ignoring higher order terms,

$$d = R - R + R \cdot \frac{r^2}{2R^2}$$

$$d \approx \frac{r^2}{2R}$$

As,

$$2d = \left(m + \frac{1}{2}\right)\lambda$$

Putting the value of 'd', we get

$$2 \cdot \frac{r^2}{2R} = \left(m + \frac{1}{2}\right)\lambda$$

$$r = \sqrt{\left(m + \frac{1}{2}\right)\lambda R}, \quad m = 0, 1, 2, \dots \text{(maxima)}$$

where:

- n is the order of the ring (1, 2, 3, ...).
- λ is the wavelength of the light used.
- R is the radius of curvature of the lens.



Newton's Rings due to
Reflected Light



Newton's Rings due to
Transmitted Light

Fig. 10: Representation of the Newton's rings due to the reflected light and transmitted light. The central ring is dark, illustrating the interference pattern where light waves reflected from the air film between a plano-convex lens and a flat glass surface are out of phase. The central ring is bright, showing an interference pattern opposite to that of reflected light. If there were no phase change, the rays reflected from two surfaces in contact would be in the same phase at the center, resulting in a bright central spot.

Diffraction:

- Diffraction is a type of interference occurring when light bends around obstacles or passes through small apertures as shown in fig.11.
- Single-slit diffraction patterns consist of central bright bands flanked by alternating dark and bright bands.
- Diffraction occurs when light passes through very small apertures or near sharp edges.
- We can get an interference pattern with a single slit provided its size is approximately λ (neither too small nor too large).
- **For example**, there is a knife which has a very sharp edge. If you look from above it, then the light gets diffracted and appears in your eyes. It is not always easy to see, but if you take two knives and place them side by side, then you will see diffraction from their edges.

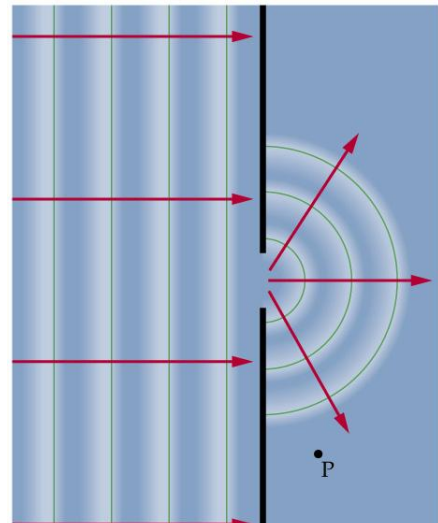


Fig. 11: Representation of the diffraction, when light bends around obstacles, producing single-slit patterns with central bright bands and alternating dark and bright bands.

Single Slit Diffraction:

A thin strip or slit as shown in fig.12, which has been cut in a metal plate or in paper. Light comes from above it and as a result, light comes out from the edges of the slit, waves emerge and they interfere with each other, as a result of which an interference pattern is formed. This pattern is visible to us near a sharp edge. There also we get dark and bright bands at other places. The point is that in diffraction we need only one source but different parts of that source can interfere with each other.

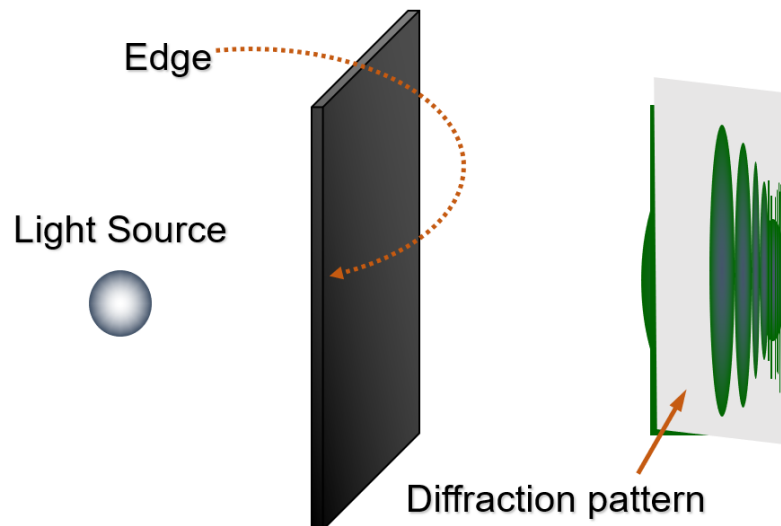


Fig. 12: Representation of the diffraction pattern from a thin slit where light waves emerging from the edges interfere, producing visible dark and bright bands near the sharp edge, showcasing interference from different parts of a single source.

Calculating Path Difference

In a single slit diffraction, a slit having width d . The phase difference between these is $w\sin\theta$, if this distance is equal to some integer wavelength, then interference will occur, otherwise not. In such a case dark fringe will be generated and then we can determine the interference pattern from this.

Conditions for dark fringes in single-slit diffraction,

$$W \sin \theta = m\lambda$$

$$m = \pm 1, \pm 2, \pm 3, \dots$$

Problem:

Light with wavelength of 511 nm forms a diffraction pattern after passing through a single slit of width 2.2×10^{-6} m. Find the angle associated with (a) the first and (b) the second dark fringe above the central bright fringe.

Solution:

1. Solving for using $m = 1$:

$$W \sin \theta = m\lambda$$

$$\theta = \sin^{-1} \left(\frac{m\lambda}{W} \right) = \sin^{-1} \left(\frac{(1)(511 \times 10^{-9} \text{ m})}{2.20 \times 10^{-6} \text{ m}} \right) = 13.4^\circ$$

2. Solving for using $m = 2$:

$$\theta = \sin^{-1} \left(\frac{m\lambda}{W} \right) = \sin^{-1} \left(\frac{(2)(511 \times 10^{-9} \text{ m})}{2.20 \times 10^{-6} \text{ m}} \right) = 27.7^\circ$$

Resolution:

Resolution refers to the ability of an optical instrument, such as a telescope or microscope, to distinguish between closely spaced objects. The phenomenon of diffraction significantly impacts resolution.

When light passes through an aperture or around the edges of an object, it bends and spreads out, causing interference patterns. The central bright spot and surrounding rings of these patterns can overlap, blurring the details of the objects being observed.

The formula for the angle θ at which the first dark fringe occurs in a diffraction pattern is given by:

$$\sin \theta = 1.22 \left(\frac{\lambda}{D} \right)$$

Rayleigh's criterion:

$$\theta_{\min} = 1.22 \frac{\lambda}{D}$$

where, λ is the wavelength of light, D is the diameter of the aperture.

- From this equation, it is evident that increasing the diameter of the aperture D results in a smaller angle θ , leading to less diffraction and better resolution.
- Conversely, using light with a shorter wavelength also enhances resolution, as shorter wavelengths diffract less.
- This also proves that if λ is less which means that if the frequency is higher then the resolution will be better accordingly, because a good telescope has a larger diameter and with this, we can see distant things and we can create difference in the close things.

Diffraction Gratings:

A diffraction grating is an optical component with a pattern of regular, closely spaced lines or slits that diffract light into several beams. These beams interfere with each other, producing a pattern of light and dark bands or lines, known as an interference pattern as shown in fig.13.

- A diffraction grating consists of multiple slits that disperse light into its component wavelengths.
- The angle of diffraction depends on the wavelength and the spacing of the slits.

The principle behind diffraction gratings relies on the interference of light waves. When light encounters the grating, it is split into multiple paths. The condition for constructive interference, where the light waves reinforce each other to create bright bands, is given by:

$$d \sin \theta = m \lambda$$

where, d is the distance between adjacent slits (grating spacing), θ is the angle of diffraction, m is the order of the maximum (an integer) and λ is the wavelength of the light.

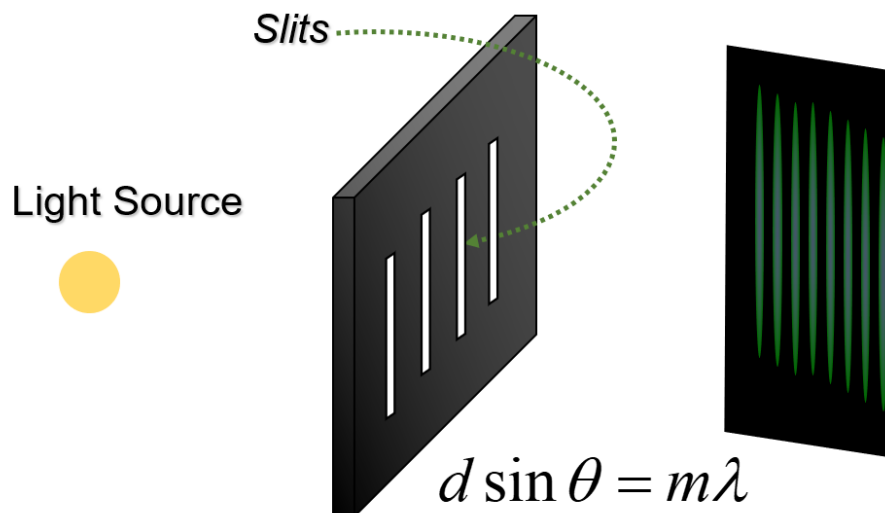


Fig. 13: Representation of the diffraction grating disperses light into beams that interfere, forming an interference pattern of light and dark bands on the screen.

Diffraction gratings can be used for various purposes, such as:

- **Analyzing Light Spectra:** By separating light into its component wavelengths, diffraction gratings are essential tools in spectrometry for analyzing the spectral composition of light from various sources.
- **Optical Instruments:** They are utilized in devices like monochromators and spectrographs to isolate specific wavelengths of light.
- **Compact Discs and DVDs:** The fine grooves on CDs and DVDs act as diffraction gratings, reflecting different colors of light and creating the characteristic rainbow pattern seen when light reflects off their surfaces.

CDs and DVDs: Practical Application of Diffraction Gratings

- CDs (Compact Discs) and DVDs (Digital Versatile Discs) are practical examples of diffraction gratings in everyday use. The surface of these discs contains tiny pits and lands arranged in a spiral track as shown in fig.14.
- When light, particularly from a laser, shines on the disc, it gets diffracted by these microscopic features. The distance between the grooves on a CD or DVD is on the order of hundreds of nanometers, close to the wavelength of visible light. As a result, the discs create interference patterns that can be read by a laser in a CD or DVD player. This diffraction and interference mechanism is what allows the digital information encoded on the disc to be read and interpreted by the device.
- A laser puts information on the CD, which is then read by a CD reader or drive inside the computer. The CD has a layer of metal over the plastic with varying thickness. A laser beam hits it, bounces back, and creates interference. Where $d \sin \theta = n \lambda$, interference is created, forming a pattern of zeros and ones. This digital information is then read by the laser.
- The interference conditions, given by $d \sin \theta = m \lambda$, explain the varying colors observed when viewing a CD or DVD under white light. Different wavelengths (colors) of light meet the interference condition at different angles, producing a spectrum of colors. On one side, you can see a bright part where information has been recorded. When you shake it, you see colors changing due to light diffraction. If you look at it under a microscope, you will see rings where a laser has put information, with a distance of about five or 6000 angstroms between them.
- As the angle of a CD or DVD changes, the conditions of interference change, resulting in different colors. The condition of interference is $d \sin \theta = n \lambda$, so the value of θ will be different for each λ . Changing the angle results in different constructive interference patterns.

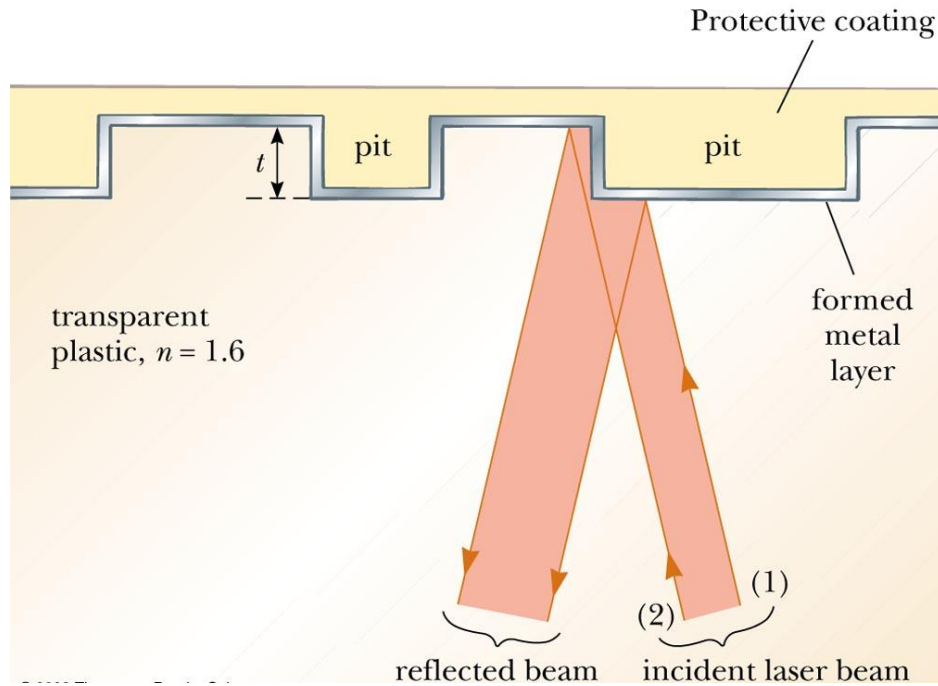


Fig. 14: Representation of the CDs and DVDs use microscopic pits and lands on their surfaces to diffract light, creating interference patterns read by lasers to encode and decode digital information.

X-ray Diffraction:

X-ray diffraction (XRD) is a powerful technique used to study the structure of crystalline materials. It involves directing X-rays at a sample and analysing the patterns produced by the diffracted rays. This technique allows scientists to determine the arrangement of atoms within a crystal and gain insights into the material's properties.

1. Principle of X-ray Diffraction

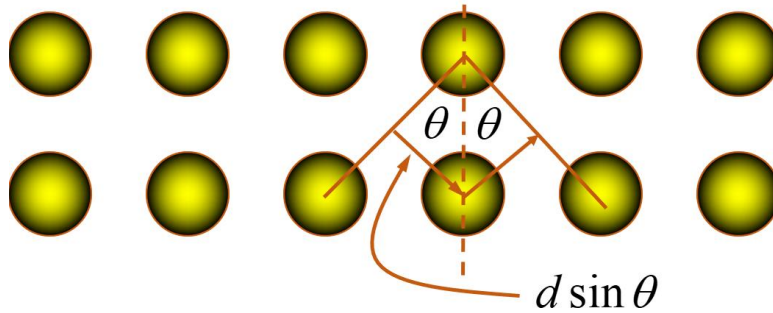
The fundamental principle of X-ray diffraction is based on the interference of X-rays scattered by the atoms in a crystal lattice. The condition for constructive interference, which results in the diffraction peaks, is given by Bragg's law:

$$2d\sin(\theta) = m\lambda$$

where:

- d is the distance between crystal planes (interplanar spacing) as shown in fig.15.
- θ is the angle of incidence.
- m is an integer (the order of the maximum).
- λ is the wavelength of the X-rays.

When X-rays strike a crystal, they are scattered by the atoms in the crystal lattice. The scattered waves interfere with each other, and constructive interference occurs when the path difference between the waves is an integer multiple of the wavelength. This results in a diffraction pattern that can be analysed to determine the crystal structure.



$$2d \sin \theta = m\lambda$$

Fig. 15: Representation of the X-ray diffraction relies on Bragg's law, $2d\sin(\theta) = m\lambda$, where d is the interplanar spacing in a crystal lattice. X-rays scattered by atoms in the lattice interfere constructively at specific angles, producing diffraction peaks that reveal the crystal's structure.

2. X-ray Crystallography

Using X-ray crystallography, we learn the properties of different salts like sodium chloride and copper sulphate. This method reveals the arrangement of atoms. However, to look inside atoms, we need Gamma rays with higher energy and smaller wavelengths.

While X-ray diffraction is useful for studying the structure of crystals and molecules, gamma rays, which have even shorter wavelengths than X-rays, can be used to explore even smaller structures, such as atomic nuclei.

To see the nucleus, we use Gamma rays, as the nucleus size is approximately 10 femtometres. Gamma rays with appropriate wavelength help us see inside the nucleus.

Conclusion

Whatever you want to see, the light used depends on the size of that thing. For smaller objects, we need smaller wavelengths and higher frequencies. This principle is essential for understanding atomic and subatomic structures. The lecture concludes with an emphasis on the importance of diffraction and interference in various fields.

Lecture #34**THE PARTICLE NATURE OF LIGHT**

A century ago, physics faced a profound question regarding light: whether light behaved as a wave or as particles, serving as a source of energy. Around 350 years ago, Newton's study of light suggested it comprised particles, based on observations of sunlight's colors. This view gained widespread acceptance due to Newton's theory. However, subsequent experiments, notably the double-slit experiment (fig 34.1), challenged this view.

Here, light passing through two narrow slits onto a screen, created an interference pattern, indicating wave-like behavior. This phenomenon arises due to variations in the time taken for light to reach the screen through different slits, resulting in wave interference and modulation of amplitude. This finding posed a significant challenge to the particle theory of light.

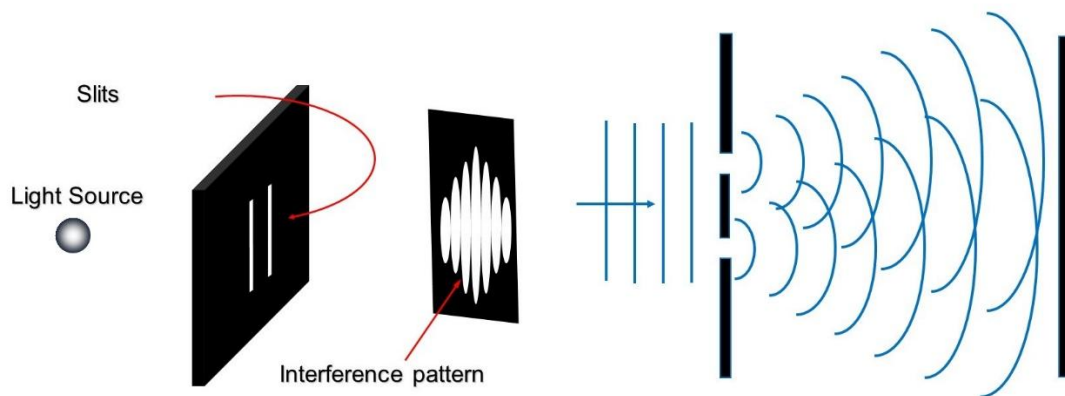


Figure 34. 1. The double-slit experiment showing light's wave-like behavior through interference patterns.

In the 19th century, Michael Faraday and James Clark Maxwell confirmed that light behaves as electromagnetic waves, as discussed in previous lectures. They demonstrated that changes in electric fields produce magnetic fields, and vice versa, forming electromagnetic waves.

Heinrich Hertz, a German scientist, further validated this theory in 1880 through experiments with transmitters and receivers, confirming the existence of these waves. Electromagnetic waves feature perpendicular electric and magnetic fields, with the electric field along the x-axis, the magnetic field along the y-axis, and the wave propagation along the z-axis. As the electric field increases, the magnetic field decreases, allowing these fields to propagate through space.

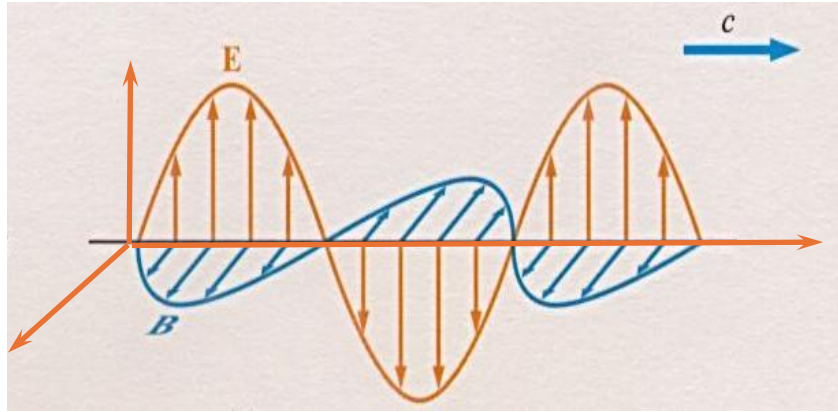


Figure 34. 2. Electromagnetic waves featuring perpendicular electric and magnetic fields.

Energy of Electromagnetic Wave

Electromagnetic fields carry energy, which can be calculated based on the product of the electric and magnetic fields as $E \times B$. The power in an electromagnetic wave is proportional to the square of the maximum intensity of either the electric field (E_0) or the magnetic field (B_0). Increasing the wave's amplitude, represented by E_0 or B_0 , results in a squared increase in energy.

$$\text{Average power per unit area} = \frac{E_{\max} B_{\max}}{2\mu_0} = \frac{E_{\max}^2}{2\mu_0 c}$$

$$\therefore \frac{E}{B} = c$$

$$\text{Average power per unit area} = \frac{c B_{\max}^2}{2\mu_0}$$

Additionally, electromagnetic waves possess momentum, despite us typically only sensing their warmth when exposed to sunlight. The linear momentum (p) is given by:

$$p = \frac{2U}{c}$$

Measurement of momentum

To detect momentum, a highly sensitive tool is required. In this setup depicted in fig 34.3, a mirror and a black disc are positioned such that when light strikes the mirror, it reflects, causing the apparatus to rotate. The rotation angle increases with higher light intensity, reflecting the energy density and momentum.

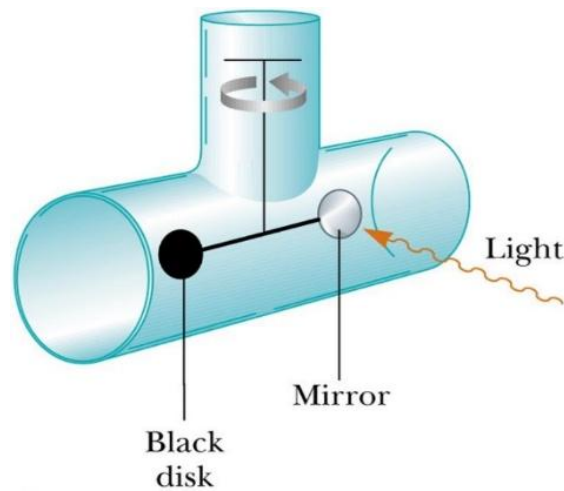


Figure 34. 3. Rotating mirror and black disc setup for momentum measurement.

Photoelectric Effect

In 1887, Heinrich Hertz conducted a groundbreaking experiment. Light illuminated a metal surface, causing electrons to be emitted and collected elsewhere, known as the photoelectric effect and the electrons that come out of the light are called photoelectrons. This phenomenon revolutionized physics, leading to further exploration.

Experimental Setup

Light of specific frequency strikes a metal plate within a vacuum chamber to ensure no air molecules interfere. Electrons are ejected from the metal upon light collision and move toward the positive terminal, connected to a battery. The resulting electron flow creates a measurable current, detectable by an ammeter.

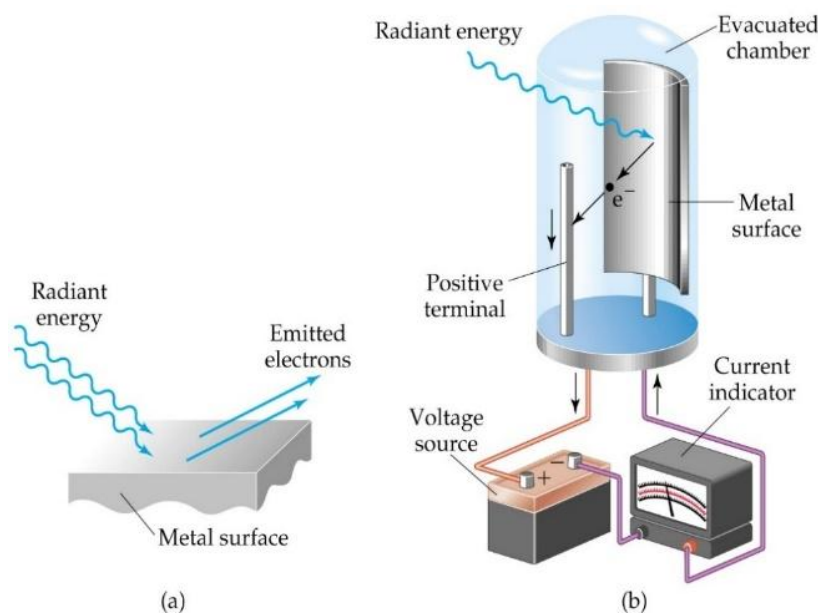


Figure 34. 4. Experimental setup for demonstrating photoelectric effect.

Conclusions

This experiment yielded significant results.

1. Firstly, the kinetic energy of electrons reaching the anode is independent of light intensity, contradicting classical theories from Newton to Maxwell. Increasing light intensity doesn't result in higher electron energy, puzzling classical theory.
2. Secondly, electrons are not emitted unless light frequency exceeds a certain threshold, regardless of intensity. Classical theory fails to explain why increasing electromagnetic field intensity doesn't cause electron emission until a specific frequency is reached.
3. The third crucial observation is that when light strikes a surface, classical theory predicts a time delay before atoms gain sufficient energy to eject electrons. However, electrons are emitted instantaneously upon light exposure, contradicting classical predictions. These discrepancies caused a major upheaval in physics, challenging classical mechanics and electromagnetic theory.

Einstein's solution to photoelectric puzzle

In 1905, Einstein proposed a groundbreaking theory, which resolved these issues but also triggered a conceptual crisis. Einstein suggested that light consists of discrete particles called quanta, each carrying a specific amount of energy. These energy quanta, or photons, are characterized by the equation:

$$E = h\nu$$

$$(\text{ or } E = \hbar\omega \text{ where } \hbar = \frac{h}{2\pi} \text{ and } \omega = 2\pi\nu)$$

$$h \text{ (Planck's constant)} = 6.626 \times 10^{-34} \text{ Joule-seconds.}$$

This equation ensures that as the frequency increases, so does the energy of the photons. Thus, Einstein's theory revolutionized our understanding of light, introducing the concept of quantized energy and laying the foundation for modern quantum mechanics.

Radio waves have lower frequency, which means that each quantum of a radio frequency wave carries less energy compared to a quantum of a gamma ray. To illustrate this concept, let's use the analogy of climbing stairs. Imagine each step on the staircase represents a quantum of energy. When you climb the stairs, you can only move up by one full step at a time. You can't take half or partial steps. Similarly, energy is quantized, meaning it comes in discrete units or "quanta." Electrons,

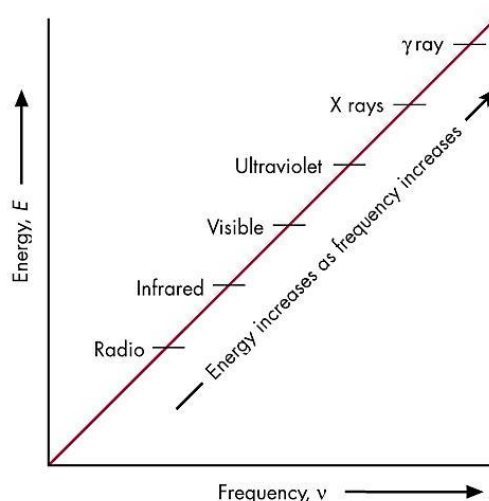


Figure 34. 5. Graph showing the direct proportionality between the energy and frequency of electromagnetic waves, illustrating that higher frequency waves have higher energy.

for instance, are only emitted from a material when the frequency of the incoming light exceeds a certain threshold frequency. Below this threshold, no electrons are emitted. However, once the threshold frequency is surpassed, the number of emitted electrons depends on the intensity of the light.

Black Body Radiation

Classical theory failed to explain the photoelectric effect and faced another challenge regarding

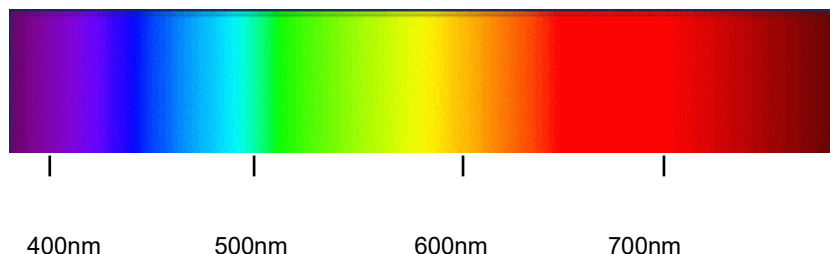


Figure 34. 6. visible light spectrum

black body radiation. When observing the frequencies emitted by a heated black body, a continuous spectrum spanning from deep blue to deep red is observed. However, heating individual atoms results in only specific colours of light being emitted. This phenomenon puzzled scientists as classical mechanics and electromagnetic theory couldn't account for it. Spectroscopic analysis of elements like hydrogen, helium, neon, and carbon revealed distinct spectral lines, such as H_α , H_β , and H_γ for hydrogen. The origin of these lines remained a mystery, leaving scientists perplexed about atomic behaviour.

Rutherford's Experiment

Rutherford's experiment challenged the Plum Pudding Model of the Atom, which proposed that electrons and protons were evenly distributed throughout. Rutherford's experiment involved bombarding atoms with alpha particles. Surprisingly, while some alpha particles passed straight through, others were deflected, and some even rebounded. This unexpected result led Rutherford to propose that atoms contained a dense central core, which he termed the nucleus. This nucleus, he reasoned, must be positively charged to repel the positively charged alpha particles. Thus, electrons orbit the nucleus much like planets orbit the sun. This revolutionary insight, confirmed around a century ago.

In the case of the simplest atom, hydrogen, a single proton forms the nucleus around which an electron orbits (see fig 34.7). This orbital motion is balanced by the attractive force of the proton and the centrifugal force exerted by the electron's motion. However, this model posed new questions, suggesting a problem that demands further exploration.

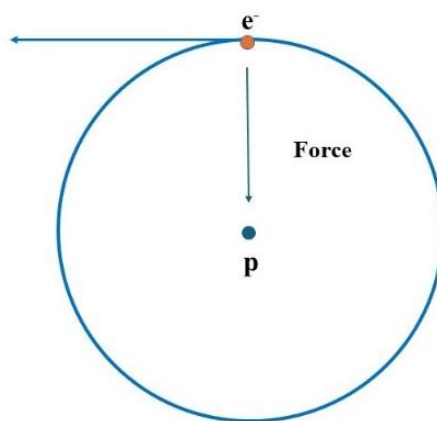


Figure 34. 7. Simplified model of a hydrogen atom with a single electron orbiting a proton, highlighting the balance between attractive and centrifugal forces.

As the electron orbits the nucleus, its velocity continuously changes due to rotation, resulting in acceleration. According to electromagnetic theory, accelerating charges emit light waves, releasing energy. This energy cannot come from the potential energy between the proton and electron, as separating opposite charges requires work. Therefore, the electron's energy loss implies it moves closer to the nucleus over time. Calculations suggest that a hydrogen atom would collapse within 10^{-8} s due to this radiation. This phenomenon applies not only to hydrogen but to all atoms, leading to electron-nucleus collisions and destruction of atom.

Bohr's Atomic Model

To prevent this destruction, Niels Bohr, a Danish scientist, proposed a hypothesis suggesting that electrons could remain stable by orbiting the nucleus in specific circular paths, where their angular momentum is quantized as n times $\hbar$ ($\hbar = h/2\pi$), with n being integers (1, 2, 3, etc.). If an electron's angular momentum follows this quantization, it will not emit radiation or lose energy, allowing the atom to maintain stability. This hypothesis was further investigated to determine its validity and implications for understanding the atomic structure.

The Bohr Atom

This atom exhibits equilibrium when the electrical force (between the nucleus (+e) and electron (-e)), balances the centrifugal force.

$$F = ma = \frac{(e)(e)}{r_n^2} = m \frac{v^2}{r_n}$$

$$\frac{e^2}{r_n^2} = \frac{mv^2}{r_n} \dots\dots\dots(1)$$

According to Niels Bohr's proposal, angular momentum is quantized as

$$L = r_n \times p = r_n p \sin 90^\circ = r_n p = mvr_n = n \frac{h}{2\pi} \equiv n\hbar$$

$$v = \frac{n\hbar}{mr_n}$$

By putting the value of v in eq (1), we get the value of r_n

$$r_n = \frac{e^2}{mv^2} = \frac{e^2 mr_n^2}{n^2 \hbar^2}$$

$$r_n = \frac{n^2 \hbar^2}{me^2}$$

From this, we deduce that the electron can orbit the nucleus without emitting radiation, provided its distance equals $\frac{n^2 \hbar^2}{me^2}$. This allows for specific orbit radii. $\frac{n^2 \hbar^2}{me^2} = 0.5 \text{ \AA}$, implying one possible value of r_n is $0.5 \text{ \AA}$, followed by subsequent values increasing by squares (4, 9, 16, etc.). Thus, the electron can occupy specific orbits but not intermediate ones.

If we add both potential and kinetic energy,

$$PE = -eV = -\frac{e^2}{r}$$

$$E_n = \frac{1}{2}mv^2 - \frac{e^2}{r_n} = \frac{1}{2}m\left(\frac{n\hbar}{mr_n}\right)^2 - \frac{e^2}{r_n} = \frac{1}{2}\frac{n^2\hbar^2}{mr_n^2} - \frac{e^2}{r_n}$$

putting the value of r_n

$$\frac{1}{2}\frac{me^4n^2\hbar^2}{n^4\hbar^4} - \frac{me^4}{n^2\hbar^2} = \frac{1}{2}\frac{me^4}{n^2\hbar^2} - \frac{me^4}{n^2\hbar^2}$$

$$E_n = -\frac{e^4m}{2n^2\hbar^2}$$

$$E_1 = -13.6 \text{ eV}$$

What does "minus" signify? It represents the energy required to remove an electron from the atom, known as ionization potential, which is 13.6 eV. Additionally, various energy levels exist, with values decreasing in multiples of 4 (13.6, 13.6/4, 13.6/9, 13.6/16, etc.).

Niels Bohr proposed that when electrons transition between these levels, light is emitted. The colour or frequency of this light is determined by the energy difference between the levels.

Consider an electron transitioning from level 3

to level 2, with the extracted energy given by $h\nu = 13.6 \left[\left(\frac{1}{2}\right)^2 - \left(\frac{1}{3}\right)^2 \right]$. Just as the atom emits light, it becomes excited in the same manner. For instance, to transition from $n = 2$ to $n = 3$, the photon's energy must satisfy $h\nu = E_3 - E_2$. Deviations from this energy level won't induce light emission.

In the hydrogen atom spectrum, depicted here, transitions from the fifth to the second level emit blue light with a wavelength of 434 nm. Conversely, the transition from level 3 to level 2 results in red light, with the lowest energy photon having a wavelength of 656 nm. The strip below showcases four different colours, each representing distinct frequencies generated by the atom. These differences arise from variations in spacing between the energy levels.

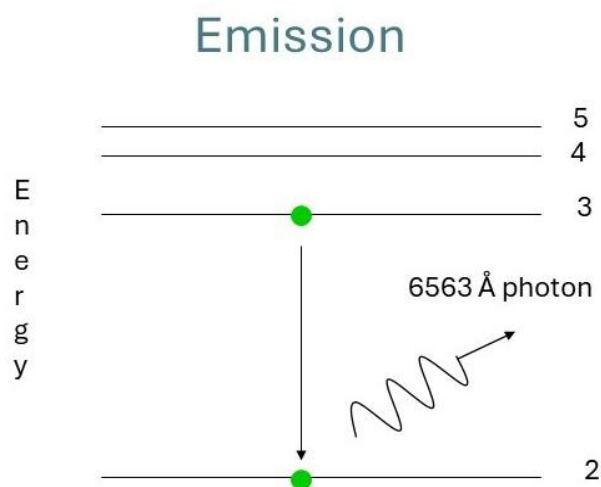
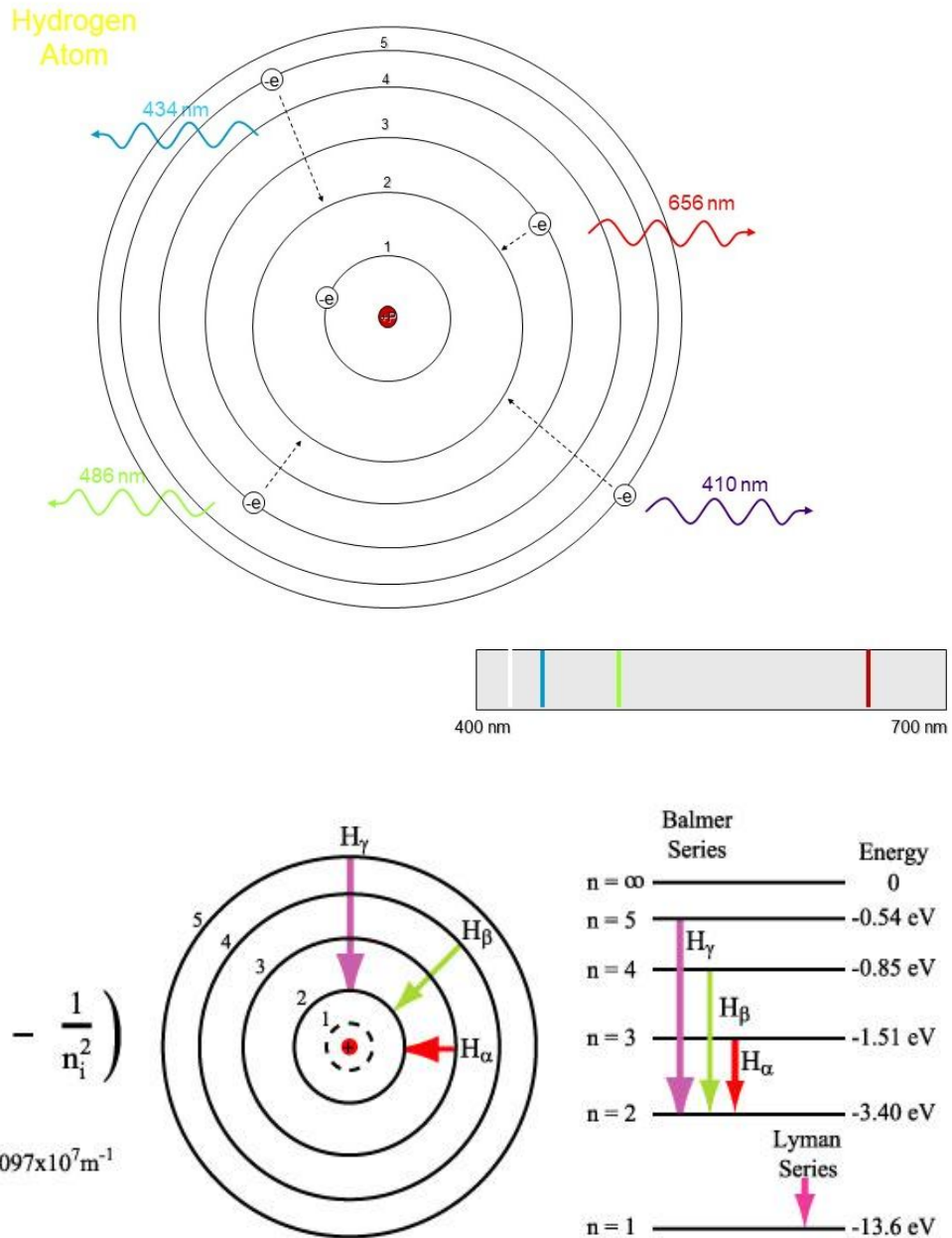


Figure 34. 8. Bohr's model: Electron transition between energy levels emitting light, with frequency determined by the energy difference.



$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

For Hydrogen, $R=1.097 \times 10^7 \text{ m}^{-1}$

Figure 34. 9. Hydrogen spectrum: transitions from higher to lower energy levels emit light at specific wavelengths, creating distinct colors.

In the hydrogen atom spectrum, transitions like H_α , H_β , and H_γ occur when the atom is excited. These transitions emit specific wavelengths of light.

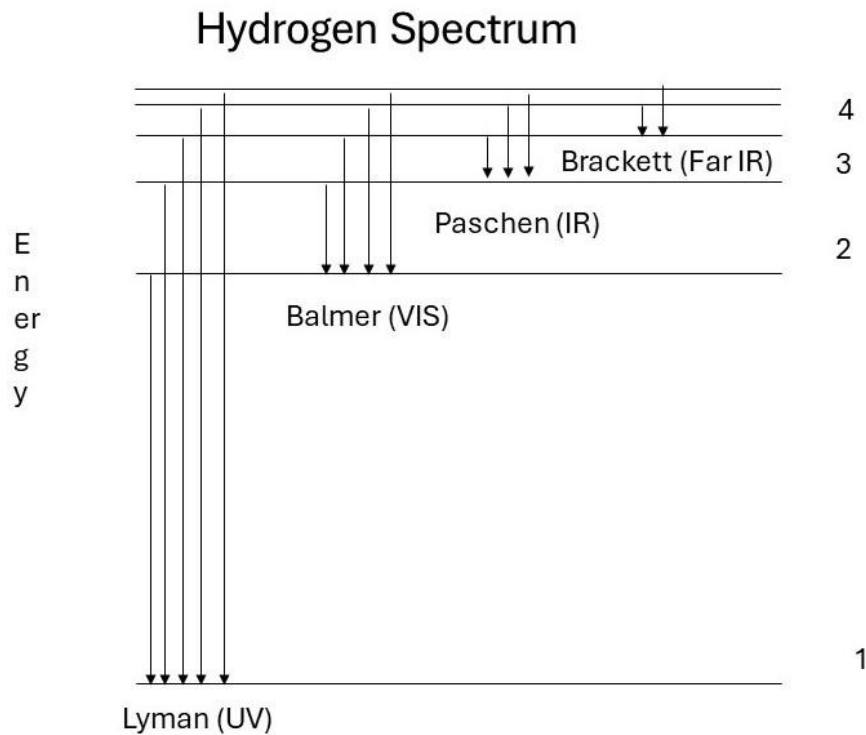


Figure 34. 10. The hydrogen spectrum is well-documented, with series such as Lyman, Balmer, Paschen, Brackett, etc., each depicting different transitions.

Fluorescence

When light strikes an object and is reflected, the frequency remains unchanged. However, this isn't always the case. In fluorescence, incident light excites atoms, causing them to temporarily transition to higher energy states before returning to lower states. Here, an ultraviolet photon collides with an electron, elevating it to a higher state. After a brief pause, the electron descends to a lower state, emitting a photon.

The emitted photon usually has a longer wavelength, hence a different color, compared to the incident ultraviolet light as shown in fig. 34.11. This color difference, known as the **Stokes shift**, is determined by the energy difference between the excited and ground states, which is influenced by the atomic structure of the material. It highlights that incoming and outgoing light can exhibit varying colors."

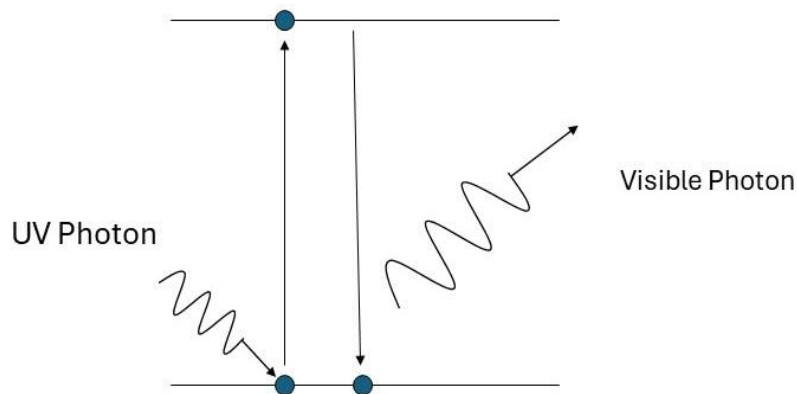


Figure 34. 11. Fluorescence: Incident light excites atoms, resulting in emission of longer wavelength photons, known as the Stokes shift.

Phosphorescence

Phosphorescence is like fluorescence but with a key difference in how long the light lasts.

1. An ultraviolet photon hits an atom and excites an electron, making it move to a higher energy level.
2. The electron then falls to a slightly lower energy level and stays there for a longer period.
3. Finally, the electron drops back to its original level, emitting light in the process.

In fluorescence, this process happens almost instantly, so the emitted light fades quickly. In phosphorescence, the electron stays in the intermediate level longer before returning to its original state. This delay means the material can glow in the dark, as it continues to emit light even after the original light source is removed. This glow can last for a much longer time, making phosphorescent materials useful for things like glow-in-the-dark items.

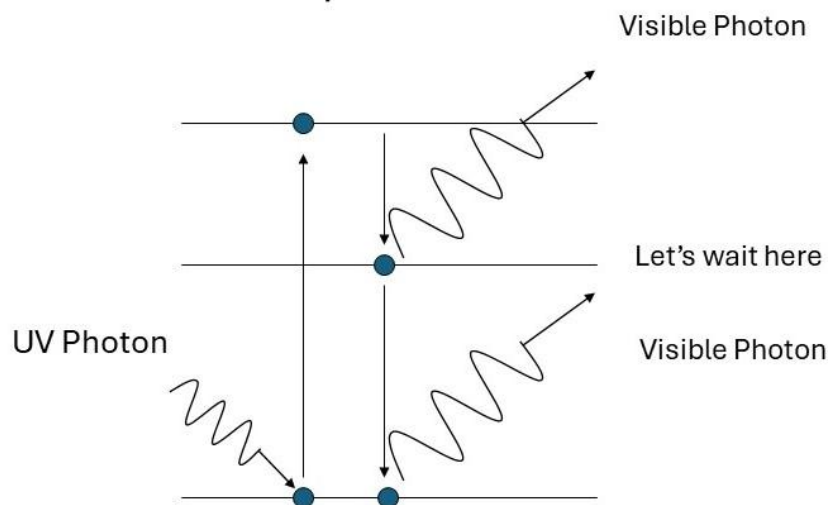


Figure 34. 12. Phosphorescence: After exposure to light, atoms release energy gradually, emitting light over an extended period.

LASER

Lasers, which are now common in many applications, work on a principle called "light amplification by stimulated emission of radiation" (LASER).

1. **Light Amplification:** This means increasing the number of photons or increasing the amplitude of EM wave. In other words, making the light brighter or stronger.
2. **Stimulated Emission:** Normally, when an atom's electron drops from a higher energy level to a lower one, it emits a photon. In stimulated emission, an incoming photon triggers the excited electron to drop to a lower energy level and emit a photon. The emitted photon has the same energy, phase, and direction as the incoming photon, creating a coherent light wave.

Working Mechanism:

- **Excitation:** Atoms in the laser material are excited to a higher energy level by an external energy source, like an electrical current or another light source.
- **Population Inversion:** More atoms are in the excited state than in the lower energy state.
- **Stimulated Emission:** When one of these excited atoms encounters a photon of the right energy, it is stimulated to emit a photon that matches the incoming one. This triggers a chain reaction where more and more photons are emitted in the same phase and direction.
- **Amplification:** These photons bounce back and forth between mirrors at each end of the laser medium, stimulating more emissions. One of the mirrors is partially transparent, allowing some of the light to escape as the laser beam.

The success of the laser has confirmed that photons are fundamental to light. Without the concept of photons, the development of lasers and many other technologies would not have been possible.

Two key points illustrate the importance of photons:

1. The photoelectric effect cannot be explained by classical theory.
2. The spectra of different atoms cannot be understood without considering photons.

There is also a third compelling argument in support of photons, which is equally significant i.e. Compton effect.

Compton Scattering

In Compton scattering, when light hits an electron, it scatters, and the outgoing light does not necessarily have the same frequency as the incoming light. This might seem like fluorescence and phosphorescence, where light excites an atom and then the emitted light has a different frequency. However, there's a key difference: in fluorescence and phosphorescence, the light transition occurs between specific atomic energy levels.

In contrast, Compton scattering involves free electrons, which can move freely and have any momentum. A bound electron in an atom can only occupy certain discrete energy levels, whereas a free electron is not restricted in this way.

Classically, when a light wave interacts with an electron, the electric field of the wave should cause the electron to oscillate and emit radiation at the same frequency as the incoming light.

However, in Compton scattering, the scattered light has a different frequency, indicating that classical theory does not fully explain the phenomenon.

Quantum Mechanical Approach: In Compton scattering, a photon with energy $h\nu$ (where ν is the frequency of blue light) collides with an electron. This collision causes the electron to gain momentum and move with increased kinetic energy. The scattered photon, now with less energy, emerges with a lower frequency than the incoming photon.

Two quantities remain conserved in this process: energy and momentum. Initially, the energy consists of the photon's energy plus the electron's rest energy, which has no kinetic component as the electron is initially at rest. Thus, the initial energy is the photon's energy.

After scattering, the total energy includes the energy of the scattered photon and the kinetic energy gained by the electron. For energy conservation to hold, the energy of the scattered photon must be less than the energy of the incoming photon, as some energy is transferred to the electron. Therefore, if a blue photon collides with an electron and a red photon is emitted, it indicates a loss of energy from the photon to the electron. This phenomenon supports the quantum theory of light and cannot be explained by classical theory alone.

Do we see photons? Where are they? How many are there, and what is their energy?

- Sunny day (outdoors): 10^{15} photons per second enter eye (2 mm pupil)
- Moonlit night (outdoors): 5×10^{10} photons/sec (6 mm pupil)
- Moonless night (clear, starry sky): 10^8 photons/sec (6 mm pupil)
- Light from dimmest naked eye star (mag 6.5): 1000 photons/sec entering eye

So far, we have presented three arguments: the photoelectric effect, atomic spectra, and the Compton effect. These arguments indicate that light consists of photons, which are quanta of energy. A photon's energy is given by $h\omega$ or $h\nu$, meaning it comes in discrete packets. There is a fourth argument related to black body radiation, which will be discussed later when we study heat.

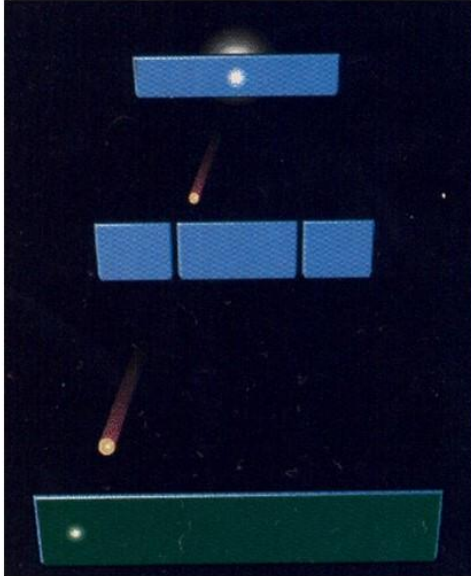
Wave-Particle Duality

Presently, we have two different theories: the particle theory, which involves photons and is associated with Einstein, and the wave theory of electromagnetic waves, related to Maxwell and Faraday. In 1924, Einstein remarked that *there are two theories of light, both indispensable and without any logical connection*. One suggests that light is composed of particles, while the other suggests it is composed of waves. This raises a dilemma: if one theory is correct, does it invalidate the other? This contradiction has led to considerable complexity in understanding the nature of light.

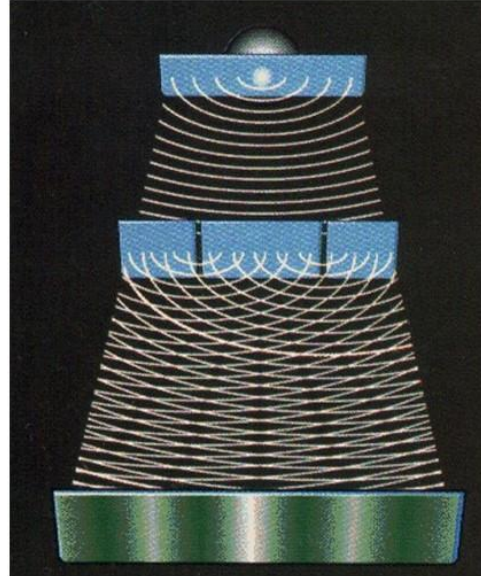
To explore this problem further, let's revisit the double-slit interference experiment. In this experiment, a light source is split into two paths that interfere with each other. We will examine this from two perspectives: light as particles and light as waves.

If we consider light as photons, individual atoms emit these photons. A photon can pass through either one slit or the other and then hit a screen, leaving a mark. If we imagine using bullets

from a rifle instead of photons, the bullets will scatter randomly with no distinct pattern, unlike waves. Waves can reinforce or cancel each other out, creating an interference pattern. This suggests that if light exhibits interference, it must be waves, contradicting the particle-based explanations of the photoelectric effect, atomic spectra, and the Compton effect.



particle?



wave?

However, this is not the whole story. While particles like photons typically don't interfere in the way waves do, photons can indeed interfere under certain conditions. The behavior of photons depends on the type of experiment performed. In some experiments, photons exhibit particle properties, such as in the photoelectric effect. In others, like the double-slit experiment, they exhibit wave properties. This dual nature is known as **wave-particle duality**.

Initially, Einstein found this concept challenging, but it is now a cornerstone of modern physics and the foundation of quantum mechanics. We will explore this further in another lecture.

Lecture 35 – GEOMETRICAL OPTICS

This lecture focuses on geometrical optics, the study of light as rays traveling in straight lines. While light is fundamentally a wave, exhibiting interference and diffraction, these wave properties are not considered in geometrical optics. Instead, we observe that light, such as from a candle, casts shadows when obstructed. Shadows form in the direction opposite to the light source, a phenomenon observable both on Earth and in the celestial bodies we see in the sky. For instance, the moon also casts its shadow on the Earth, leading to solar eclipses when the Sun's rays are blocked. This shadow has two distinct parts: a completely dark region, and the region where the darkness is partial. As the Moon orbits, these shadow regions move along with it. Similarly, lunar eclipses occur when the Sun's rays are blocked by the Earth and cast its shadow on the Moon.

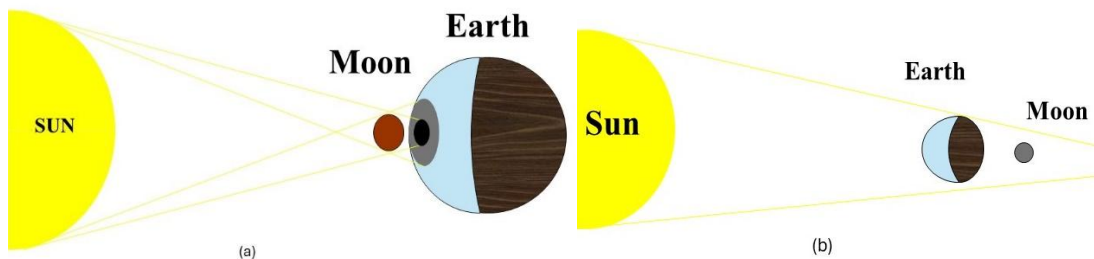


Figure 35. 1. (a) Solar eclipse (b) Lunar eclipse

Reflection and Mirrors

1. Plane Mirrors

When you look into a mirror, the image you see is known as a virtual image. It's formed by the way light rays reflect off the mirror. Imagine a candle in front of a mirror emitting light rays as shown in fig 35.2. The rays that hit the mirror reflect at the same angle. If we extend these reflected rays backward using dashed lines, they appear to converge at a point behind the mirror. This point, where the extended rays meet, is where the virtual image of the candle appears.

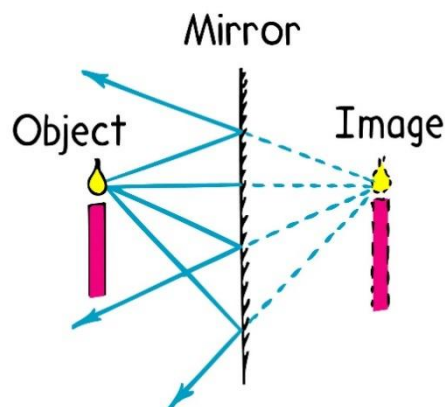


Figure 35. 2. A virtual image of a candle forms behind a mirror where reflected light rays appear to converge.

The **characteristics** of a virtual image in a plane mirror are:

- The rays appear to converge behind the mirror, although they do not physically do so.

- The image appears to be the same distance behind the mirror as the object is in front of it.
- The image is upright and the same size as the object.
- The image is laterally inverted, meaning left and right are switched.

This behavior of light is fundamental to the definition of a plane mirror, which reflects light in such a way that the virtual image seems to exist behind the mirror's surface.

The image created by a mirror is known as a virtual image because it forms where the extended rays appear to converge, but no actual object exists there. Conversely, a real image is formed when light rays converge at a point.

In a plane mirror, the angle of incidence equals the angle of reflection, meaning the light rays reflect symmetrically. This holds true for any smooth section of the mirror. However, if the mirror's surface is rough, the light rays scatter in various directions, preventing the formation of a clear image (fig 35.3).

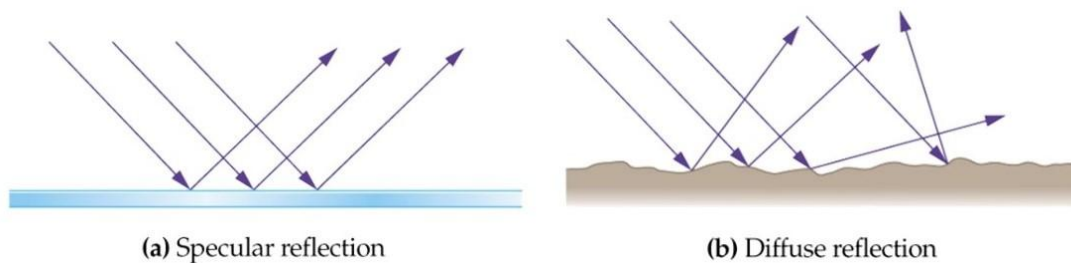


Figure 35. 3. (a) Smooth surfaces cause specular reflection with clear images, while (b) rough surfaces cause diffuse reflection, scattering light.

For a mirror to reflect light properly, its surface must be extremely smooth. Given that the wavelength of visible light ranges from 4000 to 7000 Å (or atomic lengths), the surface roughness must be significantly smaller, ideally at least eight times smaller than these wavelengths. This ensures that the light reflects accurately, maintaining the equality of the angle of incidence and the angle of reflection.

On the other hand, consider satellite dishes, which use microwaves with much larger wavelengths compared to visible light. Despite the use of a wire mesh in these dishes, they still function effectively because the large wavelengths of microwaves can reflect off a less smooth surface.

Another example of forming an image involves a source of light placed in front of a bi-convex lens which bends the light as shown in fig 35.4 (a). The light rays from the bulb pass through the lens and reach our eyes. If we trace these rays backward, they converge at a single point, forming an image.

A partially silver mirror allows light to both reflect off it and pass through it. When light from a source is directed at such a mirror, it can create images in the same location, regardless of the source's position (see fig 35.4 (c)).

If we position the same light source in different locations, its image will still be formed at the same point. For instance, fig 35.4 (a) shows light originates from the source, and when traced back, converges at a point, creating a virtual image. fig 35.4 (b) depicts identical object and image positions, while the third employs a mirror, resulting in identical mirror images. Thus, discerning between the three is challenging.

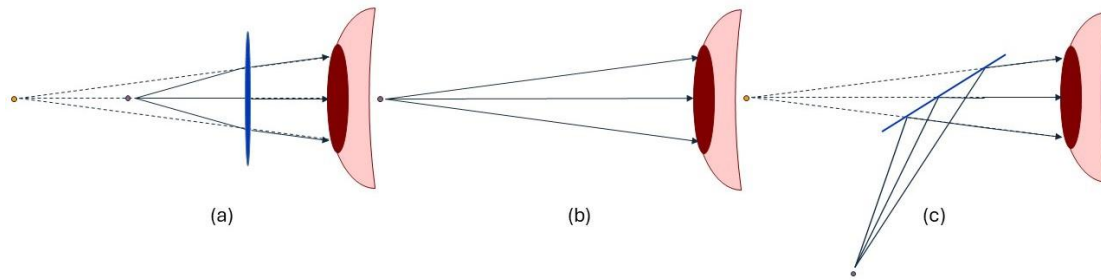


Figure 35. 4. (a) Light bending through a bi-convex lens to form an image as rays converge at a single point. (b) Identical object and image positions showing consistent image formation. (c) A partially silvered mirror creating images in the same location regardless of the source's position.

To determine if an image is real or virtual, trace the light rays backward to their apparent origin. If they seem to emanate from a point behind the mirror, it's a virtual image. It's important to note that the distance from the object to the mirror is equal to the distance from the mirror to the image.

2. Spherical Mirrors

Not all reflective surfaces are flat like glass. For example, by cutting a sphere, we can create curved mirrors that form images with different properties.

Imagine cutting a portion of a spherical object and coating its surface with reflective material illustrated in fig 35.5. Depending on whether the coating is applied to the inner or outer surface, two types of mirrors are created: **concave and convex mirrors**. Each mirror has unique reflective properties.

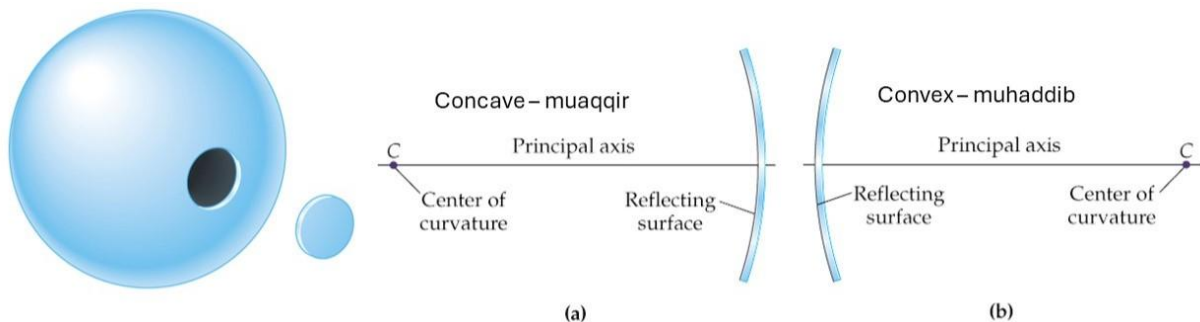


Figure 35. 5. (a) concave and (b) convex mirrors created by coating the inner and outer surfaces of a cut spherical object, respectively.

Let's focus on the concave mirror, which is relatively inexpensive. When positioned at various distances, it produces different images of surrounding lights on a screen. As the mirror is brought closer, the image clarity improves until reaching a specific position, beyond which the image becomes blurry again. This phenomenon can be explained through a diagram, where parallel lines originating from a distant source interact with the mirror.

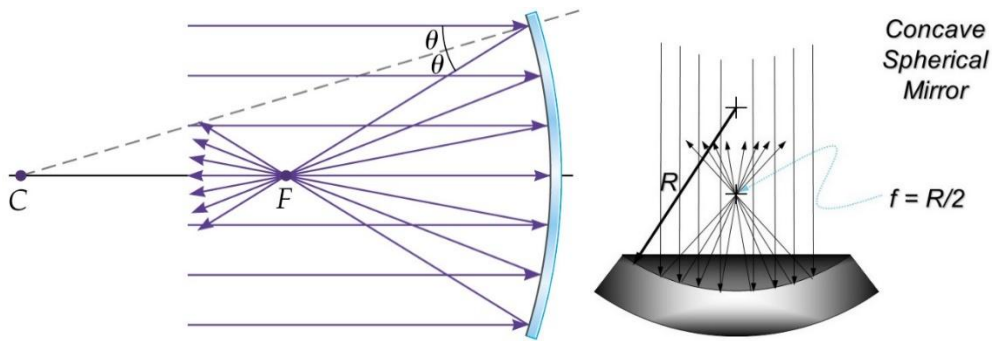


Figure 35. 6. (a) Parallel light rays converge at the focal point (F) after reflection from a concave mirror, (b) the focal point is located at $R/2$ from the center of the mirror, known as the focal length.

Parallel light rays that strike a mirror converge at a single point after reflection, known as the focal point (F) as shown in fig 35.6. For a concave mirror, the focal point is located at a distance of $R/2$ from the centre of mirror, known as the focal length.

When parallel rays strike a convex mirror, they appear to originate from a specific point behind the mirror known as the **virtual focus**. Despite seeming to originate from this point, the rays reflect off different portions of the mirror (fig 35.7). In optics, we use a convention where the focal length is negative if the focus is behind the mirror. Thus, for a convex mirror, the focal length is negative, while for a concave mirror, it's positive, at $R/2$.

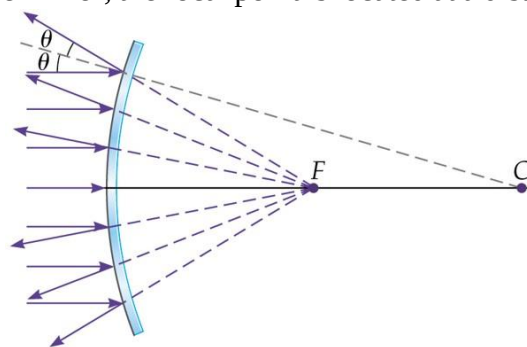


Figure 35. 7. Parallel rays striking a convex mirror appear to converge at a virtual focus point behind the mirror, reflecting off different portions of the surface.

Image formation in concave mirror

A concave mirror reflects light in a way that forms images with varying sizes. Three principal rays are considered:

- P-ray: Travels parallel to the axis, reflects through the focal point.
- F-ray: Strikes the focal point, reflects parallel to the axis.
- C-ray: Passes through the center, reflects on itself.

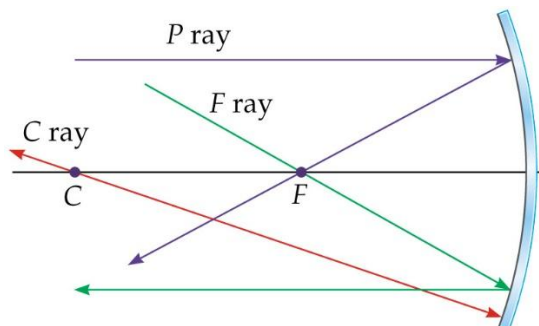


Figure 35. 8. Principal rays for a concave mirror: P-ray reflects through the focal point, F-ray reflects parallel after striking the focal point, and C-ray reflects through the center, illustrating image formation.

When an object, like a pencil with a lit tip, is placed before a concave mirror, the light rays it emits reflect off the mirror. If the object is placed beyond the center (C), the reflected rays, P

and F, intersect, forming a real axis where the image appears smaller than the object shown in fig 35.9 (a). If the object is between the center (C) and the focal point (F), the reflected rays, P and F, intersect, forming a real axis where the image appears larger than the object shown in fig 35.9 (b). This magnification property is why concave mirrors are used in applications like shaving mirrors, where an enlarged view is beneficial.

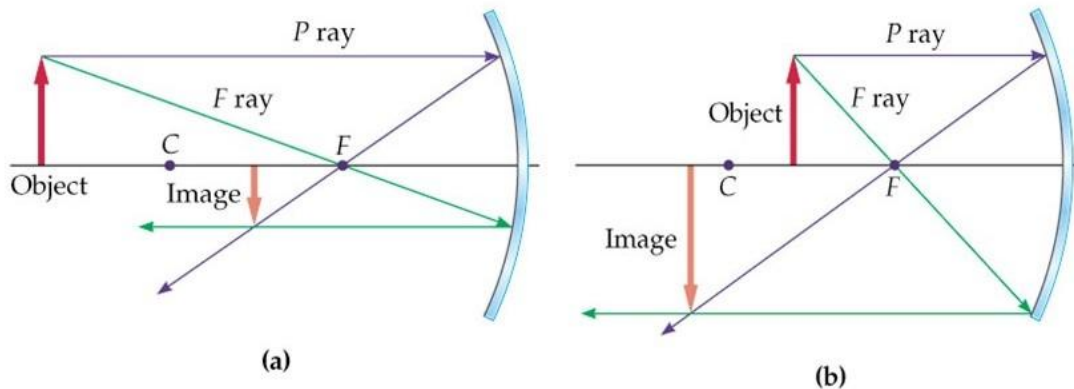
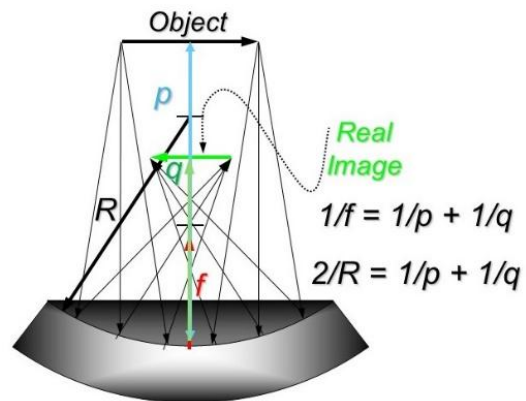


Figure 35. 9. Image formation by a concave mirror: (a) a smaller image formed when the object is placed beyond the center (C), while fig (b) illustrates a larger image formed by the object placed between the center (C) and the focal point (F), highlighting the mirror's magnification property.

In a concave mirror, an object like a pencil creates a real image. The ratio of the image size (h') to the object size (h) is known as the magnification factor. Mathematically, it's expressed as h'/h or $-q/p$, where q is the distance from the mirror to the image, and p is the distance between the object and the mirror.

Image formation in convex mirror

When an object, such as a pencil, is placed in front of a convex mirror, the reflected rays—P-rays, F-rays, and C-rays—diverge. P-rays reflect upward, C-rays reflect along the same path, and F-rays reflect parallel to the axis. Extending these rays backward, they seem to converge at a point behind the mirror, forming a diminished, upright image of the object. Unlike concave mirrors, which produce inverted images (hence the negative magnification factor), convex mirrors always create upright images. The focal length for a convex mirror is represented as $-R/2$, indicating the virtual nature of the focus.



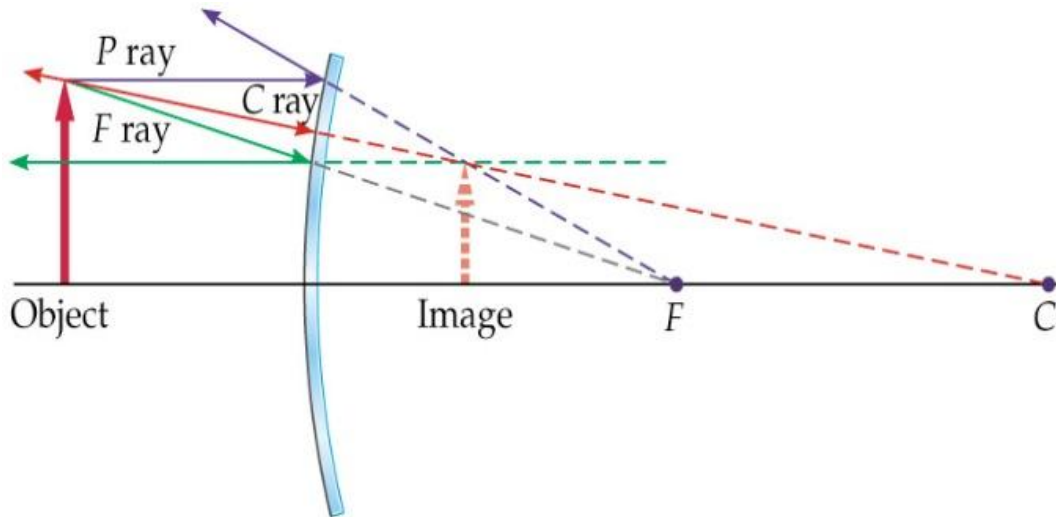
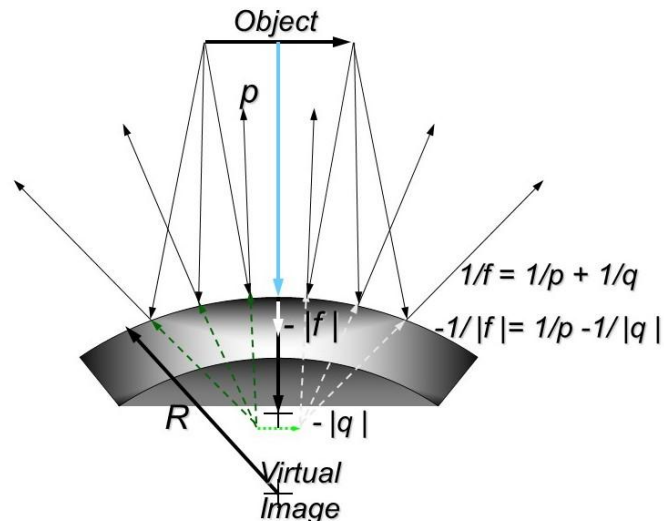


Figure 35. 10. A virtual, upright and diminished image formed by a convex mirror with diverging reflected rays appearing to converge behind the mirror.

In convex mirrors, reflected light rays diverge, creating a scattered effect. Yet, when these rays are extended backward, they seem to converge at a point, forming a virtual image. The magnification factor, (M), is the ratio of the image height to the object height, expressed as ($M = h'/h$), which is also (q/p). Unlike images formed by concave mirrors, images in convex mirrors remain upright, hence the absence of a negative sign in the magnification formula.



Refraction and Lenses

Lenses focus or disperse light due to the change in speed caused by difference in refractive index between the lens material and the surrounding medium, like air or vacuum. The refractive index (n) is defined as the ratio of the speed of light in a vacuum to its speed in the medium. When light enters a lens, it slows down and bends towards normal due to refraction. In the case of a convex lens, the light converges towards the center, while in a concave lens, it diverges.

In a convex lens, parallel light rays converge to a point, focusing the light. When a light source is positioned on the focal point of a lens, the emitted rays are parallel.

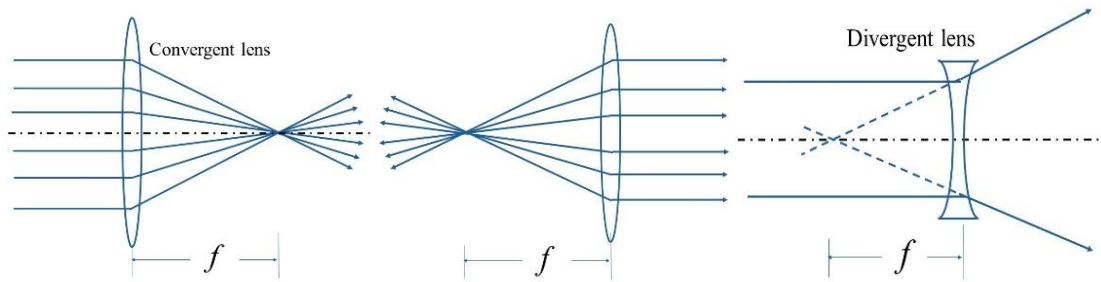


Figure 35. 11. Diagrams can help visualize how convex lenses (a) converge light to a focal point, (b) Light emitted from the focal point of a lens emerges as parallel rays, while (c) concave lenses cause light to diverge, affecting the size and focus of the resulting image.

If parallel light rays encounter a diverging (concave) lens, they scatter. Extending these diverging rays backward, they seem to meet at a virtual focal point, denoted as negative for concave lenses, while it remains positive for convex lenses.

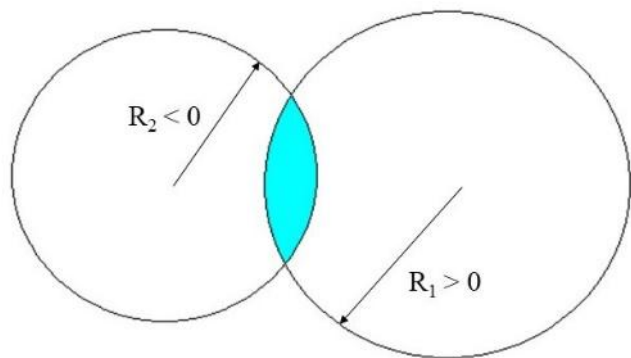
Lensmaker's formula for Spherical Single Lens

The focal length of a spherical lens is determined using the **lensmaker's formula**. It accounts for the lens's curvature on both sides, which may differ, leading to different radii of curvature (R_1 and R_2). A larger radius corresponds to a flatter lens, whereas a smaller radius indicates a more curved and thus more powerful lens.

The formula is:

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Here, n is the refractive index of lens material, and 1 is the refractive index of air. R_1 and R_2 are the radii of curvature for the lens's surfaces. R_2 is considered negative when the lens's curvature is inward from the light source's perspective. The power of a lens increases as the radii of curvature decreases. Since R_2 is negative, its magnitude contributes positively to the lens's power.



Different **lens shapes** include biconvex (curved outward on both sides), biconcave (curved inward on both sides), or a combination of convex and concave, including plano-convex (flat on one side and curved outward on the other), and plano-concave (flat on one side and curved inward on the other) configurations. Each type has unique properties affecting how light is refracted through them (fig 35.12).

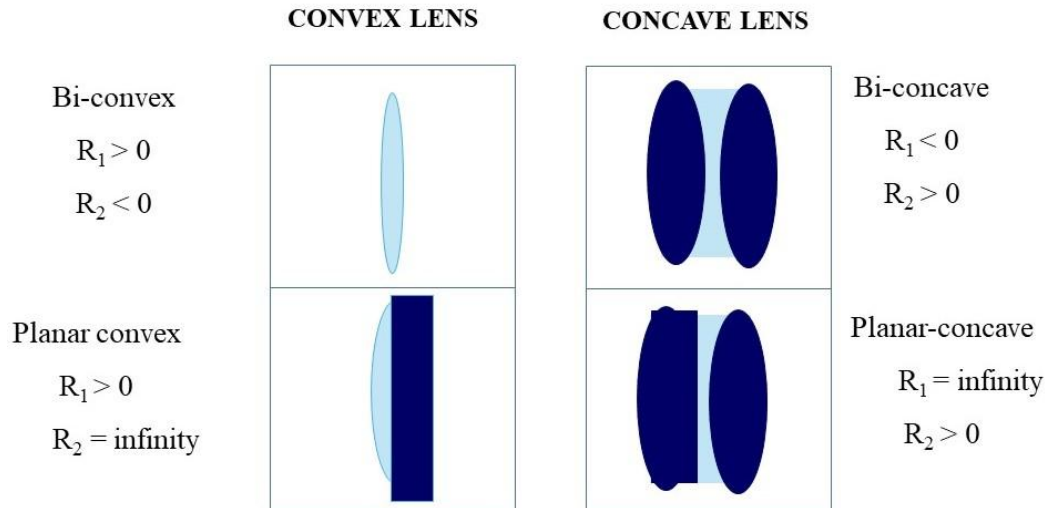


Figure 35. 12. Various lens shapes have distinct properties influencing light refraction.

Diopter

The **diopter** is the reciprocal of a lens's focal length (f) in meters, expressed as

$$\text{Diopter} = 1/f$$

The lens's power, measured in diopters, depends on its refractive index (n) and the radii of curvature (R_1 and R_2) on each side. A lens with greater power can focus or diverge light more effectively.

Image formation by convex Lens

To understand how a lens shapes light, we trace the path of light rays through the lens. For a convex or converging lens, the image formed is always **inverted**. If an object, like a pencil, is placed close to the lens, the image appears larger (magnified) and virtual because the light rays diverge and do not meet after passing through the lens indicating magnification greater than 1. As the object moves further from the lens as shown in fig 35.13, the image becomes smaller and real, as the light rays converge to a point. The magnification changes with the object's distance from the lens, affecting the size of the image produced. As the distance between the object and the lens changes, the nature of the image also changes.

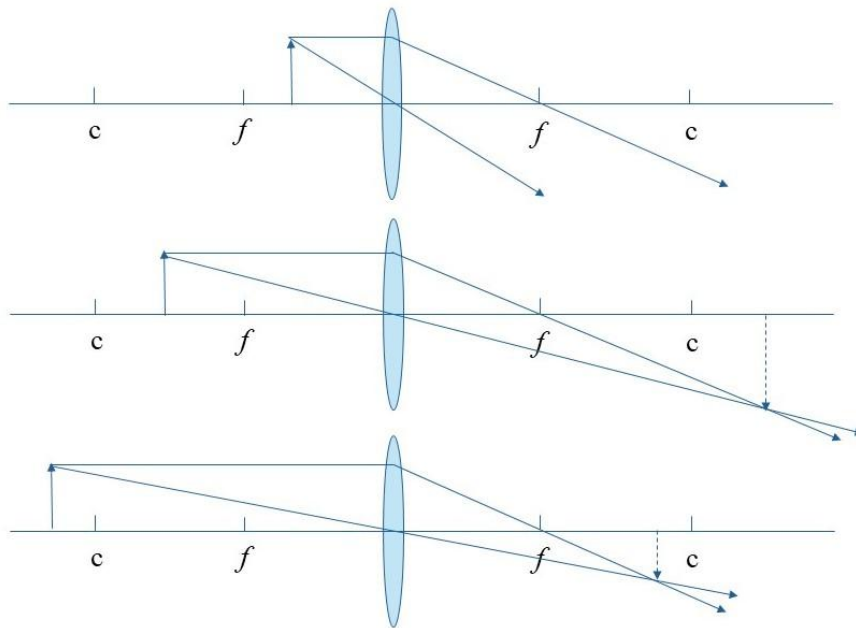
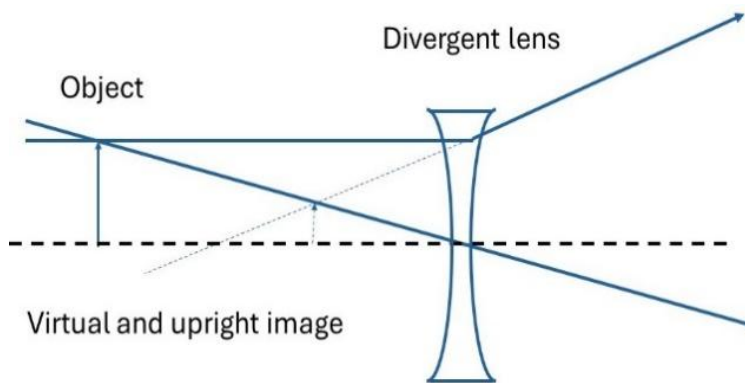


Figure 35. 13. "Figures illustrate the behavior of convex lens: close objects create larger, virtual, and magnified images, while distant objects form smaller, real images due to light ray convergence, demonstrating varying magnification and image properties.

Image formation by concave Lens

On the other hand, concave lenses produce a smaller, virtual image as they diverge from light rays. While concave lenses reduce the size of images, they are beneficial for correcting vision and other applications.



Lens Defects:

- i. A concave lens's primary function is to diverge rays coming from a distant source, such as parallel rays from the Sun. However, lens makers face challenges like **chromatic aberration** (*Aberration refers to imperfection*), where different colors focus at different points due to varying refractive indices. This means that red and blue light, for instance, will converge at slightly different focal points, leading to a blurry image. This issue is tackled by employing various techniques, including applying specialized coatings to the lenses to minimize the difference in focal points for various colors, ensuring clearer images.
- ii. **Spherical aberration** occurs when rays striking different parts of a wide lens do not converge at a single point, leading to a distorted image. This defect is particularly pronounced in wide lenses where parallel rays don't meet at one focal point. Reducing the

lens's aperture can minimize aberration, allowing parallel rays to converge more effectively and produce a clearer image.

Let's explore why do objects appear larger or smaller at different distances, and how can we calculate this?

Optical Instruments

The simple Magnifier

Consider a convex lens, when it is held close to the thumb, the thumb appears its normal size. As the lens moves away from the thumb, the size of image changes, becoming larger, until it eventually defocuses.

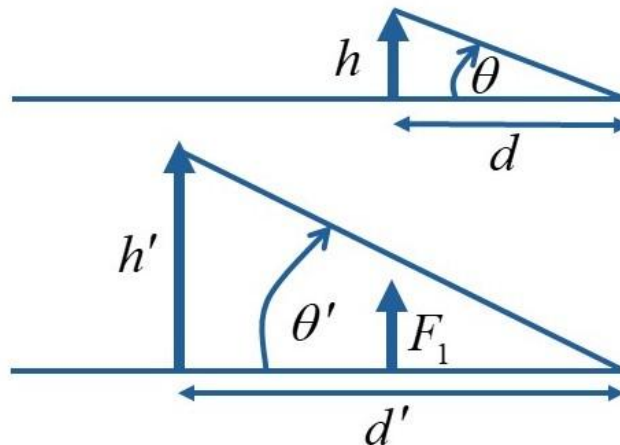


Figure 35. 14. A simple magnifier

To calculate this, we use magnification, defined as the ratio of θ' to θ .

$$m_{\theta} = \frac{\theta'}{\theta} = \frac{h / f}{h / 25\text{cm}}$$

$$m_{\theta} = \frac{25\text{cm}}{f}$$

25 cm is the typical closest distance at which the human eye can focus comfortably, about 8 inches. Closer than this, objects defocus. A shorter focal length results in greater magnification, making small objects appear larger. However, there's a limit to magnification with a single lens, as too short a focal length can cause aberrations. Thus, there are practical limitations to how much a single lens can magnify.

Compound Microscope

In a compound microscope shown in fig 35.15, magnification is achieved through two lenses: the objective and the eyepiece. The object's height is denoted as h . Light rays originating from the object pass through the objective lens, creating an intermediate image of height h' at F_1' . This image is then magnified by the eyepiece lens, resulting in a larger final image.

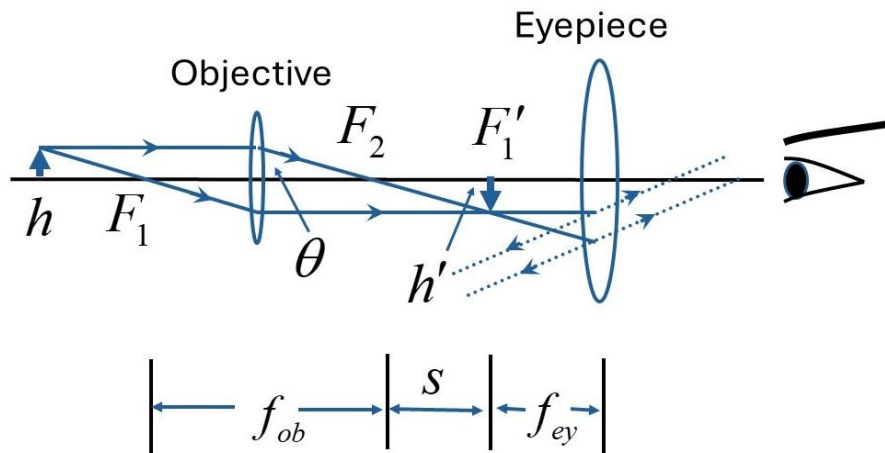


Figure 35. 15. A compound microscope, featuring multiple lenses, magnifies small objects by combining the magnification powers of its objective and eyepiece lenses.

The total magnification M of the microscope is the product of the magnifications by the objective (m_θ) and the eyepiece (m_θ) which is calculated as:

$$M = mm_\theta = -\frac{s}{f_{ob}} \frac{25cm}{f_{ey}}$$

Here, s is the tube length, f_{ob} is the focal length of the objective lens, and f_{ey} is the focal length of the eyepiece. The negative sign indicates that the intermediate image is inverted. The dual-lens system allows for greater magnification than a single lens.

Refracting Telescope

A binocular or refracting telescope can also be constructed using two lenses to enhance the apparent size of distant objects (fig 35.16). The parallel rays are focused to form a small angle since the object is far away. This angle is the size of the object divided by the distance, a very small quantity. We aim to increase this angle. The objective lens captures the object at a distance, creating an image at the focal point of size h' , which is then enlarged by the eyepiece for clearer viewing. The overall design allows for an increased angle of view, making distant objects appear closer and larger to the observer.

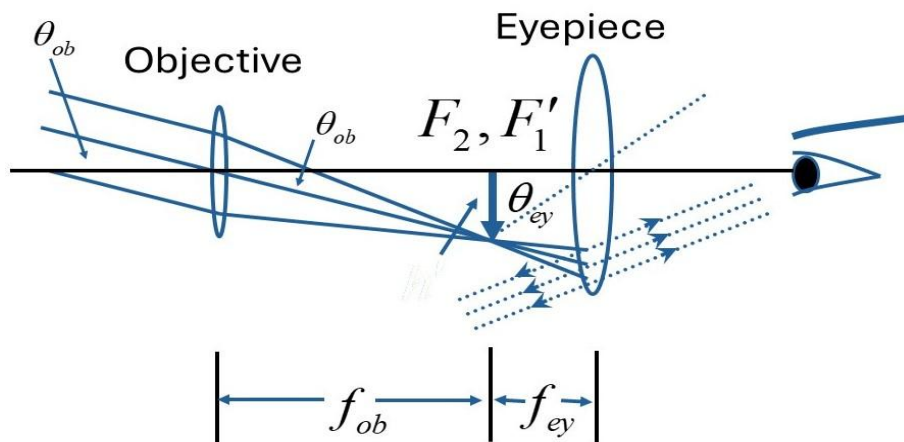


Figure 35. 16. A refracting telescope comprised of two lenses to gather and focus light from distant objects, providing magnified views of celestial bodies.

Human Eye

In a normal eye, light focuses precisely on the retina, creating a clear image. This sharp focus allows the retina to transmit electrical signals to the brain, enabling us to perceive the image. People with poor vision particularly benefit from corrective lenses.

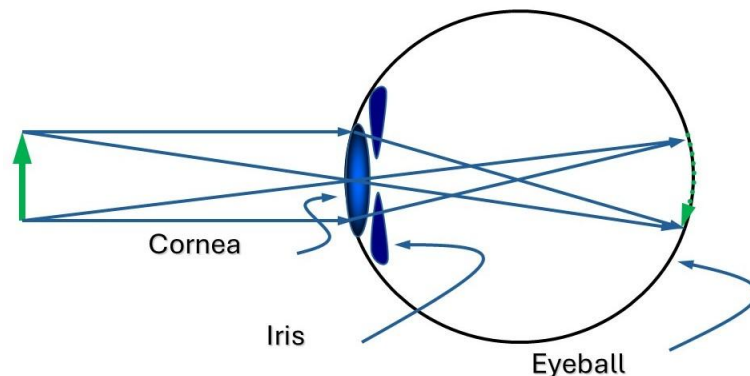


Figure 35. 17. Image formation in normal human eye.

Eye Defects and their Correction

There are two common types of vision problems:

- **Farsightedness (Hypermetropia):** Difficulty seeing near objects because the image forms behind the retina. To **correct** hypermetropia, a convex lens is used, which adjusts the focal point, so the image falls on the retina.

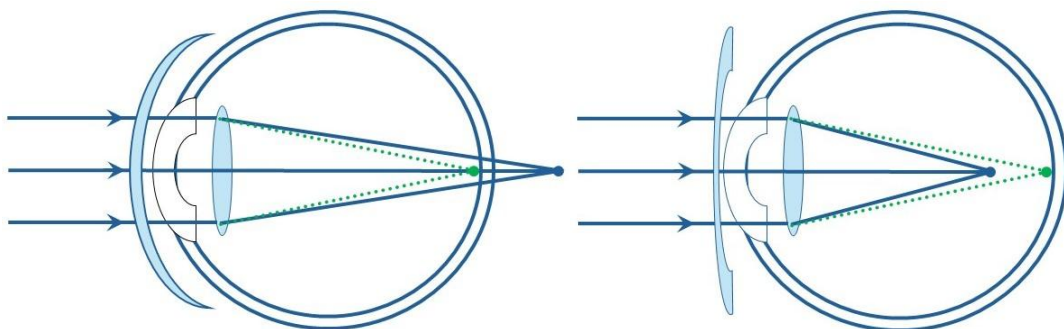


Figure 35. 18. Eye defects and their correction using spherical lenses: Hypermetropia (left), Myopia (right)

- **Nearsightedness (Myopia):** Difficulty seeing distant objects because the image forms in front of the retina. For myopia, a concave lens is employed to diverge the light rays, moving the focal point forward onto the retina. Thus, lenses help restore clear vision by compensating for the eye's focusing deficiencies.

Prism and total Internal Reflection

Prisms are fascinating optical devices that not only disperse light into its constituent colors but also enable total internal reflection. This phenomenon occurs when the angle of incidence exceeds a critical value, causing light to reflect entirely within the prism rather than passing through it. When light enters a prism and reflects internally, it can create an image that is the same size as the object if the rays are parallel. If the prism is rotated, the image can be inverted (see fig 35.19). This principle is utilized in binoculars to correct the orientation of the observed image, ensuring that it appears right-side-up to the viewer.

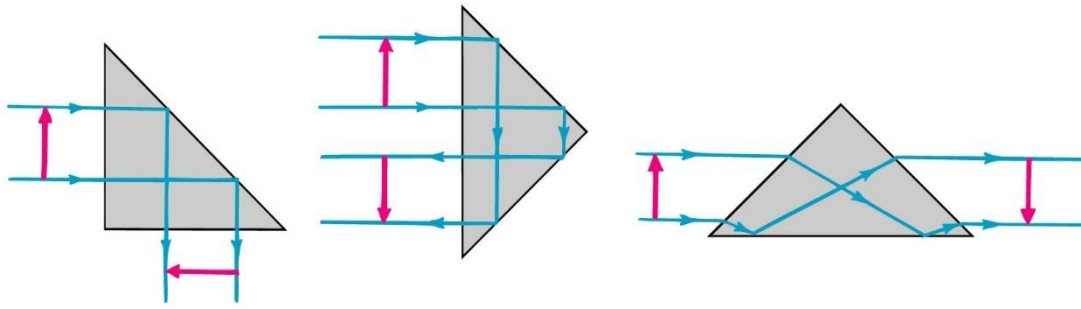


Figure 35. 19. Total internal reflection within a prism creates an image the same size as the object with parallel rays, capable of inversion when the prism is rotated.

In binoculars, prisms play a crucial role. They are used to extend the optical path without increasing the physical length of the binoculars, which keeps them compact. The prisms inside binoculars cause light to undergo total internal reflection, effectively lengthening the path the light travels before reaching the eyes. This design allows for greater magnification power while maintaining a portable size.

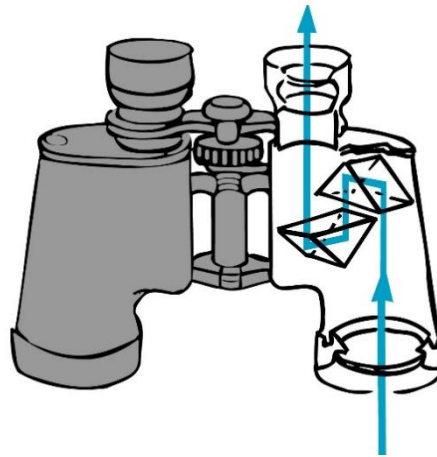


Figure 35. 20. The diagram illustrates light reflecting through multiple prisms, effectively extending the path length within a shorter physical space.

Prisms can be used as mirrors through the phenomenon of total internal reflection. When parallel rays strike the prism, it can be reflected off the internal surfaces if the angle of incidence is greater than the critical angle for the material. This reflection can effectively turn the prism into a mirror without the need for a reflective coating.

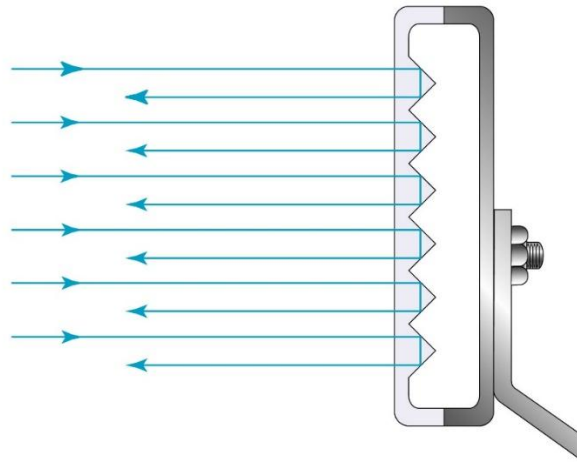


Figure 35. 21. Prisms utilizing total internal reflection to act as mirrors for parallel rays, reflecting internally without external coatings.

Advanced Optics: Gravitational Lensing

Light typically bends when it passes through materials with different refractive indices, such as air or glass. However, in space, light can also bend due to gravity, a phenomenon predicted by Einstein's general theory of relativity. This bending occurs near massive objects like stars or black holes having strong gravitational fields, creating what is known as a gravitational lens. Gravitational lensing allows us to observe distant objects like quasars, as their light is focused by these massive celestial bodies. This effect has been observed and helps astronomers study the distribution of matter in the universe.

Lecture 36 – THERMAL PHYSICS I

Introduction to Heat and Temperature:

Energy is a fundamental concept, originating from small particles like electrons and protons, as well as stars and galaxies. It can be of various forms, including chemical and atomic energy etc. **Heat** differs slightly from other forms of energy and flows from a system at high temperature to a system at low temperature. Heat has played a crucial role in human history. Heat has been pivotal in human progress. The invention of steam engines revolutionized transportation and communication, leading to the development of diesel and petrol engines, enabling long-distance travel and changing societal structures. We instinctively recognize the difference between hot and cold objects, associating heat with a higher temperature. Scientifically, **temperature** is the common property that equalizes when bodies are in contact, leading to thermal equilibrium. **Thermal equilibrium** is a state where heat transfer between two or more bodies ceases due to equal temperatures as shown in fig 36.1.

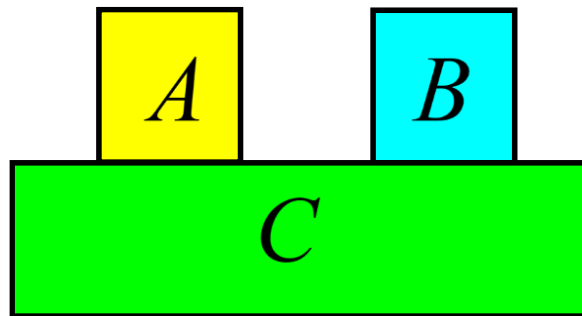


Figure 36. 1. Three bodies A, B and C are in thermal contact with each other.

Measuring Temperature

Temperature is measured using various methods. One way is by observing color changes in heated objects, such as iron turning from yellow to white as it heats. Even from a distance, like with the sun, we can infer temperature by analysing light and color.

Thermometers

Thermometers measure temperature based on the principle that *substances expand when heated*. For instance, alcohol in a thermometer expands and rises in a tube, as it gets hotter. Demonstrating with a candle, we observe the alcohol rise with heat and fall when placed in cold water, showcasing the thermometer's response to temperature changes.

Gas Thermometers: Thermometers can utilize various liquids (like mercury), but they're not limited to liquids alone. Gases also expand with heat, and if confined, their pressure increases with temperature. This principle can be demonstrated by confining hot air in a bottle using a candle, then cooling it in water, observing the pressure change. This experiment illustrates

Boyle's Law, where pressure is directly proportional to temperature (T) when volume is constant.

Constant Volume Gas Thermometer

A constant volume gas thermometer (fig 36.2) measures temperature of a gas by keeping its volume fixed and observing the pressure changes, which are calculated using the formula:

$$P = \rho \times g \times h$$

where P is pressure, ρ is the density of the liquid, g is the acceleration due to gravity, and h is the height. Here the reference level is kept fixed by raising or lowering the tube on the right side (the tube below is made of rubber or some flexible material). So, the gas volume is fixed, and the pressure is known from measuring h.

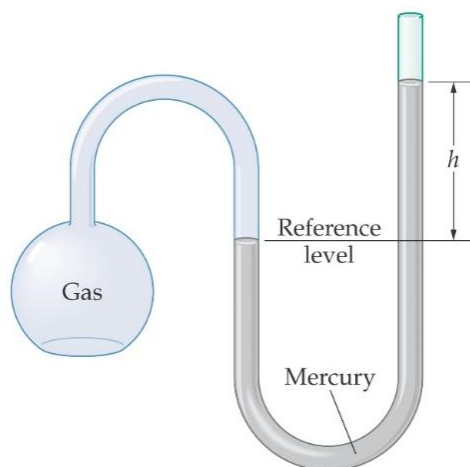


Figure 36. 2. A constant volume gas thermometer that measures temperature by maintaining a fixed gas volume and observing pressure changes using a flexible tube for height adjustment.

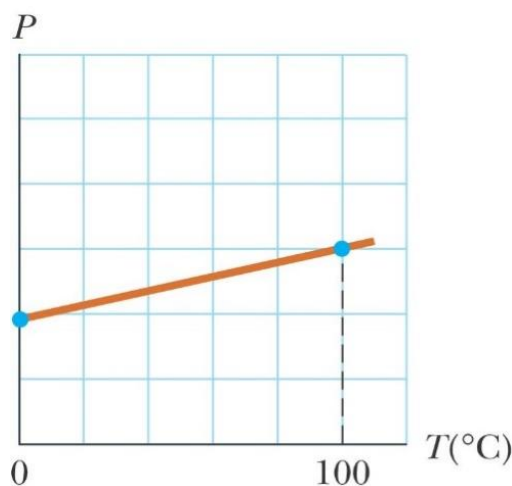


Figure 36. 3. A pressure vs. temperature graph shows a direct linear relationship, forming a straight line from 0 to 100 degrees Celsius.

Absolute Zero

There is a temperature below which it is not possible to go. In other words, if you cool and cool there comes a point after which you cannot cool anymore! If we draw P vs T graph for different gases as illustrated in fig 36.4, we observe that all the gases converge at a single point (-273.15°C) on a temperature graph, known as absolute zero (0°K or zero-degree Kelvin). This is the point where gases theoretically have zero kinetic energy.

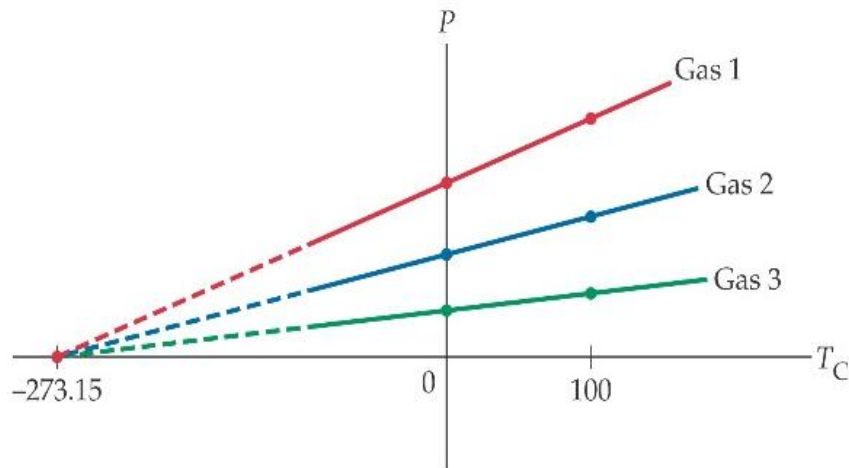


Figure 36. 4. A Pressure vs Temperature graph for different gases shows convergence at - 273.15 °C, known as absolute zero (0 K).

Temperature Scales

Centigrade, Fahrenheit, and Kelvin are three different temperature scales that use water as a reference due to its global abundance. For instance, water freezes at (0 °C), which is equivalent to (273 K) and (32 °F), and boils at (100 °C), (373 K), or (212 °F) as shown in fig 36.5 (a).

Temperature Conversion

Conversion between degrees Celsius and degrees Fahrenheit

$$T_F = \left(\frac{9}{5} F^\circ / C^\circ\right) T_C + 32^\circ F$$

Conversion between degrees Fahrenheit and degrees Celsius

$$T_C = \left(\frac{5}{9} C^\circ / F^\circ\right) (T_F - 32^\circ F)$$

Conversion between Celsius and Kelvin temperatures

$$T = T_C + 273.15$$

Temperature Extremes

While absolute zero is the lowest possible temperature, the maximum temperatures in the universe are found inside stars, reaching up to 10 million Kelvin (see fig 36.5 (b)). Star surfaces, including our Sun, range from 5000 K to 50,000 K, with hydrogen bombs producing even higher temperatures. Metals melt around 1000 °C, and gases like nitrogen, hydrogen, and helium liquefy below 0 °C. The lowest temperature achieved on Earth is approximately 10^{-7} degrees Kelvin (0.00001 K). The **lowest temperatures are achieved using cryogenic refrigeration**, approaching but not reaching absolute zero.

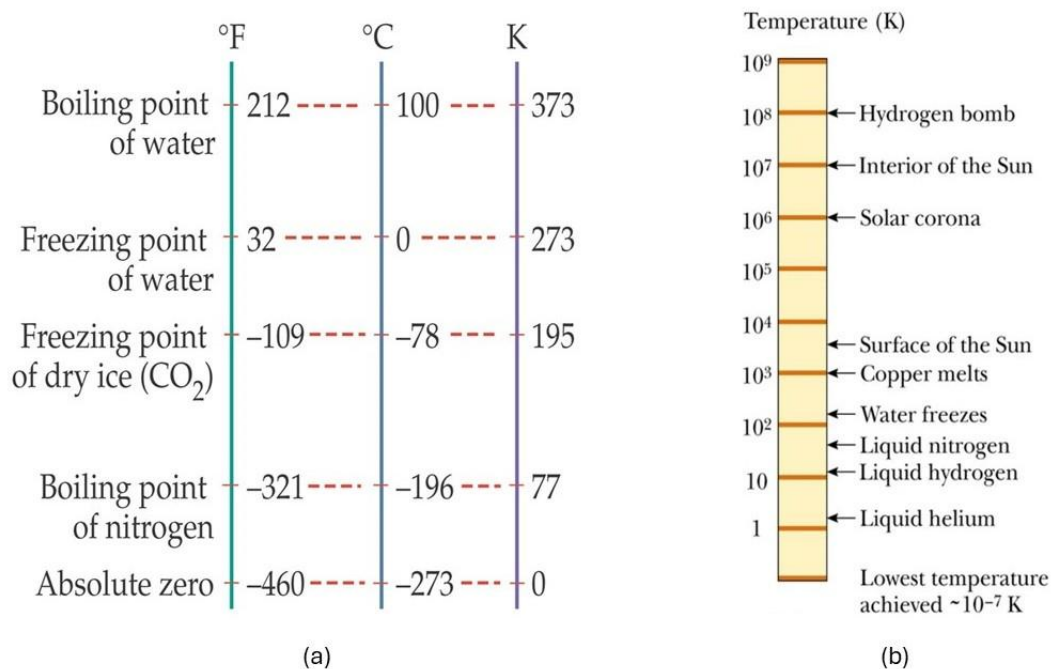


Figure 36. 5. (a) Centigrade, Fahrenheit, and Kelvin scales equivalence. (b) temperature ranges of various phenomena.

Thermometric Properties: Apart from expanding a substance and increasing gas pressure heating affects physical properties like **electrical conductivity and resistivity**, which can also be used to measure temperature. For example, a heated wire's increased resistivity indicates its temperature rise.

Joule's Experiment

A historical experiment demonstrated that *heat is a form of energy generated by work*. In the experiment, a falling weight tied to a rope moved a paddle in water, heating it as shown in fig 36.6. The greater the fall, the hotter the water became. Lifting the weight increased its potential energy, which was then converted to heat energy, not kinetic energy. This disproves the old belief that heat was a fluid called "phlogiston." Joule's experiment showed that work and heat are equivalent. Similarly, rubbing your hands together generates heat because the work done increases molecular activity.

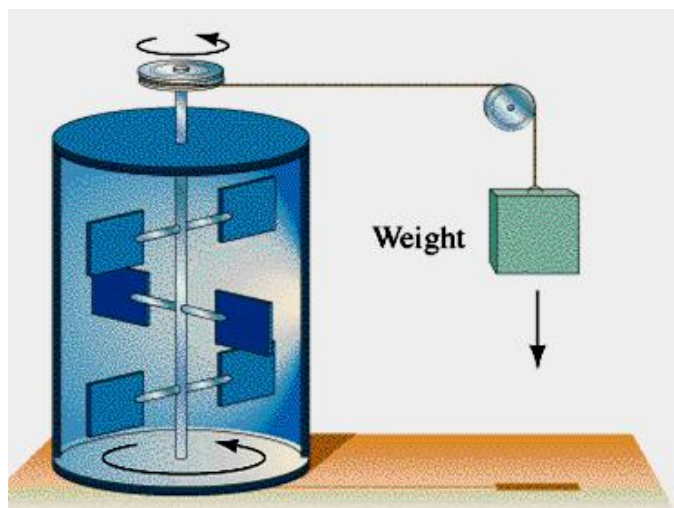


Figure 36. 6. Joule's experimental setup demonstrates that mechanical work converts into heat energy, validating the work-heat equivalence principle.

The unit of heat, the **calorie**, is defined as *the amount of heat required to raise the temperature of 1 gram of water by 1 °C*.

One kilocalorie (1 kcal) = 1000 calories

Joule's experiment also established that **1 calorie = 4.186 joules**, known as the **mechanical equivalent of heat**. This concept helps us understand that what we perceive as heat in an object is actually its internal energy.

Internal Energy:

Internal energy is a measure of the random motion of molecules within a substance, such as water or gas. The faster these molecules move, the higher the internal energy. For gases, this internal energy or kinetic energy is directly proportional to the temperature (T), described by the equation

$$\text{K.E} = \frac{3}{2}Nk_B T$$

where N is the number of molecules and k_B is the Boltzmann constant. Joule's experiments showed that work can be converted into heat, increasing temperature.

Microwave ovens are common household appliances that heat food without heating themselves. They work by emitting microwaves at frequencies that resonate with water molecules, causing them to vibrate and heat up. This selective heating allows water to warm while other materials, like a candle, remain unaffected. The microwaves' frequency is specifically tuned to the natural frequency of water molecules, enabling efficient energy transfer. This shows that understanding heat at a microscopic level reveals many other phenomena.

Thermal Expansion:

When we heat a piece of iron, it expands due to the increased vibration of its molecules, arranged in a lattice structure. Mathematically, the increase in length (ΔL) is proportional to the temperature change (ΔT). Mathematically, this relationship is expressed as

$$L = L_0 [1 + \alpha (T - T_0)]$$

where L_0 is the original length, L is the new length T_0 is the original temperature, and α is the coefficient of linear expansion.

$$L - L_0 = \alpha L_0 (T - T_0)$$

$$\Delta L = \alpha L_0 \Delta T$$

$$\alpha = \frac{\Delta L / L_0}{\Delta T}$$

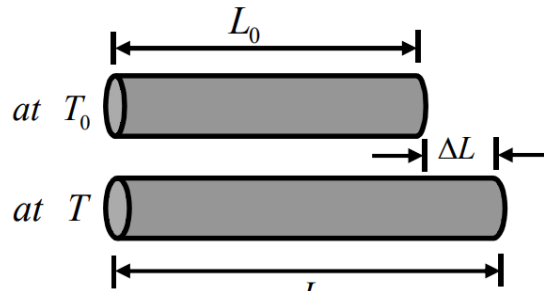


Figure 36. 7. Linear thermal expansion of an iron rod.

Thus, the **coefficient of linear expansion** α is defined as *the ratio of change in length of a material to the original length per degree change in temperature*. The units of α are per degree Celsius $^{\circ}\text{C}^{-1}$ or per Kelvin K^{-1} , as the temperature change can be measured in either Celsius or Kelvin, and the change in length is a ratio, making it dimensionless.

The same principle applies to volume expansion. The coefficient of volume expansion β relates to how much a substance expands in three dimensions. The **coefficient of volumetric expansion**, β , is defined as *the ratio of the change in volume of a material to the original volume per degree change in temperature*.

$$\beta = \frac{\Delta V / V_0}{\Delta T}$$

Relation between α and β

If we consider a cube of volume V_0 as shown in fig 36.8, length L of each side increases due to thermal expansion, and the new volume V' is:

$$\begin{aligned} V' &= (L + \Delta L)^3 = (L + \alpha L \Delta T)^3 & \therefore \text{Volume of cube} &= (\text{length})^3 \\ &= L^3 + 3\alpha L^3 \Delta T + 3\alpha^2 L^3 \Delta T^2 + \alpha^3 L^3 \Delta T^3 & (a + b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 \end{aligned}$$

If ΔT is small, we can ignore ΔT^2 and ΔT^3 . Hence, we get

$$\begin{aligned} &\approx L^3 + 3\alpha L^3 \Delta T \\ &= V_0 + 3\alpha V_0 \Delta T \\ \Rightarrow \Delta V &= 3\alpha V_0 \Delta T & \therefore \beta &= \frac{\Delta V / V_0}{\Delta T} \\ \Rightarrow \boxed{\beta = 3\alpha} \end{aligned}$$

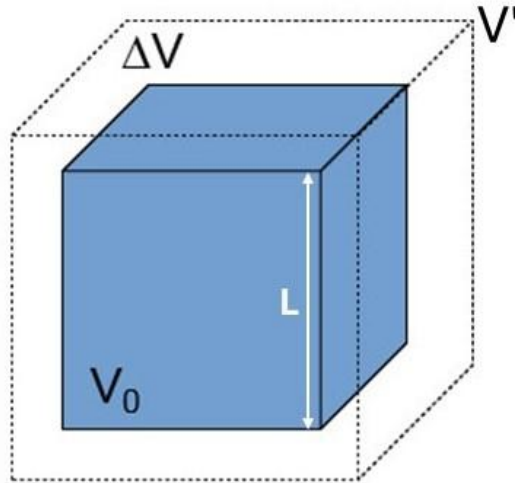


Figure 36. 8. Volume thermal expansion of a cube.

Different materials expand at different rates (fig 36.9). For example, air has a volume expansion coefficient (β) of about 0.001, while iron's is 5×10^{-4} . Materials like glass and quartz expand even less.

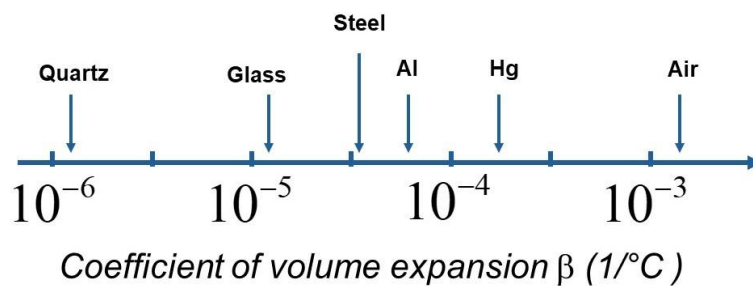


Figure 36. 9. Rates of thermal expansions of different materials.

Application of Thermal Expansion

The property of thermal expansion is utilized in thermostats as shown in fig 36.10, where a bi-metallic strip made of two different metals expands at different rates, breaking the circuit at a certain temperature (25-30 $^\circ\text{C}$) by distorting the shape to control devices like fridges or ovens.

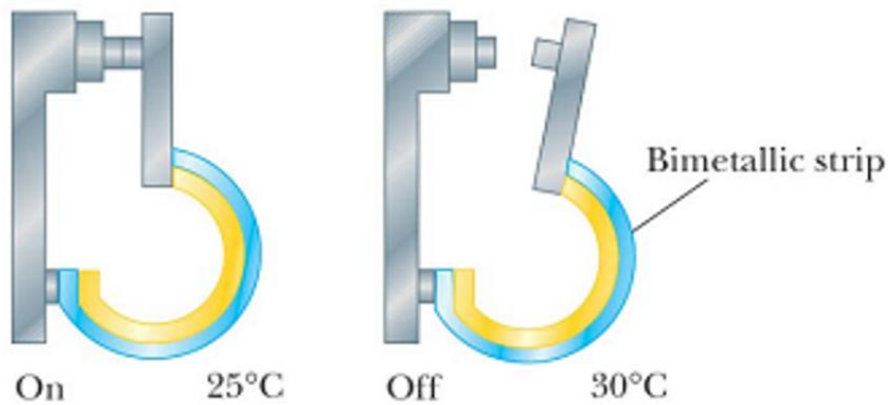


Figure 36. 10. Thermal expansion thermostat bi-metallic strip, bending at specific temperatures to break the circuit.

Anomalous Behavior of Water

When water is heated from 0 °C to 4 °C, it becomes denser, meaning its volume decreases. This anomalous behaviour is due to the complex interactions between H₂O molecules. As water temperature increases beyond 4 degrees, it behaves normally, expanding with temperature (see fig 36.11).

This property explains why, in a frozen lake, the densest water (at 4 degrees) sinks while warmer water rises. Ice, being colder than 0°C, forms on the lake's surface, while the water beneath remains at a relatively stable 4°C providing a safe environment for fish. The constant 4°C temperature below the ice ensures that fish can survive through winter. Both in summer and winter, the water at this depth maintains this temperature, providing a stable habitat. However, if the temperature fluctuates significantly, it can be detrimental to the fish, potentially leading to their demise.

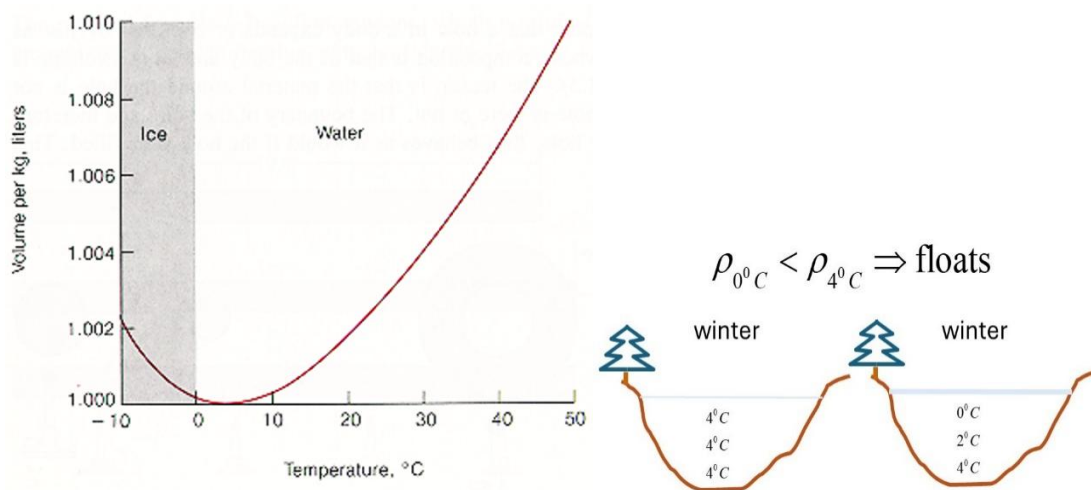


Figure 36. 11. Volume Vs Temperature graph showing anomalous Behavior of Water.

Heat Capacity

Material properties such as conductivity and resistivity are indeed affected by temperature changes. Additionally, materials can expand or contract with temperature variations. **Heat capacity** is another crucial property, representing ‘the amount of heat required to raise the temperature of 1 gram of a substance by 1 K or 1 °C’. It's mathematically defined as the heat absorbed (Q) per unit increase in temperature (ΔT), given by the formula:

$$C = \frac{Q}{\Delta T}$$

Unit of heat capacity is J/K = J/°C. Heat Q and ΔT have the same sign:

- Q is positive if ΔT is positive, heat is added to a system
- Q is negative if ΔT is negative, heat is removed from a system

Problem: A 0.5-kg block of metal with an initial temperature of 30.0 °C is dropped into a container holding 1.12 kg of water at 20.0 °C. If the final temperature of the block-water system is 20.4 °C, what is the specific heat of the metal?

Solution:

Mass of the water = $m_w = 1.12$ kg

Mass of the block = $m_b = 0.5$ kg

Specific heat of water = $c_w = 4186$ J / (kg · K)

Initial temperature of the water = $T_w = 20.0$ °C

Initial temperature of the block = $T_b = 30.0$ °C

Final equilibrium temperature = $T = 20.4$ °C

The heat lost by the metal block = $Q_{block} = m_b c_b (T_b - T)$

Heat gained by the water = $Q_{water} = m_w c_w (T - T_w)$

According to law of conservation of energy,

Heat lost = Heat gained

$$\begin{aligned} Q_{block} &= Q_{water} \\ m_b c_b (T_b - T) &= m_w c_w (T - T_w) \\ c_b &= \frac{m_w c_w (T - T_w)}{m_b (T_b - T)} \end{aligned}$$

By substituting the given values, we find that the specific heat of the metal block

$$\begin{aligned} &= \frac{(1.12 \text{ kg})[4186 \text{ J / (kg} \cdot \text{K)}](20.4^\circ\text{C} - 20.0^\circ\text{C})}{(0.500 \text{ kg})(30.0^\circ\text{C} - 20.4^\circ\text{C})} \\ &= 391 \text{ J / (kg} \cdot \text{K)} \end{aligned}$$

This method allows us to determine the specific heat capacities of various materials.

Specific Heat

Specific heat is a property that indicates a material's heat capacity, quantified in $\text{J}/(\text{kg}\cdot\text{K})$. It reflects the internal energy dynamics of the material, determining its ability to store heat. Here's a table showing the specific heats of different materials at atmospheric pressure.

Specific Heats at Atmospheric Pressure

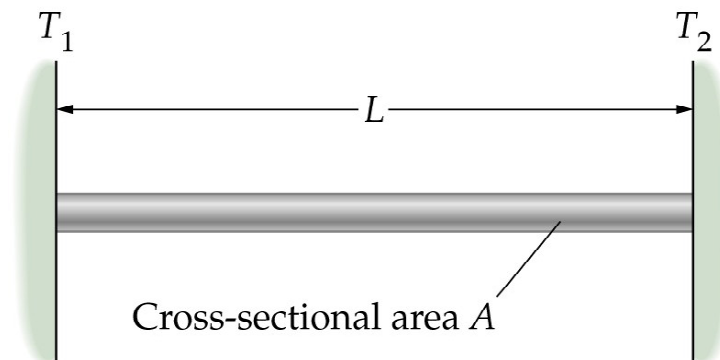
Substance	Specific heat, c [$\text{J}/(\text{kg}\cdot\text{K})$]	
Water	4186	
Ice	2090	
Steam	2010	
Beryllium	1820	
Air	1004	
Aluminum	900	
Glass	837	
Silicon	703	
Iron (steel)	448	
Copper	387	
Silver	234	
Gold	129	
Lead	128	the

like

Heat Transfer

Heat transfer occurs in three primary ways:

1. **Conduction** involves the transfer of heat through vibration and collision of atoms within a material, metal. For instance, in a metal rod heated at one end, the heat travels along the rod as atoms vibrate and collide.



2. **Convection** involves the physical movement of atoms and molecules within fluids, where heated parts move away, allowing cooler parts to take their place. For example, when a match burns, the hot air rises, cools, and then falls, creating a circulation pattern that transfers heat.
3. **Radiation** transfers heat through electromagnetic waves. A microwave oven heats water by emitting microwaves that excite the water molecules, generating heat without any physical contact.

These fundamental concepts are essential for understanding how heat interacts with different materials and environments. The principles of physics are significant due to their broad range of applications. For instance, the concepts of convection and radiation are not only fundamental to understanding heat transfer but also explain stellar phenomena. In the Sun, the core's immense temperature of 10 million degrees kelvin generates heat that rises to the surface through convection currents. Once at the surface, the heat is emitted as radiation, traveling across space to reach us. This is just one example that heat, and thermal energy play a crucial role in the universe.

Lecture 37 – THERMAL PHYSICS II

Introduction to Thermodynamics

Thermodynamics is the study of heat and how it flows. This lecture will cover the laws of thermodynamics, which are essential to physics. Despite the relatively recent discovery of atoms and molecules a century ago, the principles of thermodynamics remain unchanged and deeply rooted in our understanding of matter. Interestingly, we now interpret heat as the movement of atoms and molecules. As previously discussed, exerting work on an object increases its temperature due to the accelerated, random motion of its particles (atoms and molecules). Notably, the foundational thermodynamic laws were established without knowledge of these microscopic constituents, yet they still yield accurate predictions.

System: The specific part of the universe being studied, which can be a quantity of matter or a region in space.

Surroundings: Everything outside the system that can interact with it.

State Variables: The quantities that define the state of a thermodynamic system at equilibrium depending only on the current state of the system, not on how the system reached that state. Examples include pressure (P), volume (V), temperature (T), and internal energy (U).

Thermodynamic Variables: This is a broader term that encompasses state variables and may include other quantities used in thermodynamic analyses, such as heat (Q) and work (W). Unlike state variables, heat and work are path-dependent and not properties of the system's state.

Equation of State: The mathematical relation between these variables. Consider a gas with N atoms or molecules. The gas has a volume V, pressure P, and temperature T. These are thermodynamic variables but not independent of each other. For example, in a container with N molecules, if the volume increases, the pressure decreases (**Boyle's law:** P is inversely proportional to V) with constant temperature. Alternatively, under constant pressure, if the temperature increases, the volume increases (**Charles's law:** V is proportional to T). These relationships form the equation of state, where pressure is a function of volume and temperature, and vice versa.

$$P = f(V, T)$$

$$V = g(P, T)$$

$$T = h(P, V)$$

The equation of state for an ideal gas is given by:

$$PV = Nk_B T$$

where k_B is Boltzmann's constant, T is the absolute temperature measured in Kelvin (Kelvin = Celsius + 273.15).

Ideal Gas: An ideal gas consists of widely spaced atoms or molecules that rarely collide with each other, allowing for mostly free movement. They do collide with the container walls, but inter-particle collisions are minimal. This concept is crucial for understanding thermodynamic relationships.

Internal Energy: It is the average kinetic energy of a system's atoms and molecules (fig 37.1), which increases with temperature. It can be expressed mathematically,

$$E = \frac{3}{2} N k_B T$$

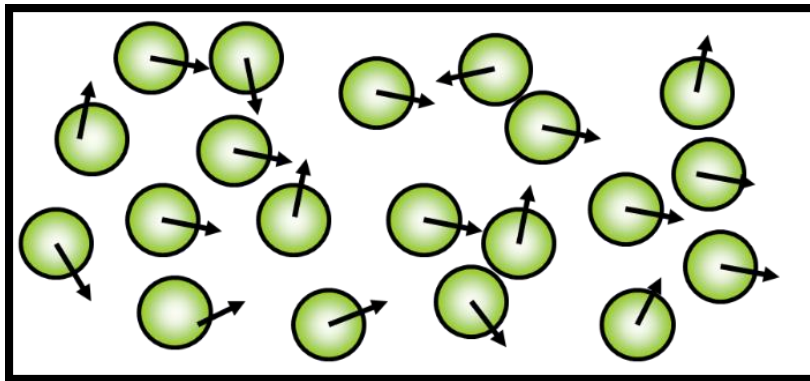


Figure 37. 1. Random motion of gas molecules contributing to the internal energy of the system.

First Law of Thermodynamics

The first law of thermodynamics—a statement of energy conservation—states that when work is done on a body and it is heated, the internal energy of the system increases. Mathematically, the increase in internal energy (ΔE) equals the heat added (ΔQ) plus the work done (ΔW), expressed as

$$\Delta E = \Delta Q + \Delta W$$

- ΔQ and ΔW are positive if heat is added to the system or work is done on a system.
- ΔQ and ΔW are negative if heat is removed from the system or work is done by a system.

This law highlights that while internal energy does not depend on the path but work and heat are path variables – their values depend on the steps leading from one state to another.

Consider a closed system with a piston, shown in fig 37.2: it's sealed except for a movable piston, containing an ideal gas. When the piston moves, it alters the system's volume without allowing gas to escape, demonstrating the relationship between work, heat, and internal energy in a controlled environment.

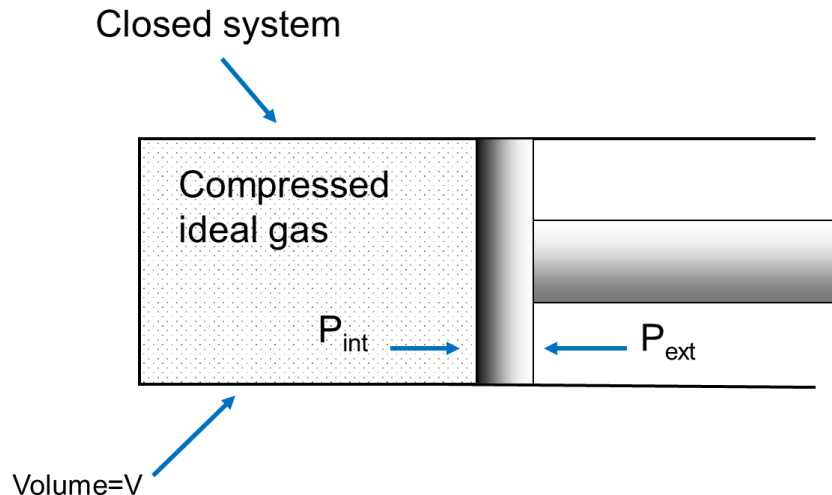


Figure 37. 2. Closed system with a movable piston, changing the volume of an ideal.

To determine the work done by a gas during expansion, If the external pressure is P_{external} , the work done by the gas on the piston, dW , is given by:

$$dw = - P_{\text{external}}. dV \quad \therefore w = F.d = (PA)d = P(Ad) = PV$$

where dV is the change in volume. The negative sign indicates that the work is done by the gas on the surroundings.

- When internal and external pressures are equal, the piston is in equilibrium.
- If $P_{\text{internal}} > P_{\text{external}}$, the gas expands, and
- If $P_{\text{internal}} < P_{\text{external}}$, it compresses.

For total work, we integrate $P_{\text{external}}.dV$ from the initial to the final volume.

$$w = - \int_{V_1}^{V_2} P_{\text{ext}} dV$$

Case I: $P_{\text{external}} = 0$ (a vacuum)

No work is done by the gas on the piston as it expands, since work is a product of force and displacement, and with zero external pressure, there's no force opposing the expansion. Consequently, if no heat is exchanged ($\Delta Q = 0$), the system's internal energy remains unchanged (so, $\Delta E = 0$). This is consistent with the first law of thermodynamics, which relates changes in internal energy to heat and work.

Case II: $P_{\text{external}} = \text{constant}$

With constant external pressure P_{external} , the work done by the gas during expansion is given by

$$W = - P_{\text{external}}. \Delta V$$

where ΔV is the change in volume (i.e., $V_{\text{final}} - V_{\text{initial}}$). The negative sign signifies that the gas is doing work on the surroundings (gas expands against the external pressure).

Case II: $P_{\text{external}} \approx P_{\text{internal}}$

Assume P_{external} and P_{internal} to be the same for calculation purposes. Using the ideal gas law ($PV = NkT$), the work done during an isothermal process (constant temperature) is calculated by integrating the pressure with respect to volume

$$P_{\text{ext}} = \frac{NkT}{V}$$

$$w = -\int_{V_1}^{V_2} \frac{NkT}{V} dV = -NkT \int_{V_1}^{V_2} \frac{dV}{V} = -NkT (\ln V_2 - \ln V_1) = -NkT \ln \frac{V_2}{V_1}$$

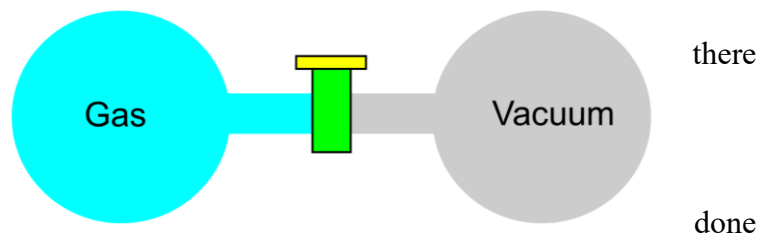
where V_1 and V_2 are the initial and final volumes, respectively.

These cases illustrate that the work done depends on the path taken to reach a particular state of the gas:

1. Free expansion: No work is done ($w = 0$).
2. Constant external pressure: Work is $-P_{\text{external}} \Delta V$.
3. Isothermal Process with $P_{\text{external}} \approx P_{\text{internal}}$: Maximum work is $w = -NkT \ln \frac{V_2}{V_1}$.

Thus, the amount of work is path-dependent, and asking for the work done by the gas without specifying the process is incorrect. The internal energy, however, is a state function and remains the same regardless of the path taken.

When a valve separating a gas from a vacuum is opened, the gas expands without doing work, as there is no external pressure to work against. Since it is an isolated system undergoing free expansion, no external work is done ($\Delta W = 0$) and no heat is



exchanged ($\Delta Q = 0$), so the internal energy remains constant ($\Delta E = \Delta Q + \Delta W = 0$). Initially, if the energy is $3/2NkT$, it stays the same after expansion.

PV diagrams

PV diagrams are helpful in visualizing this process, where the area under the curve represents the work done. For a constant pressure process (see fig. 37.3), the work done is the product of the pressure and the change in volume ($W = P \Delta V$).

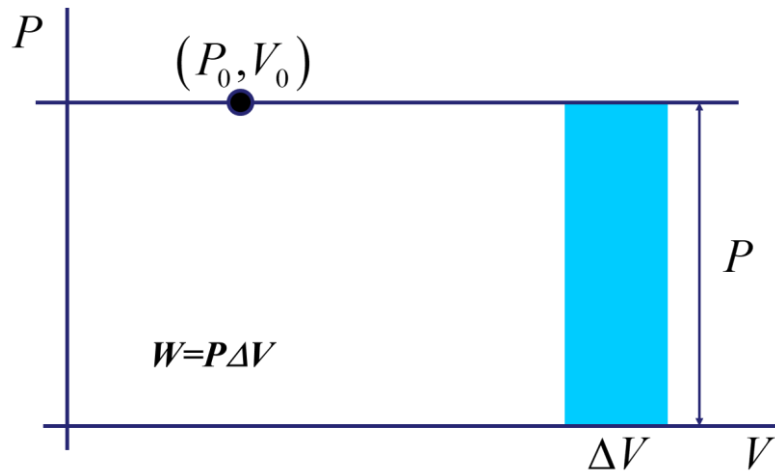


Figure 37. 3. PV diagram for constant pressure process.

Path-Dependence of Work

The work done can be different depending on the path taken. For example, if we move from an initial state P_1V_1 to a final state P_2V_2 via two different paths, the work done will vary. In an isothermal expansion, where temperature remains constant, the work done is determined as follows:

Path A: Firstly, the volume increases from V_1 to V_2 at constant pressure P_1 . Then, the pressure decreases to P_2 at constant volume V_2 , shown in fig 37.4 (path A). Since the temperature remains constant, there's no change in internal energy. Thus, the work done is pressure (starting pressure) times the change in volume.

$$W = P_1(V_2 - V_1)$$

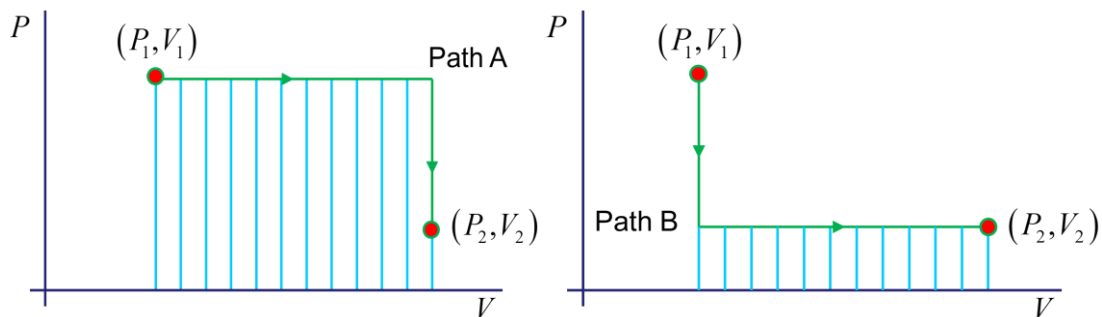


Figure 37. 4. PV diagrams showing path dependence of work

Path B: Exploring alternative paths (Path B in fig 37.4) in thermodynamics demonstrates the path-dependence of work done on or by a gas. For instance, on path B, First, decrease the pressure from P_1 to P_2 at constant volume V_1 . Then, increase the volume from V_1 to V_2 at constant pressure P_2 , the work done is

$$W = P_2(V_2 - V_1)$$

This contrasts with the previous path where the work was $W = P_1(V_2 - V_1)$. This shows that work done is indeed path-dependent, and the signs of work done on or by the gas are opposite.

Path C: For the third path (fig 37.5), which adheres to the ideal gas law $PV = NkT$, the work done is calculated as:

For an isothermal process,

$$PV = Nk_B T$$

$$dW = PdV$$

$$= Nk_B \frac{T}{V} dV$$

$$W = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{Nk_B T}{V} dV$$

$$W = Nk_B T \ln \frac{V_2}{V_1}$$

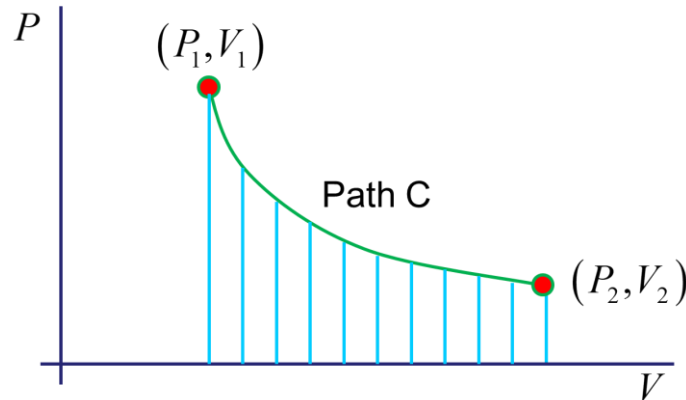


Figure 37. 5. Path C showing variable change in pressure and volume.

Specific Heat

Now, let's calculate the heat capacity of a gas, it's a measure of the amount of heat required to change the temperature of a system. There are two ways to define specific heat: constant volume (C_V) and constant pressure (C_P).

Specific Heat at Constant Volume

It (C_V) is the amount of heat required to raise the temperature of a unit mass of a substance by one degree Celsius (or one Kelvin) while keeping its volume constant. For an ideal gas at constant volume (C_V),

$$C_V = \frac{Q_V}{\Delta T}$$

$$Q_V = C_V \Delta T$$

As the volume remain constant, $dV = 0$, Hence,

$$W = 0$$

$$Q_V = \Delta E + W = \Delta E$$

$$\Delta E = C_V \Delta T$$

As $\Delta T \rightarrow 0$

$$dE = C_V dT$$

$$C_V = \frac{dE}{dT}$$

$$\therefore E = \frac{3}{2} NkT$$

$$C_V = \frac{d}{dT} \left(\frac{3}{2} NkT \right) = \frac{3}{2} Nk$$

This is a fundamental result in thermodynamics, reflecting the intrinsic energy changes within a gas due to temperature variations.

Specific Heat at Constant Pressure

C_p represents the heat capacity of a gas when it is allowed to expand under constant pressure. This means the system can do work on its surroundings as it heats up. C_p is defined as the heat absorbed Q_p divided by the change in temperature ΔT .

$$C_p = \frac{Q_p}{\Delta T}$$

$$Q_p = C_p \Delta T$$

Let's calculate the specific heat at constant pressure C_p using the first law of thermodynamics,

$$Q_p = \Delta E + W = \Delta E + P\Delta V$$

$$C_p \Delta T = \Delta E + P\Delta V$$

As $\Delta T \rightarrow 0$

$$C_p dT = dE + PdV$$

Expressing dE in terms of C_v

$$C_p dT = C_v dT + PdV$$

As $PV = Nk_B T$

Differentiating w.r.t T

$$PdV + VdP = Nk_B dT \quad \text{At const } P \text{ (} dP=0 \text{)}$$

$$PdV = Nk_B dT$$

Substitute into the first law

$$C_p dT = C_v dT + Nk_B dT$$

$$C_p = C_v + Nk_B$$

$$E = \frac{3}{2} Nk_B T$$

$$C_v = \frac{dE}{dT} = \frac{3}{2} Nk_B$$

$$C_p = \frac{3}{2} Nk_B + Nk_B$$

$$C_p = \frac{5}{2} Nk_B$$

$$C_p - C_v = \frac{5}{2} Nk_B - \frac{3}{2} Nk_B = Nk_B$$

This shows that C_p is indeed greater than C_v for an ideal gas, by an amount equal to Nk_B , which is the number of particles multiplied by Boltzmann's constant. The difference arises because at constant pressure, the gas does work on the surroundings as it expands, requiring additional energy input to raise the temperature by the same amount compared to a constant volume scenario.

The behavior of an ideal gas can become complex when considering real molecules that have rotational and vibrational modes. For diatomic or polyatomic molecules, such as water (H₂O), carbon dioxide (CO₂), or hydrogen (H₂), their rotational motions contribute to their energy and, consequently, to their specific heat capacities displayed in fig 37.6.

The specific heat at constant volume C_V for a monatomic gas is $\frac{3}{2} nK$, accounting for translational motion along the x, y, and z axes. For diatomic or polyatomic gases, which can rotate, additional degrees of freedom contribute to the specific heat. For example, a diatomic molecule such as CO₂ has two additional rotational degrees of freedom.

$$K = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{1}{2}I_x\omega_x^2 + \frac{1}{2}I_y\omega_y^2$$

$$E = 5 \times \left(\frac{1}{2} Nk_B T \right) = \frac{5}{2} Nk_B T$$

$$C_V = \frac{5}{2} Nk_B$$

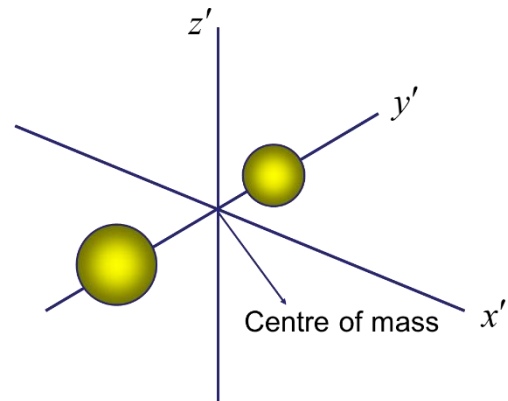


Figure 37. 6. Diatomic molecule having translational and rotational degrees of freedom.

Therefore, molecules that can rotate have a higher specific heat.

Adiabatic Process

So far, we've discussed gas expansions under isothermal (constant temperature) and constant pressure conditions. The third type of expansion is adiabatic, where no heat is exchanged with the surroundings ($dQ = 0$). An isolated system executing adiabatic expansion is shown in fig 37.7.

$$dQ = dE + dW = 0 \quad \text{or} \quad dE = -dW$$

This means any work done by the gas reduces its internal energy. For an adiabatic process, the relationship between pressure and volume is given by

$$C_V dT + PdV = 0$$

$$C_V dT + Nk_B T \frac{dV}{V} = 0 \quad \therefore P = \frac{Nk_B T}{V}$$

$$\frac{dT}{T} + \frac{Nk_B}{C_V} \frac{dV}{V} = 0 \dots \dots \dots (1)$$

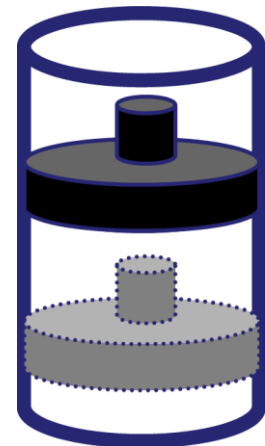


Figure 37. 7. An isolated system in which no heat enters or leave.

$$\text{As } C_p - C_v = Nk_B$$

$$\frac{Nk_B}{C_v} = \frac{C_p - C_v}{C_v} = \frac{C_p}{C_v} - 1$$

$$\text{Define } \frac{C_p}{C_v} \equiv \gamma$$

$$\frac{Nk_B}{C_v} = \gamma - 1, \text{ put in eq (1) and integrate}$$

$$\frac{dT}{T} + (\gamma - 1) \frac{dV}{V} = 0$$

$$\ln T + (\gamma - 1) \ln V = \text{Constant}$$

$$\ln(TV^{\gamma-1}) = \text{Constant}$$

$$TV^{\gamma-1} = \text{Constant}$$

$$\frac{PV}{Nk_B} V^{\gamma-1} = \text{Constant}$$

$$\therefore PV = Nk_B T \rightarrow T = \frac{PV}{Nk_B}$$

$$PV^\gamma = \text{Constant}$$

For an ideal gas undergoing an adiabatic expansion from an initial state P_1V_1 to a final state P_2V_2 , the work done is:

$$dQ = dE + dW = 0$$

$$C_v dT + dW = 0 \Rightarrow dW = -C_v dT$$

$$W_{\text{adiabatic}} = dW = -\int C_v dT = -C_v \Delta T$$

$$W_{\text{adiabatic}} = -C_v \Delta T = C_v (T_1 - T_2)$$

$$W_{\text{adiabatic}} = C_v \left(\frac{PV_1}{Nk_B} - \frac{PV_2}{Nk_B} \right) = \frac{PV_1 - PV_2}{\gamma - 1}$$

This is a direct application of the first law of thermodynamics, which is fundamentally the law of conservation of energy. It states that the change in internal energy of a system is equal to the heat added to the system plus the work done on the system.

Second Law of Thermodynamics

The second law of thermodynamics asserts that *there can be no process whose only final result is to transfer thermal energy from a cooler object to a hotter object*. This principle implies that it's impossible to extract work from a system solely by transferring heat from a colder to a hotter body.

Applications of Second Law of Thermodynamics

This principle has numerous applications, particularly in engines. Fig 37.8 shows a **steam engine** in which water is heated in a boiler until it becomes steam. This steam then expands and pushes a piston, which in turn drives a wheel, converting the steam's thermal energy into mechanical work. After doing work, the steam is condensed back into water, which can be heated again, creating a cycle.

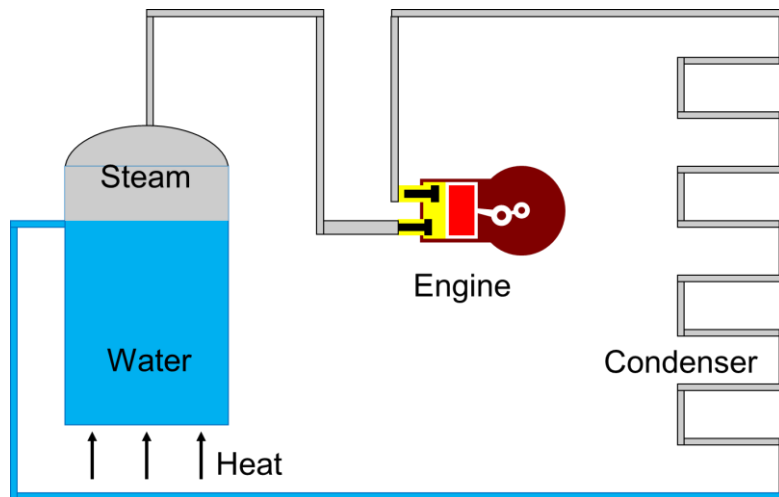


Figure 37. 8. Steam engine

Similarly, **internal combustion engines** (fig 37.9) in motorcycles and cars function on the same basic principle. Gas enters the engine, is compressed, and ignited with a spark (ignition stroke). The piston then moves out (power stroke) and finally, the exhaust stroke occurs.

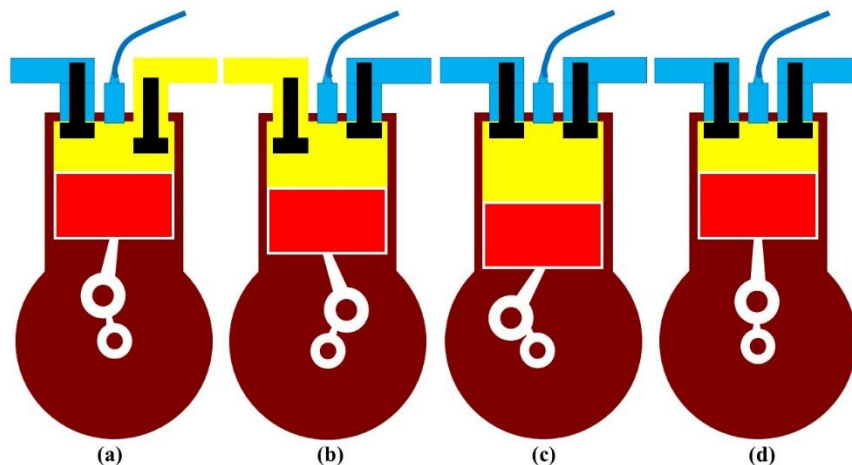


Figure 37. 9. Internal combustion engine in vehicles demonstrating (a) compression stroke (b) ignition stroke (c) power stroke (d) exhaust stroke

These engines operate between a high temperature (over 100°C for steam, higher for diesel or petrol during ignition) and a low temperature ($30\text{--}40^{\circ}\text{C}$ for exhaust release). Heat, Q_h , is taken from the high temperature reservoir T_h , allowing the engine to perform work. Not all heat converts to work; some is rejected as excess heat, Q_c , to the environment, called the cold reservoir at temperature T_c . Fig 37.10 demonstrates the schematics of heat engine.

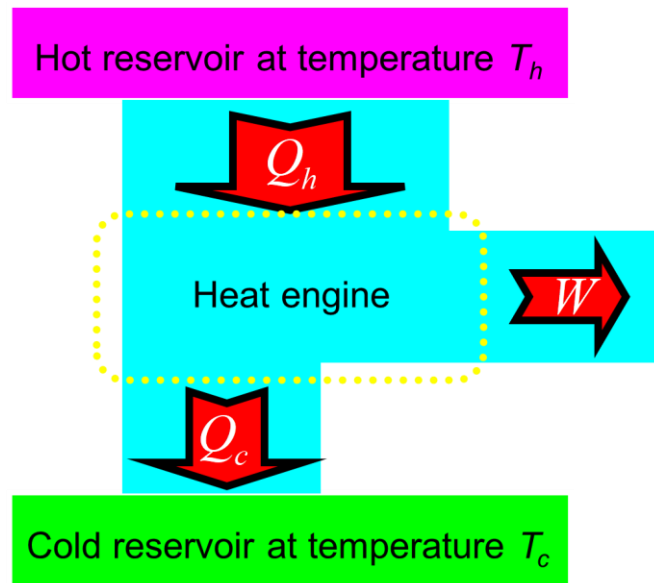


Figure 37. 10. Working mechanism of heat engine.

This follows the first law of thermodynamics:

$$Q = \Delta E + W$$

$$W = Q_{in,h} - Q_{out,c}$$

Efficiency (ϵ) measures the work obtained from heat input: mathematically defined as the work output (W) divided by the heat input (Q_{in}).

$$\epsilon = \frac{W}{Q_{in,h}} = \frac{Q_{in,h} - Q_{out,c}}{Q_{in,h}} = 1 - \frac{Q_{out,c}}{Q_{in,h}}$$

Since $Q_{out,c}$ is proportional to the lower temperature (T_c) and $Q_{in,h}$ to the higher temperature (T_h), efficiency is:

$$\epsilon = 1 - \frac{T_c}{T_h}$$

This ensures efficiency is always less than 1, making 100% efficiency impossible.

Refrigerator

A refrigerator operates in reverse compared to a heat engine. While a heat engine converts heat from a high temperature to work and expels some waste heat, a refrigerator removes heat from a low temperature area (like a water bottle inside it) and rejects it at a higher temperature. This process requires external energy, usually electricity.

In a refrigerator, work (W) is input, and heat (Q_c) is removed from inside and rejected as Q_h outside (fig 37.11).

The efficiency of a refrigerator is measured by its coefficient of performance, which is the amount of heat extracted per unit of work input. The second law of thermodynamics states that

a refrigerator cannot operate without external work; it is impossible for a refrigerator to transfer heat from a cold to hot body without additional energy input. Thus, its efficiency will always be less than 1.

Entropy

Entropy, a concept from the second law of thermodynamics, measures disorder in a system. Mathematically, if you add a small amount of heat (dQ) to a system at temperature (T), the increase in entropy (dS) is:

$$ds = \frac{dQ}{T}$$

Entropy of an ideal gas

For an ideal gas, the state equation is $PV = nKT$. When heat is added, the entropy change can be derived from the first law of thermodynamics:

$$dQ = dE + dW = dE + PdV$$

Substituting P from the ideal gas law,

$$dQ = C_v dT + Nk_B T \frac{dV}{V}$$

$$dS = \frac{dQ}{T} = C_v \frac{dT}{T} + Nk_B \frac{dV}{V}$$

To find the total change in entropy ΔS when the temperature changes from T_1 to T_2 , we integrate:

$$\Delta S = \int \frac{dQ}{T} = C_v \ln \frac{T_2}{T_1} + Nk_B \ln \frac{V_2}{V_1}$$

Physically, as energy is added to a system and temperature increases, molecules move faster (more randomly), increasing disorder and entropy. More order means less entropy, and more disorder means higher entropy.

The second law of thermodynamics states that entropy always increases in an isolated system. Over time, any isolated system, whether gas, liquid, or solid, will naturally become more disordered. For example, imagine gathering all the atoms and molecules scattered in a room and confining them in small box. This action decreases entropy because the system becomes more ordered. However, if you open the box and allow the atoms and molecules to disperse throughout the room, entropy increases due to the rise in disorder. This concept also links entropy to the flow of time. We never observe atoms and molecules spontaneously gathering

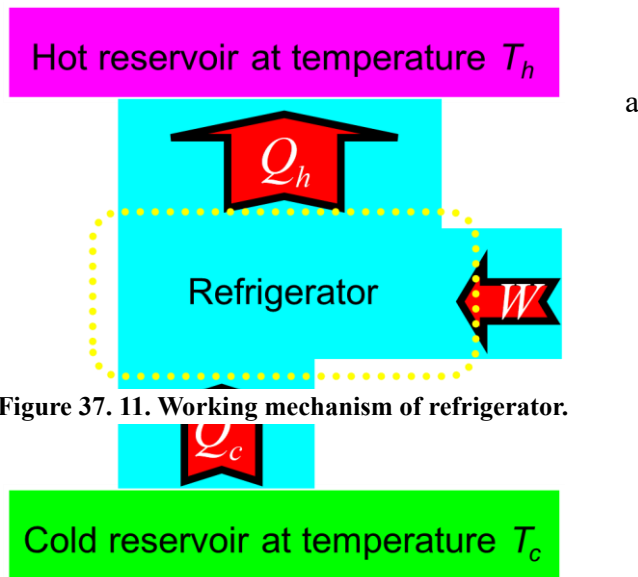


Figure 37. 11. Working mechanism of refrigerator.

in one place, which would decrease entropy. Thus, the increase in entropy is a measure of time's forward progression.

Lecture 38 – THERMAL PHYSICS III

This is the final lecture on heat. Let's review key concepts from the last two lectures. Imagine a hot piece of iron and a cold piece of iron. The atoms in the hot piece are vibrating rapidly, exhibiting more random thermal energy than the cold piece. When combined, heat transfers from the hot piece to the cold piece, not the other way around. This illustrates the second law of thermodynamics, which states that work can only occur when one body is hot and the other is cold, with heat always moving from hot to cold.

Entropy

Entropy is a measure of disorder. When energy transfers from a hot body to a cold one, the system's entropy increases. The change in entropy, dS , is defined as the heat added, dQ , divided by the temperature T ,

$$ds = \frac{dQ}{T}$$

In the previous lecture, we learned to calculate the increase in entropy for an ideal gas. To find the total change in entropy, we integrate from the initial temperature (T_1) to the final temperature (T_2):

$$\Delta S = \int_{T_1}^{T_2} \frac{dQ}{T} = C_v \ln \frac{T_2}{T_1} + Nk_B \ln \frac{V_2}{V_1} \dots\dots\dots(1)$$

Conditions for Entropy Increase:

If the initial and final temperatures or volumes are equal ($T_1 = T_2$ or $V_1 = V_2$), the entropy change is zero because the logarithm of 1 is zero. Thus, entropy increases only when $T_2 > T_1$ or $V_2 > V_1$, or both. The above equation shows that entropy increases with higher temperatures or volumes.

Examples of Increase in Entropy

1. **Matchsticks:** Entropy means disorder, to illustrate this, consider matchsticks in the box. Initially, they are neatly arranged 38.1 (a)), representing low entropy. When scattered, they become disordered, shows increasing entropy (fig 38.1 (b)).
2. **Ink in Water:** Another example is dropping ink into water. Initially, the ink is concentrated (low entropy). As it spreads, the disorder increases, raising entropy.
3. **Molecules in a Box:** In the figure 38.2, molecules are confined to one side of a box. They move rapidly, bouncing off walls. When the divider is removed, they spread out, increasing entropy without adding energy showing how entropy naturally increases.

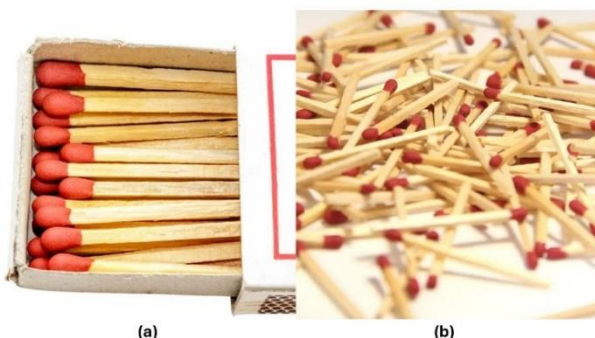


Figure 38. 1. (a) Organized matchsticks (b) scattered matchsticks representing increasing disorder.

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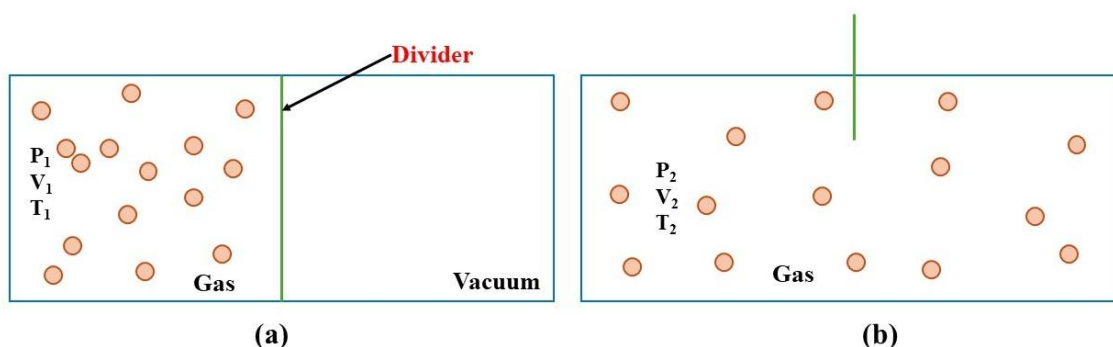


Figure 38. 2. (a) Molecules confined to one side of a box, (b) molecules spread out, illustrating increasing entropy.

The second law of thermodynamics states that in an isolated system, entropy can increase or remain constant but never decrease. For example, when the partition in a box of confined molecules is removed, atoms and molecules spread out, increasing entropy without adding energy. This is reflected as $\ln \frac{V_2}{V_1}$ in the eq (1) where V_2 is greater than V_1 , leading to increased entropy.

Entropy and States of Matter

Now, let's apply entropy in different states of matter:

1. **Solids:** Solids have the least entropy because their atoms are fixed in place, only vibrating slightly around their positions. This high level of order means low entropy.
2. **Liquids:** Liquids have higher entropy than solids because their atoms or molecules can move more freely. The increased movement and ability to flow result in more disorder.
3. **Gases:** Gases have the highest entropy as their atoms or molecules move freely and rapidly in all directions. The high degree of movement and spread leads to the greatest disorder.

A better understanding of these states helps us see how entropy varies with the degree of molecular movement and disorder.

Heat of Transformation

Latent Heat of Fusion

Let's consider a piece of ice. When heat is added to it, its entropy increases as the molecules become more active. If we continue to add heat, the ice will start to melt because the kinetic energy overcomes the intermolecular forces holding the molecules together. This process requires energy known as the **latent heat of fusion (L_F)** or **latent heat of melting**, which is the **heat absorbed per unit mass of the material to change its state from solid to liquid**.

As the ice melts, the temperature remains constant at 0°C (273.15 K) until all the ice has melted. The heat required to melt m grams of ice is:

$$Q = mL_F$$

Latent Heat of Vaporization

Similarly, heating water increases its temperature until it reaches 100°C, where it begins to boil and undergoes a phase transition from liquid to gas. The heat required for this transition is called the **latent heat of vaporization (L_V)**, which is **defined as the heat needed to vaporize a unit mass of the material**.

$$Q = mL_V$$

How much heat must be added to change ice to water to steam?

1. Heat required to rise the temperature of 1 gram of ice by 1°C = 0.5 cal/g

Heat required to rise the temperature of m grams of ice is: $\Delta Q = c_i m \Delta T$

2. Heat required melt ice = 80 cal/g ∴ calculated as $\Delta Q = mL_F$
3. As we heat the water, its temperature rises. Heat required to rise the temperature of 1 gram of water by 1°C = 1 cal/g = 4.18 J ∴ $\Delta Q = c_w m \Delta T$
4. Heat required vaporize water = 540 cal/g ∴ $\Delta Q = mL_V$
5. Heating steam further requires 0.5 cal/ (g °C). ∴ $\Delta Q = c_s m \Delta T$

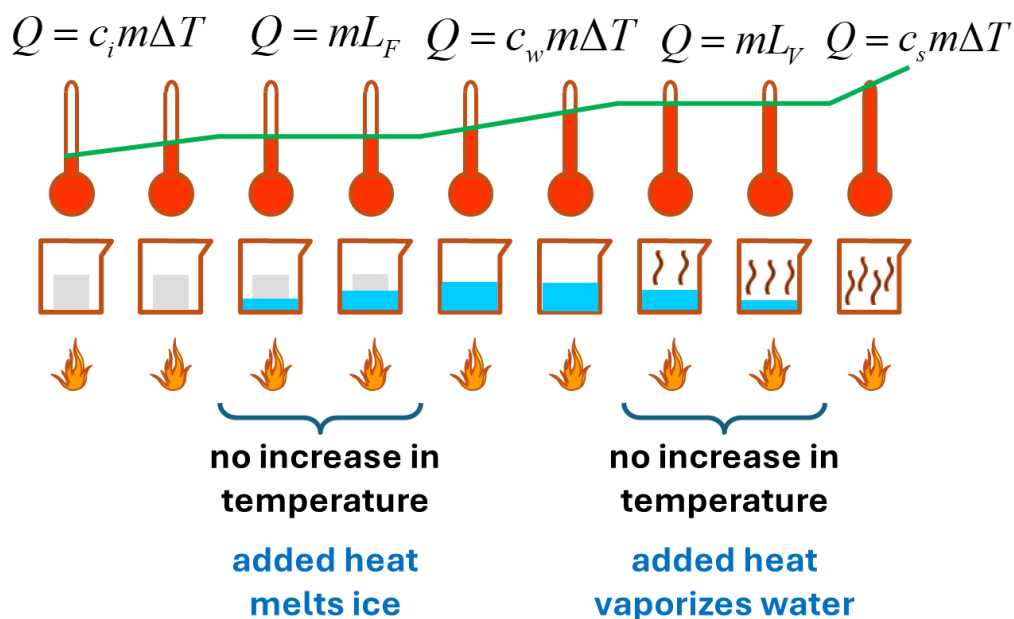


Figure 38. 3. Heat Stages: Ice to steam—temperature rise, melting, boiling, and continued heating.

The process of heating ice involves increasing its temperature to 0 °C, where it melts. The temperature remains constant until all the ice melts, then rises to 100 °C, where water boils and turns to steam. Continued heating increases the steam's temperature (see fig 38.3).

A graph represented in fig 38.4 shows the specific heat of ice, the phase transition at 0 °C, the heating of water, the phase transition at 100 °C, and the heating of steam. If we heat steam excessively, the water molecules dissociate into hydrogen and oxygen atoms. Further heating causes ionization, where electrons separate from protons, requiring several thousand degrees Celsius.

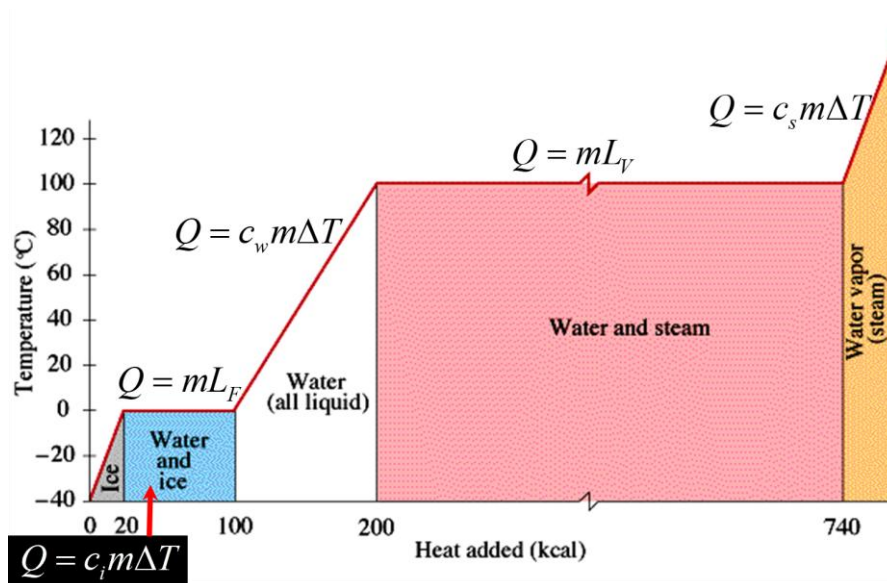


Figure 38. 4. Heating curve illustrating the transitions of ice at -40°C into hot steam in five steps.

Other Types of Phase Transitions

1. There are various types of phase transitions in different materials. Besides gas, liquid, and solid transitions, consider a permanent magnet. When heated beyond a certain point, it loses its magnetism because the magnetic domains become disordered, increasing entropy. This is a **phase transition between magnetized and un-magnetized states**.

2. **Evaporation** is another phase transition. In a liquid, molecules move to temperature. As temperature rises, molecular motion increases. Fast-moving molecules can escape the liquid, turning into vapor as displayed 38.5. For example, water in a glass decreases over time as some molecules evaporate. The temperature of the liquid



Figure 38. 5. Water molecules with high enough KE due to random intermolecular collisions can break free from the surface and escape.

- reflects its average random kinetic energy. When fast-moving molecules leave, the average kinetic energy decreases, cooling the remaining liquid. This principle explains why water stored in porous pots stays cool; evaporation lowers the average kinetic energy, reducing the temperature. Evaporation causes cooling, explaining why we use fans in the summer. Fans help evaporate sweat, which cools the body by removing fast-moving molecules, thus lowering temperature and average kinetic energy.
3. Now, let's discuss **boiling**. Water boils at 100°C at standard atmospheric pressure. However, this boiling point changes with pressure. In a pressure cooker, water can boil at higher temperatures, like 105°C or 110°C . Conversely, at higher altitudes, water boils at lower temperatures, such as 95°C or 98°C , due to reduced atmospheric pressure.

For a molecule to escape from a liquid, it must possess enough energy to overcome the attractive forces of other molecules. This principle underlies evaporation and boiling, showcasing how molecular energy and external conditions affect phase transitions.

Statistical Thermodynamics

Statistical thermodynamics is the *microscopic* or *molecular* approach to understand and predict the macroscopic phenomena and the calculation of macroscopic properties from the properties of individual molecules. For example, knowing the molecules in a system (like oxygen, carbon, hydrogen) allows us to determine properties such as pressure, entropy, boiling point, and melting point.

In contrast, **classical thermodynamics** doesn't consider individual molecules but focuses on macroscopic properties, like the energy required to change the temperature or state of a substance.

Statistical thermodynamics emerged about 100 years after classical thermodynamics and offers a deeper understanding of heat. As substances are heated, their atoms and molecules exhibit more random motion and vibration, which we interpret as heat. Understanding randomness is crucial, as it applies to many systems.

To better understand this concept, think about randomness: variations are always present in any system like cars moving at different speeds on a road.

To create a histogram (fig 38.6) of car speeds, we divide the speed range into intervals (bins) and count the number of cars in each bin. For example, if there are 11 cars with speeds between 1 and 2 miles per hour. The height of each bar in the histogram corresponds to the number of cars within each speed interval.

Average speed can be calculated as:

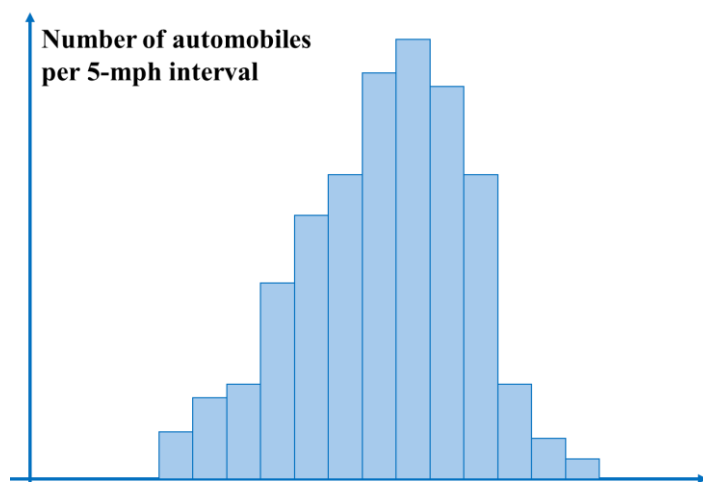


Figure 38. 6. Histogram of Car Speeds: bar height represents the number of cars in each speed interval.

$$N = \sum_i n(v_i)$$

$$\bar{v} = \frac{\text{Sum of all observations}}{\text{Total number of all observations}} = \frac{\sum_i v_i n(v_i)}{\sum_i n(v_i)} \quad \text{Where } n(v_i) \text{ is the number of cars moving with velocity } v_i.$$

with velocity v_i .

The **probability** $P(v_i)$ of finding a car with speed v_i represents the likelihood that a randomly selected car from a given set will have a velocity exactly equal to v_i . We can express it as:

$$P(v_i) \equiv \frac{n(v_i)}{\sum_i n(v_i)} = \frac{n(v_i)}{N}$$

Where N is the total number of cars. Using this probability, the average speed can be expressed as:

$$\bar{v} = \sum_i v_i P(v_i)$$

This approach allows us to statistically analyze and understand the distribution of car speeds based on empirical data gathered from the histogram. To calculate averages, we consider that some cars move faster and some slower than the average speed. This concept applies to gases as well, where different molecules move at different velocities, but there is an average speed.

Mean Free Path

When examining gas molecules, they often pass by each other but occasionally collide. **The average distance between collisions is called the mean free path (λ).**

To calculate this, we need the average speed of the molecules v and the number of molecules per unit volume n . Two molecules collide when their distance is equal to the sum of their radii ($d = r_1 + r_2$), shown in fig 38.7 (a).

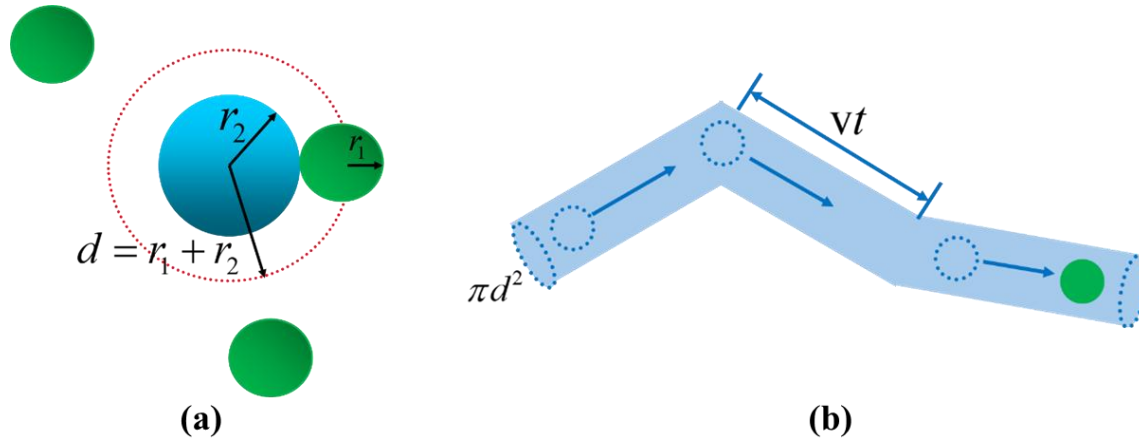


Figure 38. 7. (a) Molecules collide when the $d = r_1 + r_2$ (b) In a tube, mean free path λ is calculated using tube dimensions and mean free time t , where area = πd^2 and length = vt .

Imagine a tube, depicted in fig 38.7 (b), where molecules are moving with the average speed v . The area and length of the tube covered in time t are πd^2 and vt . t is the average time it takes for a molecule to collide with another after a collision known as mean free time. We can calculate the mean free path λ using the formula:

$$\begin{aligned} \lambda(n_v \pi d^2 vt) &\approx vt & \therefore d &= R_1 + R_2 \\ \therefore \lambda &\approx \frac{1}{n_v \pi d^2} \\ \lambda &= \frac{1}{\sqrt{2} n_v \pi d^2} \quad (\text{more accurate}) \end{aligned}$$

Where n_v is the number of molecules per unit volume ($V = \pi d^2 vt$) or number density, d is the effective diameter used to determine collision.

In denser gases, the mean free path is shorter because collisions are more frequent. Conversely, in less dense gases, the mean free path is longer, meaning collisions are less frequent and the distance between them is greater.

In very dilute gases, where collisions are rare, we refer to them as ideal gases. Before discussing ideal gases, let's define a mole. A **mole** is the amount of gas (or any other substance) containing 6.02×10^{23} molecules, known as Avogadro's number (N_A). Different gases have different types of atoms with varying masses. The **molar mass** (M) is the mass of one mole of a gas, measured in grams. The **number of moles** (n) is the number of atoms divided by Avogadro's number ($n = N / N_A$).

The ideal gas law can be written as

$$PV = Nk_B T$$

where P is the pressure, V is the volume, n is the number of moles, k is the Boltzmann constant, and T is the temperature. Alternatively, using the universal gas constant R , it is

$$PV = nRT$$

where $R = 8.315 \text{ J/(mol K)}$ and n is the number of moles in a gas. It's crucial to use kelvin for temperature because pressure varies linearly with it as illustrated in fig 38.8, converging zero at absolute zero.

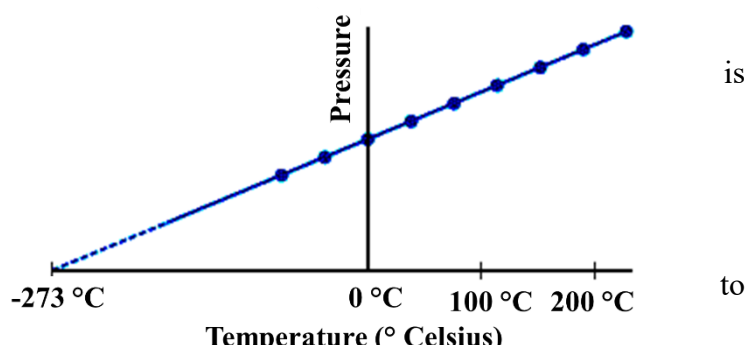


Figure 38. 8. Graph illustrating linear relation between temp in kelvin and pressure, converging to zero at absolute zero.

In an ideal gas, molecules move independently without collisions, and their velocities vary. The speed distribution graph (fig 38.9 (a)) shows this randomness: some move fast, others slow, with an average speed. The distribution of molecular speeds changes with temperature: in a cold gas, molecules move slower, while in a hot gas, they move faster. This results in different speed distributions for different temperatures as shown in fig 38.9 (b). For example, at $T = 80 \text{ K}$, the distribution peak is lower compared to $T = 300 \text{ K}$, where molecules move faster and collide more frequently, pushing against the container walls.

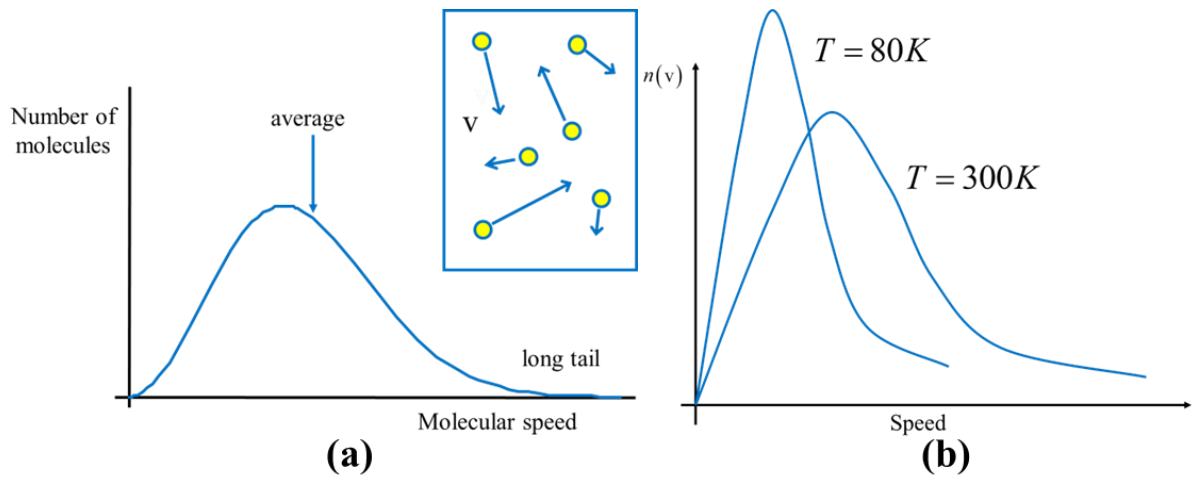


Figure 38. 9. (a) Speed distribution for the molecules of a gas. (b) A comparison of the speed distribution at two different temperatures, with lower average speeds at lower temperatures, but equal areas due to the same total number of molecules.

Pressure in a Gas

Pressure in a gas is created by molecules colliding with the walls of a container. When a molecule hits the wall, its momentum changes direction, creating a force according to Newton's law (force equals the rate of change of momentum).

To calculate this pressure, consider the number of molecules N per unit volume V and their average speed v_x . The distance a molecule travels in time Δt is $(v_x \Delta t)$. The volume it covers is $V_x \Delta t A$, where A is the area of the wall. Only half the molecules move in the positive x -direction, so the number of collisions is:

$$\text{Molecules that hit the wall} = \frac{1}{2} \frac{N}{V} v_x \Delta t A$$

$$\text{The change in momentum for one molecule} = mv_x - (-mv_x) = 2mv_x$$

where m is the mass. The total change in momentum is:

$$\begin{aligned} \Delta p &= (2mv_x) \times \left(\frac{1}{2} \frac{N}{V} v_x \Delta t A \right) \\ &= \frac{N}{V} mv_x^2 A \Delta t \\ P &= \frac{F}{A} = \frac{1}{A} \frac{\Delta p}{\Delta t} = \frac{N}{V} mv_x^2 \\ PV &= Nm v_x^2 = 2N \left(\frac{1}{2} mv_x^2 \right)_{av} \end{aligned}$$

This formula shows how pressure relates to the average kinetic energy of gas molecules (through $\left(\frac{1}{2} mv_x^2 \right)$ and their number density $\left(\frac{N}{V} \right)$. It helps us understand how molecular collisions create pressure in a gas.

The Molecular Interpretation of Temperature

Considering the kinetic theory of gases, the average kinetic energy per molecule can be related to the temperature as:

$$PV = Nk_B T = 2N \left(\frac{1}{2} m v_x^2 \right)_{av}$$

$$\left(\frac{1}{2} m v_x^2 \right)_{av} = \frac{1}{2} k_B T$$

Considering all directions (x, y, z), the average velocity squared is:

$$(v_x^2)_{av} = (v_y^2)_{av} = (v_z^2)_{av}$$

$$(v^2)_{av} = (v_x^2)_{av} + (v_y^2)_{av} + (v_z^2)_{av} = 3(v_x^2)_{av}$$

The average kinetic energy of a molecule is:

$$K_{av} = \left(\frac{1}{2} m v^2 \right)_{av} = \frac{3}{2} k_B T$$

Thus, the total kinetic energy of N molecules is:

$$K = N \left(\frac{1}{2} m v^2 \right)_{av} = \frac{3}{2} N k_B T$$

This shows that the average kinetic energy of a molecule is proportional to the temperature T. The root mean square velocity v_{rms} is:

$$(v^2)_{av} = \frac{3k_B T}{m}$$

$$v_{rms} = \sqrt{(v^2)_{av}} = \sqrt{\frac{3k_B T}{m}}$$

This means the mean squared speed is proportional to the temperature, and as temperature increases, the average speed of molecules increases. The root mean square speed is equal to $\sqrt{T}$.

In this lecture, we explored how statistical thermodynamics, specifically kinetic theory simplifies complex problems. In physics, solving interactions between multiple bodies (like the Sun, Earth, and Moon) is challenging. With more bodies, it becomes even harder, often requiring computers. However, statistical thermodynamics handles many-body problems more easily because it focuses on average properties rather than detailed interactions. For example, when calculating pressure, we consider the average effect of many molecules, not the specific actions of each one. This approach allows us to ask different, more manageable questions, making statistical thermodynamics a powerful tool.

Lecture_39

Let's dig into the significant changes that have shaped our world. When Alexander the Great, made his mark on the inhabited regions of the world, it brought about tremendous changes. History has seen the rise and fall of countries, sultanates, and capitals over thousands of years. However, in my opinion, the revolution that occurred in the year 1905 stands out distinctly.

The Imaginary Revolution

This revolutionary thought troubled many minds for a long time, presenting a strange yet fascinating picture. It was certainly an imaginary revolution, but its implications were profound. This revolution was about breaking the atom and harnessing infinite amounts of energy from it. This discovery revealed how energy is released from the sun and stars, leading to the development of atomic power plants and even the atomic bomb.

This breakthrough allowed us to uncover the secrets hidden inside the atom. On one hand, the atom is the smallest unit, while on the other, it is as if the universe itself started from an atom. Einstein contributed significantly to this understanding. From the Theory of Relativity, we learned that before Newton's time, people had a specific understanding of time and space.

The Concept of Space-Time

Previously, time was perceived as a flowing river, moving in one direction with a constant flow. Space was considered something that could only be measured, and it was viewed as a fixed entity. However, Einstein changed this picture, introducing the concept of space-time. He explained that space and time are not separate but rather a single, interconnected entity known as space-time.

The Nature of Time

Physicists are concerned with how we can measure time and space. Understanding time involves recognizing periodic processes. For example, a pendulum swings from one side to the other, and this periodic motion can be used to measure time.

However, pendulums are not perfect clocks. Factors like temperature changes can alter the length of the pendulum, affecting its time period. A more accurate clock is the Earth's rotation, which completes one revolution in 24 hours. The Earth's orbit around the sun, completing one revolution in a year, also serves as a reliable timekeeper. Even more precise are atomic clocks, which measure time with such accuracy that we can observe the slowing rotation of the Earth.

Measuring Space

To measure space, we can use various methods. One way is to use standardized lengths, although creating identical units of measure can be challenging due to slight differences. Modern methods use atomic properties to determine length accurately. The distance between atoms can be used to measure total length.

Inertial Frames

To grasp Einstein's Theory of Special Relativity, we must first understand the concept of an inertial frame. Imagine sitting in a train that moves smoothly without any vibrations or noise. In such a situation, you cannot tell if the train is stationary or moving. This describes an inertial frame—a frame of reference where there is no acceleration.

Newton's First Law of Motion states that an object will remain at rest or continue in uniform motion unless acted upon by an external force. This law holds true in an inertial frame. For example, if a body is at rest or moving with constant velocity, it will continue to do so in the absence of external forces.

Non-Inertial Frames

If the train you are in starts to accelerate or decelerate, you will feel a force acting on you, indicating a non-inertial frame. In such frames, external forces cause changes in motion. Newton defined an inertial frame as one where his First Law is valid. In non-inertial frames, forces such as acceleration can affect objects' motion.

The Earth as an Inertial Frame

While we often consider the ground as a reference frame, it is not truly inertial. The Earth rotates on its axis and orbits the sun, causing slight acceleration. Newton approximated the Earth as an inertial frame for practical purposes, noting that objects on Earth experience minimal acceleration.

The Universal Frame and Time keeping

Newton also proposed a universal frame, fixed with respect to distant stars, providing an exact inertial frame. He imagined a clock in the universe that could tell the exact time, with innumerable reference frames and inertial frames.

Observations and Events in Different Frames

In any frame of reference, we can observe events. For example, a person traveling in a car or on a ship experiences different reference frame. Observations made in these frames are considered events. For instance, striking a match could be an event, and its occurrence can be noted in different frames of reference.

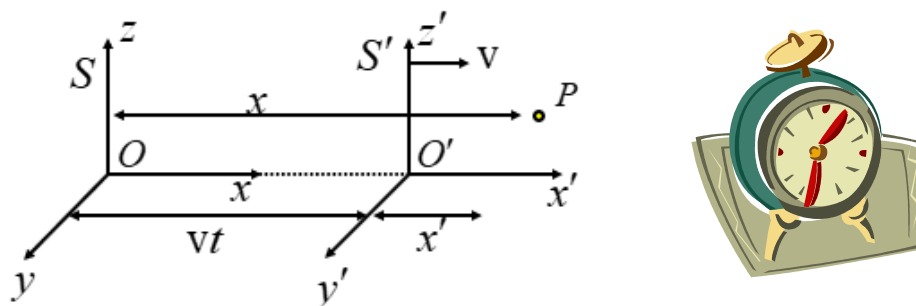


Figure 1.1: An event at time $t = t'$ in S frame of reference.

In one frame, as shown in figure 1.1, we describe an event by specifying its coordinates, such as (x, y, z, t) . The same event in another frame will have different spatial coordinates but the same time coordinate, according to Newton's view of absolute time. If we denote the second frame's coordinates with primes, we get transformations like:

$$\begin{aligned}x' &= x - vt, \\y' &= y, \\z' &= z, \\t' &= t\end{aligned}$$

These transformations are known as Galilean transformations, which give common sense results. Here is an example of a rod, as shown in figure 1.2, that is at rest, lies on x-axis.



Figure 1.2: A rod AB is placed along x-axis.

$$\text{length in S-frame} = x_B - x_A$$

$$\text{length in S'-frame} = x'_B - x'_A$$

Galilean transformation gives:

$$x'_B = x_B - vt_B \text{ and } x'_A = x_A - vt_A$$

Thus, length in S'-frame

$$x'_B - x'_A = x_B - x_A - v(t_B - t_A)$$

Since the two end points, A and B are measured at the same instant $t_B = t_A$,

$$x'_B - x'_A = x_B - x_A$$

$$\text{length in S-frame} = \text{length in S'-frame}$$

This shows that the length of the object remains the same in both frames. Similarly, the velocities and accelerations also transform predictably.

Velocity transformation

Galilean transformation of position along x-axis is

$$x' = x - vt$$

differentiate wrt time:

$$\frac{dx'}{dt} = \frac{dx}{dt} - v \frac{dt}{dt}$$

$$\frac{dx'}{dt} = \frac{dx}{dt} - v \quad \text{eq(1)}$$

we know

$$t = t'$$

$$\Rightarrow \frac{dx'}{dt} = \frac{dx'}{dt'}$$

eq(1) becomes

$$\frac{dx'}{dt'} = \frac{dx}{dt} - v$$

similarly, y and z components are

$$\frac{dy'}{dt'} = \frac{dy}{dt}, \quad \frac{dz'}{dt'} = \frac{dz}{dt}$$

$$\frac{dx'}{dt'} = u'_x = \text{the x-component of the velocity measured in } S' \text{-frame}$$

$$\frac{dx}{dt} = u_x = \text{the x-component of the velocity measured in } S \text{-frame}$$

Therefore, the velocity transformation is

$$u'_x = u_x - v \quad u'_y = u_y \quad u'_z = u_z$$

In general, the vector form is

$$\vec{u}' = \vec{u} - \vec{v}$$

Acceleration transformation

Galilean transformation of velocity along x-axis is

$$u'_x = u_x - v$$

differentiate wrt time:

$$\frac{du'_x}{dt'} = \frac{d}{dt}(u_x - v)$$

$$\frac{du'_x}{dt'} = \frac{du_x}{dt} - \frac{dv}{dt}$$

$$\frac{du'_x}{dt'} = \frac{du_x}{dt} \quad \because \frac{dv}{dt} = 0 \text{ because } v = \text{constant}$$

similarly, y and z components are

$$\frac{du'_y}{dt'} = \frac{du_y}{dt}, \quad \frac{du'_z}{dt'} = \frac{du_z}{dt}$$

Therefore, the acceleration transformation is

$$a'_x = a_x, \quad a'_y = a_y, \quad a'_z = a_z$$

In general, the vector form is

$$\vec{a}' = \vec{a},$$

Thus, the acceleration in the S' frame is the same as in the S frame, and Newton's laws apply equally in all inertial frames.

Newton's second law: $\vec{F} = m\vec{a}$ in S-frame

But in S'-frame: $\vec{F}' = m'\vec{a}'$

But: $m' = m \quad \vec{a} = \vec{a}'$

$$\therefore \vec{F}' = m'\vec{a}' = m\vec{a} = \vec{F}$$

This is known as the principle of Newtonian relativity, which states that the laws of physics are the same in all inertial frames.

The Speed of Light and Ether Theory

Now, let's consider the speed of light. Suppose you measure the speed of light in a laboratory and find it to be 3×10^{10} centi-meters per second or 3×10^8 meters per second. If you place a light source on a moving car, according to Newtonian mechanics, the speed of light should be affected by the car's speed. Scientists once believed that light, like sound, needed a medium to travel through, called ether.

However, the Michelson-Morley experiment, which aimed to detect the presence of ether, failed to find any evidence of it. This experiment showed that the speed of light is constant, regardless of the motion of the source or the observer. This finding puzzled scientists because it contradicted the notion of ether.

The Dawn of New Physics

Einstein's journey into the realm of relativity began with deep questions about the nature of electromagnetism and light. He wondered why Maxwell's equations, which describe electromagnetic waves, would hold true in one frame and not another. This curiosity led to the development of his two fundamental postulates of Special Relativity, which have been tested and validated countless times. Let's delve into the concepts that revolutionized our understanding of space and time.

The Two Pillars of Special Relativity

Einstein proposed two main principles:

1. **The Principle of Relativity:** The laws of physics are the same in all inertial frames of reference. This means that no matter how fast you are moving, if you are not accelerating, the same physical laws apply to you as to someone who is stationary.
2. **The Constancy of the Speed of Light:** The speed of light in a vacuum is constant and independent of the motion of the light source or the observer. This means that no matter how fast you are moving, you will always measure the speed of light as 3×10^{10} centi-meters per second or 3×10^8 meters per second.

These principles seem straightforward, but they lead to some surprising and counterintuitive results. These two postulates led to a profound shift in our understanding of space and time. They implied that measurements of space and time are relative to the observer's frame of reference. This realization paved the way for the development of Special Relativity, fundamentally changing our perception of the universe.

Synchronizing Clocks in Different Frames

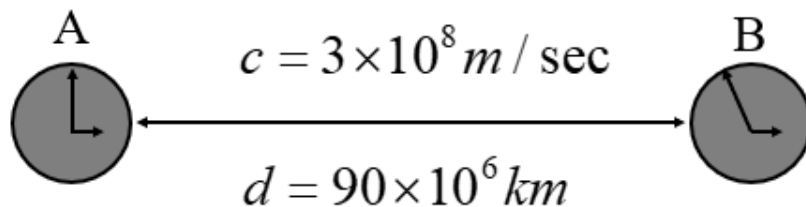


Figure 1.3: Two clocks in a single frame of reference which are far apart.

To understand how time and space are related, let's consider synchronizing clocks, in a single frame of reference. Suppose we have two clocks, one at point A and another at point B, 90 million kilo-meters apart, as shown in figure 1.3. If a light signal is sent from B to A, it will take 5 minutes to travel this distance (since light travels at 3×10^8 meters per second).

$$\text{Time taken by in reaching from B to A} = t = \frac{d}{c}$$

$$t = \frac{90 \times 10^6 \text{ km}}{3 \times 10^8 \text{ m/sec}}$$

or

$$t = \frac{9 \times 10^{10} \text{ m}}{3 \times 10^8 \text{ m/sec}} = 300 \text{ sec} = 5 \text{ min}$$

When the light signal reaches clock A, it shows 3:00 PM. Since it took 5 minutes for the light to travel, clock B must have shown 2:55 PM when the light left. Thus, we can synchronize the clocks by accounting for the time it takes for light to travel between them. This method ensures that all clocks in the frame show the same time, considering the finite speed of light.

Measuring Time in Moving Frames

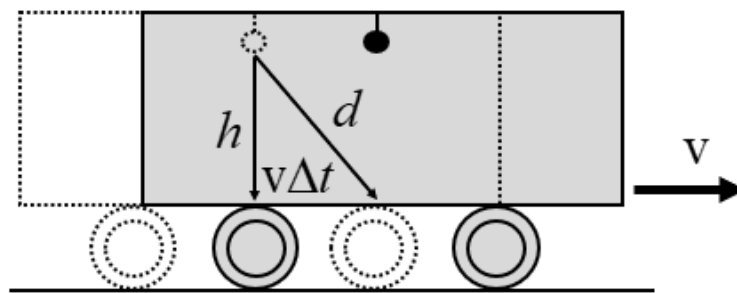


Figure 1.4: A moving train with a flashlight fixed at ceiling pointing to the floor.

Now, imagine you are on a train moving smoothly and rapidly, as shown in figure 1.4. You have a flashlight pointing from the ceiling to the floor. If the distance from the ceiling to the floor is h , and the speed of light is c , the time t it takes for the light to travel from the ceiling to the floor, according to your clock, is:

$$\text{Time it take the light to reach at floor} = \Delta t' = \frac{h}{c} \quad \text{eq(i)}$$

However, an observer standing on the ground sees the train moving. For this observer, the light also has to cover the horizontal distance that the train travels while the light is moving from the ceiling to the floor. If the train's speed is v and the light takes time Δt , the horizontal distance covered is $v\Delta t$.

The total distance the light travels, according to the ground observer, forms the hypotenuse of a right triangle with sides h and $v\Delta t$. Using the Pythagorean theorem:

$$d^2 = h^2 + (v\Delta t)^2$$

$$d = \sqrt{h^2 + (v\Delta t)^2}$$

as we know $d = c\Delta t$

$$c\Delta t = \sqrt{h^2 + (v\Delta t)^2}$$

$$\Delta t = \frac{\sqrt{h^2 + (v\Delta t)^2}}{c}$$

Solving for Δt , taking square on both sides

$$(c\Delta t)^2 = h^2 + (v\Delta t)^2$$

$$(c\Delta t)^2 - (v\Delta t)^2 = h^2$$

$$(\Delta t)^2(c^2 - v^2) = h^2$$

$$(\Delta t)^2 = \frac{h^2}{(c^2 - v^2)}$$

taking squareroot on both sides

$$\Delta t = \frac{h}{\sqrt{c^2 - v^2}}$$

taking c^2 common

$$\Delta t = \frac{h}{\sqrt{c^2 \{1 - (v^2 / c^2)\}}}$$

$$\Delta t = \frac{h}{c\sqrt{1 - (v^2 / c^2)}}$$

or

$$\Delta t = \frac{h}{c} \frac{1}{\sqrt{1 - (v / c)^2}}$$

using eq(i)

$$\Delta t = \Delta t' \frac{1}{\sqrt{1 - (v / c)^2}}$$

Therefore

$$\Delta t' = \Delta t \sqrt{1 - (v / c)^2}$$

$$\text{or } \Delta t' = \frac{\Delta t}{\gamma} \quad \text{where } \gamma = \frac{1}{\sqrt{1 - (v / c)^2}}$$

$$\text{for } v < c, \gamma > 1 \Rightarrow \Delta t' < \Delta t$$

This time Δt is longer than the time t measured by the observer on the train. The difference is due to the fact that the light has to cover more distance when the train is moving.

Gamma (γ) is always greater than or equal to 1. When the speed v is much less than the speed of light c , γ is very close to 1, and we do not notice time dilation in our everyday experiences. However, as v approaches c , γ increases significantly, and time dilation becomes noticeable.

For example, if you travel in a fast rocket, the time measured on your clock (moving clock) would appear to run slower compared to a clock on Earth (stationary clock). This effect is more pronounced at speeds close to the speed of light.

Example:

A muon is traveling through the laboratory at three-fifths the speed of the light. How long does it last?

Solution:

$$\text{Speed of muon} = v = \frac{3}{5}c$$

$$\Delta t' = \frac{\Delta t}{\gamma} \quad \text{where } \gamma = \frac{1}{\sqrt{1 - (v/c)^2}}$$

lets calculate γ

$$\gamma = \frac{1}{\sqrt{1 - (v/c)^2}}$$

$$\gamma = \frac{1}{\sqrt{1 - \left(\frac{3c}{5c}\right)^2}}$$

$$\gamma = \frac{1}{\sqrt{1 - (3/5)^2}} = \frac{5}{4}$$

So, it lives longer (than at rest) by a factor of 5/4

The Twin Paradox: An Intriguing Comparison

To grasp the nuances of Special Relativity, let's delve into the Twin Paradox, a fascinating thought experiment that challenges our understanding of time and motion.

The Setup

Imagine two identical twins, A and B, born at the same time. Twin A remains on Earth while Twin B embarks on a journey into space, traveling at speeds close to that of light. After many years, Twin B returns to Earth and finds that Twin A has aged significantly, while Twin B has aged much less.

The Paradox

At first glance, this seems paradoxical. According to the Theory of Relativity, all inertial frames are equivalent. Thus, Twin A could argue that Twin B should age faster, and vice versa. However, in practice, the traveling twin (B) is younger upon return.

Breaking the Symmetry

The resolution lies in understanding that the traveling twin (B) was not in a purely inertial frame. Twin B experienced acceleration and deceleration phases during the journey, breaking the symmetry of inertial frames. Thus, Twin B's frame was not equivalent to Twin A's purely inertial frame on Earth.

Lorentz Contraction: Objects Shrink in Motion

Now, let's explore another fundamental aspect of Special Relativity: Lorentz Contraction. This phenomenon states that objects moving at high speeds appear shorter in the direction of motion to an outside observer.

The Thought Experiment

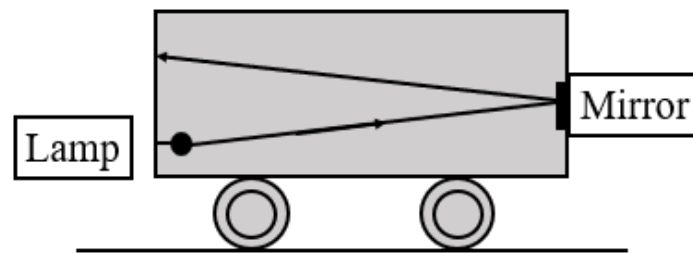


Figure 1.5: A high-speed train having a flashlight and mirror at its ends.

Imagine you're on a high-speed train with a flashlight, as shown in figure 1.5. You stand at one end of the train car, and a mirror is placed at the other end. You shine the light towards the mirror. According to your frame of reference, the light travels a distance $2\Delta x'$ (to the mirror and back) in a time $2\Delta x'/c$, as shown in figure 1.6.

$\Delta x' =$ length of the car with respect to car observer

$$\Delta t' = \frac{2\Delta x'}{c} \quad \text{eq(I)}$$

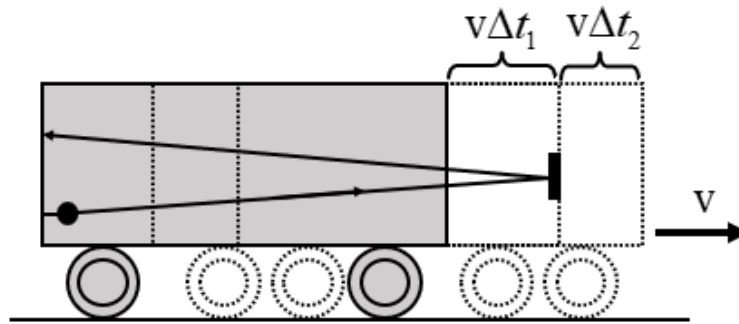


Figure 1.6: In fast-speed train light travels an extra distance $2\Delta x'$

Δx = length of the car according to ground observer

Δt_1 = time for the signal to reach the front end

Δt_2 = return time

However, for an observer on the ground, the train is moving. Thus, they perceive the light traveling a longer distance because the mirror is moving away from the light when it is emitted and moving towards the light on its return journey.

Mathematical Explanation

For the ground observer, the total distance covered by light includes the forward distance (increased by the train's movement) and the backward distance (decreased by the train's movement).

Let Δt_1 be the time for the light to reach the mirror and Δt_2 be the time for the light to return.

1. For the forward journey, the distance covered by light is $\Delta x + v\Delta t_1$:

$$c\Delta t_1 = \Delta x + v\Delta t_1$$

solving for Δt_1

$$c\Delta t_1 - v\Delta t_1 = \Delta x$$

$$\Delta t_1(c - v) = \Delta x$$

$$\Delta t_1 = \frac{\Delta x}{(c - v)}$$

2. For the return journey, the distance covered by light is $\Delta x - v\Delta t_2$:

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$$c\Delta t_2 = \Delta x - v\Delta t_2$$

solving for Δt_2

$$c\Delta t_2 + v\Delta t_2 = \Delta x$$

$$\Delta t_2(c + v) = \Delta x$$

$$\Delta t_2 = \frac{\Delta x}{(c + v)}$$

The total time Δt for the round trip is:

$$\Delta t = \Delta t_1 + \Delta t_2 = \frac{\Delta x}{(c - v)} + \frac{\Delta x}{(c + v)}$$

$$\Delta t = \Delta x \left(\frac{1}{c - v} + \frac{1}{c + v} \right)$$

$$\Delta t = \Delta x \left(\frac{c + v + c - v}{c^2 - v^2} \right)$$

$$\Delta t = \Delta x \left(\frac{c + c}{c^2 - v^2} \right)$$

$$\Delta t = \Delta x \frac{2c}{c^2(1 - v^2 / c^2)}$$

$$\Delta t = \frac{2\Delta x / c}{(1 - v^2 / c^2)} \quad \text{eq(II) (w.r.t ground observer)}$$

From time dilation formula:

$$\Delta t' = \sqrt{1 - v^2 / c^2} \Delta t$$

using eq(II)

$$\Delta t' = \sqrt{1 - v^2 / c^2} \left(\frac{2\Delta x / c}{(1 - v^2 / c^2)} \right)$$

$$\Delta t' = \sqrt{1 - v^2 / c^2} \left(\frac{2\Delta x / c}{\sqrt{(1 - v^2 / c^2)^2}} \right)$$

$$\Delta t' = \sqrt{1 - v^2 / c^2} \left(\frac{2\Delta x / c}{\sqrt{(1 - v^2 / c^2)} \sqrt{(1 - v^2 / c^2)}} \right)$$

$$\Delta t' = \frac{2\Delta x / c}{\sqrt{1 - v^2 / c^2}}$$

$$\text{using eq(I)} \quad \therefore \Delta t' = \frac{2\Delta x'}{c}$$

$$\frac{2\Delta x'}{c} = \frac{2\Delta x / c}{\sqrt{1 - v^2 / c^2}}$$

$$\Delta x' = \frac{\Delta x}{\sqrt{1 - v^2 / c^2}}$$

Conclusion of Lorentz Contraction

The length of the moving object (train) appears contracted by a factor of $\sqrt{1 - v^2 / c^2}$. For example, a one-meter rod moving at high speed will appear shorter to a stationary observer. The faster the object moves, the greater the contraction.

Practical Implications and Observations

In our daily lives, the effects of Special Relativity are negligible because the speeds involved are much less than the speed of light. However, in high-energy physics and astronomy, these effects are significant. For instance, particles accelerated to near-light speeds in colliders exhibit both time dilation and length contraction, validating Einstein's predictions.

The Evolution of Transformations: Galilean vs. Lorentz

In classical mechanics, we often use Galilean transformations to relate the coordinates of events between different reference frames. However, these transformations fall short when dealing with speeds close to the speed of light. In Einstein's theory of relativity, Galilean transformations are replaced by Lorentz transformations, which accurately describe the relationships between coordinates in different frames at high velocities.

Galilean Transformations

Galilean transformations work well for everyday speeds. They assume that time is absolute and the same for all observers, while positions and velocities transform simply with the relative motion between frames.

The Need for Lorentz Transformations

When dealing with velocities approaching the speed of light, the Galilean transformations are no longer valid. Instead, we use Lorentz transformations, which account for the relativistic effects predicted by Einstein's theory of relativity. These transformations show how time and space coordinates of an event change between two frames moving at a constant velocity relative to each other.

Lorentz Transformations

Consider two frames of reference, as shown in figure 1.7, one is with coordinates (x, y, z, t) and other has coordinates (x', y', z', t') , moving with relative speed of v , as given below.

Now, an event E occurs. Let's calculate Lorentz transformation from moving frame to stationary and vice versa.

where:

- x and t are the coordinates of the event in the original frame.
- x' and t' are the coordinates in the moving frame.
- v is the relative velocity between the frames.
- c is the speed of light.
- γ (gamma) is the Lorentz factor, defined as $\gamma = \frac{1}{\sqrt{1 - v^2 / c^2}}$.

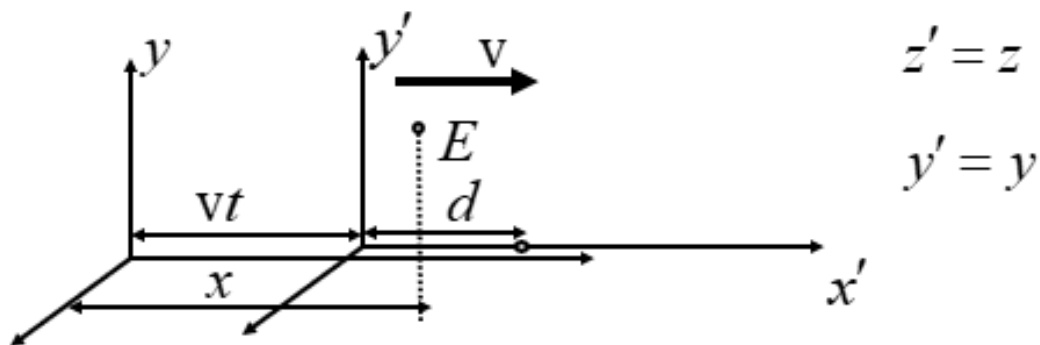


Figure 1.7: For an event E, frame of reference with prime coordinates has a relative speed v .

$$x = d + vt$$

But

$$d = \frac{x'}{\gamma}$$

$$\Rightarrow x' = \gamma(x - vt)$$

because x' is moving with respect to O –frame

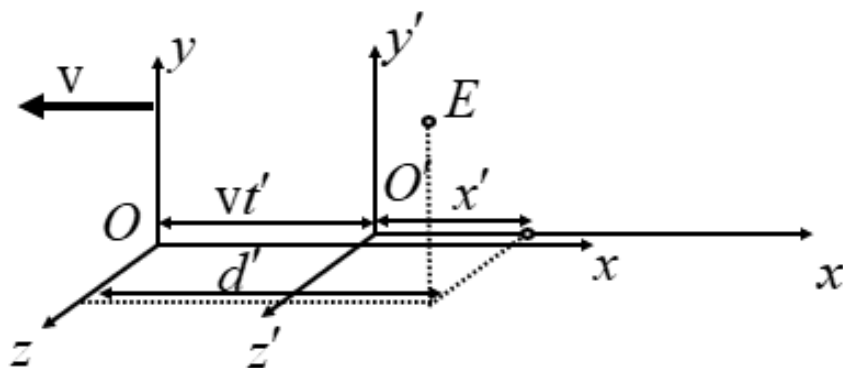


Figure 1.8: For an event E, frame of reference with coordinates (x, y, z, t) has a relative speed v .

$$x' = d' - vt'$$

But

$$d' = \frac{x}{\gamma}$$

$$\Rightarrow x = \gamma(x' + vt')$$

$$\therefore x = \gamma(x' + vt')$$

$$x = \gamma(\gamma(x - vt) + vt')$$

$$x = \gamma^2(x - vt) + \gamma vt'$$

$$x = \gamma^2 x - \gamma^2 vt + \gamma vt'$$

dividing with γ^2 on both sides, we get

$$\frac{x}{\gamma^2} = x - vt + \frac{vt'}{\gamma}$$

$$x(1 - v^2/c^2) = x - vt + \frac{vt'}{\gamma} \quad \therefore \gamma^2 = \frac{1}{1 - v^2/c^2}$$

$$x - xv^2/c^2 = x - vt + \frac{vt'}{\gamma}$$

$$-xv^2/c^2 = -vt + \frac{vt'}{\gamma}$$

$$vt - xv^2/c^2 = \frac{vt'}{\gamma}$$

$$v(t - xv/c^2) = \frac{vt'}{\gamma}$$

$$(t - xv/c^2) = \frac{t'}{\gamma}$$

$$t' = \gamma(t - xv/c^2)$$

$$\therefore t' = \gamma\left(t - \frac{v}{c^2}x\right)$$

So, collectively, all Lorentzian equations are given below:

$$x' = \gamma(x - vt) \quad x = \gamma(x' + vt)$$

$$y' = y \quad y = y'$$

$$z' = z \quad z = z'$$

$$t' = \gamma\left(t - \frac{v}{c^2}x\right) \quad t = \gamma\left(t' + \frac{v}{c^2}x'\right)$$

Simultaneity:

Let's test this Lorentz transformation for a specific situation. Consider we have clocks in S-frame (stationary) and in S'-frame (moving), as shown in figure 1.9. Now at $t=0$ check the value of t' .

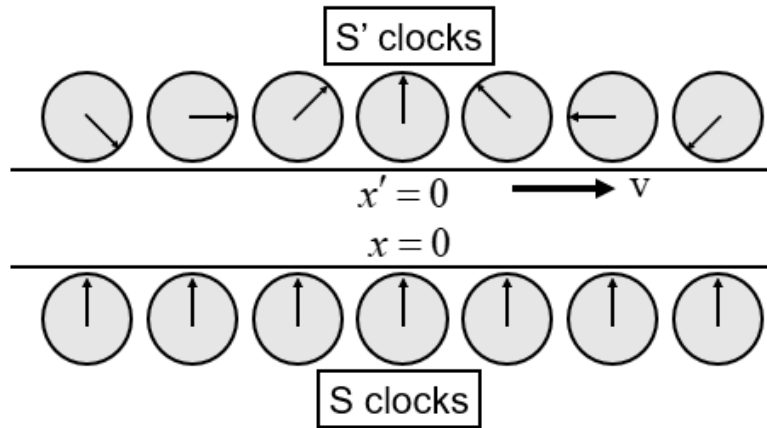


Figure 1.9: For a fixed position, clocks for two frames of references are presented.

As we know

$$t' = \gamma \left(t - \frac{v}{c^2} x \right)$$

put $t=0$

$$t' = \gamma \left(0 - \frac{v}{c^2} x \right)$$

$$t' = -\gamma \frac{v}{c^2} x$$

Clearly, t is not equal to t' .

Time Dilation and Length Contraction

The Lorentz transformations reveal that time and space are intertwined, and their measurements depend on the observer's frame of reference. This intertwining leads to two significant effects: time dilation and length contraction.

Time Dilation

Time dilation refers to the phenomenon where time appears to pass slower for an object moving relative to an observer's frame.

From the Lorentz transformation, if we consider an event occurring at the same location in the moving frame ($\Delta x = 0$), the relationship between time intervals in the two frames can be calculated.

We know

$$t = \gamma \left(t' + \frac{v}{c^2} x' \right)$$
$$\Rightarrow \Delta t' = \gamma \left(\Delta t - \frac{v}{c^2} \Delta x \right)$$

putting $\Delta x = 0$, we get

$$\Delta t' = \gamma \Delta t$$

This equation shows that the time interval t' in the moving frame is longer than the time interval t in the stationary frame, indicating that moving clocks run slower.

Length Contraction

Length contraction states that objects moving relative to an observer will appear shortened along the direction of motion.

If we measure the length of an object moving with velocity v , the contracted length $\Delta x'$ can be calculated as:

We know

$$x' = \gamma (x - vt)$$
$$\Rightarrow \Delta x' = \gamma (\Delta x - v \Delta t)$$

putting $\Delta t = 0$, we get

$$\Delta x' = \gamma \Delta x$$

Here, Δx is the length of the object in its rest frame. As the speed increases, γ increases, making $\Delta x'$ shorter.

Einstein velocity addition rule:

The Lorentz transformation also helps us relate the velocities of objects in different frames. If a particle moves with velocity u in frame S , its velocity u' in frame S' moving with velocity v relative to S is derived below.

Suppose a particle moves a distance dx in a time dt . Its velocity u is then

$$u = \frac{dx}{dt} \text{ (in S-frame)}$$

In S' -frame, meanwhile, it has moved a distance

$$dx' = \gamma (dx - v dt)$$

in time

$$dt' = \gamma \left(dt - \frac{v}{c^2} dx \right)$$

The velocity in S'-frame is therefore

$$u' = \frac{dx'}{dt'}$$

putting values of dx' and dt' , we get

$$u' = \frac{\gamma(dx - vdt)}{\gamma \left(dt - \frac{v}{c^2} dx \right)}$$

$$u' = \frac{dx - vdt}{dt - \frac{v}{c^2} dx}$$

$$u' = \frac{dt(dx/dt - v)}{dt \left(1 - \frac{v}{c^2} dx/dt \right)}$$

$$u' = \frac{dx/dt - v}{1 - \frac{v}{c^2} dx/dt}$$

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}} \quad \text{eq(a)} \quad \therefore u = dx/dt$$

lets calculate for u , we have

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

$$u' \left(1 - \frac{uv}{c^2} \right) = u - v$$

$$u' - u' \frac{uv}{c^2} = u - v$$

$$u' + v = u + u' \frac{uv}{c^2}$$

$$u' + v = u \left(1 + \frac{u'v}{c^2} \right)$$

$$\frac{u' + v}{\left(1 + \frac{u'v}{c^2} \right)} = u$$

or

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}} \quad \text{eq(b)}$$

Eq(a) and eq(b) are Einstein velocity addition rule.

Example:

Suppose, instead of particle with speed u we have light itself then u becomes c . What will be its speed in prime coordinates?

Solution:

If $u = c$ then $u' = ?$

Using the Einstein velocity addition rule, we get

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

putting $u = c$

$$u' = \frac{c - v}{1 - \frac{cv}{c^2}}$$

$$u' = \frac{c - v}{1 - \frac{v}{c}}$$

$$u' = c \frac{c - v}{c - v} = c$$

$$\Rightarrow u' = c$$

Speed of light same in all reference frames

Example:

Suppose, an ordinary particle of speed u and relative speed of v have very low speed as compared to speed of light c . What will be its speed in prime coordinates?

Solution:

If $u, v \ll c$ then $u' = ?$

Using the Einstein velocity addition rule, we get

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}} \rightarrow u - v \quad \therefore uv \ll c^2 \text{ so } 1 - \frac{uv}{c^2} \approx 1$$

$$u' = u - v$$

For ordinary speeds, Einstein velocity addition rule reduces to Galilean velocity addition rule

The Famous Equation: $E=mc^2$

Einstein's most famous equation, $E=mc^2$, illustrates the equivalence of mass and energy. It implies that mass can be converted into energy and vice versa. This principle is fundamental to nuclear reactions and particle physics, where tiny amounts of mass convert into vast amounts of energy.

Lecture_40

Introduction

Physics is a field where expressing ideas using the least number of words or formulas is highly appreciated. If an idea is interesting and can be explained concisely, it stands out. In this book, we will delve into the fascinating theory of Special Relativity, as introduced by Albert Einstein. Special Relativity revolves around two main postulates which we will explore in detail, along with their implications.

Key Postulates of Special Relativity

1. The Speed of Light is Constant

The first postulate of Special Relativity is that light travels at the same speed in every inertial frame of reference, regardless of the motion of the light source or the observer. Whether the frame is moving, stationary, or running very fast in any direction, the speed of light remains constant.

2. The Laws of Physics are the Same in All Inertial Frames

The second postulate states that the laws of physics have the same form in every inertial frame of reference. Whether you are on Earth, on the moon, or in a spaceship, the physical laws you discover and verify remain consistent.

These two postulates, although simple and brief, have profound implications and have led to significant discoveries.

Time Dilation and Length Contraction

Time Dilation

One of the consequences of Special Relativity is time dilation. If an observer is moving very fast relative to you in an inertial frame, all the clocks in that observer's frame will appear to be running slower from your perspective. This means that everything in that frame seems to happen more slowly.

Lorentz / Length Contraction

Another consequence is length contraction. If an object is moving towards or away from you, its length will appear shorter. If the original length is L , the observed length L' is given by:

$$L' = \frac{L}{\gamma}$$

where γ (gamma) is the Lorentz factor defined as:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Here, v is the relative velocity of the moving frame, and c is the speed of light.

Lorentz Transformation

Understanding Frames of Reference

Consider two frames of reference, as shown in fig 40.1: S and S' . The origin of S is at O , and the origin of S' is at O' . If something is at a certain place at a certain time, it has coordinates in space and time. For simplicity, let's consider one-dimensional space with coordinates (t, x) in frame S and (t', x') in frame S' .

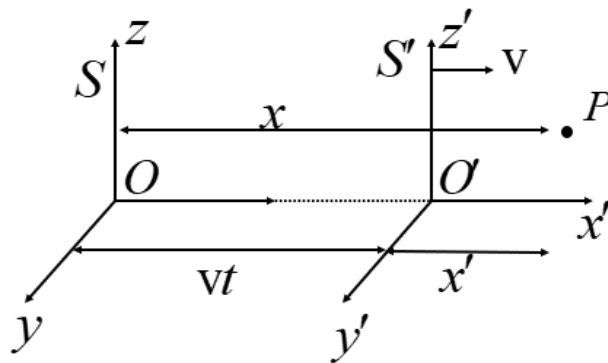


Figure 40.1: Two frame of references with origin O and O'

Lorentz Transformation Equations

The relationship between these coordinates is given by the Lorentz transformation. Without deriving it fully here, the transformation equations are:

$$x' = \gamma(x - vt) \quad x = \gamma(x' + vt')$$

$$t' = \gamma\left(t - \frac{v}{c^2}x\right) \quad t = \gamma\left(t' + \frac{v}{c^2}x'\right)$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

These equations show how space and time coordinates transform between two inertial frames moving relative to each other at a constant velocity v .

Invariant Interval

The distance between two events in spacetime is defined by the interval, which remains invariant under Lorentz transformation. For events separated by Δt in time and Δx in space, the interval I is given by:

$$I = c^2 \Delta t^2 - \Delta x^2$$

This interval remains the same in all inertial frames, ensuring consistency of physical laws.

Consequences of Invariant Interval

To understand the significance, consider a light pulse emitted in frame S' . The light pulse spreads at the speed of light c . In frame S , the same pulse is observed moving at c . The interval for this light pulse is zero:

$$I' = (c\Delta t')^2 - (\Delta x')^2$$

Putting $\Delta t'$ and $\Delta x'$ from Lorentz transformation

$$= c^2 \gamma^2 \left(\Delta t - \frac{v}{c^2} \Delta x \right)^2 - \gamma^2 (\Delta x - v\Delta t)^2$$

$$\because (a-b)^2 = a^2 + b^2 - 2ab$$

$$= c^2 \gamma^2 \left[(\Delta t)^2 + \left(\frac{v}{c^2} \Delta x \right)^2 - 2\Delta t \frac{v}{c^2} \Delta x \right] - \gamma^2 \left[(\Delta x)^2 + (v\Delta t)^2 - 2\Delta x v\Delta t \right]$$

$$= \gamma^2 \left[c^2 (\Delta t)^2 + c^2 \left(\frac{v}{c^2} \Delta x \right)^2 - c^2 2\Delta t \frac{v}{c^2} \Delta x \right] - \gamma^2 \left[(\Delta x)^2 + (v\Delta t)^2 - 2\Delta x v\Delta t \right]$$

$$= \gamma^2 \left[(c\Delta t)^2 + \frac{v^2}{c^2} (\Delta x)^2 - 2v\Delta t\Delta x \right] - \gamma^2 \left[(\Delta x)^2 + (v\Delta t)^2 - 2\Delta x v\Delta t \right]$$

taking γ^2 common

$$= \gamma^2 \left[(c\Delta t)^2 + (\Delta x)^2 \frac{v^2}{c^2} - 2v\Delta t\Delta x - (\Delta x)^2 - (v\Delta t)^2 + 2v\Delta x\Delta t \right]$$

$$\begin{aligned} I' &= \gamma^2 \left[(c\Delta t)^2 + (\Delta x)^2 \frac{v^2}{c^2} - \cancel{2v\Delta t\Delta x} - (\Delta x)^2 - (v\Delta t)^2 + \cancel{2v\Delta t\Delta x} \right] \\ &= \frac{1}{(1-v^2/c^2)} \left[(c^2 - v^2)(\Delta t)^2 - (\Delta x)^2 (1 - v^2/c^2) \right] \because \gamma^2 = \frac{1}{(1-v^2/c^2)} \\ &= \frac{1}{(1-v^2/c^2)} \left[c^2 (1 - v^2/c^2)(\Delta t)^2 - (\Delta x)^2 (1 - v^2/c^2) \right] \\ &= \frac{(1-v^2/c^2)}{(1-v^2/c^2)} \left[(c\Delta t)^2 - (\Delta x)^2 \right] \\ &= (c\Delta t)^2 - (\Delta x)^2 = I \end{aligned}$$

Thus,

$$(c\Delta t)^2 - (\Delta x)^2 = (c\Delta t')^2 - (\Delta x')^2$$

Time-Like and Space-Like Intervals

Time-Like Intervals

When the time difference between two events is greater than the spatial distance ($c^2\Delta t^2 > \Delta x^2$), the interval is time-like. This means the events are separated more by time than by space.

Space-Like Intervals

When the spatial distance is greater than the time difference ($\Delta x^2 > c^2\Delta t^2$), the interval is space-like. This means the events are separated more by space than by time.

Consistency Across Frames

If an interval is time-like or space-like in one frame, it remains so in all frames, ensuring the consistency of cause and effect.

World Lines and Light Cones

World Lines

A world line represents the trajectory of a particle through spacetime, as shown in fig 40.2. For a particle at rest, the world line is vertical, indicating no change in spatial position over time. For a moving particle, the world line is slanted.

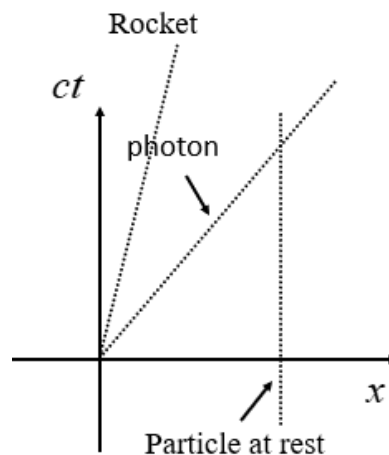


Figure 40.2: A world line graph where position on x-axis and ct on y-axis.

Light Cones

The light cone represents the possible positions of a light pulse emanating from an event, as shown in fig 40.3. The past light cone includes all events that could have influenced the event, while the future light cone includes all events that the event could influence.

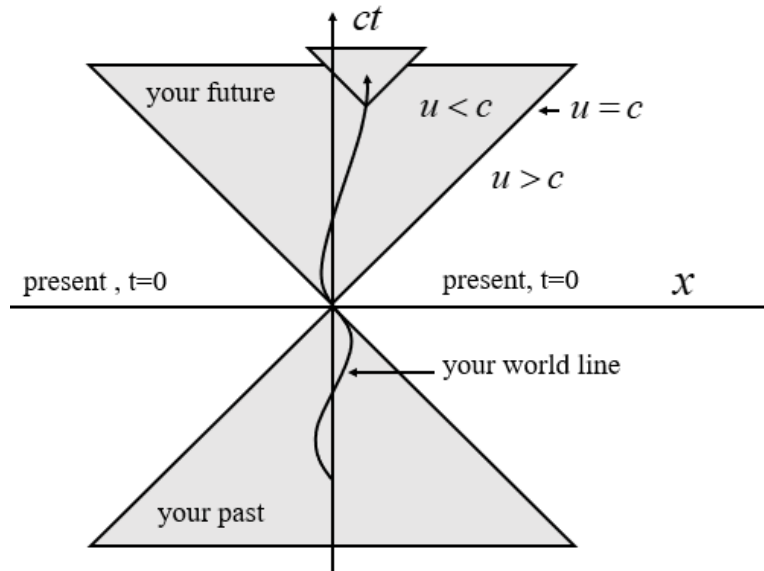


Figure 40.3: The light cone that represents future, past and present.

Particle Movement

A particle can only move within its light cone, as it cannot travel faster than the speed of light. This constraint shapes the particle's past, present, and future possibilities.

Preservation of Causality

Cause and Effect

In Special Relativity, the order of cause and effect is preserved. The invariant interval ensures that if one event causes another, this relationship remains consistent in all frames. For instance, if a bullet is fired or a ball is thrown, as shown in fig 40.4, and later hits a target, this sequence remains the same in all frames.

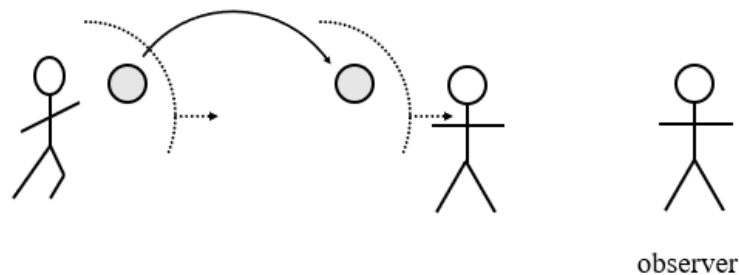


Figure 40.4: A ball is thrown from one man to another with an observer

No Reversal of Order

It is impossible for the order of events to reverse in any frame. The invariance of the interval guarantees that cause always precedes effect, maintaining the logical sequence of events.

The Doppler Effect and Relativity

In previous discussions, we have talked about sound and the Doppler effect. The Doppler effect is observed when there is a change in the frequency of sound as the source moves relative to an observer. For instance, if a car is honking its horn while moving, a person standing still will hear the frequency change as the car approaches and then passes by. This change in frequency is called the Doppler effect.

The Doppler Effect for Light

The Doppler effect also applies to light, but here, we must use relativity to understand it fully. Imagine a vehicle moving towards a person with its headlights on, as shown in fig 40.5. The frequency of the light from the headlights, as seen by a person standing still, will change. We denote the frequency in the source rest frame (the frame of the moving vehicle) as ν_0 , and its time period as T_0 , so:

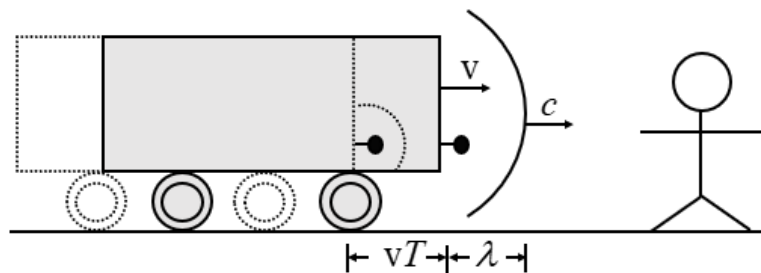


Figure 40.5: A moving vehicle that generates a light pulse

$$\nu_0 = \frac{1}{T_0} = \text{frequency measured in source rest frame}$$

ν = frequency measured with respect to observer = ?

λ = distance between two successive crests

Now, we want to determine the frequency that the person standing still (the observer) will see. Consider that one pulse of light is emitted from the source, and then another pulse is emitted after a time T_0 . The distance between two successive pulses or crests of the light wave is λ . The speed of light is c , so the distance covered by the light in time T_0 is cT_0 .

In the frame of the observer, the vehicle covers a distance vT_0 in the same time period, where v is the velocity of the vehicle. Thus, the distance λ between successive pulses of light is given by:

$$\lambda = c \cdot T_0 - v \cdot T_0 = T_0(c - v)$$

From this, we can determine the frequency ν that the observer will see:

$$\nu = \frac{c}{\lambda} = \frac{c}{T_0(c - v)}$$

However, T_0 is the time period measured in the rest frame of the source. The observer measures the time period T , which is related to T_0 through the time dilation formula in relativity:

$$T = \gamma T_0 = \frac{T_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Therefore, the observed frequency ν becomes:

$$\begin{aligned} \nu &= \frac{c}{\lambda} = \frac{c}{T(c - v)} \\ \nu &= \frac{c}{(c - v) \frac{T_0}{\sqrt{1 - \frac{v^2}{c^2}}}} \\ \nu &= \frac{c}{(c - v) T_0} \sqrt{1 - \frac{v^2}{c^2}} \\ \nu &= \left(\frac{c}{c - v} \right) \frac{\sqrt{c^2 - v^2}}{c} \nu_0 \\ \nu &= \left(\frac{c}{c - v} \right) \frac{\sqrt{(c + v)(c - v)}}{c} \nu_0 \quad \because a^2 - b^2 = (a + b)(a - b) \\ \nu &= \frac{\sqrt{(c + v)(c - v)}}{\sqrt{(c - v)(c - v)}} \nu_0 \quad \because a - b = \sqrt{(a - b)^2} = \sqrt{(a - b)(a - b)} \\ \Rightarrow \nu &= \sqrt{\frac{c + v}{c - v}} \nu_0 \end{aligned}$$

Applications of the Doppler Effect in Relativity

This Doppler effect for light has many practical applications. For example, police use Doppler radar to measure the speed of vehicles. An electromagnetic pulse (often of a lower frequency than visible light) is emitted from the radar gun, hits the moving vehicle, and reflects back. The change in frequency of the reflected pulse allows the radar to calculate the speed of the vehicle.

Another interesting application is in astronomy. By observing the light from distant stars and galaxies, scientists can determine their velocities. If a star is moving away from us, the light we receive is redshifted (its frequency decreases). If it is moving towards us, the light is blue shifted (its frequency increases). This observation has led to significant discoveries about the universe.

Edwin Hubble and the Expanding Universe

About 100 years ago, Edwin Hubble observed that light from distant galaxies is redshifted, indicating that these galaxies are moving away from us. The farther away a galaxy is, the faster it is receding. This observation led to the hypothesis that the universe began with a massive explosion, now known as the Big Bang, and has been expanding ever since.

Energy and Momentum in Special Relativity

In special relativity, the concepts of energy and momentum are intertwined with space and time. According to Newton's laws, momentum is defined as the product of mass and velocity ($p=mv$). In non-relativistic situations, momentum is conserved in collisions.

However, in relativistic contexts, where speeds are close to the speed of light, this definition of momentum does not hold. Instead, relativistic momentum is defined as:

$$p = \gamma mv$$

Where,

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

This ensures that momentum is conserved even at high velocities. To understand this, consider two particles colliding in a non-relativistic context, as shown in fig 40.6. Before the collision, particle A has velocity u_A and particle B has velocity u_B . After the collision, their velocities change to u_C and u_D , respectively. In S-frame, conservation of momentum gives:

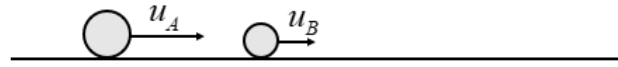
$$m_A u_A + m_B u_B = m_C u_C + m_D u_D$$

and

$$m_A + m_B = m_C + m_D$$

Consider the collision of two particles...

Before collision



Collision



After collision

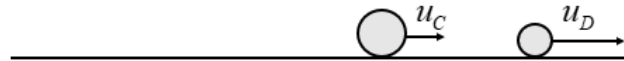


Figure 40.6: Collision between two particles in non-relativistic speed

In S'-frame which is moving with speed v with respect to S-frame. Using Galilean velocity addition rule, we get:

$$u' = u - v \quad \text{or} \quad u = u' + v$$

Conservation of momentum becomes

$$m_A(u'_A + v) + m_B(u'_B + v) = m_C(u'_C + v) + m_D(u'_D + v)$$

$$m_A u'_A + m_B u'_B = m_C u'_C + m_D u'_D$$

Now using Einstein velocity addition rule

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}} \quad u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

Conservation of momentum becomes

$$\begin{aligned} m_A \left(\frac{u'_A + v}{1 + u'_A v/c^2} \right) + m_B \left(\frac{u'_B + v}{1 + u'_B v/c^2} \right) \\ = m_C \left(\frac{u'_C + v}{1 + u'_C v/c^2} \right) + m_D \left(\frac{u'_D + v}{1 + u'_D v/c^2} \right) \end{aligned}$$

Now momentum is not conserved. So, momentum with old definition will not be conserved relativistically. Can we find some suitable new definition of momentum? However, in relativistic mechanics, we must adjust the definition of momentum to ensure it remains conserved at high velocities.

The new definition must have two properties

- 1) It must yield a law of conservation of momentum that satisfies the principle of relativity.
- 2) At low speeds, the new definition must reduce to

$$\vec{p} = m\vec{u}$$

Relativistic momentum

The relativistic formula for the momentum of a particle of mass m moving with velocity $\vec{u}$ is

$$\vec{p} = \frac{m\vec{u}}{\sqrt{1-u^2/c^2}} = \gamma m\vec{u}$$

In S-frame, conservation of momentum implies that

$$m_A \gamma_A u_A + m_B \gamma_B u_B = m_C \gamma_C u_C + m_D \gamma_D u_D$$

In S'-frame, conservation of momentum becomes

$$m_A \left(\frac{1}{\gamma} \gamma'_A u'_A + \gamma_A v \right) + m_B \left(\frac{1}{\gamma} \gamma'_B u'_B + \gamma_B v \right) = m_C \left(\frac{1}{\gamma} \gamma'_C u'_C + \gamma_C v \right) + m_D \left(\frac{1}{\gamma} \gamma'_D u'_D + \gamma_D v \right)$$

$$m_A \gamma'_A u'_A + m_B \gamma'_B u'_B = m_C \gamma'_C u'_C + m_D \gamma'_D u'_D$$

Thus, momentum is conserved in both S and S' frames

In the limit of low speeds:

$$\vec{p} = \frac{m\vec{u}}{\sqrt{1-u^2/c^2}} \rightarrow m\vec{u} \text{ as } u/c \rightarrow 0$$

Relativistic Energy

Just as momentum is redefined in relativity, so is energy. The relativistic kinetic energy K is not simply

$$K = \frac{1}{2} mu^2$$

as in Newtonian mechanics.

Let's calculate the expression for energy.

$$K = \int F dx$$

$$K = \int \frac{dp}{dt} dx \quad \because F = \frac{dp}{dt}$$

$$K = \int \frac{dx}{dt} dp$$

$$K = \int u dp \quad \text{eq(1)}$$

we know

$$p = m \left(\frac{u}{\sqrt{1 - u^2 / c^2}} \right)$$

so,

$$dp = md \left(\frac{u}{\sqrt{1 - u^2 / c^2}} \right)$$

Using product rule of derivative,

$$dp = \frac{mdu}{\sqrt{1 - u^2 / c^2}} + \frac{m(u^2 / c^2) du}{(1 - u^2 / c^2)^{3/2}} = \frac{mudu}{(1 - u^2 / c^2)^{3/2}}$$

Substituting dp in eq(1), we get

$$K = \int_0^u u dp$$

$$K = m \int_0^u \frac{udu}{(1 - u^2 / c^2)^{3/2}}$$

$$K = \frac{mc^2}{\sqrt{1 - u^2 / c^2}} - mc^2$$

Or,

$$K = E - E_0$$

Where total relativistic energy E is defined as

$$E = \frac{mc^2}{\sqrt{1 - u^2 / c^2}}$$

and rest energy E_0 is defined as

$$E_0 = mc^2$$

In an isolated system of particles, the total relativistic energy remains constant.

In the limit of low speeds:

$$K = \frac{mc^2}{\sqrt{1-u^2/c^2}} - mc^2$$

$$\frac{1}{\sqrt{1-u^2/c^2}} = (1-u^2/c^2)^{-1/2} = 1 + \frac{u^2}{2c^2} + \dots$$

$$K = mc^2 \left(1 + \frac{u^2}{2c^2} + \dots \right) - mc^2 \rightarrow \frac{1}{2}mu^2$$

$$\text{as } u/c \rightarrow 0$$

Thus, relativistic kinetic energy expression reduced into old kinetic energy expression for low speeds.

From

$$E = \frac{mc^2}{\sqrt{1-u^2/c^2}} = \gamma mc^2 \quad \text{and} \quad \vec{p} = \frac{m\vec{u}}{\sqrt{1-u^2/c^2}} = \gamma m\vec{u}$$

We get

$$\vec{u} = \frac{\vec{p}}{\gamma m} = \frac{\vec{p}c^2}{E}$$

$$\text{or } pc = E \left(\frac{u}{c} \right)$$

$$\begin{aligned} (pc)^2 &= E^2 \left(\frac{u}{c} \right)^2 = \gamma^2 m^2 c^4 (1 - 1/\gamma^2) \\ &= \gamma^2 m^2 c^4 - m^2 c^4 = E^2 - m^2 c^4 \end{aligned}$$

$$E^2 = p^2 c^2 + m^2 c^4$$

A particle may have energy and momentum even when it has no mass

$$E = pc$$

$$\text{or } c = \frac{pc^2}{E}$$

Rest Energy

Even when a particle is at rest ($v = 0$), it possesses energy due to its mass, known as rest energy: $E_0 = mc^2$

This equation reveals that mass can be converted into energy, and vice versa, a concept fundamental to nuclear physics and cosmology.

Nuclear Fission and Fusion

In nuclear reactions, such as fission (splitting of heavy atoms) and fusion (combining of light atoms), mass is converted into energy. These principal powers nuclear reactors and atomic bombs.

Particle-Antiparticle Creation

Energy can also create particles. For example, a high-energy photon can produce an electron and its antimatter counterpart, the positron. This particle creation is observed in high-energy physics experiments.

Lecture#41

The Reality of Matter: A Physics Perspective

This lecture aims to elucidate the debate about the nature of matter and light, specifically focusing on whether they exist as particles or waves. This exploration is rooted in physics, a field known for its logical approach to understanding and explaining the world through arguments and experiments.

Introduction

For centuries, humanity has pondered the nature of reality, leading to profound questions about the fundamental makeup of the universe. This lecture will delve into one such debate: the reality of matter, particularly through the lens of physics. The focus will be on the dual nature of light and matter, exploring the concepts of particles and waves.

The Nature of Light and Matter

Physics is a logical discipline, where theories and concepts are proven or disproven through systematic experimentation. However, some findings in physics might seem counterintuitive in our everyday experience. For instance, if someone tells you that matter exists in a lump sum form or as individual particles, you might accept it without question. But if the same person claims that matter exists as waves, this might seem nonsensical. Yet, this is the very crux of the debate we will explore.

The Dual Nature of Light

In previous discussions, we explored the question of whether light is composed of particles or waves. Experiments have shown that light exhibits properties of both. When light interacts with matter, it often behaves as if it consists of particles, known as photons. Photons are packets of energy.

The Photoelectric Effect

The photoelectric effect is a phenomenon where light striking a metal surface causes the ejection of electrons. Here's a simplified explanation:

1. **Incident Light:** Light of a certain frequency (ν) hits a metal plate.

2. **Electron Ejection:** The light energy is absorbed by electrons in the metal, causing them to be ejected.
3. **Current Flow:** These ejected electrons move towards a collecting plate, creating a current in the circuit.

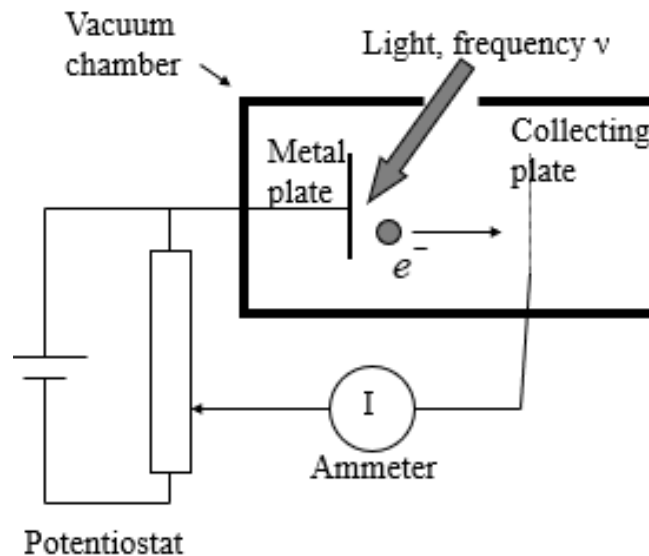


Fig 41.1: Schematic of the photoelectric effect experiment: Light of frequency ν is incident on a metal plate inside a vacuum chamber, causing the emission of electrons (e^-). These electrons are collected by a collecting plate, and the resulting current is measured by an ammeter. A potentiostat is used to control the potential difference between the metal and collecting plates.

Increasing the voltage using a potentiometer increases the energy of the ejected electrons. This observation leads to the conclusion that the energy of photons is proportional to their frequency.

$$E = h\nu$$

Here, E is the energy of the photon, h is Planck's constant, and ν is the frequency of the light.

Einstein's Photon Theory

Albert Einstein proposed that light consists of photons, and the energy of each photon is proportional to its frequency. This was revolutionary because it explained why increasing light intensity alone does not cause electron ejection unless the light has a certain minimum frequency. This minimum frequency corresponds to the work function of the metal.

Work Function (W)

The work function is the minimum energy required to eject an electron from the metal surface.

$$K_{\max} = h\nu - W$$

This equation explains that the kinetic energy of the ejected electron ($K_{\max}$) is equal to the energy of the incoming photon minus the work function.

Compton Scattering

Another critical experiment proving the particle nature of light is Compton scattering. This experiment involves X-rays colliding with a target material, resulting in a change in the wavelength of the scattered X-rays.

1. **Incident Photon:** An X-ray photon strikes an electron in the target.
2. **Scattering:** The photon transfers some of its energy to the electron, causing the electron to recoil.
3. **Shift in Wavelength:** The scattered photon has a longer wavelength (lower energy) than the incident photon.

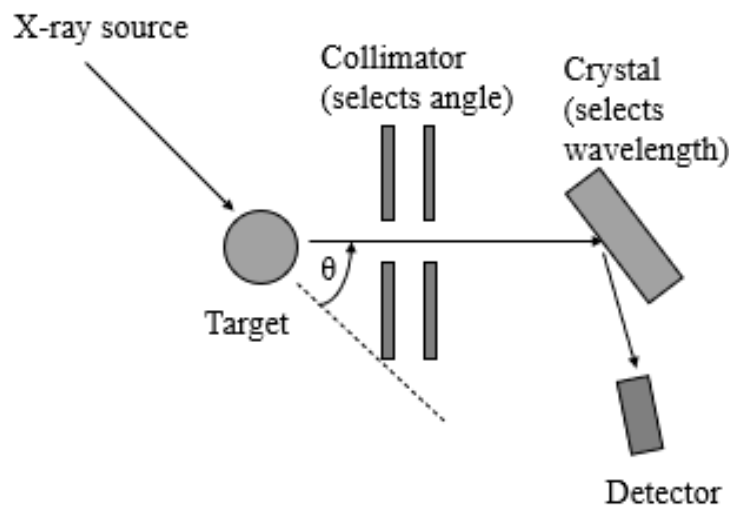


Fig 41.2: Schematic of an X-ray diffraction experiment: X-rays from an X-ray source are directed onto a target. The collimator selects the angle (θ) of the X-ray beam, and a crystal is used to select specific wavelengths of the diffracted X-rays. The intensity of the diffracted X-rays is measured by a detector.

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

Here, λ is the wavelength of the scattered photon, λ is the initial wavelength, h is Planck's constant, m_e is the electron mass, c is the speed of light, and θ is the scattering angle.

Result: peak in scattered radiation shifts to longer wavelength than source. Amount depends on θ (but not on the target material).

Classical picture: Re-radiation is at *same frequency and wavelength* as incident radiation. Change in wavelength of scattered light is completely unexpected classically

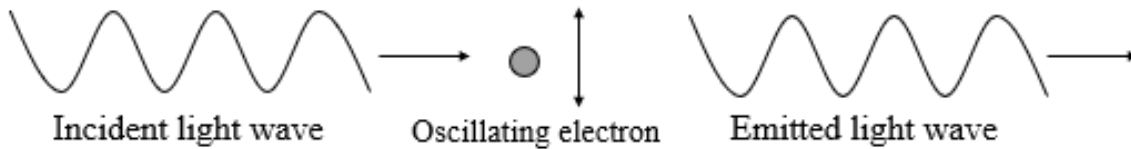


Fig 41.3: Representation of light scattering: An incident light wave interacts with an electron, causing it to oscillate. The oscillating electron then emits a light wave, resulting in scattered light.

Compton's explanation: "billiard ball" collisions between particles of light (X-ray photons) and electrons in the material.

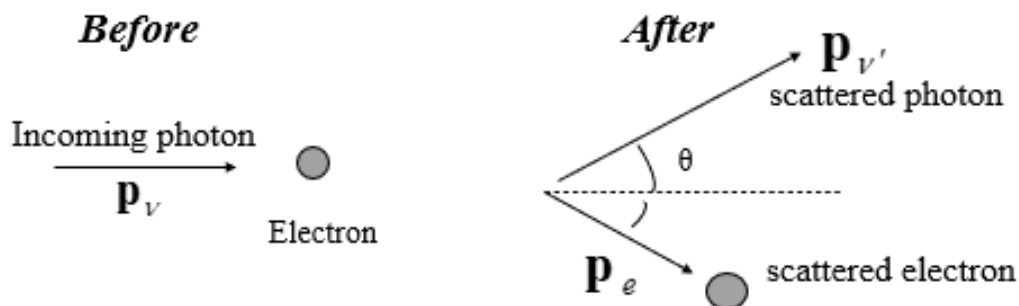


Fig 41.4: Diagram of Compton Scattering: Before the collision, an incoming photon with momentum p_v interacts with a stationary electron. After the collision, the photon is scattered with momentum $p_{v'}$, and the electron is scattered with momentum p_e in a different direction.

Conservation of energy

$$h\nu + m_e c^2 = h\nu' + (p_e^2 c^2 + m_e^2 c^4)^{1/2}$$

Conservation of momentum

$$\mathbf{p}_\nu = \mathbf{p}_{\nu'} + \mathbf{p}_e$$

$$\frac{h}{\lambda} \hat{\mathbf{i}} = \mathbf{p}_{\nu'} + \mathbf{p}_e$$

From this Compton derived the change in wavelength

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

$$= \lambda_c (1 - \cos \theta) \geq 0$$

$$\lambda_c = \text{Compton wavelength} = \frac{h}{m_e c} = 2.4 \times 10^{-12} \text{ m}$$

Relativity and Photon Properties

Relativity also plays a crucial role in understanding photons. According to Einstein's theory of relativity, the energy (E) and momentum (p) of a photon are related.

Relativistic Energy-Momentum Relation:

$$E^2 = (pc)^2 + (m_e c^2)^2$$

For a photon, since its rest mass (m) is zero: $E = pc$ and $c = \lambda \nu$

Given that the momentum p is related to the wavelength (λ) of the photon:

$$p = \frac{h}{\lambda} = \frac{h\nu}{c}$$

$$E = \hbar \omega \quad p = \hbar k \quad \omega = 2\pi \nu$$

$$k = \frac{2\pi}{\lambda} \quad \hbar \equiv \frac{h}{2\pi}$$

This indicates that the energy of a photon is inversely proportional to its wavelength.

The dual nature of light and matter as both particles and waves is a cornerstone of modern physics. Experiments like the photoelectric effect and Compton scattering provide substantial evidence for the particle nature of light, leading to the understanding that light can behave as discrete packets of energy called photons.

The exploration of these concepts not only enriches our understanding of physics but also offers profound insights into the nature of reality. This duality challenges our everyday notions and encourages us to view the universe through a more nuanced lens.

By adhering to the principles of logic and experimentation, physics continues to unveil the intricate and often surprising fabric of the cosmos.

The Reality of Wave-Particle Duality: Understanding Light and Matter

This book explores the complex nature of light and matter, focusing on the intriguing concept of wave-particle duality. The discussion is grounded in physics, a field characterized by its logical approach and reliance on experiments to prove or disprove theories.

Wave-Particle Duality

The concept of wave-particle duality explains that light and matter exhibit both wave-like and particle-like properties. This duality is essential for understanding phenomena such as interference and diffraction.

Interference and Diffraction

If light were only particles, we could not explain interference and diffraction. For example, when light from two sources meets, we can observe areas of constructive interference (where light intensity is doubled) and destructive interference (where light intensity is nullified). This behavior is characteristic of waves.

De Broglie Hypothesis

The concept of matter waves was first introduced by the French scientist Prince Louis de Broglie in the 1920s. De Broglie proposed that, like light, particles of matter also exhibit wave-like properties. This idea is encapsulated in the de Broglie hypothesis, which states that the wavelength (λ) of a particle is given by:

$$\lambda = \frac{h}{p}$$

Here, h is Planck's constant (6.63×10^{-34} Js) and p is the particle's momentum ($p=mv$, where m is mass and v is velocity).

Example 1: Cricket Ball

Let's calculate the de Broglie wavelength of a cricket ball with a mass of 0.5 kg moving at 2 m/s:

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{0.5 \times 2} = 6.63 \times 10^{-34} \text{ m}$$

This wavelength (10^{-34}) is *extremely small* compared to the size of the cricket ball ($\sim 10^{-2}$ m). For macroscopic objects like cricket balls, the wave-like behavior is negligible.

Example 2: Electron with Kinetic Energy of 50 eV

Now consider an electron with a kinetic energy of 50 eV. The relationship between momentum and kinetic energy is:

$$K = \frac{p^2}{2m_e} = \frac{h^2}{2m_e \lambda^2} \Rightarrow \lambda = \frac{h}{\sqrt{2m_e K}} = 1.7 \times 10^{-10} \text{ m}$$

This wavelength (1.7×10^{-10} m) is comparable to the size of an atom ($\sim 10^{-10}$ m). Therefore, for electrons, the wave nature becomes significant and can lead to observable phenomena, such as diffraction and interference.

Key Takeaway

- For macroscopic objects (e.g., cricket balls), the de Broglie wavelength is negligibly small, and wave-like properties are irrelevant.
- For microscopic particles (e.g., electrons), the de Broglie wavelength is significant and comparable to atomic dimensions, making their wave-like behavior observable.

Electron Diffraction Experiment (1927)

This experiment demonstrates how electrons behave like waves and can exhibit interference patterns, a key idea in quantum mechanics.

The Setup:

- Electrons are accelerated using a voltage, which determines their energy.
- These electrons are directed onto a crystal, which acts as a reflective surface.
- The electrons collide with the crystal and are reflected at various angles.

Key Concept:

The behavior of the reflected electrons is analysed by measuring the **intensity** of the reflected waves at different angles. To understand this, we consider the **path difference** between the incoming and reflected electron waves.

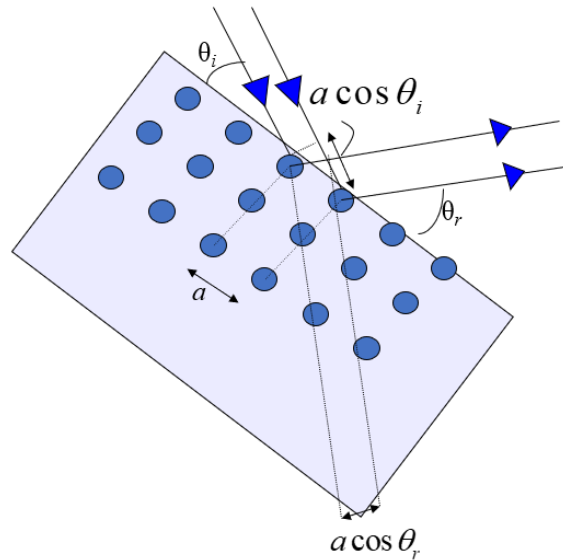


Fig 41.5: Diagram illustrating Bragg's diffraction from a crystal lattice. Incident waves at an angle θ_i interact with lattice planes separated by distance a . The path difference between reflected waves at an angle θ_r is determined by the geometry, enabling constructive interference when the Bragg condition $2a\cos\theta=n\lambda$ is satisfied, where n is an integer and λ is the wavelength of the incident wave.

Path Difference:

The path difference between the incoming and reflected waves is given by:

$$\Delta\text{Path} = a(\cos\theta_r - \cos\theta_i)$$

Where:

- a is the distance between atomic layers in the crystal.
- θ_i is the angle of incidence.
- θ_r is the angle of reflection.

Interference:

- When the path difference (ΔPath) is an integer multiple of the wavelength (λ), i.e., $\Delta\text{Path} = n\lambda$ (where $n=1,2,3,\dots$), **constructive interference** occurs, and the intensity of the reflected electrons is maximized.
- When the path difference is a half-integer multiple of the wavelength ($\Delta\text{Path} = (n+0.5)\lambda$), **destructive interference** occurs, and the intensity is minimized.

Observations:

- The wavelength of electrons is given by the **de Broglie equation**: $\lambda = h/p = h/mv$. Here, h is Planck's constant, m is the mass of the electron, and v is its velocity.
- By changing the accelerating voltage, the energy of the electrons (and hence their wavelength) can be controlled.
- When the electrons reflect off the crystal, their interference creates a **diffraction pattern**, which depends on the wavelength of the electrons and the angles of incidence and reflection.

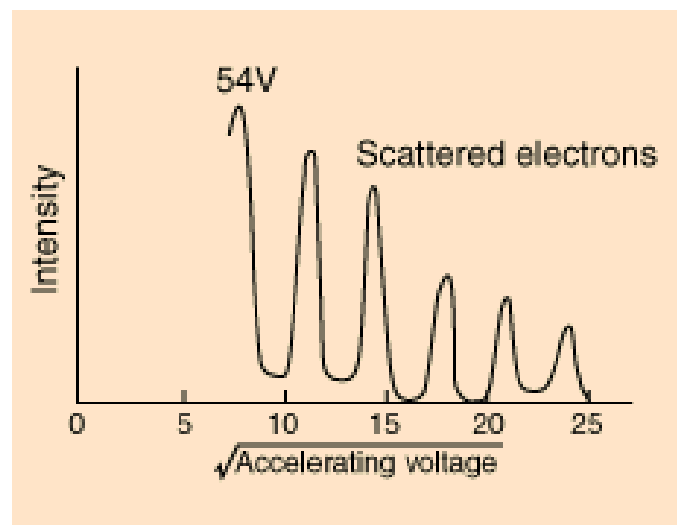


Fig 41.6: Graph showing the variation of intensity of scattered electrons as a function of the square root of the accelerating voltage. The prominent peak at 54V corresponds to resonance, indicative of inelastic scattering, where electrons lose energy by exciting atoms. This behavior is associated with electron diffraction and the quantized nature of energy exchange in atomic systems.

Evidence for Wave Nature:

- If we fix the observation angle (θ_r) and vary the accelerating voltage, the intensity of the reflected electrons shows oscillations: it increases, decreases, and increases again.

- These oscillations indicate the presence of an interference pattern, proving that electrons behave like waves.

Double-Slit Experiment with Particles

This experiment, originally done with light, demonstrates the wave-like behavior of particles such as electrons and helium atoms through interference patterns.

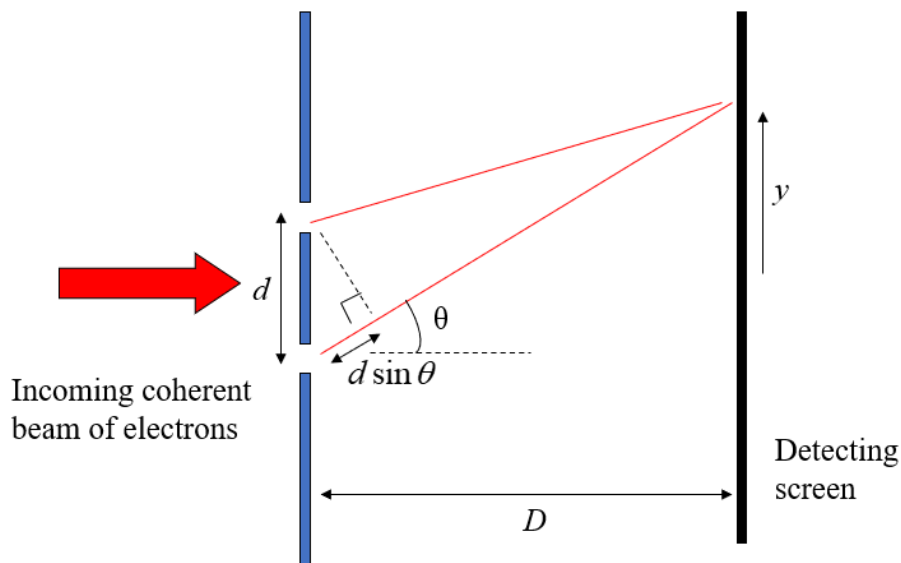


Fig 41.7: Schematic of a double-slit electron diffraction experiment. A coherent beam of electrons passes through two slits separated by distance d , producing an interference pattern on the detecting screen placed at a distance D . The angle θ relates to the path difference $d \sin \theta$, which determines the constructive and destructive interference on the screen at position y .

The Setup:

- A beam of particles (e.g., electrons or helium atoms) is directed toward a plate with **two very small holes (slits)**.
- After passing through the slits, the particles hit a screen, where their positions are recorded using a detector.
- Since particles like electrons are charged, the detector can measure their positions by counting their charges.

Key Idea: Path Difference and Interference

When particles pass through the slits, they follow different paths. The **path difference** between the two paths is:

$$\Delta \text{Path} = d \sin \theta$$

Where:

- d is the distance between the two slits.
- θ is the angle of deflection.

If particles behave like waves, interference occurs based on the following conditions:

- **Constructive interference** (bright spots): $\Delta \text{Path} = n\lambda$ ($n=1,2,3,\dots$).
- **Destructive interference** (dark spots): $\Delta \text{Path} = (n+0.5)\lambda$.

This creates a pattern of alternating bright and dark regions on the screen, indicating where more or fewer particles are detected.

Simplifying the Math:

When the screen is far away from the slits, θ is very small, so $\sin\theta \approx \theta$. The angular position of the interference maxima is:

$$\begin{aligned}\theta &\approx \frac{n\lambda}{d} \\ \Rightarrow \Delta\theta &\approx \frac{\lambda}{d}\end{aligned}$$

The position of the maxima on the screen (y) is related to θ by:

$$y = D \tan \theta \approx D\theta$$

Substituting for θ , we get:

$$\begin{aligned}\Delta y &\approx D\Delta\theta \\ \Rightarrow \Delta y &= \frac{\lambda D}{d}\end{aligned}$$

Where:

- D is the distance from the slits to the screen.
- λ is the wavelength of the particles.

- n indicates the order of the maxima.

Example: Helium Atoms

For helium atoms:

- Distance between the slits (d): $8\text{ }\mu\text{m}$.
- Distance to the screen (D): 64 cm .
- De Broglie wavelength (λ): 10^{-10} .

Using $\Delta y = \lambda D / d$, the calculated spacing between maxima is about $8.2\text{ }\mu\text{m}$, which matches experimental observations ($8.2\text{--}8.4\text{ }\mu\text{m}$). This consistency confirms that helium atoms, like electrons and light, exhibit interference.

Universal Wave Behavior:

This experiment shows that **wave-like interference** is a universal phenomenon:

- All particles, regardless of size (e.g., electrons, helium atoms, or heavier particles), have a wavelength determined by the **de Broglie relation**: $\lambda = h/p = h/mv$ (Where p is the momentum, m is mass, and v is velocity.)
- Even larger particles exhibit interference, though their wavelengths are much smaller.

A Strange Observation: Single-Particle Interference

The most surprising result is that **a single particle can interfere with itself**:

- If only one particle is sent at a time, an interference pattern still forms after many repetitions.
- If one slit is blocked or if an attempt is made to measure which slit the particle passes through, the interference pattern disappears.

This suggests that particles behave as waves when unobserved but as particles when measured.

Observation itself changes their behavior, a phenomenon that highlights the mysterious nature of quantum mechanics.

Interference and Observation: A Strange Connection

Modern experiments show that a single particle, whether it's a photon, electron, or helium atom, can **interfere with itself** as a wave. However, this wave-like behavior disappears the moment

we try to **observe or measure the particle's path**. This phenomenon might seem strange, but it reveals a fundamental principle of quantum mechanics: **the act of measurement changes the state of a system**.

Why Does Measurement Change the System?

Let's break it down with an analogy:

Imagine you're blindfolded and trying to locate two flowers or balls (let's say lilies) in front of you. You don't know exactly where they are, so you stretch out your hand or use a stick to feel around. When you touch a flower, you change its position or bend it slightly. No matter how gently you touch it, your action will alter its original state, even if only a little. The same principle applies in the quantum world.

Now think of the particles:

- If you want to determine the position of an electron, you need to interact with it somehow. For instance, you might use a **photon** (a particle of light) to detect it.
- However, photons carry energy and momentum. When a photon interacts with the electron to "observe" it, this interaction transfers some energy or momentum to the electron, changing its behavior.

Heisenberg's Uncertainty Principle

In the 1920s, **Werner Heisenberg** introduced a fundamental principle in quantum mechanics known as the **Uncertainty Principle**. This principle states:

- **You cannot precisely know both the position and the momentum (speed $\times$ mass) of a particle at the same time.**
 - If you pinpoint the position of a particle, its momentum becomes uncertain.
 - If you know the momentum exactly, you lose information about its position.

This isn't because of a limitation in our measurement tools; it's a fundamental property of nature itself.

Why Does This Happen?

Imagine an experiment where an electron passes through a tiny hole (a slit) and hits a screen with a detector:

1. When the electron passes through the hole, its position becomes relatively well-defined (we know it's somewhere within the hole).
2. However, as a result of this confinement, the electron's **momentum in the perpendicular direction (y-direction)** becomes uncertain. This is because confining the particle's position increases the range of possible momenta, as explained by quantum mechanics.

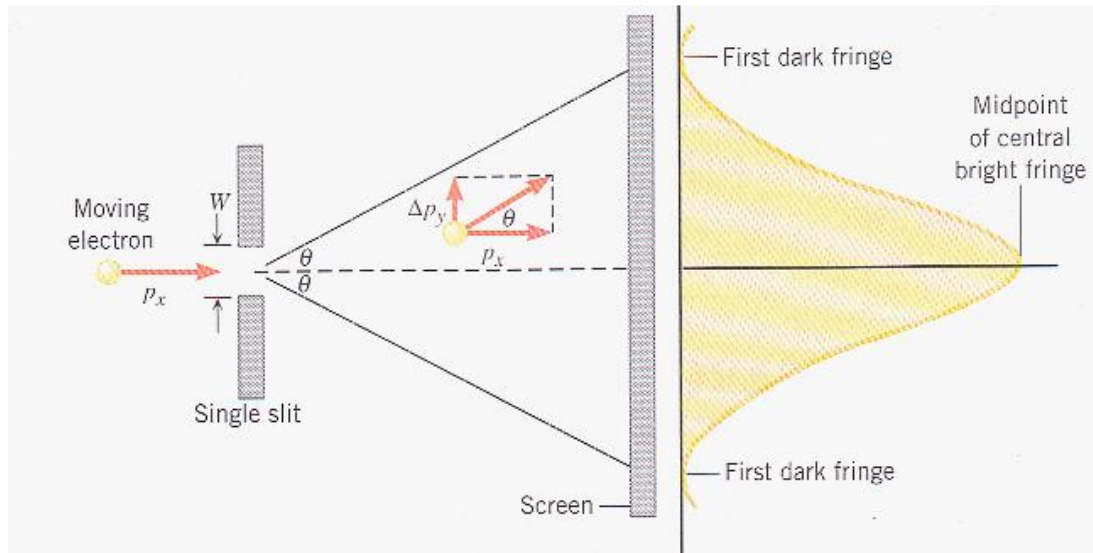


Fig 41.8: Diagram illustrating single-slit diffraction of moving electrons. An electron with momentum p_x passes through a slit of width W , resulting in an angular spread θ due to diffraction. The uncertainty in transverse momentum, Δp_y , corresponds to the spread of the interference pattern on the screen, which includes a central bright fringe and adjacent dark fringes. This setup demonstrates the uncertainty principle.

$$\sin \theta = \frac{\lambda}{W} \Rightarrow \theta \approx \frac{\lambda}{W}$$

But, on the other hand,

$$\tan \theta = \frac{\Delta p_y}{p_x} \Rightarrow \theta \approx \frac{\Delta p_y}{p_x}$$

$$\frac{\Delta p_y}{h / \lambda} = \frac{\lambda}{W}$$

$W = \text{uncertainty in } y \text{ position} = \Delta y$

$$\therefore \Delta p_y \Delta y \approx h$$

Here:

- Δy : Uncertainty in the position of the particle in the y-direction.

- Δp_y : Uncertainty in the particle's momentum in the y-direction.
- h : Planck's constant, a tiny but universal number.

This equation tells us that the more precisely we know one value (e.g., Δy), the less precisely we can know the other (Δp_y).

Heisenberg's Microscope

This is an **imaginary thought experiment** designed to illustrate the **Uncertainty Principle** in action.

1. **The Setup:** Imagine using a microscope to observe a particle. A photon (light particle) comes from a source, hits the particle, and scatters. The scattered photon then reaches the microscope lens and forms an image of the particle on a screen.
2. **The Challenge:**
 - To locate the particle's position accurately (Δy), we need photons with a very small wavelength (λ) because shorter wavelengths provide better resolution (the ability to distinguish small details).
 - However, shorter wavelengths mean photons carry more energy and momentum (since $p = h / \lambda$, where h is Planck's constant). When such a high-energy photon collides with the particle, it disturbs the particle's momentum significantly (Δp_y).

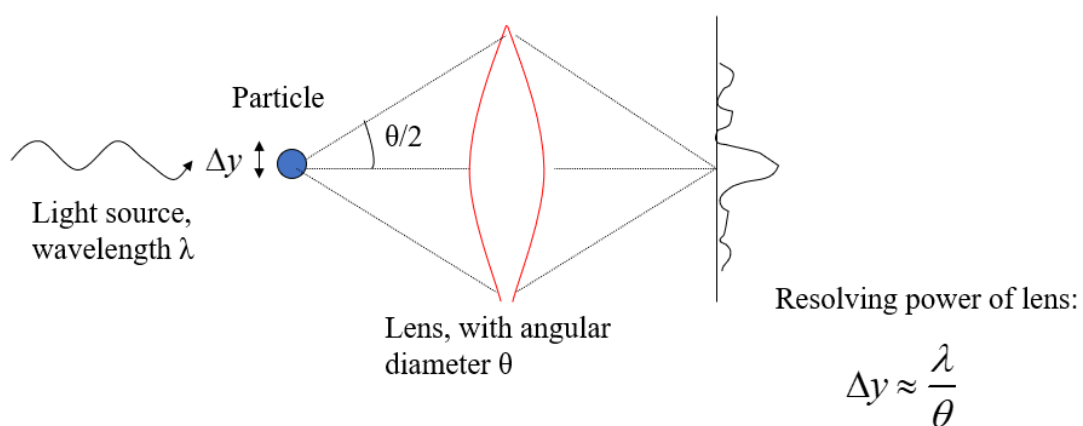


Fig 41.9: Schematic representation of Heisenberg's microscope. A light source emits photons with wavelength λ , which scatter off a particle. The scattered light is focused by a lens with angular aperture θ . The uncertainty in the particle's position, Δy , is related to the resolving power of the lens: $\Delta y \approx \lambda \theta$. This demonstrates the relationship between measurement precision and the wavelength of the probing light.

Photons transfer momentum to the particle when they scatter.

Magnitude of p is the same before and after the collision.

Uncertainty in **photon** y-momentum
= Uncertainty in **particle** y-momentum

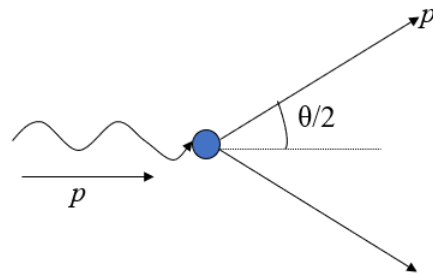
$$-p \sin(\theta/2) \leq p_y \leq p \sin(\theta/2)$$

Small angle approximation

$$\Delta p_y = 2p \sin(\theta/2) \approx p\theta$$

de Broglie relation gives $p = h/\lambda$ and so $\Delta p_y \approx \frac{h\theta}{\lambda}$

From before $\Delta y \approx \frac{\lambda}{\theta}$ hence $\Delta p_y \Delta y \approx h$



Key Insights

1. **Uncertainty Is Fundamental:** Even if we don't actively measure the particle, its position and momentum inherently have uncertainties. This is not due to measurement errors but is a property of nature.
2. **Energy and Time:** A similar uncertainty exists between **energy** (ΔE) and **time** (Δt):

$$\Delta E \Delta t \geq \hbar / 2$$

This means that the more precisely we measure energy, the less precisely we know the time it's associated with, and vice versa.

Why Is This Important?

1. The Uncertainty Principle laid the foundation of **quantum mechanics**, which replaced classical mechanics as the framework for understanding the behavior of particles at very small scales.

2. Quantum mechanics is essential for modern technologies:

- **Transistors** (used in computers and smartphones)
- **Semiconductors**
- **Lasers**
- And much more.

Without the development of quantum mechanics, much of the technology we rely on today would not exist.

Lecture# 42

Introduction to Quantum Mechanics**Introduction**

Quantum mechanics is a branch of physics that studies the behavior of atoms and particles smaller than atoms. It explains phenomena that classical physics, which works well for larger objects, cannot describe. This chapter will introduce the need for quantum mechanics, its fundamental concepts, and its implications.

Scale of the Universe: Big and Small

Let us first understand what we mean by 'big' and 'small.' Humans are about 1–2 meters tall. The Earth is roughly 10^7 meters across, while our galaxy is 10^{21} meters across. On the smaller side, cells are about 10^{-5} meters, DNA molecules 10^{-6} meters, and atoms 10^{-10} meters. At the atomic scale and smaller, the rules of classical physics no longer apply.

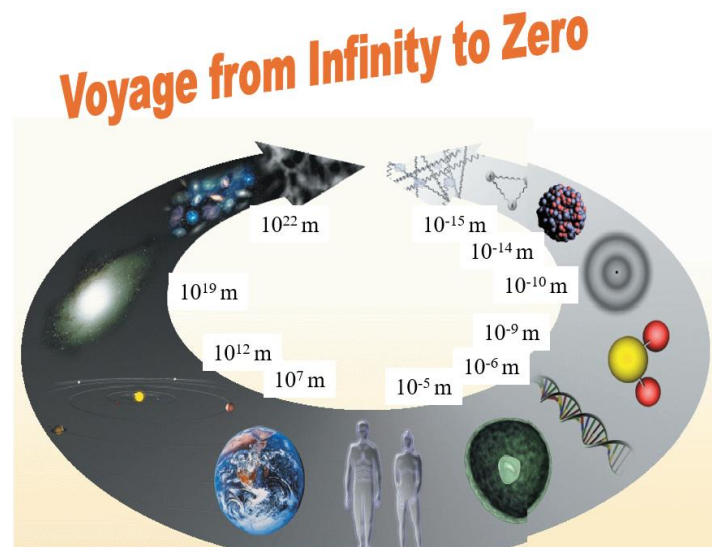


Fig 42.1: A journey through the universe, from the largest scales like galaxies (10^{22} m) to the tiniest particles (10^{-15} m), showing how everything is connected across different sizes.

Quantization in Quantum Mechanics

Quantum mechanics is based on the idea that energy comes in fixed packets or bundles, which are called *quanta*. Imagine you are climbing a ladder: you must climb step by step.

You can't be at a height of 1.5 steps or 2.3 steps, it's either one step, two steps, etc. Similarly, in quantum mechanics, energy levels are like steps. You can only gain or lose energy in specific amounts, not in between.

For example, when an electron in an atom absorbs energy, it can only absorb certain quantities of energy—like moving up specific steps on the ladder. The electron cannot exist in a state with an energy level in between these steps. When it drops back down, it releases energy in the same fixed quantity, in discrete packets (quanta).

Similarly, molecules, like water (H₂O) or carbon dioxide (CO₂), also have energy in specific quantities. For instance, a carbon dioxide molecule cannot oscillate with an energy of 0.23 units. Its energy must be in certain fixed amounts, like 1 unit, 2 units, or 3 units.

Quantum mechanics tells us that energy is quantized, meaning it exists only in specific, discrete amounts, not in any arbitrary value.

Why Do We Need Quantum Mechanics?

Classical physics, including Newton's laws of motion, Maxwell's equations for electromagnetism, and thermodynamics, successfully explained most phenomena by the 19th century. However, three significant problems remained unsolved:

1. **Blackbody Radiation:** Heated objects emit electromagnetic radiation. Classical physics predicted an incorrect distribution of energy, known as the ultraviolet catastrophe.
2. **Photoelectric Effect:** Light shining on metal ejects electrons, but only if the light's frequency exceeds a certain threshold. This could not be explained by classical theories.
3. **Atomic Spectra:** Atoms emit light at specific frequencies, forming a discrete spectrum that classical physics failed to explain.

Planck's Hypothesis

In 1900, Max Planck proposed that energy is quantized and comes in packets called 'quanta.' The energy (E) of a quantum is proportional to its frequency (ν):

$$E = h\nu$$

where h is Planck's constant (6.626×10^{-34} J·s). This idea solved the blackbody radiation problem by correctly predicting the energy distribution.

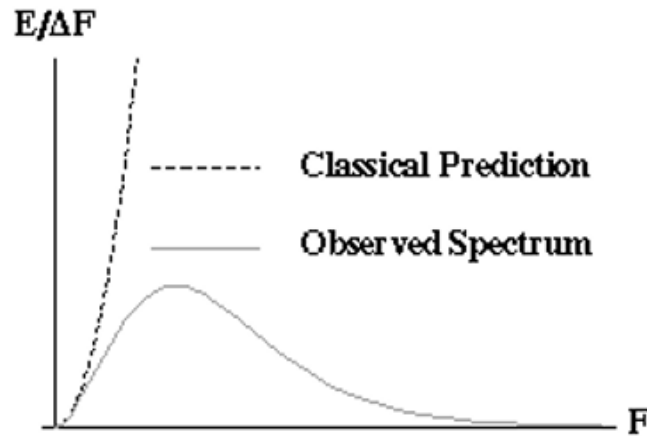


Fig 42.2: Comparison of the classical prediction and the observed spectrum, showing how classical physics fails to explain the actual behavior at higher frequencies, leading to the ultraviolet catastrophe

This hypothesis resolved a significant issue in classical physics concerning blackbody radiation. Classical theories predicted infinite energy at high frequencies, which did not align with experimental data. Planck's quantization of energy matched observations, explaining why energy levels are distributed as they are and laying the foundation for quantum mechanics.

Einstein and the Photoelectric Effect

In 1905, Albert Einstein expanded on Planck's idea, proposing that light itself is made of quanta, called photons. Each photon's energy is given by $E = h\nu$. The photoelectric effect showed that electrons are ejected only if the photon's energy exceeds the material's threshold energy.

Einstein's contribution to the photoelectric effect explained how light interacts with matter on a quantum level. He proposed that light is composed of discrete packets of energy called photons, building on Planck's earlier ideas about quantization.

The photoelectric effect occurs when light shines on a metal surface and ejects electrons. Classical physics predicted that the intensity of light (its brightness) should determine whether electrons are emitted, but this was incorrect. Instead, Einstein showed that only the light's frequency mattered. If the frequency of the light was above a certain threshold, photons carried enough energy ($E = h\nu$) to eject electrons from the metal. If the frequency was too low, no electrons were emitted, regardless of intensity.

This insight not only explained the photoelectric effect but also demonstrated the particle-like nature of light, a key concept in quantum mechanics. It earned Einstein the Nobel Prize in 1921 and helped establish the foundation of quantum theory.

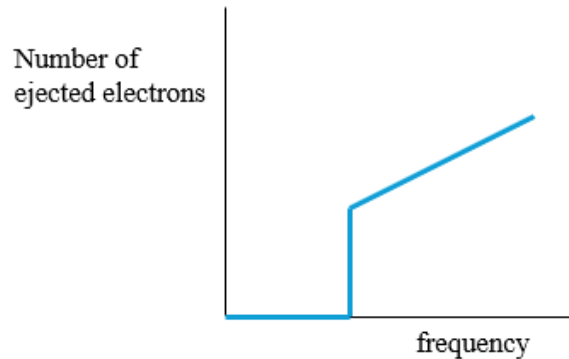


Fig 42.3: The graph shows how the number of ejected electrons depends on the frequency of light, with no electrons ejected below a threshold frequency and an increase above it, demonstrating the photoelectric effect

Mathematically,

Energy: $E = pc = h\nu$ (and $c = \lambda \nu$)

Momentum: $p = \frac{h}{\lambda}$

Also commonly written as:

$E = \hbar \omega$ where $\omega = 2\pi\nu$

$p = \hbar k$ where $k = \frac{2\pi}{\lambda}$, $\hbar \equiv \frac{h}{2\pi}$

Wave-Particle Duality

Louis de Broglie proposed in 1923 that matter, like light, has both wave and particle properties. The wavelength (λ) of a particle is related to its momentum (p):

de Broglie relation $\lambda = \frac{h}{p}$ ← Planck's constant
 de Broglie wavelength $h = 6.63 \times 10^{-34} \text{ Js}$

Particles of momentum $p = mv$ have associated

$$\text{wavelength, } \lambda = \frac{h}{mv}.$$

This wave-particle duality was confirmed experimentally in the Davisson-Germer experiment, which showed that electrons produce interference patterns like waves.

Wave-Particle Duality and Bohr's Atomic Model

Matter can behave both as particles and waves, a concept known as *wave-particle duality*.

Niels Bohr used this idea to take the first step toward solving the mystery of the atom.

However, his solution was not complete at the time, but it marked a significant breakthrough.

The Problem with Classical Physics

In classical physics, if an electron orbits a proton, its velocity is constantly changing direction. This means the electron is accelerating. According to Maxwell's theory, any accelerating charge emits electromagnetic radiation, losing energy in the process. As the electron loses energy, it would spiral into the nucleus, causing the atom to collapse. Classical physics could not explain why this didn't happen, leaving the structure of the atom unresolved.

Bohr's Solution

In 1923, Bohr proposed that the electron behaves like a wave and can only orbit the nucleus if an integer number of *de Broglie waves* fit perfectly within its orbit, as shown in fig 42.4.

Imagine the electron as a wave wrapped around the nucleus in a circular shape. If the wave doesn't "close" properly, the orbit is not allowed.

When the wave forms a closed loop, the atom becomes stable, and the electron can stay in that orbit without losing energy. This idea introduced the concept of quantized orbits, where only certain energy levels are allowed for the electron.

This insight was a major step toward understanding atomic structure and laid the foundation for modern quantum mechanics.

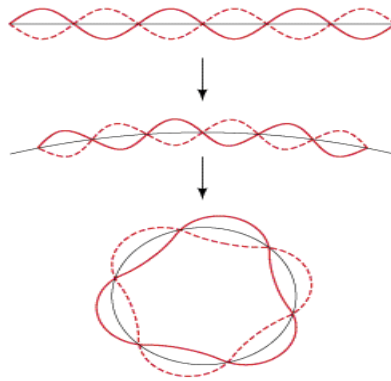


Fig 42.4: This figure illustrates a visualization from the Bohr model of the atom. It shows how an electron's wave can be represented. At the top, the electron behaves like a wave along a straight path. As we move down, the wave shortens and becomes more complex, eventually forming a closed loop, representing the electron orbiting the nucleus in a circular motion. This helps explain the quantized nature of electron orbits in the Bohr model.

Heisenberg's Uncertainty Principle

Imagine measuring the length of an object. Even with the best tools, errors can arise due to external factors like temperature or imperfections in the instrument. These errors can be minimized but never eliminated entirely. Similarly, when measuring tiny particles like electrons, there is always an inherent uncertainty.

Werner Heisenberg stated that it is impossible to measure both the position (Δx) and momentum (Δp) of a particle with perfect accuracy. The principle is expressed as:

$$\Delta x \Delta p_x \geq \hbar / 2$$

$$\Delta y \Delta p_y \geq \hbar / 2$$

$$\Delta z \Delta p_z \geq \hbar / 2$$

where $\hbar$ is the reduced Planck's constant. This principle challenges classical concepts of precision and determinism.

Exploring the Uncertainty Principle in a Confined System

Imagine a single particle, like a ball, trapped between two walls, as shown in fig 42.5. The ball moves back and forth, colliding with the walls endlessly because there is no friction or energy loss in this ideal setup. In classical physics, we might imagine stopping the ball completely, making it stationary, which means its velocity and momentum would be zero.

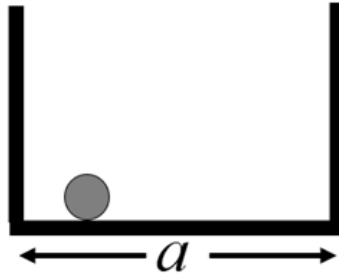


Fig 42.5: This figure represents the particle in a well problem in quantum mechanics. The box has width a , and a particle confined within this space. The particle is restricted to move only inside the box, which illustrates the concept of quantized energy levels that arise due to the confinement.

However, in quantum mechanics, the **Uncertainty Principle** makes this impossible.

The Role of the Uncertainty Principle

The Uncertainty Principle states that we cannot simultaneously know both the exact position (Δx) and exact momentum (Δp) of a particle. These uncertainties are related by the inequality:

$$\Delta x \Delta p \geq \hbar / 2$$

$\therefore a$ is the length of the box

$$\Delta p a \approx \frac{\hbar}{2}$$

$$\Delta p \approx \frac{\hbar}{2a} \quad \text{eq(1)}$$

As we know

$$\text{Kinetic energy} = \frac{1}{2} m v^2$$

$$\therefore p = m v \text{ or } v = \frac{p}{m}$$

$$\text{Kinetic energy} = \frac{1}{2} m \left(\frac{p}{m}\right)^2$$

$$\text{Kinetic energy} = \frac{p^2}{2m}$$

Similarly, we can write

$$\text{Kinetic energy} = \frac{(\Delta p)^2}{2m}$$

using eq(1), we get

$$\text{Kinetic energy} \geq \frac{\hbar^2}{8ma^2}$$

Why the Ball Can't Be Stationary

In classical physics, a stationary ball has zero kinetic and potential energy, and we can pinpoint its position. But in quantum mechanics:

1. If the particle's momentum is exactly zero ($\Delta p=0$), its position becomes completely uncertain, violating the Uncertainty Principle.
2. This means the particle cannot remain perfectly still; there is always some "quantum motion" due to this fundamental limitation.

The Uncertainty Principle demonstrates that even in simple systems, like a ball confined between two walls, the behavior of particles is fundamentally different in quantum mechanics than in classical physics. Particles always exhibit a degree of uncertainty in their position and momentum, preventing perfect stillness or precision.

Application of the Uncertainty Principle to a Harmonic Oscillator

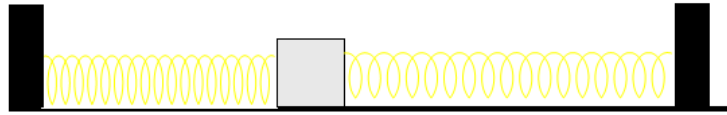


Fig 42.6: This figure illustrates a mass-spring system, a common example of a harmonic oscillator. As the spring compresses and stretches, the mass moves back and forth, demonstrating simple harmonic motion.

The Heisenberg Uncertainty Principle also applies to systems like a **harmonic oscillator**, a fundamental concept in physics. A common example of a harmonic oscillator is a mass attached to a spring as shown in fig 42.6, oscillating back and forth due to the restoring force of the spring.

Classical View of a Harmonic Oscillator

In classical physics, it is possible to envision a scenario where the mass comes to rest at the equilibrium position, eliminating all kinetic energy. In this state, the potential energy is also minimized, and the system is stationary. This view assumes that both the position and velocity of the mass can be known precisely.

Quantum View of a Harmonic Oscillator

In quantum mechanics, however, this stationary state is not possible due to the **Uncertainty Principle**:

$$\Delta x \Delta p \geq \hbar / 2$$

Here, Δx is the uncertainty in position, and Δp is the uncertainty in momentum (related to velocity).

- If we attempt to eliminate all kinetic energy, making the momentum (p) exactly zero ($\Delta p = 0$), the uncertainty in position (Δx) becomes infinite. This means the particle's position cannot be precisely determined, violating the principle.
- Conversely, confining the mass to a very specific position increases the uncertainty in its momentum, ensuring that some motion always exists.

Minimum Energy

The quantum harmonic oscillator inherently possesses a **minimum energy** also called the **zero-point energy**, even in its lowest energy state. This energy is given by:

Minimum energy (classical) = 0

Minimum energy (quantum) = $\frac{1}{2}\hbar\omega$ where $\omega = \sqrt{\frac{k}{m}}$

where ω is the angular frequency of the oscillator. The zero-point energy reflects the fact that the particle cannot be completely stationary due to the Uncertainty Principle.

In both confined systems, like a particle between two walls, and oscillatory systems, like the harmonic oscillator, quantum mechanics imposes fundamental limits on how precisely we can define a particle's position and momentum. These limits give rise to phenomena such as zero-point energy and ensure that no system can be perfectly still, regardless of classical expectations.

The Energy-Time Uncertainty Principle

The **energy-time uncertainty principle** is a fundamental concept in quantum mechanics. It is expressed as:

$$\Delta E \Delta t \geq \hbar / 2$$

where ΔE represents the uncertainty in energy measurement, and Δt represents the uncertainty in the time interval over which the energy is measured.

Explanation of the Principle

This principle implies that:

1. If we want to measure the energy of a particle with absolute certainty ($\Delta E=0$), we would require an infinitely long time interval ($\Delta t \rightarrow \infty$).
2. In reality, measurements are made over finite time intervals (Δt), resulting in an inherent uncertainty in the energy measurement (ΔE).
3. The product of these uncertainties is always greater than or equal to $\hbar/2$, highlighting a fundamental limit in nature.

Implications of the Energy-Time Uncertainty Principle

This principle explains several quantum phenomena that cannot be understood using classical physics.

Quantum Tunnelling

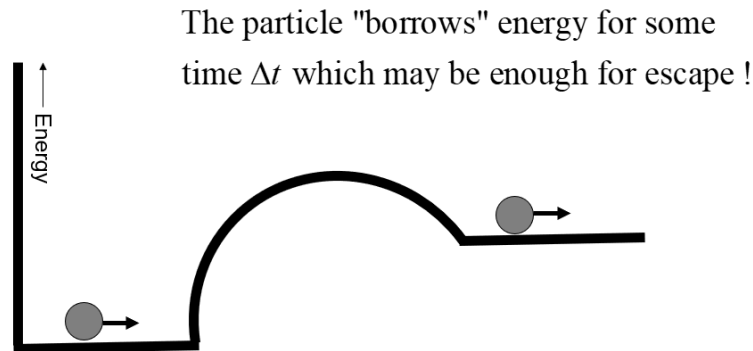


Fig 42.7: Illustration of quantum tunneling: A particle 'borrows' energy temporarily to escape a potential barrier it couldn't classically overcome.

One significant consequence is **quantum tunnelling**, where particles can "tunnel" through energy barriers, as shown in fig 42.7, that would be insurmountable according to classical physics.

- For a very short time (Δt), the particle can "borrow" energy (ΔE) from the uncertainty relationship, allowing it to cross the barrier.
- This process does not violate the conservation of energy because the borrowed energy is "returned" after the particle has tunnelled.

Real-World Applications

Quantum tunneling has profound implications, such as:

- **Nuclear Fusion in Stars:** Tunneling enables protons in the cores of stars to overcome the Coulomb barrier, leading to fusion reactions that power stars.
- **Radioactive Decay:** Tunneling allows particles trapped within a nucleus to escape, resulting in radioactive decay.

The energy-time uncertainty principle reveals a fascinating interplay between energy and time in the quantum world. It not only imposes limits on precision but also enables processes like quantum tunneling, which have critical applications in explaining natural phenomena and technological advancements.

Quantum States and Wave Functions

Probability and Two-State Systems in Quantum Mechanics

Quantum mechanics fundamentally differs from classical mechanics by introducing a probabilistic understanding of particles. The likelihood of finding a particle at a specific location or in a particular state is described by a **probability distribution**, a concept that challenges classical notions of certainty.

In classical mechanics, we can predict the exact state of a system if all initial conditions are known. However, in quantum mechanics this is not possible.

Two-State Systems

A two-state system is one where a particle, such as an electron, can exist in only two possible states. For example, the spin of an electron can be either **up** or **down**, analogous to a coin having either heads or tails.

Representation of States

To describe the spin states:

$|\uparrow\rangle$ is a Dirac "ket" that denotes an electron with spin up.

$|\downarrow\rangle$ is a Dirac "ket" that denotes an electron with spin down.

Example Calculation

Suppose the state of the electron is given by:

$$|\Psi\rangle = \sqrt{\frac{2}{3}}|\uparrow\rangle + \sqrt{\frac{1}{3}}|\downarrow\rangle \quad \because \text{denotes an electron with Dirac ket } |\rangle$$

$$P(\uparrow) = \frac{2}{3} \text{ and } P(\downarrow) = \frac{1}{3}$$

Where

$P(\uparrow)$ means probability of having an electron in spin-up state

$P(\downarrow)$ means probability of having an electron in spin-up state

More generally a state can also be written as : $|\Psi\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle$

with probabilities as $P(\uparrow) = |c_1|^2$ and $P(\downarrow) = |c_2|^2$.

$|\Psi\rangle = c_1|\uparrow\rangle + c_2|\downarrow\rangle$ is called a *quantum state* c_1 and c_2 are called *quantum amplitudes*.

If a_1 and a_2 are the amplitudes of the two possibilities for a particular event to take place, then amplitude for the total event is:

$$A = a_1 + a_2.$$

The probability for the event to occur is given by

$$P = |A|^2 = |a_1 + a_2|^2$$

In daily experience we add *probabilities*:

$$P = P_1 + P_2$$

But in quantum mechanics we add *amplitudes*:

$$\begin{aligned} P = |a_1 + a_2|^2 &= a_1^* a_1 + a_2^* a_2 + a_1^* a_2 + a_2^* a_1 \\ &= P_1 + P_2 + a_1^* a_2 + a_2^* a_1 \end{aligned}$$

Two-state systems, such as the spin of an electron, are fundamental examples that demonstrate how quantum states can be probabilistically distributed, leading to measurable outcomes when observed.

The Schrödinger Equation

- In 1926 Schrödinger proposed a wave equation that describes the way matter waves change in space and time.
- Schrödinger's wave equation is a key element in quantum mechanics.
- Schrödinger's wave equation is solved for the *wave function*, Ψ
- The wavefunction $\Psi(x, t)$ of a particle is the amplitude to be at position x at time t .
- The probability of finding the particle at position x between x and $x + dx$ (at time t) is $|\Psi(x, t)|^2 dx$.
- $\Psi(x, t)$ is determined by solving the "Schrodinger equation".

What does it mean?

- Whole subject in itself
- We will look at how can be interpreted
- ψ is the amplitude of the matter wave.
- The end of certainty!

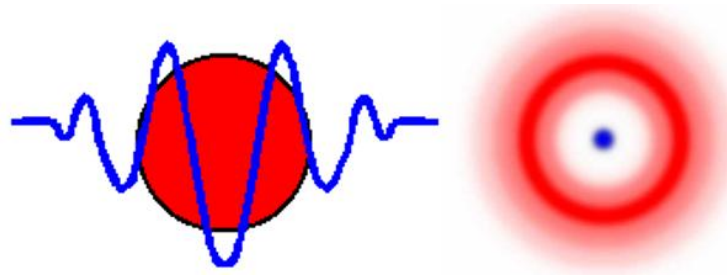


Fig 42.8: Visualization of quantum behavior: On the left, a particle exhibits wave-like oscillations; on the right, the probability distribution of its location is shown as a red intensity pattern

Double-

Slit Experiment: Evidence of Wave-Particle Duality

The Experiment Setup

In the famous double-slit experiment, electrons are passed through two small slits in a metal plate, and their positions are observed on a screen behind the slits. The surprising outcome is the formation of an interference pattern, similar to what is seen when light passes through slits. This result provides direct evidence that electrons—and matter in general—exhibit wave-like behavior.

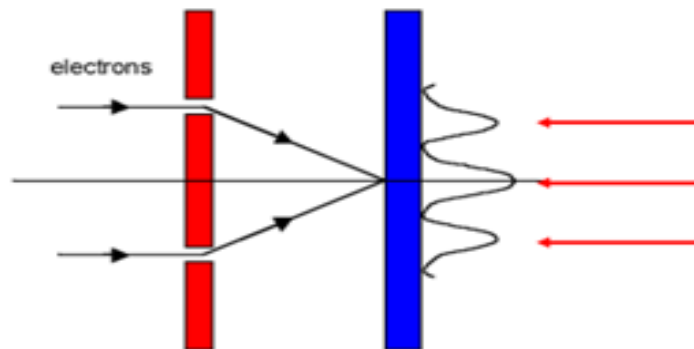


Fig 42.9: Electron double-slit experiment: Electrons pass through two slits (red), creating an interference pattern (blue) that shows their wave-like nature.

Observing the Interference Pattern

- On the screen, the interference pattern appears as alternating regions of high intensity (bright spots) and low intensity (dark spots).

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- The high-intensity regions correspond to areas where the value of $|\psi|^2$, the square of the wave function, is large, indicating a higher probability of finding an electron.
- Conversely, the low-intensity regions correspond to areas where $|\psi|^2$ is small, meaning the probability of finding an electron is low.

If a detector is placed on the screen and moved across the interference pattern, it will record fluctuating probabilities: high at some points and low at others. This pattern is evidence that the electrons behave like waves and interfere with each other.

The Wave Nature of Electrons

- The double-slit experiment confirms the wave-particle duality of matter, a cornerstone of quantum mechanics.
- Electrons, which are particles, also behave like waves. These waves can interfere constructively (creating maxima) or destructively (creating minima).

Self-Interference and the de Broglie Wavelength

- When the number of electrons passing through the slits is reduced to the point that only one electron passes through at a time, the interference pattern still forms over time.
- This suggests that an individual electron behaves as a wave and interferes with itself as it passes through both slits simultaneously.
- The wave-like behavior is determined by the electron's de Broglie wavelength, which is related to its momentum:

Wave-Particle Duality and Measurement

If we attempt to measure which slit the electron passes through:

- The act of measurement disrupts the wave function.
- The interference pattern disappears, and the electrons behave purely as particles.

This highlights a fundamental principle of quantum mechanics: the outcome depends on whether or not we observe the system.

Key Takeaways

1. Wave-Particle Duality: Electrons and other particles behave as both particles and waves.
2. Interference Pattern: The wave nature of electrons causes them to interfere, creating observable patterns.
3. Measurement Effect: Attempting to observe the particle's path destroys the interference pattern, emphasizing the probabilistic nature of quantum mechanics.
4. de Broglie Wavelength: The wavelength of matter waves depends on the particle's momentum and determines its wave-like properties.

This experiment profoundly altered our understanding of the nature of matter and laid the foundation for the principles of quantum mechanics.

Understanding the Electron Cloud and Probability Distribution in the Atom

The Role of the Wave Function in Locating Electrons

In quantum mechanics, the **wave function** (ψ) provides information about the probability distribution of an electron in an atom. While it is impossible to pinpoint the exact location of an electron due to the **Heisenberg Uncertainty Principle**, the wave function helps us determine where the electron is most likely to be found.

Electron Probability Distribution in a Hydrogen Atom

Consider a hydrogen atom, which consists of a **proton** at the center and an electron surrounding it. The electron's position is described by a probability distribution:

- Near the proton, the probability of finding the electron is highest.
- As we move farther from the proton, the probability decreases.

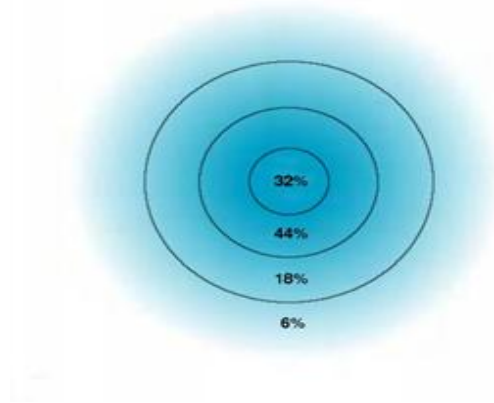


Fig 42.10: This figure illustrates a series of concentric circles, each representing different percentage values (32%, 44%, 18%, and 6%), as a probability distribution of finding an electron at various distances from a central point, illustrating the likelihood of an electron's location in a quantum system.

For example, in a hydrogen atom:

- The probability of finding the electron in the **first region (closest to the proton)** is approximately **32%**.
- In the next outer region, the probability increases to **44%**, indicating that this is the most likely region for the electron to be found.
- Beyond that, the probabilities decrease to **18%**, **6%**, and so on.

While it is possible for the electron to be very far from the proton, the chances of this are extremely small.

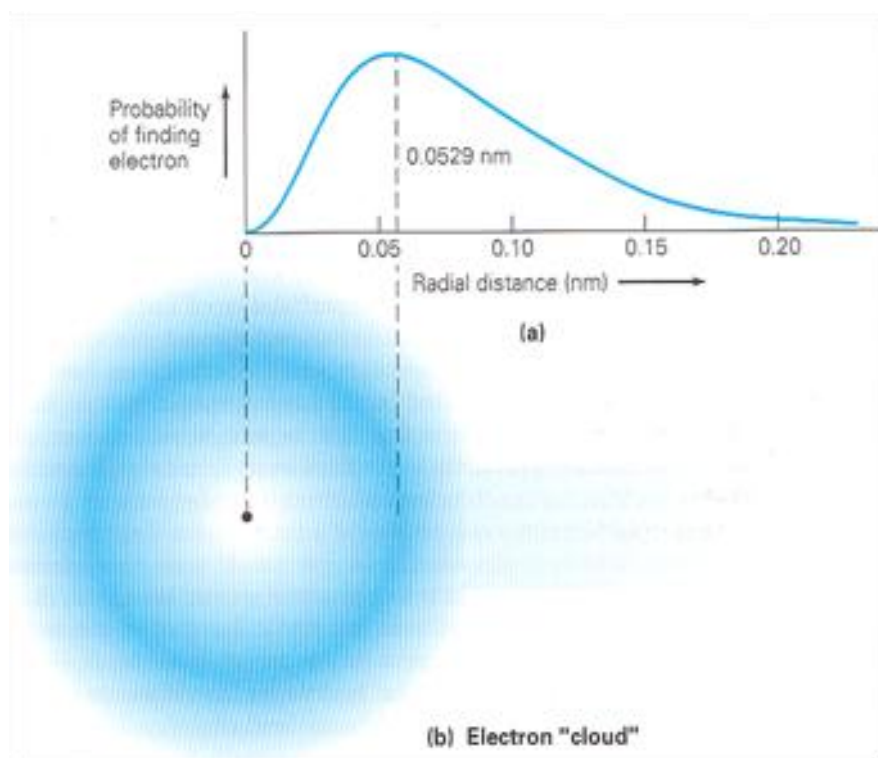


Fig 42.11: The figure shows (a) the probability distribution of finding an electron as a function of radial distance from the nucleus, peaking at 0.0529 nm, and (b) the corresponding electron 'cloud,' representing the regions where the electron is most likely to be found

The 1s State and Electron Cloud

The **1s state** (or the ground state of hydrogen) represents the lowest energy level where the electron is most likely to be found. In this state:

- The **angular momentum** of the electron ($L=0$) means it has no rotational motion around the nucleus.
- The electron forms a **cloud-like distribution** around the nucleus.

This electron cloud is denser closer to the nucleus and becomes thinner as we move farther away, reflecting the changing probabilities of the electron's location.

A New Picture of the Atom

This modern understanding of the atom differs significantly from the classical **Bohr model**, in which the electron was thought to revolve around the nucleus in fixed orbits with defined

angular momentum.

In quantum mechanics:

- The nucleus (proton) forms the dense core of the atom.
- Surrounding the nucleus is a **probability cloud** of electrons, resembling a smoke-like structure.
- The **density of the cloud** indicates the likelihood of the electron's presence at any given location.

Some Insights

1. **Electron Cloud:** The electron does not follow a specific path but exists as a cloud of probabilities around the nucleus.
2. **1s State:** In this ground state, the electron has the highest probability of being found near the nucleus.
3. **Probabilistic Nature:** The wave function provides probabilities, not certainties, about the electron's location.
4. **Modern Atomic Model:** This probabilistic approach to understanding the atom contrasts with earlier deterministic models like the Bohr model.

This concept of the electron cloud revolutionized our understanding of atomic structure and laid the foundation for modern quantum chemistry.

Exploring P States and Higher Orbitals in the Atom

Understanding P States

Beyond the simplest atomic state, the 1s state, we encounter more complex electron distributions associated with higher angular momentum. These are described as P states:

- In P states, the angular momentum quantum number ($l=1$) indicates the presence of angular momentum.
- There are three distinct forms of P states, corresponding to the magnetic quantum number (m_l), which can take values of -1 , 0 , and $+1$.

- These states represent different orientations of the electron's probability cloud in space, resulting in characteristic "dumbbell-shaped" orbital structures.

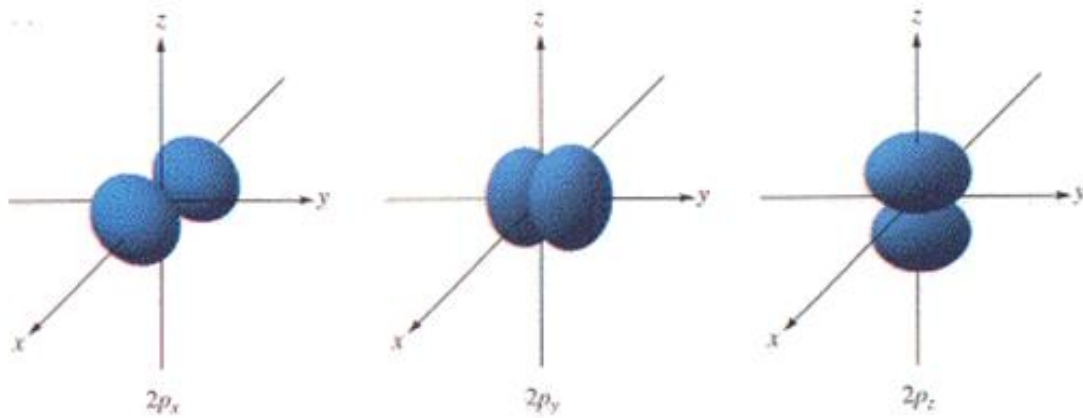


Fig 42.12: The figure shows the shapes and orientations of the three 2p atomic orbitals (2p_x, 2p_y, and 2p_z), each aligned along a different axis (x, y, and z) in three-dimensional space.

D Orbitals and Beyond

Electrons can also occupy d orbitals, which correspond to an angular momentum quantum number of $l=2$:

- These orbitals exhibit even more intricate shapes, such as "cloverleaf" patterns, reflecting their complex angular momentum distributions.
- As the value of l increases, orbitals become progressively more complex, corresponding to f orbitals ($l=3$) and higher states.

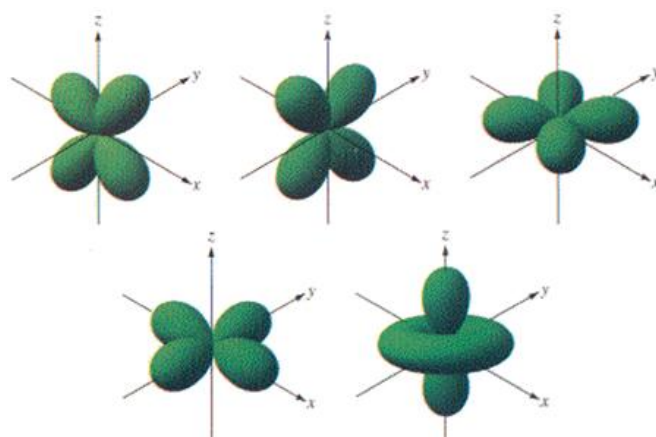


Fig 42.13: Shapes of d-orbitals: Different orientations of d-orbitals (d_{xy} , d_{xz} , d_{yz} , $d_{x^2-y^2}$, and d_{z^2}) in three-dimensional space, showing their unique symmetrical shapes around the nucleus.

The Role of the Schrödinger Equation

The Schrödinger equation serves as the foundation for understanding these orbital shapes and their associated energies. By solving this equation for the hydrogen atom, physicists determine:

1. **Energy Levels:** Each orbital has a specific energy, which depends on the principal quantum number (n) and angular momentum quantum number (l).
2. **Wave Functions:** The mathematical solutions to the Schrödinger equation (the wave functions) describe the probability distributions of electrons in various orbitals.
3. **Orbital Shapes:** These wave functions, when visualized, reveal the characteristic shapes of s, p, d, and higher orbitals.

The Complex Distribution of Electrons

Electrons do not merely "revolve" around the nucleus in simple circular paths, as suggested by earlier models like Bohr's. Instead:

- They occupy regions of space with distinct probability distributions.
- Each orbital corresponds to a unique wave function, which determines the likelihood of finding an electron in a particular region.
- These distributions become increasingly complex with higher angular momentum.

Some Insights

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1. P States: Electrons in P orbitals exhibit angular momentum and occupy dumbbell-shaped probability regions.
2. Higher Orbitals: D and f orbitals display even more intricate structures due to their higher angular momentum.
3. Schrödinger Equation: This mathematical framework explains the energy and shapes of orbitals, forming the foundation of quantum mechanics.
4. Modern View of Electrons: Unlike earlier models, quantum mechanics reveals electrons as existing within complex, three-dimensional probability clouds rather than simple orbits.

This modern understanding of atomic orbitals provides profound insights into chemical bonding, molecular shapes, and the fundamental principles governing atomic interactions. For deeper exploration, studying the mathematical solutions of the Schrödinger equation in quantum mechanics texts is highly recommended.

Identical particles are indistinguishable.
If particles A and B are interchanged then
 $|\Psi(1,2)|^2 = |\Psi(2,1)|^2 \Rightarrow$
either $\Psi(1,2) = +\Psi(2,1)$ BOSONS
or $\Psi(1,2) = -\Psi(2,1)$ FERMIONS

Note: $\Psi(1,1) = -\Psi(1,1) \therefore \Psi(1,1) = 0$

Lecture 43

Introduction to Atomic Physics

Introduction

Thousands of years ago, people were curious about the smallest possible unit of matter. It was only with the advent of modern science, specifically with the discovery of the principles of quantum mechanics about 100 years ago, that we gained a complete understanding of the atom. By combining predictions and observations, we can determine the accuracy of our current understanding of the atom.

To put this in historical context, around 2,500 years ago, a Greek philosopher named Democritus pondered the following question: If I have an apple, and I cut it in half, and keep dividing each half into smaller and smaller parts, will there ever be a point where I can't divide it any further, and will there be a smallest possible unit left? Democritus called this smallest unit an "Atomos." As shown in Fig 43.1.

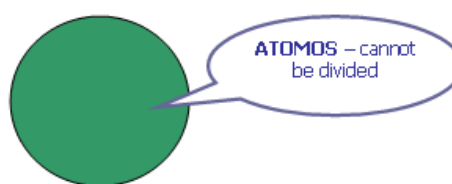


Fig 43. 1: An illustration depicting an atom, the smallest particle in the universe.

Ikhwan al Safa once raised a question about atoms, wondering what happens between them. They named the unknown space between atoms as empty space. After much debate, it was decided that matter comprises of atoms and empty space between the atoms. This led to the concept of the indescribable existence of atoms.

Lavoisier, a French chemist, noted that in all chemical reactions, the total mass of reactants before and after the reaction remains the same. It was concluded that whenever you burn something, the weight of the total material does not change. This is known as the law of conservation of matter.

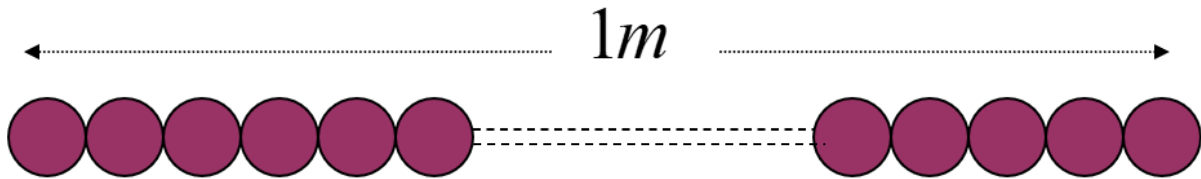
John Dalton made a significant contribution to the understanding of atoms and matter. In 1803, he laid down some basic principles related to atoms and matter. Dalton stated that every element is made of similar atoms, and you can differentiate between one element, and another based on the weight of their atoms. He also stated that when atoms of different elements combine to form compounds, their ratio is either one and two, one and three, or two and five - simple proportions. When molecules are formed, the properties of that pure matter will neither be related to one element nor to another. For instance, when hydrogen and oxygen gases combine, water is formed, which is liquid and in a molten state - H_2O .

Avogadro's Hypothesis, propounded by Avogadro, is a significant principle in chemistry. It states that if two gases have the same number of molecules, the same pressure, and temperature, then the number of molecules in each gas is equal. For example, if there are equal numbers of hydrogen and oxygen molecules in two gases, the pressure and temperature being the same, then the number of molecules in each gas will be equal.

Assuming that the system has the same number of molecules, if we take 1 kilogram of hydrogen gas, there will be Avogadro's number of molecules present in it, which is $N_0 = 6.023 \times 10^{26}$

molecules. Hydrogen gas consists of hydrogen atoms, whereas hydrogen molecules consist of two hydrogen atoms. Thus, the mass of hydrogen molecules will be twice that of hydrogen atoms. Similarly, if oxygen is present, the mass will be 16 times greater, as oxygen is 16 times heavier than hydrogen. This principle is fundamental, and we can use it to determine the size of atoms and

their mass. If we imagine a solid, we can picture the atoms in it as small spheres or tablets. If the diameter of each sphere is $2r$, then the radius is r , and the number of atoms we can fit in one meter is $1/2r$. We can also calculate the volume of each atom based on the density and atomic weight, which is A/ρ .



In a cube of $1m$ side we then have $(1/2r)^3$ atoms.

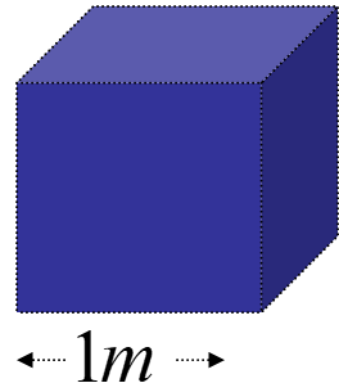
In 1kg .atom we have 6×10^{26} atoms and each atom occupies a volume $(A/\rho) \text{ m}^3$.

(where A = atomic weight and ρ = density).

$$\therefore N_0 = (1/2r)^3 \times A/\rho$$

$$\Rightarrow r = \frac{1}{2} \left(\frac{A}{\rho N_0} \right)^{1/3} \quad r_{Ag} \approx r_{Be} \approx 10^{-10} \text{ m}$$

ATOMS ARE THEREFORE MOSTLY THE SAME SIZE !!



In 1985, J.J. Thomson proposed a model of an atom called the "Plum Pudding" model. According to this model, the atom is made up of a positively charged material with negatively charged electrons scattered throughout it as shown in Fig 43.2. Thomson's analogy likened the atom to a pudding, with the positive charges spread evenly throughout, and the negatively charged electrons scattered like plums throughout the pudding. This model assumed that if the atom were a sphere, the positive and negative charges would be equally distributed. However, this model was proven incorrect by Ernest Rutherford.

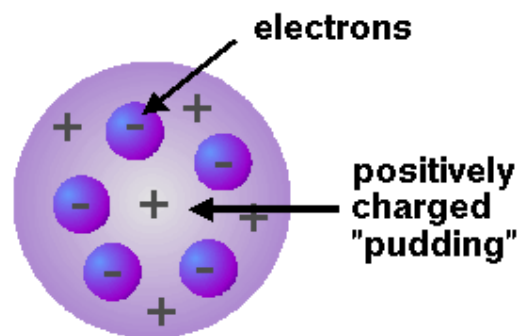


Fig 43. 2: The atom is made up of a positively charged material with negatively charged electrons scattered throughout it, called Plum Pudding Model.

Rutherford's α Scattering Experiment

In 1911, Rutherford conducted an important experiment. In this experiment, alpha particles were released into a chamber, and they collided with a plate of gold as shown in Fig 43.3. Some alpha particles went straight, some went towards the left, some towards the right, but some of them returned after the collision. As alpha particles are heavy and contain two protons and two neutrons, they are small but heavy particle. When they collided with the gold foil, it was observed that some particles either flew out without any change, while others came out at a very low angle as shown in Fig 43.4. Their deflection was very small, but Rutherford was surprised by this fact. He was surprised because when a heavy body collides with a light body, the heavy body never goes back, and the light body moves forward. However, if a heavy body collides with another heavy body, it is possible that it will go back after the collision.

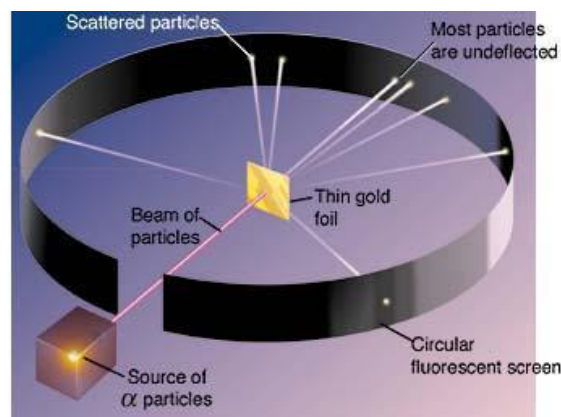
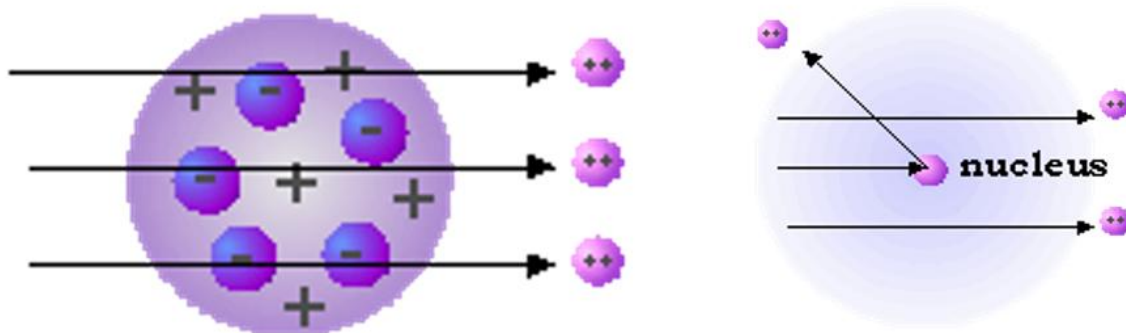


Fig 43. 4: Rutherford's α Scattering Experiment, demonstrating the deflection of alpha particles by a thin gold foil, leading to the discovery of the atomic nucleus.

Fig 43. 3: The experiment showed that while most particles passed through the foil, some were deflected at large angles, leading to the groundbreaking discovery of the atomic nucleus. This result suggested that atoms consist of a dense, positively charged core surrounded by a cloud of electrons.



This experiment proved that there is a small but heavy body somewhere in the atom. Is there such a nucleus, a heavy nucleus, with which this alpha particle would collide and go back? Thus, we now know that the plum pudding model is incorrect. It cannot be that the protons are spread everywhere in the atom along with the electrons. In fact, the shape of the atom is that

there is a center in the middle, and the electrons revolve around it. This center is very small, and the electrons are far away from it.

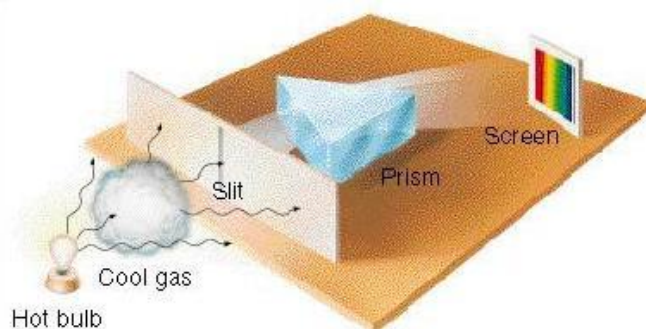


Fig 43. 5: Accelerating electrons emit electromagnetic radiation, leading to a decrease in the atom's radius and potential atomic collapse. This was resolved by the discovery that electrons emit radiation at specific frequencies as observed in absorption spectrum.

Let me give you an idea of the size of an atom. If we imagine the center of an atom as big as a cricket ball, the nearest electron would be about 8 to 10 kilometers away from it. Therefore, there is a considerable amount of empty space inside an atom, making it mostly empty. Rutherford presented an empty cloud picture of the atom, but there is a significant flaw in this picture that no one has been able to remove. If there is any point around which the electron rotates, it accelerates, which means its velocity (the sum of its speed) keeps changing, even if its speed remains constant. This acceleration causes electromagnetic radiation to come out from it, as Maxwell had proven a long time ago that whenever a charge accelerates, it emits electromagnetic radiation as shown in Fig 43.5.

When an atom accelerates, it produces radiation, which is related to its speed. As a result, the atom's radius decreases, and its electron collides with the nucleus, leading to the atom's destruction. This problem had no solution until it was discovered that if the electron rotates, the light it emits can have any frequency. However, the radiation coming out of an atom is only of a few specific colors, as seen in an absorbed spectrum experiment. This means that an atom



Fig 43. 6: The emission spectrum displays distinct wavelengths of light emitted by an element, while the absorption spectrum shows the specific wavelengths absorbed by the element, leaving dark lines in the continuous spectrum.

can only absorb frequencies of a certain wavelength as shown in Fig 43.6. When you heat a gas, such as hydrogen or oxygen, the colors and frequencies that emerge are also limited, creating an emission spectrum as shown in Fig 43.7.

At the dawn of the 20th century, the Rutherford model of the atom faced a major challenge. The model failed to account for both absorption and emission, and the light emitted from it was strong enough to destroy the atom. This problem had no solution at the time.

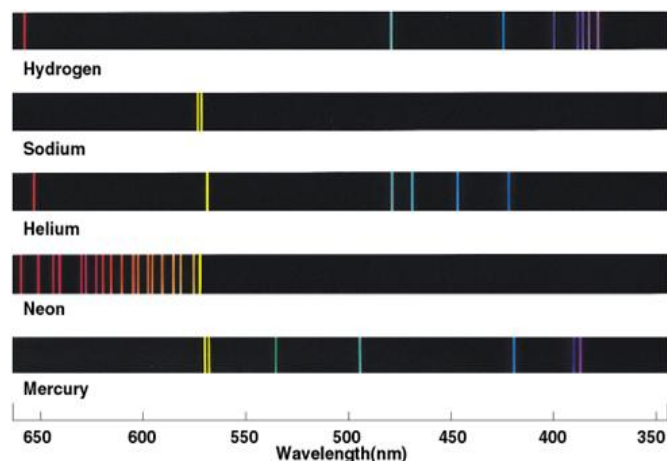


Fig 43. 7: Emission and absorption spectra of hydrogen, sodium, helium, neon, and mercury. The figure compares the characteristic spectral lines emitted and absorbed by each element, highlighting their unique spectral fingerprints and demonstrating how each element's emission and absorption spectra differ.

Bohr's Theory

Niels Bohr proposed a theory about atoms, which states that the energies of atoms are quantized and that transitions between states only occur when they move from one state to another without stopping at any intermediate state as shown in Fig 43.8.

According to Bohr's theory, when an atom absorbs light or energy, it is because a photon collides with it, causing it to move from a lower state to a higher state. The energy of the absorbed photon is exactly equal to the difference between the two states. Similarly, when light is emitted from an atom, it is because the atom moves from a higher state to a lower state.

Furthermore, the electron states are responsible for the movement between higher and lower states. If the wavelength of the light is

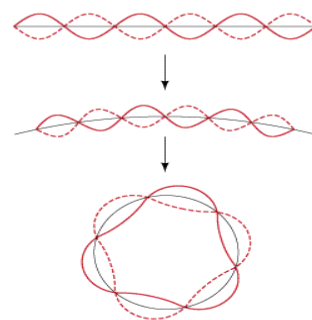


Fig 43. 8: Here is how a standing wave can be bent into a circle. The radius of the circle is just one of the Bohr orbits

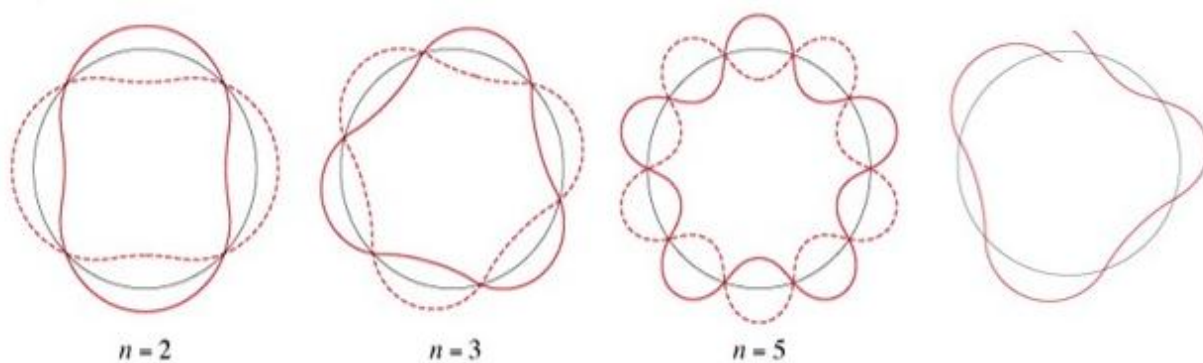


Fig 43. 9: The electron states are responsible for the movement between higher and lower states. If the wavelength is not exactly correct, wave interference makes the wave disappear. So only integral numbers of wavelengths can interfere constructively.

not precisely correct, the wave interference will cause the wave to disappear. Therefore, only integral numbers of wavelengths can interfere constructively as shown in Fig 43.9.

Bohr's Hypothesis

- Electrons can only take discrete energies (energy is related to radius of the orbit)
- Electrons can jump between different orbits due to the absorption or emission of photons.
- Dark lines in the absorption spectra are due to photons being absorbed.
- Bright lines in the emission spectra are due to photons being emitted.
- The electron has a wavelength and forms standing waves in its orbit around the nucleus.
- An integral number of electron wavelengths must fit into the circumference of the circular orbit.

$$n\lambda = 2\pi r$$

$$p = mv = \frac{h}{\lambda}$$

$$= \frac{h}{(2\pi r / n)} = \frac{n\hbar}{r} \quad n = 1, 2, 3, \dots$$

$$L = rmv = n\hbar$$

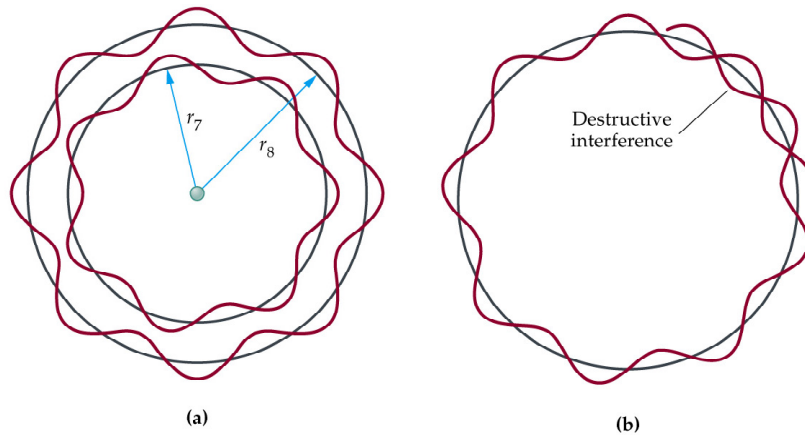


Fig 43. 10: The electron has a wavelength and forms standing waves in its orbit around the nucleus (a). If the wavelength is not exactly correct, wave interference makes the wave disappear (b).

Bohr Orbits

There are two forces that act on an electron: the electrostatic force, which pulls it closer to the proton, and the centrifugal force, which pushes it outwards as shown in Fig 43.11. If the electron is in a state of rest, meaning it's not falling towards the nucleus or running away from it, then these two forces should be equal.

We also know that angular momentum is quantized, and the value is directly proportional to n^2 . Here, n can be 1, 2, 3, and so on, and n^2 can be 1, 4, 9, 16, and so on, which shows that the electron can be one unit away from the nucleus or four, nine, sixteen units away and so on.

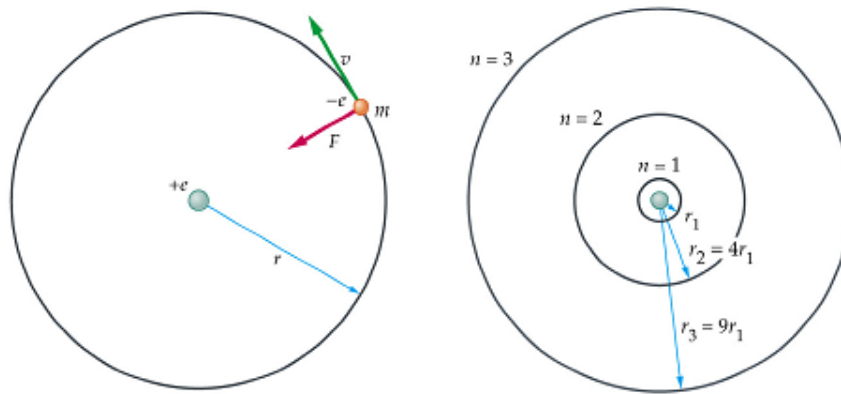


Fig 43. 11: Bohr's orbit illustrating the two forces acting on an electron: the electrostatic force pulling it towards the proton and the centrifugal force pushing it outward. In a stable orbit, these forces are balanced. The figure also shows that angular momentum is quantized, with permissible values proportional to n^2 , where n is a positive integer. This quantization indicates that the electron can occupy discrete orbits at distances corresponding to n^2 units from the nucleus.

$$\frac{mv_n^2}{r} = k \frac{e^2}{r^2}$$

$$L_n = n\hbar$$

$$v_n = \frac{n\hbar}{mr_n}$$

$$r_n = \left(\frac{\hbar^2}{mke^2} \right) n^2 \quad n = 1, 2, 3, \dots$$

The radius of the electron in different states is directly proportional to n^2 , which is called the Bohr Orbit. The energy of these different orbits will also be different.

- For $n = 1$ the electron orbit is closest to the nucleus, $r_n = a_0 \equiv 0.53 \times 10^{-8} \text{ cm}$ (the Bohr radius).
- For higher n , $r_n = a_0 n^2$.

Note that the electron is smaller in orbits farther from the nucleus.

$$v_n = \frac{ke^2}{n\hbar}$$

If we calculate the energy, we can see that the electron orbit and the one closest to the nucleus (where $a=1$) will have the smallest radius orbit, which is approximately half an angstrom. If the electron is in $n=2$, then the radius will be four times bigger, and if it is in $n=3$, then the radius will be nine times bigger, and so on.

Energy of a Bohr Orbit

As the distance between an electron and the nucleus increases, its speed decreases. This change in distance results in two forms of energy - potential and kinetic. The negative sign in the potential energy equation represents the fact that energy is needed to separate positively and negatively charged particles.

$$\begin{aligned}
 E &= K + U = \frac{1}{2}mv^2 + U \\
 &= \frac{1}{2}\left(\frac{ke^2}{r}\right) - \frac{ke^2}{r} = -\frac{ke^2}{2r} \\
 \therefore E_n &= -\left(\frac{ke^2}{2}\right)\left(\frac{mke^2}{\hbar^2}\right)\frac{1}{n^2} = \\
 &= -\left(\frac{mk^2e^4}{2\hbar^2}\right)\frac{1}{n^2} = -(13.6 \text{ eV})\frac{1}{n^2}
 \end{aligned}$$

In the case of removing an electron from a hydrogen atom, 13.6 electron volts are required when the electron is in the $n = 1$ state, and 4 times as much energy is needed when the electron is in the $n = 2$ state. The kinetic energy is lower than the potential energy, leading to a negative total energy.

The Spectrum of Hydrogen

In this picture, you can see the energy plotted against the principal quantum number, n . The lowest energy level occurs for $n = -R_H$, followed by $-1/4 R_H$ and so on as shown in Fig 43.12. These different levels represent the various states of the hydrogen atom. According to Bohr, transitions can occur between these different levels. If a lot of energy is given to the electron of a hydrogen atom, it can transition from the $n = 1$ state to the $n = 2$ state. The amount of energy required can be calculated by the difference in energy between the two states. For instance, the energy for $n = 1$ is -13.6 and for $n = 2$, it is $-13.6/4$. If a photon with the exact amount of energy required for the transition comes into contact with the electron, the photon is absorbed, and the atom moves from one state to another. Collisions between two hydrogen atoms can also cause excitations. However, the most common way is through light absorption and emission. Many different transitions are possible, resulting in various frequencies being emitted from the hydrogen atom. Each atom has its own unique frequency or fingerprint. Bohr's model has been successful in understanding the spectral lines of hydrogen. It is based on the

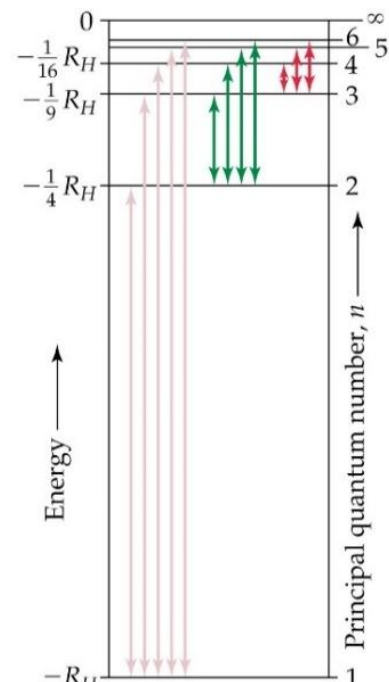


Fig 43. 12: The spectrum of hydrogen energy levels represents the various states of the hydrogen atom.

hypothesis that the delay should be complete, and the number of waves can be one, two, three, four, etc. Quantum mechanics has since been developed, and it has confirmed the same principles that Bohr predicted. When the Schrödinger equation is solved for the wave function, we can get all the energy levels of the hydrogen atom and other atoms, which are the same as predicted by Bohr's model.

- The principal quantum number $n = 1, 2, 3 \dots$
- The orbital angular momentum quantum number l , where $L = \sqrt{l(l+1)}\hbar$ and $l = 0, 1, 2, 3 \dots, n-1$.
- The magnetic quantum number m_l where $m_l = -l, \dots, -2, -1, 0, 1, 2, \dots, l$ ($L_z = m_l \hbar$)
- The electron spin quantum number m_s where $m_s = -\frac{1}{2}, \frac{1}{2}$

I would like to find a solution to a problem without solving the Schrödinger equation. This equation yields a function with two parts: one part is called the radial wave function, which is related only to the distance from the center as shown in Fig 43.13. In this graph, you can see a picture of the square of the radial wave function, which shows the probability of finding an electron at distance R . The highest probability is at a_0 .

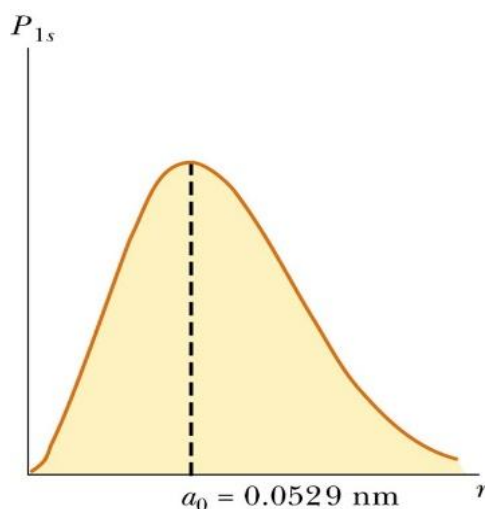


Fig 43. 13: The probability per unit length of finding the electron versus distance from the nucleus for the hydrogen atom in the 1s (ground) state. Note that the graph has its maximum value when r equals the first Bohr radius, a_0 .

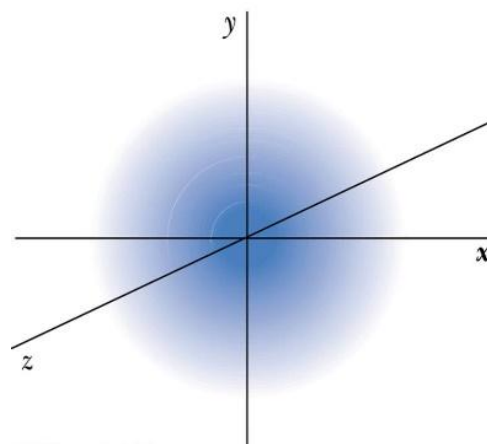


Fig 43. 14: The spherical electron cloud for the hydrogen atom in its 1s state.

When the electron is in the symmetric state, which is equal to 0 or in the s state, it is spread around the nucleus or proton with spherical symmetry. The highest density of the electron cloud is at a_0 . As you move away from the atom, the probability of finding an electron decreases as shown in Fig 43.14. This is different from classical physics where an electron rotates only in a radius of one Bohr radius. We can compare this to smoke - it is spread everywhere, but where it is denser, there is a higher probability of finding an electron. This information is specifically

about the s-state. Similarly, we have $l = 1$, $l = 2$, and $l = 3$ states, with the $l = 1$ state also known as the P state.

- There are three p -orbitals p_x , p_y , and p_z .
- The three p -orbitals lie along the x -, y -, and z -axes of a Cartesian system as shown in Fig 43.15.
- The letters correspond to allowed values of m_l of -1, 0, and +1.
- The orbitals are dumbbell shaped.
- As n increases, the p -orbitals get larger.
- All p -orbitals have a node at the nucleus.

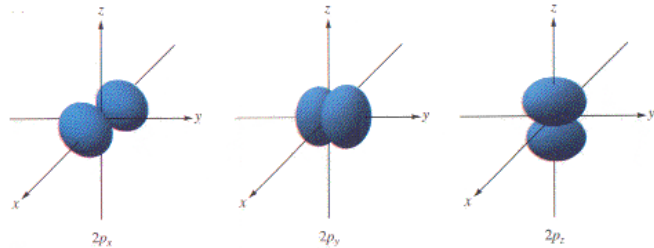


Fig 43. 15: Illustration of the three p -orbitals (p_x , p_y , p_z), aligned along the Cartesian axes. These dumbbell-shaped orbitals correspond to m_l values of -1, 0, and +1, get larger with increasing n , and have a node at the nucleus.

It has maxima on three different axes, and D state is more complicated than that. All these things were for the hydrogen atom.

Electron Spin and the Pauli Exclusion Principle

The hydrogen atom is the simplest atom, consisting of only one electron rotating around the nucleus. However, when an atom has more than one electron, a new phenomenon occurs. As mentioned in my last lecture, electrons are fermions, meaning that if two electrons are exchanged, the function changes its sign. This property of electrons leads to the Pauli exclusion principle, which states that two electrons cannot be in the same quantum state and therefore cannot occupy the same point in space. However, an exception to this rule is allowed for electrons with opposite spins, allowing them to be identified as distinct particles as shown in Fig 43.16. This principle dictates that only one electron with a specific spin can occupy an orbital at a given time. If another electron is added to the same orbital, it must have a different spin. This is how the Pauli exclusion principle works, and it helps to explain the behavior of different atoms.

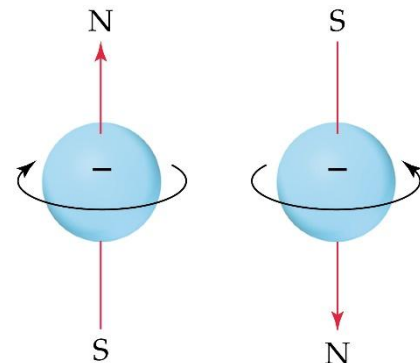


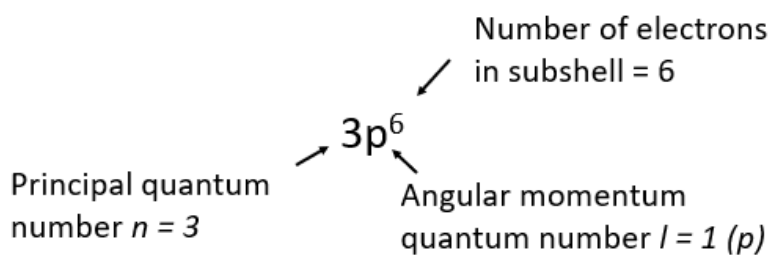
Fig 43. 16: As an electron moves in its orbit about the nucleus, its spin can be either (a) up or (b) down.

n	Possible Values of l	Subshell Designation	Possible Values of m_l	Number of Orbitals in Subshell	Total Number of Orbitals in Shell
1	0	1s	0	1	1
2	0	2s	0	1	4
	1	2p	1, 0, -1	3	
3	0	3s	0	1	
	1	3p	1, 0, -1	3	9
	2	3d	2, 1, 0, -1, -2	5	
4	0	4s	0	1	
	1	4p	1, 0, -1	3	16
	2	4d	2, 1, 0, -1, -2	5	
	3	4f	3, 2, 1, 0, -1, -2, -3	7	

Table 43. 1: Table illustrating the relationship between orbitals and quantum numbers. It details how the principal quantum number (n), angular momentum quantum number (l), magnetic quantum number (m_l), define the unique states of electrons within an atom's orbitals.

Helium has two protons in its nucleus and one or two neutrons depending on its isotope. It also has two electrons, each with opposite spin. For Helium, $n = 1$, $l = 0$, and $m_l = 0$, completing one shell.

To make Lithium neutral, one more electron is needed, which goes into the $n = 2$ state where $l = 0$ or 1 , starting a second shell. By adding more electrons, more complex atoms are formed. The process continues with Hydrogen, Helium, Lithium, Beryllium, Boron, Carbon, and so on. When the second shell is filled, eight electrons can come in. There are two possible values of l for $n = 2$, either s-state or P-state. For $l = 0$, two electrons can come in, one with an up spin and one with a down spin. For $l = 1$, there are three possible m_l values (-1, 0, 1), and for each one, two electrons can come in, one up and one down. This means that 2 electrons come from the s-state and 6 come from the P-state. This process can be repeated for $n = 3$, 4, and so on, to make electronic configurations for all elements in the periodic table.



Electronic configuration refers to the arrangement of electrons in the shells of an atom. The configuration is determined by the principal quantum number, which is the energy level of the electron. For instance, if the principal quantum number is 3 and the angular momentum quantum number is 1 (P state), then the subshell can hold up to 6 electrons. Using this method, we can create the electronic structures of various atoms.

For example, oxygen has 8 protons, 8 neutrons, and 8 electrons, while phosphorus has 15 protons, 16 neutrons, and 15 electrons, and zinc has 30 protons, 35 neutrons, and 30 electrons. The structure of an atom consists of a nucleus surrounded by S and P electrons, followed by outer shells.

The Pauli exclusion principle ensures that the electrons occupy the lowest available energy levels, and each shell has a maximum number of electrons it can hold. The $n = 1$ shell is

completely filled with a maximum of two electrons, and no electron can enter it from outside. If the Pauli exclusion principle is eliminated, all atoms except hydrogen and helium will come to an $n = 1$ state, causing a significant change in the world.

Subatomic particles in some atoms

16	31	65
O	P	Zn
8	15	30
8 p⁺	15 p⁺	30 p⁺
8 n	16 n	35 n
8 e⁻	15 e⁻	30 e⁻

Lecture 44

Introduction to Nuclear Physics

The Misunderstanding of Nuclear Physics

Many people think that nuclear physics is only about making bombs. This idea comes from the fact that about 60 years ago, nuclear bombs were dropped on Hiroshima and Nagasaki, causing the deaths of around 200,000 innocent people. However, nuclear physics has many other applications. In this chapter, we will explore the true nature of nuclear physics. We aim to understand the centre of the atom and the wonders it holds. This chapter is the beginning of our journey into nuclear physics.

Discovery of the Atomic Nucleus

In 1911, Ernest Rutherford discovered that the atom has a centre, known as the nucleus. Before this, it was believed that an atom contained both positive and negative charges. The negative charges, called electrons, were already known. When we rub certain materials, they become charged because of the electrons. To balance these negative charges, there must be positive charges, which we call protons.

Earlier, the idea was that these positive and negative charges were mixed together, like a pudding. This was known as the plum pudding model. However, when Rutherford conducted his famous gold foil experiment, he found that some alpha particles, which are very heavy, bounced back after hitting the foil. This proved that atoms have a dense centre, or nucleus, with electrons far away from it.

Structure of the Atom

The new understanding is that an atom has a nucleus at its centre, with electrons orbiting around it as shown in Fig 44.1. This is similar to how planets orbit the Sun. However, according to quantum mechanics, this is not entirely accurate. Instead, electrons are like clouds of probability, surrounding the nucleus. Inside the nucleus, we find protons and neutrons.

Protons and neutrons are not elementary particles; they are made of even smaller particles, which we will discuss later. The two basic particles in nuclear physics are the proton and

neutron. The mass of a proton is approximately 1.67×10^{-27} kg, and a neutron is slightly heavier, with a mass of 1.67×10^{-27} kg. Protons are positively charged, while neutrons are neutral.

Units of Measurement in Nuclear Physics

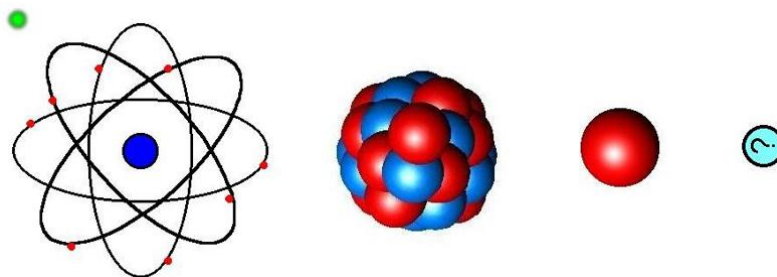


Fig 44. 1: Modern understanding of the atom, with a dense nucleus containing protons and neutrons, surrounded by a cloud of electrons. This model replaces the earlier plum pudding model.

Using standard units like kilograms can be inconvenient in nuclear physics. Instead, we use Einstein's famous equation, $E = mc^2$, to convert mass into energy units. For example, the mass of a proton is $938 \text{ MeV}/c^2$, and a neutron is $940 \text{ MeV}/c^2$. Electrons are much lighter, with a mass of about $0.5 \text{ MeV}/c^2$.

Another unit used in nuclear physics is the fermi, which is equal to 10^{-15} meters. This unit helps measure the size of nuclei, which is around 1.2 fermis.

Identifying Different Nuclei

Nuclei are identified by the number of protons (denoted by Z) and the total number of protons and neutrons (denoted by A). The atomic number (Z) represents the number of protons, while the mass number (A) is the sum of protons and neutrons. Each nucleus has a chemical symbol based on its proton count.

Isotopes

Isotopes are versions of an element with the same number of protons but varying numbers of neutrons. For example, hydrogen has one proton and no neutrons, deuterium has one proton and one neutron, and tritium has one proton and two neutrons as shown in Fig 44.2. These are all isotopes of hydrogen, meaning they have the same number of protons but different numbers of neutrons.

ISOTOPES OF HYDROGEN

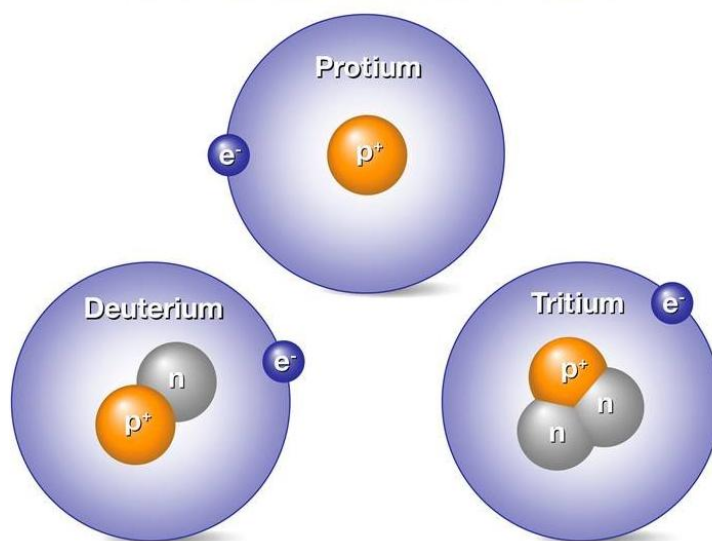


Fig 44. 2: Illustration of hydrogen's three isotopes: protium (one proton), deuterium (one proton, one neutron), and tritium (one proton, two neutrons), highlighting their differences in atomic mass and composition.

Despite their structural differences, all these isotopes of hydrogen exhibit identical chemical properties. This uniformity arises because the number of protons determines an element's chemical behaviour, such as its ability to form water (H_2O), regardless of the number of neutrons. Similarly, carbon exists in several isotopic forms: Carbon-12, Carbon-13, and Carbon-14. Carbon-12, the most abundant form, has six protons and six neutrons. Carbon-13 has an additional neutron, while Carbon-14 contains two extra neutrons. This variability in neutron count among isotopes contributes to different physical properties and behaviours, such as radioactive decay rates.

Imaging the Nucleus

The nucleus, though minute, holds significant mysteries. Unlike atoms, whose structure is observable with light microscopy, the nucleus poses a challenge due to the wavelength limitations of light. Light, being a wave, cannot resolve objects smaller than its wavelength, rendering nuclei invisible to optical microscopes.

To overcome this, scientists employ electron accelerators. Despite electrons being particles, they exhibit wave-like properties described by a wavelength (λ) inversely proportional to their momentum (p). By accelerating electrons to high speeds, their wavelengths shorten enough to interact with nuclei.

Probing the Nucleus

In an electron accelerator, electrons collide with nuclei. These collisions scatter electrons in detectable patterns, revealing crucial details about nuclear structure and composition. The size of a nucleus, approximated by the formula $r = r_0 A^{1/3}$, depends on the number of protons (Z) and neutrons (N). Here, $r_0 = 1.2 \text{ fm}$ represents the nuclear radius constant, reflecting the compact yet dense nature of nuclei.

$$r \approx r_0 A^{1/3} \quad \text{where } r_0 = 1.2 \text{ fm}$$

$$\text{note: } V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi r_0^3 A$$

$$\therefore \frac{A}{V} = \frac{3}{4\pi r_0^3}$$

This is the number of nucleons per cubic fermi.

Understanding Nuclear Density

Unlike everyday objects with defined boundaries, nuclei exhibit a nearly uniform density throughout. This means that regardless of the element—whether lead, gold, or platinum—the volume occupied per nucleus remains relatively consistent. This phenomenon highlights the unique density distribution within nuclei, where high-density regions blend seamlessly into surrounding space, rather than abruptly ending as in macroscopic objects.

The Nature of Nuclear Force

Inside the nucleus of an atom, there are protons and neutrons. Protons are positively charged, and like charges repel each other. This means that protons naturally push each other away. However, the nucleus remains stable because of a force that is much stronger than the repulsive force between protons. This force is known as the strong nuclear force.

The strong nuclear force is incredibly powerful but has a very short range. If it weren't for this

force, the nucleus would fall apart because the repulsion between protons would be too strong to keep them together. To understand how strong this force is, consider that it takes about 8 million electron volts (MeV) to remove a nucleon (a proton or neutron) from the nucleus, whether the nucleus is small or large. This uniformity indicates the strong nuclear force is very powerful but only acts over very short distances.

Range of the Nuclear Force

The range of the nuclear force is critical in understanding nuclear stability. If the force had a long range, it would pull all nuclei towards each other, making it extremely difficult to remove a nucleon from a larger nucleus. However, since it takes the same amount of energy to remove a nucleon from both small and large nuclei (about 8 MeV), it indicates that the force has a short range, typically between one and two fermis (1 fermi = 10^{-15} meters). This range is much smaller than the size of even the largest nuclei, which are around 10 fermis in diameter.

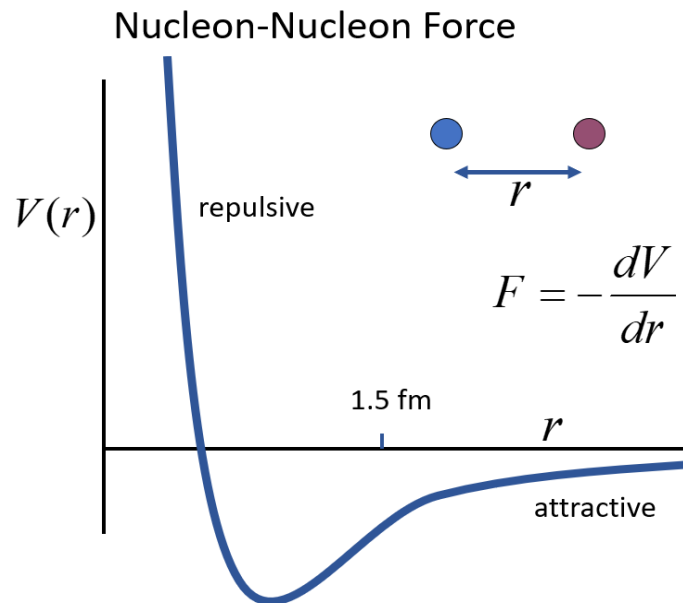


Fig 44. 3: The strong nuclear force between nucleons is attractive but short range; the Coulomb force between protons is long range, but repulsive.

The nuclear force has two main components: an attractive force that pulls nucleons together when they are within a certain range, and a repulsive force that pushes them apart if they get too close as shown in Fig 44.3. This dual nature ensures that the nucleus remains stable, with nucleons packed closely but not too tightly together.

The Yukawa Theory

In 1935, Japanese scientist Hideki Yukawa proposed a theory to explain the nuclear force. He suggested that the force between nucleons (protons and neutrons) is mediated by an exchange of particles he called mesons. According to Yukawa, a proton or neutron sends a meson to another nucleon, and this exchange creates an attractive force, similar to two people playing catch with a ball.

Yukawa's theory was revolutionary because it provided a way to quantify the range of the nuclear force using principles of quantum mechanics, specifically Heisenberg's uncertainty principle. The uncertainty principle states that the product of the uncertainty in energy (ΔE) and the uncertainty in time (Δt) is approximately equal to Planck's constant (h). This means that for a very short time, the conservation of energy can be violated, allowing for the temporary existence of particles like mesons.

Calculating the Range of the Force

Using Heisenberg's uncertainty principle, we can estimate the range of the nuclear force. The strong nuclear force is transmitted between a proton and neutron by the creation and exchange of a pion as shown in Fig 44.4. If a meson with mass m is exchanged between nucleons, the uncertainty in energy ΔE can be approximated by mc^2 , where c is the speed of light. The uncertainty in time Δt is then

$\Delta t \approx \frac{\hbar}{2mc^2}$. The distance d that the meson can

travel in this time is given by $c \times \Delta t$, which leads to $d = \hbar/(mc)$.

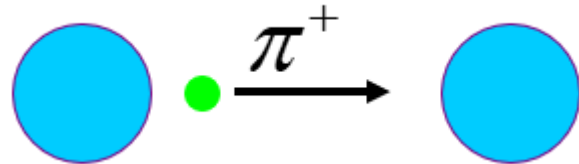


Fig 44. 4: The strong nuclear force is transmitted between a proton and neutron by the creation and exchange of a pion. The pion is created through a temporary violation of conservation of mass-energy and travels from the proton to the neutron and is recaptured. It is not directly observable and is called a virtual particle.

$$\Delta E \Delta t \approx \hbar / 2$$

$$\Delta E = \text{Minimum energy of pion} \approx mc^2$$

$$\therefore \Delta t \approx \frac{\hbar}{2mc^2}$$

$$\text{distance travelled by pion} \approx c \Delta t \approx \frac{\hbar}{2mc}$$

$$\text{Put } \frac{\hbar}{2mc} \approx 1.2 \text{ fm (range of nuclear force)}$$

$$\Rightarrow mc^2 \approx 124 \text{ MeV} / c^2$$

If we substitute typical values, we find that the range of the nuclear force is about 1.2 fermis. Yukawa predicted that the mass of the meson would be around 124 MeV/ c^2 , and later experiments confirmed the existence of mesons with masses in this range, validating his theory.

Nuclear Fission and Energy Release

One of the most important applications of nuclear physics is the process of nuclear fission, where a large nucleus splits into two or more smaller nuclei as shown in Fig 44.5. When this happens, the total mass of the resulting nuclei is less than the mass of the original nucleus. This mass difference, when multiplied by c^2 (Einstein's famous equation $E = mc^2$), gives the amount of energy released in the fission process.

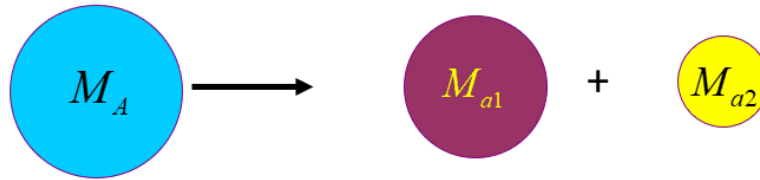


Fig 44. 5: A large atomic nucleus splits into two smaller nuclei upon neutron impact with releasing energy.

$$Q = (M_A - M_{a1} - M_{a2})c^2 = \text{energy released}$$

For example, when a uranium nucleus undergoes fission, it splits into smaller nuclei such as barium and krypton, along with a few free neutrons. The combined mass of these products is less than the original uranium nucleus, and the missing mass is converted into a significant amount of energy. This energy release is the principle behind nuclear power plants and atomic bombs.

Introduction to Binding Energy

When we talk about the nucleus of an atom, one important concept to understand is binding energy. Binding energy is the energy that holds the nucleus together. If we look at different nuclei, we can measure how much binding energy they have. Iron has one of the highest binding energies, around 8.6 or 8.7 MeV (million electron volts) per nucleon as shown in Fig 44.6. In comparison, uranium has a binding energy about 7% lower. Some nuclei are stable, meaning they will stay the same for millions of years, while others are unstable and will eventually decay.

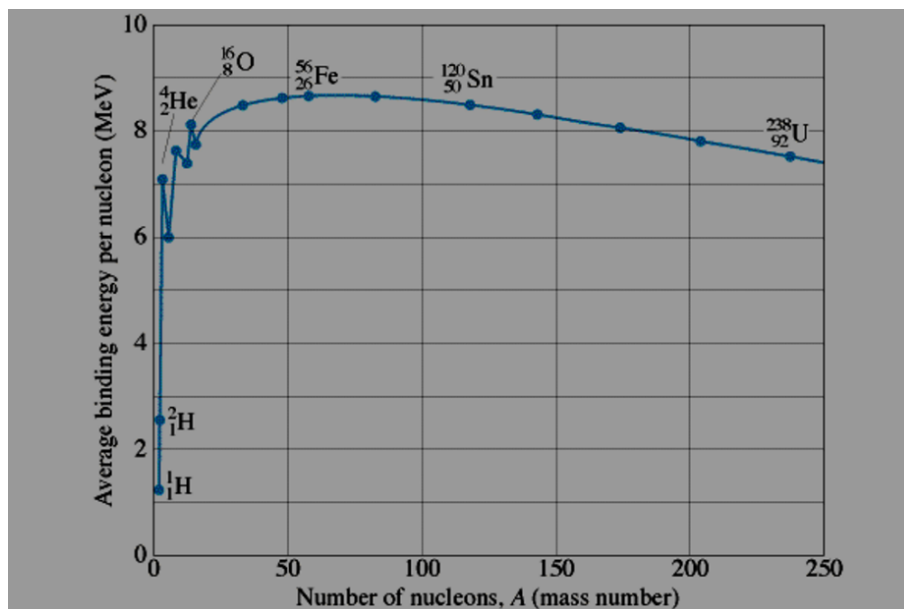


Fig 44. 6: The curve of binding energy, showing how fusion and fission can result in the release of nuclear energy.

Nuclear Decay and Stability

The stability of a nucleus depends on its protons and neutrons. When we talk about unstable nuclei, they decay over time. We cannot predict exactly when a particular nucleus will decay, but we can talk about the decay of many nuclei in statistical terms. If we have a large number of unstable nuclei, the number those decays over a certain time period is proportional to the number of remaining nuclei.

This can be written as:

$$\Delta N \propto N$$

$$\Delta N \propto \Delta t$$

Here, ΔN is the number of decaying nuclei, N is the number of nuclei present, and Δt is the time interval. Combining these, we get:

$$\Delta N = -\lambda N \Delta t$$

The negative sign indicates that the number of nuclei is decreasing over time. For very small-time intervals, this equation can be written as:

$$\frac{dN}{dt} = -\lambda N$$

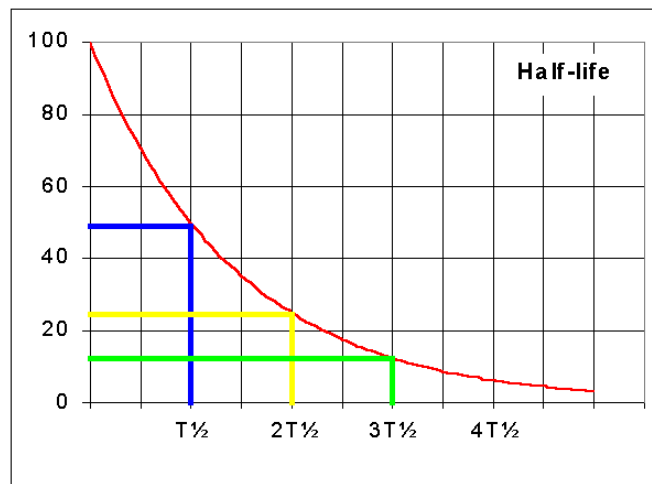
The solution to this differential equation is:

$$N(t) = N_0 e^{-\lambda t}$$

Here, N_0 is the initial number of nuclei, and λ is the decay constant. The decay constant λ determines how quickly the number of nuclei decreases. The greater λ , the more radioactive the sample.

Half-Life

The concept of half-life is used to describe how long it takes for half of a sample of radioactive material to decay. The half-life ($T_{1/2}$) is the time required for half of the nuclei in a sample to decay as shown in Fig 44.7. This can be derived from the decay equation:



$$N(t) = \frac{N_0}{2}$$

Substituting this into the decay equation, we get:

$$\frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$$

Fig 44. 7: Plot of the exponential decay law for radioactive nuclei. The vertical axis represents the number of radioactive nuclei present at any time t, and the horizontal axis is time. The parameter $T_{1/2}$ is the half-life of the sample.

Solving for $T_{1/2}$, we get:

$$T_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$$

Stability of Nuclei

The stability of a nucleus depends on the number of protons (Z) and neutrons (N) it has. For small nuclei, a roughly equal number of protons and neutrons is ideal. For example, the most stable isotope of oxygen, oxygen-16, has 8 protons and 8 neutrons. Adding more neutrons makes the nucleus less stable. Heavy elements like lead (Z = 82) have more neutrons than protons. Lead-208, for instance, has 82 protons and 126 neutrons. If we add one more neutron to lead-208, it becomes unstable. This shows that the balance between protons and neutrons is crucial for stability.

No stable nuclei exist naturally with more than 82 protons. When the proton number exceeds 92, no stable nuclei are found at all. This is because the repulsive force between protons becomes too great to be balanced by the strong nuclear force.

Examples of Half-Life

Different substances have very different half-lives. For example, polonium has a half-life of just 1.64×10^{-4} seconds, meaning it decays almost instantly. Krypton has a half-life of 3.16 minutes, while strontium's is 28.5 years. Radium has a half-life of 1600 years, carbon-14's is 5730 years, and uranium-238 has a half-life of 4.5 billion years. The amount of uranium-238 on Earth has decreased by half since the planet formed about 4.5 billion years ago.

Practical Application: Radon Decay

Radon gas decaying to polonium. Radon-222 decays to polonium-218 by emitting an alpha particle. The half-life of radon-222 is 3.8 days. If we start with 20,000 radon atoms, after 3.8 days, 10,000 radon atoms will be left, and 10,000 polonium atoms will have been produced. After another 3.8 days, 5,000 radon atoms will remain, and the number of polonium atoms will have increased.

Problem Example: Silicon Decay

Handouts PHY101

In a lab there is 2 micrograms of pure silicon-31, which has a half-life of 2.5 hours. How many silicon-31 nuclei we have initially and how many are left after 4 hours.

Solution:

First, we need to calculate the number of nuclei in 2 micrograms of silicon-31. Silicon-31 has an atomic mass of 31 grams per mole, and Avogadro's number (the number of atoms in a mole) is 6.02×10^{23} . Therefore, in 31 grams of silicon-31, there are 6.02×10^{23} atoms. In 2 micrograms (2×10^{-6} grams), the number of atoms is:

$$N_0 = \left(\frac{2 \times 10^{-6} \text{ g}}{31 \text{ g/mol}} \right) \times 6.02 \times 10^{23} \text{ atoms/mol} = 3.89 \times 10^{16} \text{ nuclei}$$

Next, we calculate the decay constant λ using the half-life formula: Half-life = 2.5 h = 9000 s

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{9000} = 7.7 \times 10^{-5} \text{ s}^{-1}$$

Now, we use the decay equation to find the number of nuclei remaining after 4 hours:

$$N(t) = N_0 e^{-\lambda t}$$

$$t = 4h = 14400 \text{ s}$$

$$N(t) = 3.89 \times 10^{16} e^{(-7.7 \times 10^{-5} \times 14400)}$$

$$N(t) = 3.89 \times 10^{16} \times 0.348 = 1.35 \times 10^{16} \text{ nuclei}$$

So, after 4 hours, there are approximately 1.35×10^{16} silicon-31 nuclei remaining from the initial 3.89×10^{16} nuclei.

Introduction to Radioactive Decay

Radioactive decay is a fundamental concept in nuclear physics. It helps us understand how old certain objects are, such as ancient artifacts, and even how long-ago animals or humans died. This understanding is based on the Law of Radioactive Decay. This law allows us to determine the age of an object or remains by analysing the radioactive elements present in them.

The Role of Carbon Dating

One of the key applications of radioactive decay is carbon dating. This process is essential for dating ancient biological materials. Here's how it works:

Carbon-12 and Carbon-14

Living organisms absorb carbon from their environment, which includes two types of carbon atoms: carbon-12 and carbon-14. Carbon-12 is stable and does not change over time, while carbon-14 is radioactive and decays at a known rate.

How Carbon-14 Decays

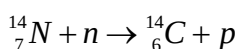
When an organism dies, it stops absorbing carbon. The amount of carbon-12 in its remains stays the same, but the amount of carbon-14 decreases over time. By measuring the ratio of carbon-14 to carbon-12 in a sample, scientists can estimate how long it has been since the organism died. The half-life of carbon-14 is about 5730 years, meaning that half of the carbon-14 in a sample decays over this period.

When a living organism dies, CO_2 no longer absorbed. Thus ratio of carbon 14:12 decreases by half every 5730 years.

Measure activity, $N = N_0 e^{-\lambda t}$ or

activity $A = A_0 e^{-\lambda t}$ with $A_0 = 0.23 \text{ Bq/g}$

Amount of isotopes in atmosphere approximately constant, despite a half-life of 5730 y.



Example of Carbon Dating

A dead body was found in the mountains. Scientists analysed a sample from the body and found that the ratio of carbon-14 to carbon-12 had decreased to half of what it would be in a living organism. This indicated that the person had died approximately 5300 years ago. This calculation uses the decay constant of carbon-14, which is $1.21 \times 10^{-4} \text{ Bq/g}$.

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{5730} = 1.21 \times 10^{-4} \text{ y}^{-1}$$

$$0.121 = 0.23 e^{-1.21 \times 10^{-4} \times t}$$

$$\ln \frac{0.121}{0.23} = -1.21 \times 10^{-4} \times t$$

$$t = 5300 \text{ years}$$

Continuous Production of Carbon-14

You might wonder why carbon-14 doesn't keep decreasing universally. The answer is that carbon-14 is continuously produced in the atmosphere when cosmic rays collide with nitrogen atoms. This ongoing production keeps the amount of carbon-14 in the environment relatively constant.

Dating Older Objects with Uranium-238

For dating objects older than what carbon dating can handle, scientists use other radioactive elements like uranium-238. Uranium-238 has a much longer half-life of 4.5 billion years, making it suitable for dating very ancient geological formations and artifacts.

Types of Radioactive Decay

There are three primary types of radioactive decay: alpha decay, beta decay, and gamma decay as shown in Fig 44.8. Each type involves different particles being emitted from the nucleus.

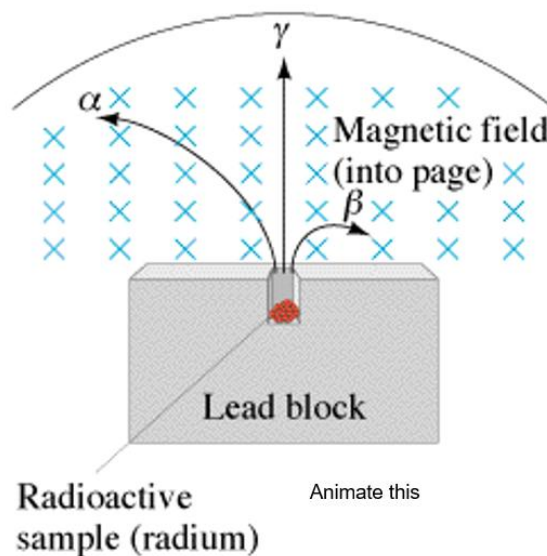


Fig 44.8: The radiation from a radioactive source, such as radium, can be separated into three components using a magnetic field to deflect the charged particles. The detector array at the right records the events. The gamma ray isn't deflected by the magnetic field.

Alpha Decay

Alpha particles consist of two protons and two neutrons. In alpha decay, a large nucleus releases an alpha particle as shown in Fig 44.9. Alpha particles are relatively large and positively charged. They can be ejected from the nucleus when the nucleus is unstable.

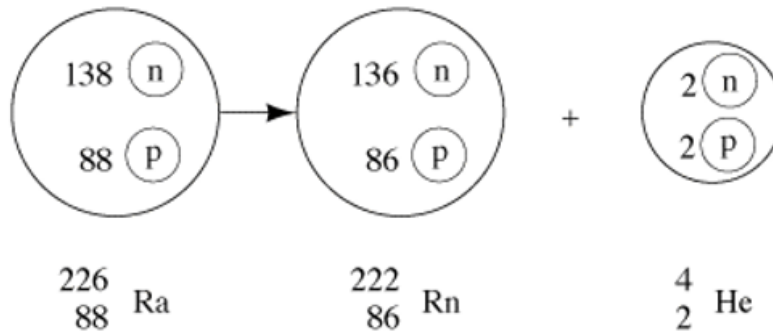


Fig 44. 9: An alpha particle, composed of two protons and two neutrons, is emitted during alpha decay from a large, unstable nucleus. These relatively large, positively charged particles can be expelled from the nucleus due to instability.

$${}^A_Z X \rightarrow {}^{A-4}_{Z-2} X' + {}^4_2 \text{He}$$

$$Q = (M_{\text{parent}} - M_{\text{daughter}} - M_{\alpha})c^2$$

- The unstable nuclei simply cannot overcome the proton repulsion indefinitely.
- We saw from the binding energy graph that helium-4 nuclei are very stable.
- Inside big nuclei, two protons and two neutrons seem to form a little gang that hangs together.

Beta Decay

Beta particles are electrons or positrons (the antimatter counterpart of electrons). In beta decay, a neutron in the nucleus transforms into a proton, emitting an electron or a positron. Beta particles are much smaller than alpha particles and can penetrate materials more deeply.

$$n \rightarrow p + e^{-} + \bar{\nu}$$

$${}^A_Z X \rightarrow {}^A_{Z+1} X' + e^{-} + \bar{\nu}$$

$${}^A_Z X \rightarrow {}^A_{Z-1} X' + e^{+} + \nu$$

- Involves the **weak nuclear force**
- Neutron turns into a proton plus electron plus anti-neutrino
- Proton turns into a neutron plus positron plus neutrino

Gamma Decay

Gamma rays are high-energy photons. They are emitted when a nucleus transitions from a higher energy state to a lower one, releasing energy in the form of gamma radiation. Gamma rays have no mass and no charge, making them highly penetrating.

Gamma decay involves the emission of gamma photons when the nucleus changes energy levels. This is similar to how electrons in an atom emit photons when they transition between energy levels, but the energy involved in gamma decay is much higher, typically in the range of millions of electron volts (MeV).

In-Depth Look at Beta Decay

In beta decay, the emitted beta particles are actually electrons or positrons, which were not initially understood to be such. Today, we know that these particles are fundamental components of the process.

Beta decay is particularly important because it is involved in many natural processes, including the heat and light we receive from the sun.

How Beta Decay Works

In beta decay, a neutron transforms into a proton, an electron, and a third particle called a neutrino. Wolfgang Pauli, a German scientist, theorized the existence of the neutrino to account for the missing momentum and energy during beta decay. Neutrinos are neutral and interact very little with matter, making them difficult to detect.

The Role of the Weak Nuclear Force

Beta decay is governed by the weak nuclear force. This force is weaker and slower than the strong nuclear force but is crucial for many natural processes, including the reactions that power the sun. If the weak force and beta decay did not occur, there would be no heat or light from the sun, and life as we know it would not exist.

Example of Beta Decay

If a neutron is placed in a magnetic field, it can transform into a proton and an electron. The electron will move in one direction and the proton in the opposite direction. However, this transformation does not create a balanced momentum, indicating the presence of a third

particle, the neutrino. This process helps maintain the balance and transformation of particles within stars, leading to the production of energy.

Origin of Nuclei

The nuclei of elements found on Earth were not formed at the beginning of the universe. Initially, only hydrogen atoms existed. Over time, through processes occurring within stars, heavier elements like iron, lead, and aluminium were formed.

Formation of Elements in Stars

Inside stars, hydrogen atoms collide and fuse to form heavier elements. When stars explode in supernovae, they scatter these elements across the universe. Gravity then causes these elements to coalesce, forming planets and other celestial bodies. Our Earth and everything on it, including us, are composed of elements formed inside stars.

Summary

The lecture emphasizes that nuclear physics extends beyond weaponry, focusing on unravelling the atom's fundamental nature and natural phenomena. It begins with Ernest Rutherford's 1911 discovery of the atomic nucleus, overturning the previous "plum pudding" model and revealing protons and neutrons at the atom's core. Electrons were described as existing in probability clouds surrounding the nucleus. The study of isotopes, distinguished by neutron numbers, illustrates nuclear structure's complexity, probed using high-energy electron accelerators. The lecture delves into nuclear forces: the strong force binding protons and neutrons via pion exchange, and the short range dictated by quantum principles. It explores decay processes, crucial for dating organic remains like carbon-14 dating, which revealed the age of a 203-year-old specimen as 5300 years. Beta decay, involving electron or positron emission, and gamma decay, emitting high-energy photons, underscore quantum mechanics' role in nuclear processes. Finally, heavy element formation in stars underscores nuclear physics' cosmic implications, enriching our understanding of the universe's elemental composition and evolution.

Lecture 45

Physics of the Sun

Exploring the Physics of Our Sun

Today, we will delve into a fascinating and vital topic: the physics of our Sun. Our Sun, like other stars, is a colossal sphere of gas, but it holds a special place in our universe due to its proximity and crucial role in sustaining life on Earth. Understanding the Sun provides us insights not only about our own star but also about the billions of other stars scattered across the universe.

Basic Facts About the Sun

1. **Mass:** The Sun's mass is approximately 2×10^{30} kg, which is 333,000 times the mass of Earth. We determine the Sun's mass using Newton's Law of Gravitation. When we observe the gravitational interactions between the Sun, Earth, and other celestial bodies like the Moon or other planets, we can calculate the Sun's mass.
2. **Diameter:** The Sun's diameter is about 109 times that of Earth, approximately 1,3920,000 km.
3. **Age:** The Sun is around 4.6 billion years old. We determine this age through various scientific methods, including the study of solar and planetary formation.
4. **Rotation:** The Sun's rotation period varies between its equator and poles which is 25 days at equator, and 36 at poles (at surface). This difference is because the Sun is not a solid body but a sphere of gas. As a result, different parts of the Sun rotate at different speeds.
5. **Temperature:** At its core, the Sun reaches temperatures of about 15 million degrees Celsius. Its surface temperature, however, is much cooler at around 5770 degrees Celsius. We measure these temperatures using black body radiation principles, where the colour and energy spectrum of the Sun's light provide clues about its temperature.
6. **Density:** The Sun's density is about eight times that of gold in its core but averages to about 1.5 times that of water overall. This average density is calculated by dividing the Sun's mass by its volume.
7. **Composition:** The Sun is composed mainly of hydrogen (72%) and helium (25%), with the remaining 3% consisting of heavier elements like iron and carbon. We determine

the Sun's composition by analysing the spectrum of sunlight, which shows absorption lines corresponding to various elements.

8. **Luminosity:** The luminosity of the sun is 3.83×10^{27} joules per second. The Sun emits approximately 4×10^{24} kilowatts of power. To put this in perspective, it would take billions of power plants like the Tarbela Dam (which generates about 5000 MW) to match the Sun's output.

Source of the Sun's Energy

For a long time, scientists speculated about the source of the Sun's immense energy. Early theories suggested chemical reactions or gravitational contraction as possible sources. However, calculations showed that these processes could only sustain the Sun for a few million years, which contradicts the observed age of the Sun. The breakthrough came with Einstein's famous equation, $E=mc^2$, which revealed that mass could be converted into energy. This principle underlies the process of nuclear fusion occurring in the Sun's core, where hydrogen nuclei combine to form helium, releasing vast amounts of energy in the process.

Nuclear Fusion

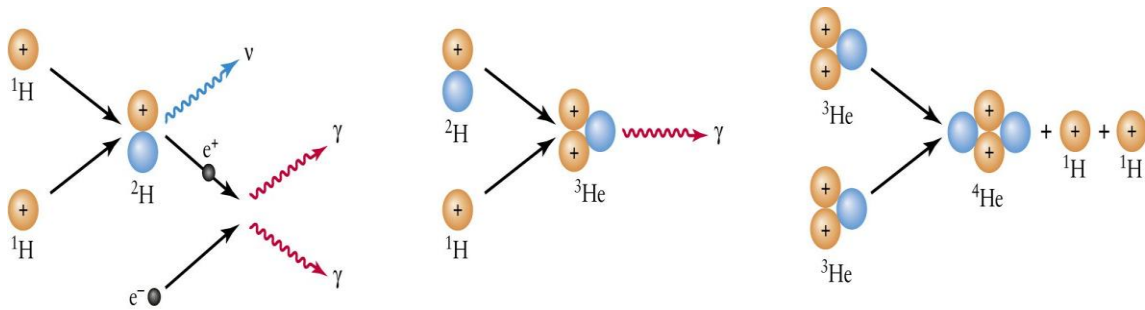
The Sun generates energy through a process called nuclear fusion. This process involves the fusion of hydrogen nuclei (protons) to form helium nuclei. However, it's essential to understand that the mass of four protons is slightly more than the mass of a helium nucleus, which consists of two protons and two neutrons. This difference in mass results from the nuclear binding energy that holds the nucleus together.

When four protons fuse to form a helium nucleus, energy is released in the form of photons, which we see as light. This process of fusion is what powers the Sun and enables it to emit vast amounts of energy.

Steps of the Fusion Process

1. **Initial Collision:** Two protons collide to form a deuterium nucleus (one proton and one neutron), a positron, and a neutrino as shown in Fig 45.1. The positron quickly encounters an electron, and they annihilate each other, producing gamma rays.
2. **Formation of Helium-3:** The deuterium nucleus collides with another proton, forming helium-3 (two protons and one neutron) and emitting a gamma ray.

3. **Creation of Helium-4:** Two helium-3 nuclei collide to form a helium-4 nucleus (two protons and two neutrons) and release two protons.



a Step 1:

- Two protons (hydrogen nuclei, ^1H) collide.
- One of the protons changes into a neutron (shown in blue), a neutral, nearly massless neutrino (ν), and a positron (e^+), a positively charged electron.
- The proton and neutron form a hydrogen isotope (^2H).
- The positron encounters an ordinary electron (e^-), annihilating both particles and converting them into gamma-ray photons (γ).

b Step 2:

- The ^2H nucleus from the first step collides with a third proton.
- A helium isotope (^3He) is formed and another gamma-ray photon is released.

c Step 3:

- Two ^3He nuclei collide.
- A different helium isotope with two protons and two neutrons (^4He) is formed and two protons are released.

Fig 45. 1: Steps of the fusion process: initial proton collision forming deuterium, formation of helium-3, and creation of helium-4, with gamma rays and protons released during the reactions.

This entire sequence releases energy because the mass of the resulting particles is less than the mass of the initial four protons. The difference in mass is converted into energy according to Einstein's equation, $E=mc^2$.

Nuclear Fusion in the Sun

- Stellar core temperatures are greater than the ~ 8 million K required for hydrogen fusion
- Amount of repulsion increases with more positive charge, so fusion of larger nuclei requires increasingly higher temperatures & densities (100 million K for helium fusion)
- Thus, stars fuse hydrogen first.
- Reaction efficiency ~ 0.008 . Sun can last ~ 10 billion years on core hydrogen fusion

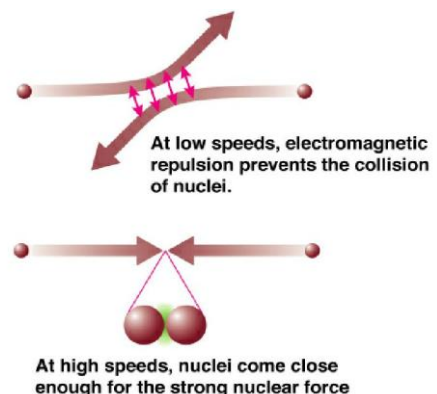


Fig 45. 2: The immense pressure and temperature at the Sun's core provide the necessary conditions for overcoming the electrostatic repulsion between nuclei, enabling fusion reactions.

Why Can't We Replicate Fusion on Earth?

Achieving nuclear fusion on Earth is challenging because it requires extremely high temperatures and pressures to overcome the repulsive force between protons. In the Sun, temperatures reach about 15 million degrees Celsius at the core, allowing fusion to occur. On Earth, creating and maintaining such conditions is currently beyond our technological capabilities.

Structure of the Sun

The Sun can be divided into three main regions:

1. **Core:** The core is the innermost part of the Sun as shown in Fig 45.3, where nuclear fusion occurs. The immense pressure and temperature in the core allow hydrogen nuclei to overcome their repulsive forces and fuse into helium, releasing energy.
2. **Radiation Zone:** Surrounding the core, the radiation zone is where energy from the core is transported outward by photons. However, this energy transfer is slow because photons constantly collide with particles, changing direction and taking a long time to reach the surface. It can take up to 170,000 years for a photon to travel from the core to the surface of the Sun.
3. **Convection Zone:** The outermost layer of the Sun's interior is the convection zone. Here, hot plasma rises to the surface, cools, and then sinks back down in a cyclic process called convection. This process helps transport energy from the inner layers to the Sun's surface.

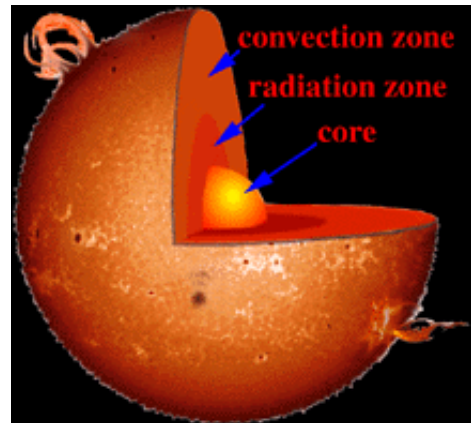


Fig 45. 3: Structure of the Sun's Interior.

Solar Granulation

Solar granulation refers to the pattern of small, bright cells, called granules, visible on the Sun's surface. These granules are caused by convection currents of hot plasma rising from the interior. Each granule is typically about 1,000 kilometres in diameter and lasts for several minutes before dissipating.

Surface and Atmosphere

The Sun's surface, known as the photosphere, as shown in Fig 45.4, has a temperature of about 5770 degrees Celsius. Above the photosphere is the corona, a region with much higher temperatures, reaching up to a million degrees Celsius. The corona is visible during a solar eclipse and is studied using spectroscopic analysis to understand its properties.



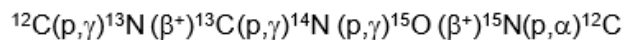
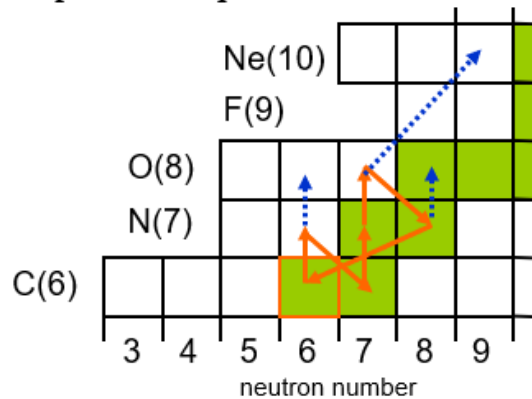
Fig 45. 4: Outermost layer of Sun – a low-density cloud of plasma; visible during solar eclipses (as above). Hotter than most of the inner layers. Average temperature 1 million K, but can reach 3 million K

The Sun's Energy Output

The Sun emits a tremendous amount of energy, converting about four million tons of hydrogen into helium every second. This fusion process releases 25 million electron volts of energy and produces neutrinos, which are particles that rarely interact with matter and escape the Sun quickly.

The CNO Cycle

In addition to the proton-proton chain reaction described above, the Sun also uses the CNO cycle (Carbon-Nitrogen-Oxygen cycle) to fuse hydrogen into helium. This cycle involves carbon, nitrogen, and oxygen as catalysts to speed up the fusion process. The presence of these elements is crucial for the CNO cycle to occur.

Example for a sequence of reactions in stars: the CN cycle

Net effect: $4p \rightarrow \alpha + 2e^+ + 2\nu_e$

But requires C or N as catalysts (second and later generation star)

Need reaction rates to:

- establish existence of cycle
- determine total rate of cycle

Fig 45. 5: A sequence of nuclear reactions occurring in stars, known as the Carbon-Nitrogen (CN) cycle. This cycle involves the fusion of hydrogen into helium with carbon, nitrogen, and oxygen acting as catalysts, playing a crucial role in energy production in stars more massive than the Sun.

The Sun generates energy through nuclear fusion, where hydrogen nuclei (protons) combine to form helium. This process releases a tremendous amount of energy in the form of light and heat. Each second, the sun turns 4 million tons of hydrogen into energy. Energy generation is from fusion is



Mass Comparison and Nuclear Binding

If you compare the mass of four protons with the mass of a helium nucleus, which consists of two protons and two neutrons, you'll find that four protons weigh more. This may seem strange because protons are lighter than neutrons. However, the mass of the helium nucleus is less due to the nuclear binding energy that holds it together. This binding energy is what's released during fusion.

Energy Output and the Sun's Lifespan

The Sun emits vast amounts of energy, converting approximately four million tons of hydrogen into helium every second. This fusion process releases 25 million electron volts of energy per reaction. Given the current rate of hydrogen consumption, the Sun can continue this process for about 10 billion years in total, with approximately 5 billion years remaining.

Reaction Efficiency

The reaction efficiency of the fusion process in the Sun is about 8%. This means that not all the hydrogen gets converted into energy; some mass is lost as energy in the form of photons.

Hydrostatic Equilibrium in the Sun

The Sun is in a state of equilibrium due to the balance between two opposing forces: gravity pulling inward and pressure pushing outward as shown in Fig 45.6. This balance prevents the Sun from collapsing under its gravity or exploding due to excessive internal pressure.

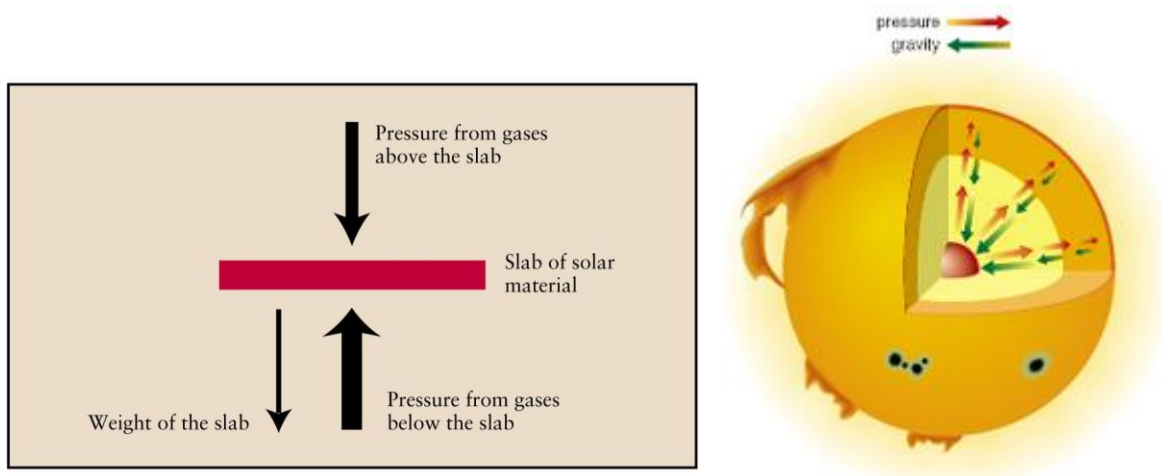


Fig 45. 6: The Sun maintains equilibrium through the balance of two opposing forces: gravity pulling inward and pressure pushing outward. This balance prevents the Sun from collapsing under its own gravity or exploding due to excessive internal pressure.

Calculating the Forces

To understand this equilibrium, imagine a thin shell within the Sun. The mass of this shell experiences gravitational pull inward and pressure from the gas pushing outward as shown in Fig 45.7. By calculating the forces acting on this shell, we can understand how the Sun maintains its structure.

1. **Gravitational Force:** The gravitational force acting on the shell is determined by the mass inside the shell and its distance from the center.
2. **Pressure Force:** The pressure from the gas pushes the shell outward, counteracting gravity.

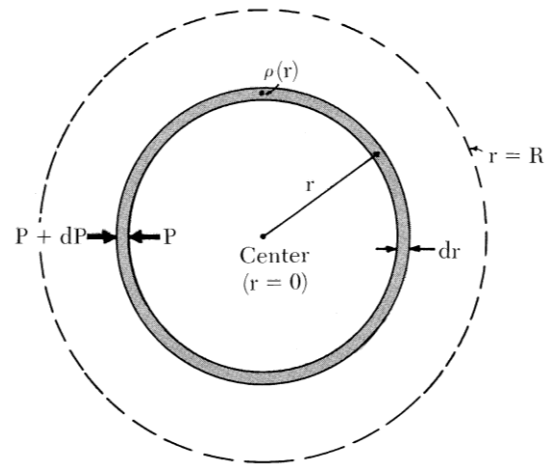


Fig 45. 7: A thin shell within the Sun experiences inward gravitational pull and outward gas pressure. Balancing these forces explains the Sun's stability, preventing expansion or collapse.

When these forces are balanced, the Sun remains stable. If the pressure were too high, the Sun would expand, and if gravity were too strong, it would collapse.

$$dF = -\frac{GM(r) \times \rho(r) 4\pi r^2 dr}{r^2}$$

$$dF = dP \times 4\pi r^2 dr$$

$$\therefore dP = -\frac{GM(r)\rho(r)dr}{r^2}$$

$$\text{or, } \frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

Similarly,

$$M(r) = \int_0^r \rho(r') 4\pi r'^2 dr'$$

Knowing $\rho(r)$ gives the mass distribution and pressure.

$$\text{Model: assume } \rho \approx \frac{M}{(4\pi/3)R^3} \approx 1.4 \text{ gm cm}^{-3}$$

$$P_{\text{centre}} \approx \frac{GM\rho}{R} \approx 3 \times 10^9 \text{ atmospheres}$$

The Sun's Core and Neutrinos

Photons generated in the core of the Sun take about 170,000 years to reach the surface due to constant collisions with particles. However, neutrinos, which are also produced in the fusion process, can escape from the core to the surface in just a few seconds. These neutrinos provide valuable information about the Sun's core.

Detecting Neutrinos

Scientists have developed methods to detect neutrinos and measure their energy, which helps us understand the conditions inside the Sun. Neutrinos interact very weakly with matter, making them difficult to detect, but advancements in technology have made it possible.

Helioseismology: The Study of Solar Waves

Helioseismology is the science of studying waves on the Sun. This field emerged in 1965 when scientists noticed that different parts of the Sun's surface move at different speeds. By 1975, the first concepts of helioseismology were developed, and today, it's a well-established science. Another way to study the Sun's interior is through helioseismology, the science of studying sound waves traveling through the Sun. The Sun's surface moves up and down, similar to how the Earth's surface moves during an earthquake. By analysing these movements, scientists can infer details about the Sun's internal structure.

The Bouncing Sun

It might surprise you to learn that the surface of the Sun bounces up and down, much like the surface of the Earth during a flood. Obviously, we can't place a meter on the Sun to measure this movement directly. Instead, we use a technique called the Doppler effect to study these changes.

The Doppler Effect

The Doppler effect helps us observe how the Sun's surface moves. When the Sun's surface moves upwards, the frequency of the light (or photons) it emits increases. When the surface moves downwards, the frequency decreases. By analysing these frequency changes, we can map the heat waves on the Sun's surface and understand their intensity.

The Importance of Solar Energy

The Sun emits both heat and light, which travel a long distance to reach us on Earth. Without this energy, life on Earth would be impossible. The Sun acts as a black body, which means it absorbs and emits radiation efficiently.

Black Body Radiation

Black body radiation occurs when the atoms and molecules inside a heated body move, creating electromagnetic waves. These waves are what we perceive as radiation. This process can be described by the Stefan-Boltzmann Law, which states that the power radiated by a black body is proportional to the fourth power of its temperature.

Stefan-Boltzmann Law

The Stefan-Boltzmann Law helps us calculate the radiation emitted by an object. For example, a sphere with a diameter of 10 centimetres at 20 degrees Celsius (or 293 Kelvin) would radiate 418 watts per square meter. If you include the area, the output is around 13 watts. Similarly, your body radiates about 100 watts of energy due to radiation.

Thermal Radiation from a Solid

Calculate the power radiated by a 10-cm-diameter sphere of metal at room temperature (20°C).

$$\begin{aligned}P &= \sigma T^4 && \text{(power radiated per area)} \\&= (5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4})(293\text{K})^4 \\&= 418 \text{ W/m}^2 \\ \text{Area: } A &= 4\pi r^2 = 4\pi(5 \times 10^{-2} \text{ m})^2 = 3.14 \times 10^{-2} \text{ m}^2 \\ \text{Power radiated} &= P \times A = (418 \text{ W/m}^2)(3.14 \times 10^{-2} \text{ m}^2) \\&= 13 \text{ Watts}\end{aligned}$$

Radiation from the Sun and the Earth

Just like the Sun, the Earth also radiates energy. We receive energy from the Sun, and the Earth emits this energy back into space. The balance between these incoming and outgoing energies determines the Earth's temperature.

Surface Temperature of the Earth

To understand the balance of energy, we define three quantities:

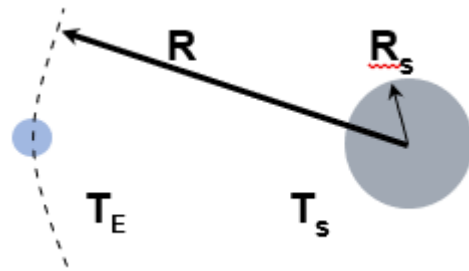
1. **Solar Flux at the Sun's Surface (P_s):** This is the energy emitted by the Sun.
2. **Solar Flux at the Earth (P_R):** This is the energy received by the Earth from the Sun.

3. **Ground Flux:** This is the energy emitted by the Earth.

$$P_s = \text{Sun's flux at its surface} = \sigma T_s^4$$

$$P_R = \text{Sun's flux at earth} = \sigma T^4 \times \left(\frac{R_s}{R}\right)^2$$

$$J_E = \text{Earth's flux at its surface} = \sigma T_E^4$$



Power absorbed from sun = Power radiated by earth

$$P_R \times \pi R_E^2 = P_E \times 4\pi R_E^2$$

$$P_R = 4P_E$$

$$\sigma T_s^4 \times \left(\frac{R_s}{R}\right)^2 = 4\sigma T_E^4$$

$$\therefore R_s = 7 \times 10^8 \text{ metres}$$

$$R = 1.5 \times 10^{11} \text{ metres}$$

$$T_s = 5800^\circ \text{ K}$$

$$T_E = \left(\frac{R_s}{2R}\right)^{1/2} T_s = 280 \text{ K}$$

Fig 45. 8: the star's radius (R_s), the distance from the star to the planet R , and the effective temperatures of the star (T_s) and the planet (T_E). These relationships help in understanding the equilibrium temperature of a planet based on its distance from the star and the star's properties.

Using these quantities, we can calculate the Earth's equilibrium temperature. The energy received from the Sun must equal the energy emitted by the Earth for the temperature to remain stable. If there is a difference, the Earth's temperature will change.

The Average Temperature of Earth

By calculating the solar flux and the ground flux, we find that the Earth's equilibrium temperature is around 280 degrees Kelvin (or 7 degrees Celsius). However, due to the reflection of sunlight and other factors, the actual average surface temperature of the Earth is around 290 degrees Kelvin (or 13 degrees Celsius). About 30% of the sun's radiation reflects off our atmosphere.

The Greenhouse Effect and Global Warming

The temperature of the Earth is crucial because it relates to global warming, a major issue today. When the Sun's heat reaches the Earth, some of it is trapped by greenhouse gases, like

carbon dioxide, in the atmosphere. This trapping of heat raises the Earth's temperature.

Greenhouse Gases and Their Impact

Carbon dioxide is a significant greenhouse gas. It has always been present in the atmosphere, but its concentration has increased due to human activities like burning fossil fuels. These increased levels of carbon dioxide make it harder for heat to escape from the Earth's surface, leading to a rise in temperature.

Molecular Vibrations and Quantum Mechanics

Each molecule, including carbon dioxide, has its own vibrations. In quantum mechanics, these vibrations correspond to specific energy levels. A molecule can only exist in these discrete energy levels as shown in Fig 45.9, much like how you can only sit or stand and not hover halfway in between. These vibrational levels are crucial in understanding how greenhouse gases interact with radiation.

Energy levels of an oscillator:

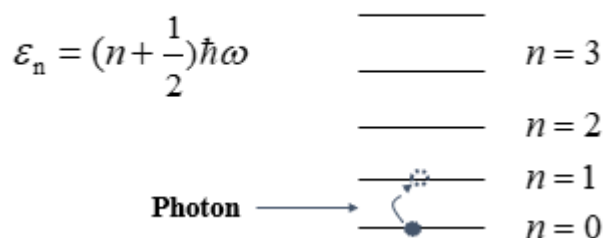


Fig 45. 9: Each molecule, including carbon dioxide, has specific vibrations corresponding to discrete energy levels in quantum mechanics. These levels are essential for understanding how greenhouse gases interact with radiation.

The Greenhouse Effect

When radiation from the Sun reaches the Earth, it tries to escape back into space. However, gases like carbon dioxide and methane in the atmosphere absorb this radiation. This absorption causes these molecules to vibrate, trapping heat and warming the Earth. These gases are called greenhouse gases because their effect is similar to the way a greenhouse traps heat.

- The main greenhouse gases are CO₂, CH₄, H₂O. (N₂, O₂ and Ar are transparent to solar and earth radiation.)
- CO₂ is now 360 ppm of the atmosphere, up from 227 ppm in 1750, before the industrial revolution. Plants turn H₂O + CO₂ into O₂ plus organics.
- Some estimates predict a 3 to 10 °C rise in the earth's surface temperature over the next 100 years due to the increased CO₂ greenhouse effect.

Methane and Carbon Dioxide

Methane (CH₄) is another significant greenhouse gas. It, like carbon dioxide, has molecular vibrations that fall within the same frequency range as the radiation trying to escape Earth. This means that methane also traps heat, contributing to global warming.

The Impact of Greenhouse Gases

The trapped heat causes the Earth's temperature to rise. Estimates suggest that the Earth's temperature could increase by 5 to 10 degrees Celsius in the next 50-100 years if we do not take action to reduce greenhouse gas emissions. This temperature rise could have severe consequences, such as melting ice caps and rising sea levels. To protect our planet, we must control emissions of carbon dioxide and methane.

The Future of the Sun

The Sun will not last forever. It consumes hydrogen through nuclear fusion to produce energy. Currently, this fusion process occurs in the Sun's core, where temperatures are about 8 million degrees Kelvin. However, in about five billion years, the Sun will have exhausted most of its hydrogen.

The Helium Phase

As hydrogen runs out, the Sun will begin to fuse helium. This process requires even higher temperatures and pressures. The fusion of helium will give the Sun a new lease on life, but it will also cause significant changes. The Sun will expand into a Red Giant, becoming 100 times its current size. This expansion will likely engulf nearby planets, including Earth.

The Red Giant Phase

In the Red Giant phase, the Sun's core will be primarily helium, and the outer layers will expand dramatically. This phase will last until the helium in the core is exhausted. As the core contracts and heats up, it will eventually ignite helium fusion, producing carbon and oxygen.

The Supergiant Phase and White Dwarf

After the helium is depleted, the Sun will enter the Supergiant phase. It will expand even more, up to 300 times its current size. Eventually, the Sun will shed its outer layers, leaving behind a hot core called a white dwarf. This white dwarf will be incredibly dense; a small amount of its material would weigh as much as the Earth. Over time, the white dwarf will cool and fade.

The Ultimate Fate of Stars

Not all stars end as white dwarfs. Massive stars, much heavier than our Sun, can end in more dramatic ways. When their fuel is exhausted, they collapse under their own gravity, creating a black hole. A black hole has such strong gravitational pull that not even light can escape from it.

Understanding the Sun with Physics

Studying the Sun requires knowledge from various branches of physics:

- **Classical Mechanics:** Used to understand the balance of forces within the Sun.
- **Thermodynamics:** Helps us understand the heat transfer and energy production in the Sun.
- **Statistical Mechanics:** Describes the behavior of particles in the Sun's hot interior.
- **Nuclear Physics and Quantum Mechanics:** Explain the fusion reactions that power the Sun.

Applying Physics to the Sun

By applying these branches of physics, we can understand the Sun's structure, energy production, and eventual fate. For example, Newton's laws explain the gravitational forces, while thermodynamics and statistical mechanics describe the movement of particles and energy distribution.

